

Catalysis **10** and Catalytic Reactors

It isn't that they can't see the solution. It is that they can't see the problem.

—G. K. Chesterton

Overview. The objectives of this chapter are to develop an understanding of catalysts, reaction mechanisms, and catalytic reactor design. Specifically, after reading this chapter one should be able to

- Define a catalyst and describe its properties.
- Describe the steps in a catalytic reaction and in chemical vapor deposition (CVD).
- Suggest a mechanism and apply the concept of a rate-limiting step to derive a rate law.
- Use nonlinear regression to determine the rate law and rate-law parameters that best fit the data.
- Use the rate law parameters to design PBRs and fluidized CSTRs.
- Describe the analogy between catalytic steps and CVD in microelectronic fabrication.
- Analyze catalyst decay and reactors that can be used to help offset decay.

The various sections of this chapter roughly correspond to these objectives.

10.1 Catalysts

Catalysts have been used by humankind for over 2000 years.¹ The first observed uses of catalysts were in the making of wine, cheese, and bread. It was found that it was always necessary to add small amounts of the previous

¹ S. T. Oyama and G. A. Somorjai, *J. Chem. Educ.*, 65, 765 (1986).

batch to make the current batch. However, it wasn't until 1835 that Berzelius began to tie together observations of earlier chemists by suggesting that small amounts of a foreign substance could greatly affect the course of chemical reactions. This mysterious force attributed to the substance was called *catalytic*. In 1894, Ostwald expanded Berzelius's explanation by stating that catalysts were substances that accelerate the rate of chemical reactions without being consumed during the reaction. During the 180 years since Berzelius's work, catalysts have come to play a major economic role in the world market. In the United States alone, sales of process catalysts will reach over \$20 billion by 2018, the major uses being in petroleum refining and in chemical production.

10.1.1 Definitions

A *catalyst* is a substance that affects the rate of a reaction but emerges from the process unchanged. A catalyst usually changes a reaction rate by promoting a different molecular path ("mechanism") for the reaction. For example, gaseous hydrogen and oxygen are virtually inert at room temperature, but react rapidly when exposed to platinum. The reaction coordinate (cf. Chapter 3) shown in Figure 10-1 is a measure of the progress along the reaction path as H_2 and O_2 approach each other and pass over the activation energy barrier to form H_2O . *Catalysis is the occurrence, study, and use of catalysts and catalytic processes.* Commercial chemical catalysts are immensely important. Approximately one-third of the material gross national product of the United States involves a catalytic process somewhere between raw material and finished product.² The development and use of catalysts is a major part of the constant search for new ways of increasing product yield and selectivity from chemical reactions. Because a catalyst makes it possible to obtain an end product by a different pathway with a lower energy barrier, it can affect both the yield and the selectivity.

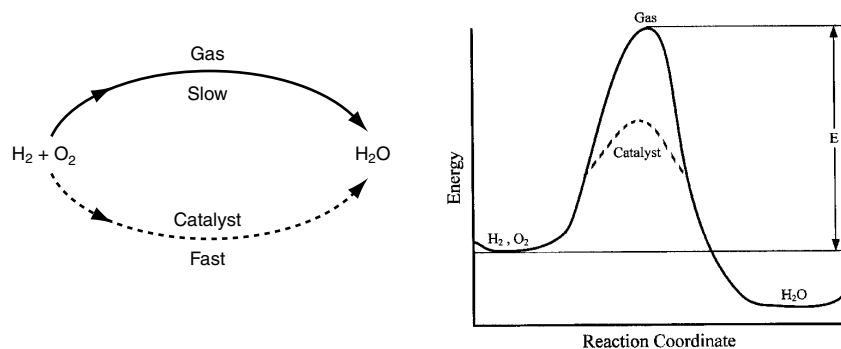


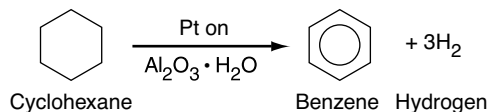
Figure 10-1 Different reaction paths.

Catalysts can accelerate the reaction rate but cannot change the equilibrium.

Normally when we talk about a catalyst, we mean one that speeds up a reaction, although strictly speaking, a catalyst can either accelerate or slow the formation of a particular product species. *A catalyst changes only the rate of a reaction; it does not affect the equilibrium.*

² V. Haensel and R. L. Burwell, Jr., *Sci. Am.*, 225(10), 46.

The 2007 Nobel Prize for Chemistry was awarded to Gerhard Ertl for his pioneering work on heterogeneous catalytic reactions. A *heterogeneous catalytic reaction* involves more than one phase; usually the catalyst is a solid and the reactants and products are in liquid or gaseous form. One example is the production of benzene, which is mostly manufactured today from the dehydrogenation of cyclohexane (obtained from the distillation of petroleum crude oil) using platinum-on-alumina as the catalyst:



The simple and complete separation of the fluid product mixture from the solid catalyst makes heterogeneous catalysis economically attractive, especially because many catalysts are quite valuable and their reuse is demanded.

A heterogeneous catalytic reaction occurs at or very near the fluid–solid interface. The principles that govern heterogeneous catalytic reactions can be applied to both catalytic and noncatalytic fluid–solid reactions. The two other types of heterogeneous reactions involve gas–liquid and gas–liquid–solid systems. Reactions between gases and liquids are usually mass-transfer limited.

10.1.2 Catalyst Properties

Ten grams of this catalyst possess more surface area than a U.S. football field

Catalyst types:

- Porous
- Molecular sieves
- Monolithic
- Supported
- Unsupported

Typical zeolite catalyst

High selectivity to para-xylene

Because a catalytic reaction occurs at the fluid–solid interface, a large interfacial area is almost always essential in attaining a significant reaction rate. In many catalysts, this area is provided by an inner porous structure (i.e., the solid contains many fine pores, and the surface of these pores supplies the area needed for the high rate of reaction), see Figures 10-4(b) and 10-9. The area possessed by some porous catalysis materials is surprisingly large. A typical silica-alumina cracking catalyst has a pore volume of 0.6 cm³/g and an average pore radius of 4 nm. The corresponding surface area is 300 m²/g of these *porous catalysts*. Examples include the Raney nickel used in the hydrogenation of vegetable and animal oils, platinum-on-alumina used in the reforming of petroleum naphthas to obtain higher octane ratings, and promoted iron used in ammonia synthesis. Sometimes pores are so small that they will admit small molecules but prevent large ones from entering. Materials with this type of pore are called *molecular sieves*, and they may be derived from natural substances such as certain clays and zeolites, or they may be totally synthetic, such as some crystalline aluminosilicates (see Figure 10-2). These sieves can form the basis for quite selective catalysts; the pores can control the residence time of various molecules near the catalytically active surface to a degree that essentially allows *only* the desired molecules to react. One example of the high selectivity of zeolite catalysts is the formation of para-xylene from toluene and methane shown in Figure 10-2(b).³ Here, benzene and toluene enter through the zeolite pore and react on the interior surface to form a mixture of ortho-, meta-, and para-xylenes. However, the size of the pore mouth is such that only

³ R. I. Masel, *Chemical Kinetics and Catalysis* (New York: Wiley Interscience, 2001), p. 741.

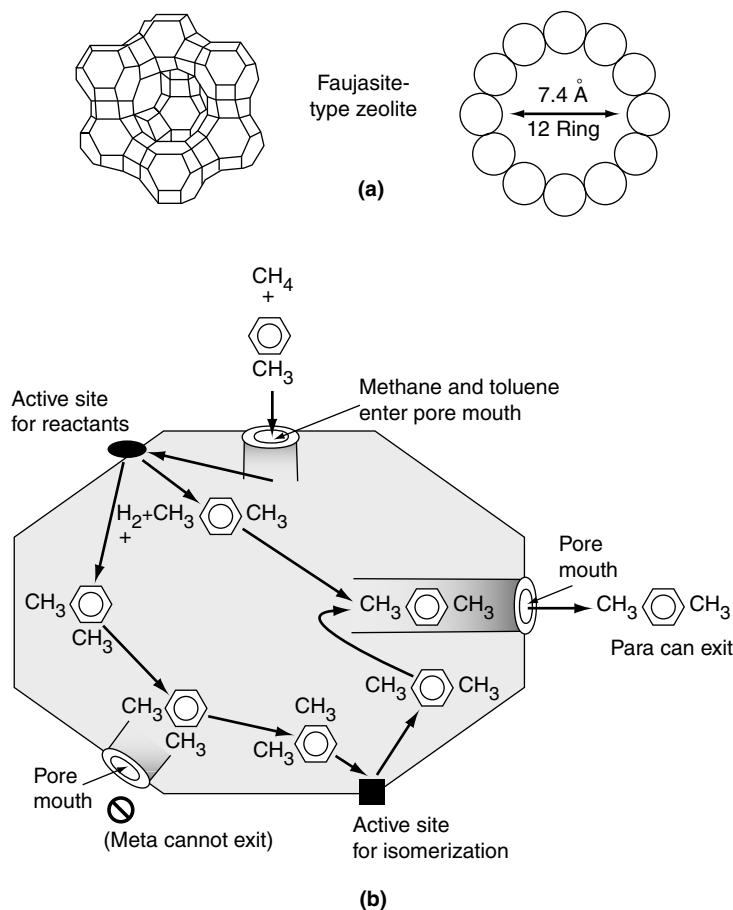


Figure 10-2 (a) Framework structures and (b) pore cross sections of two types of zeolites. (a) Faujasite-type zeolite has a three-dimensional channel system with pores at least 7.4 Å in diameter. A pore is formed by 12 oxygen atoms in a ring. (b) Schematic of reaction CH_4 and $\text{C}_6\text{H}_5\text{CH}_3$. (Note that the size of the pore mouth and the interior of the zeolite are not to scale.) [(a) from N. Y. Chen and T. F. Degnan, *Chem. Eng. Prog.*, 84(2), 33 (1988). Reproduced by permission of the American Institute of Chemical Engineers. Copyright © 1988 AIChE. All rights reserved.]

para-xylene can exit through the pore mouth, as meta- and ortho- xylene with their methyl group on the side cannot fit through the pore mouth. There are interior sites that can isomerize ortho-xylene and meta-xylene to para-xylene. Hence, we have a very high selectivity to form para-xylene.

In some cases a catalyst consists of minute particles of an active material dispersed over a less-active substance called a *support*. The active material is frequently a pure metal or metal alloy. Such catalysts are called *supported catalysts*, as distinguished from *unsupported catalysts*. Catalysts can also have small amounts of active ingredients added called *promoters*, which increase their activity. Examples of supported catalysts are the packed-bed catalytic converter in an automobile, the platinum-on-alumina catalyst used in petroleum reforming, and the vanadium pentoxide on silica used to oxidize sulfur dioxide in manufacturing sulfuric acid. On the other hand, platinum gauze for ammonia oxidation, promoted iron for ammonia synthesis, and the silica–alumina dehydrogenation catalyst used in butadiene manufacture typify unsupported catalysts.

10.1.3 Catalytic Gas-Solid Interactions

For the moment, let us focus our attention on gas-phase reactions catalyzed by solid surfaces. For a catalytic reaction to occur, at least one and frequently all of the reactants must become attached to the surface. This attachment is known as *adsorption* and takes place by two different processes: physical adsorption and chemisorption. *Physical adsorption* is similar to condensation. The process is exothermic, and the heat of adsorption is relatively small, being on the order of 1 to 15 kcal/mol. The forces of attraction between the gas molecules and the solid surface are weak. These van der Waals forces consist of interaction between permanent dipoles, between a permanent dipole and an induced dipole, and/or between neutral atoms and molecules. The amount of gas physically adsorbed decreases rapidly with increasing temperature, and above its critical temperature only very small amounts of a substance are physically adsorbed.

The type of adsorption that affects the rate of a chemical reaction is *chemisorption*. Here, the adsorbed atoms or molecules are held to the surface by valence forces of the same type as those that occur between bonded atoms in molecules. As a result, the electronic structure of the chemisorbed molecule is perturbed significantly, causing it to be extremely reactive. Interaction with the catalyst causes bonds of the adsorbed reactant to be stretched, making them easier to break.

Figure 10-3 shows the bonding from the adsorption of ethylene on a platinum surface to form chemisorbed ethylidyne. Like physical adsorption, chemisorption is an exothermic process, but the heats of adsorption are

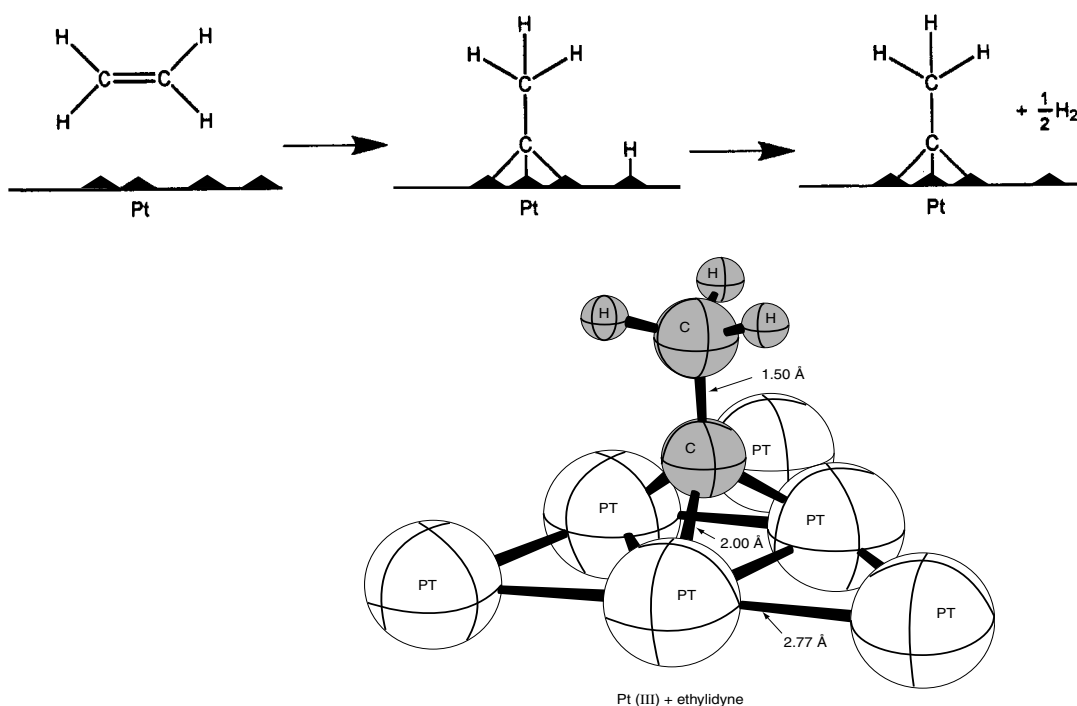


Figure 10-3 Ethylidyne chemisorbed on platinum. (Adapted from G. A. Somorjai, *Introduction to Surface Chemistry and Catalysis*. © 1994 John Wiley & Sons, Inc. Reprinted by permission of John Wiley & Sons, Inc. All rights reserved.)

Chemisorption on active sites is what catalyzes the reaction.

generally of the same magnitude as the heat of a chemical reaction (i.e., 40 to 400 kJ/mol). If a catalytic reaction involves chemisorption, it must be carried out within the temperature range where chemisorption of the reactants is appreciable.

In a landmark contribution to catalytic theory, Taylor suggested that a reaction is not catalyzed over the entire solid surface but only at certain *active sites* or *centers*.⁴ He visualized these sites as unsaturated atoms in the solids that resulted from surface irregularities, dislocations, edges of crystals, and cracks along grain boundaries. Other investigators have taken exception to this definition, pointing out that other properties of the solid surface are also important. The active sites can also be thought of as places where highly reactive intermediates (i.e., chemisorbed species) are stabilized long enough to react. This stabilization of a reactive intermediate is key in the design of any catalyst. Consequently, for our purposes we will define an *active site* as a *point on the catalyst surface that can form strong chemical bonds with an adsorbed atom or molecule*.

One parameter used to quantify the activity of a catalyst is the *turnover frequency (TOF)*, f . It is the number of molecules reacting per active site per second at the conditions of the experiment. When a metal catalyst such as platinum is deposited on a support, the metal atoms are considered active sites. The TOFs for a number of reactions are shown in *Professional Reference Shelf R10.1*. The *dispersion*, D , of the catalyst is the fraction of the metal atoms deposited on a catalyst that are *on* the surface.

10.1.4 Classification of Catalysts

One common way to classify catalysts is in terms of the type of reaction they catalyze.

Table 10-1 gives a list of representative reactions and their corresponding catalysts. Further discussion of each of these reaction classes and the materials that catalyze them can be found on the CRE Web site's *Professional Reference Shelf R10.1*.

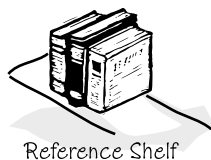


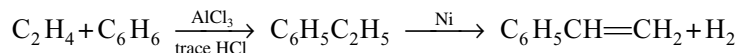
TABLE 10-1 TYPES OF REACTIONS AND REPRESENTATIVE CATALYSTS

<i>Reaction</i>	<i>Catalysts</i>
1. Halogenation–dehalogenation	CuCl ₂ , AgCl, Pd
2. Hydration–dehydration	Al ₂ O ₃ , MgO
3. Alkylation–dealkylation	AlCl ₃ , Pd, Zeolites
4. Hydrogenation–dehydrogenation	Co, Pt, Cr ₂ O ₃ , Ni
5. Oxidation	Cu, Ag, Ni, V ₂ O ₅
6. Isomerization	AlCl ₃ , Pt/Al ₂ O ₃ , Zeolites

If, for example, we were to form styrene from an equimolar mixture of ethylene and benzene, we could first carry out an alkylation reaction to form

⁴ H. S. Taylor, *Proc. R. Soc. London*, A108, 105 (1928).

ethyl benzene, which is then dehydrogenated to form styrene. We need both an alkylation catalyst and a dehydrogenation catalyst:



10.2 Steps in a Catalytic Reaction

A photograph of different types and sizes of catalysts is shown in Figure 10-4a. A schematic diagram of a tubular reactor packed with catalytic pellets is shown in Figure 10-4b. The overall process by which heterogeneous catalytic reactions proceed can be broken down into the sequence of individual steps shown in Table 10-2 and pictured in Figure 10-5 for an isomerization reaction.



Figure 10-4a Catalyst particles of different shapes (spheres, cylinders) and sizes (0.1 cm to 1 cm). (Photo courtesy of the Engelhard Corporation.)

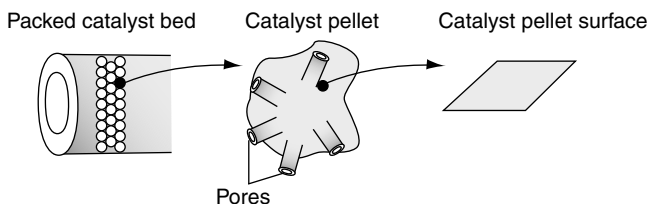


Figure 10-4b Catalytic packed-bed reactor—schematic.

Each step in Table 10-2 is shown schematically in Figure 10-5.

A reaction takes place *on* the surface, but the species involved in the reaction must get *to* and *from* the surface.

The overall rate of reaction is limited by the rate of the slowest step in the sequence. When the diffusion steps (1, 2, 6, and 7 in Table 10-2) are very fast compared with the surface reaction-rate steps (3, 4, and 5), the concentrations in the immediate vicinity of the active sites are indistinguishable from those in the bulk fluid. In this situation, the transport or diffusion steps do not affect the overall rate of the reaction. In other situations, if the reaction steps are very fast compared with the diffusion steps, mass transport does affect the reaction rate. In systems where diffusion from the bulk gas or liquid to the external catalyst surface or to the mouths of catalyst pores affects the rate, i.e., steps 1 and 7, changing the flow conditions past the catalyst should change the overall reaction rate (see Chapter 14). Once inside the porous catalysts, on the other hand, diffusion within the catalyst pores, i.e., steps 2 and 6, may limit the rate of reaction and, as a result, the overall rate will be unaffected by external flow conditions even though diffusion affects the overall reaction rate (see Chapter 15).

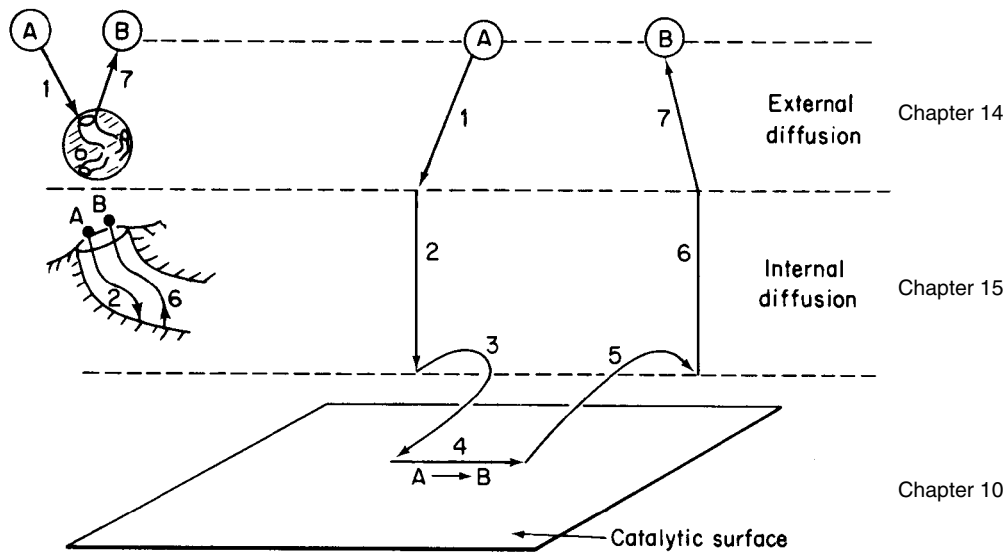


Figure 10-5 Steps in a heterogeneous catalytic reaction.

TABLE 10-2 STEPS IN A CATALYTIC REACTION

1. Mass transfer (diffusion) of the reactant(s) (e.g., species A) from the bulk fluid to the external surface of the catalyst pellet
2. Diffusion of the reactant from the pore mouth through the catalyst pores to the immediate vicinity of the internal catalytic surface
3. Adsorption of reactant A onto the catalyst surface
4. Reaction on the surface of the catalyst (e.g., $A \longrightarrow B$)
5. Desorption of the products (e.g., B) from the surface
6. Diffusion of the products from the interior of the pellet to the pore mouth at the external surface
7. Mass transfer of the products from the external pellet surface to the bulk fluid

There are many variations of the situation described in Table 10-2. Sometimes, of course, two reactants are necessary for a reaction to occur, and both of these may undergo the steps listed above. Other reactions between two substances may have only one of them adsorbed.

With this introduction, we are ready to treat individually the steps involved in catalytic reactions. In this chapter, only steps 3, 4, and 5—i.e., adsorption, surface reaction, and desorption—are considered as we assume that the diffusion steps (1, 2, 6, and 7) are very fast so that the overall reaction rate is not affected by mass transfer in any fashion. Further treatment of the effects involving diffusion limitations is provided in Chapters 14 and 15.

Where Are We Heading?[†] As we saw in Chapter 7, one of the tasks of a chemical reaction engineer is to analyze rate data and to develop a rate law that can be used in reactor design. Rate laws in heterogeneous catalysis seldom

In this chapter we focus on:
 3. Adsorption
 4. Surface reaction
 5. Desorption

[†] “If you don’t know where you are going, you’ll probably wind up someplace else.”
 Yogi Berra

An algorithm

follow power-law models and hence are inherently more difficult to formulate from the data. To develop an in-depth understanding and insight as to how the rate laws are formed from heterogeneous catalytic data, we are going to proceed in somewhat of a reverse manner than what is normally done in industry when one is asked to develop a rate law. That is, we will postulate catalytic mechanisms and *then* derive rate laws for the various mechanisms. The mechanism will typically have an adsorption step, a surface reaction step, and a desorption step, one of which is usually rate-limiting. Suggesting mechanisms and rate-limiting steps is not the first thing we normally do when presented with data. However, by deriving equations for different mechanisms, we will observe the various forms of the rate law one can have in heterogeneous catalysis. Knowing the different forms that catalytic rate equations can take, it will be easier to view the trends in the data and deduce the appropriate rate law. This deduction is usually what is done first in industry before a mechanism is proposed. Knowing the form of the rate law, one can then numerically evaluate the rate-law parameters and postulate a reaction mechanism and rate-limiting step that are consistent with the rate data. Finally, we use the rate law to design catalytic reactors. This procedure is shown in Figure 10-6. The dashed lines represent feedback to obtain new data in specific regions (e.g., concentrations, temperature) to evaluate the rate-law parameters more precisely or to differentiate between reaction mechanisms.

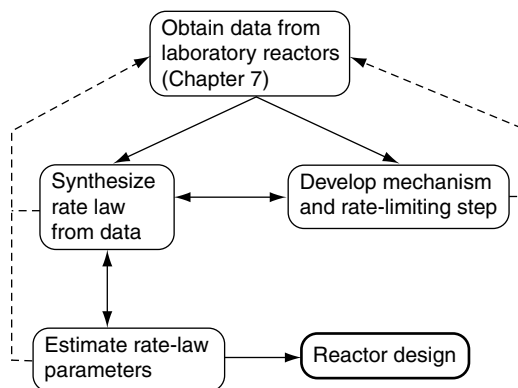


Figure 10-6 Collecting information for catalytic reactor design.

Chance Card:
Do not pass go
proceed directly to
Section 10.2.3.



We will discuss each of the steps shown in Figure 10-5 and Table 10-2. As mentioned earlier, this chapter focuses on Steps 3, 4, and 5 (the adsorption, surface reaction and desorption steps) by assuming that Steps 1, 2, 6, and 7 are very rapid. Consequently, to understand when this assumption is valid, we shall give a quick overview of Steps 1, 2, 6, and 7. These steps involve diffusion of the reactants to and within the catalyst pellet. While these diffusion steps are covered in detail in Chapters 14 and 15, it is worthwhile to give a brief description of these two mass-transfer steps to better understand the entire sequence of steps. If you have had the core course in *Mass Transfer* or *Transport Phenomena* you can skip Sections 10.2.1 and 10.2.2, and go directly to Section 10.2.3.

10.2.1 Step 1 Overview: Diffusion from the Bulk to the External Surface of the Catalyst

External and internal mass transfer in catalysis are covered in detail in Chapters 14 and 15.

For the moment let's assume that the transport of A from the bulk fluid to the external surface of the catalyst is the slowest step in the sequence. We lump all the resistance to transfer from the bulk fluid to the surface in the mass-transfer boundary layer surrounding the pellet. In this step the reactant A, which is at a bulk concentration C_{Ab} must travel (diffuse) through the boundary layer of thickness δ to the external surface of the pellet where the concentration is C_{As} , as shown in Figure 10-7. The rate of transfer (and hence rate of reaction, $-r'_A$) for this slowest step is

$$\text{Rate} = k_C (C_{Ab} - C_{As})$$

where the mass-transfer coefficient, k_C , is a function of the hydrodynamic conditions, namely the fluid velocity, U , and the particle diameter, D_p .

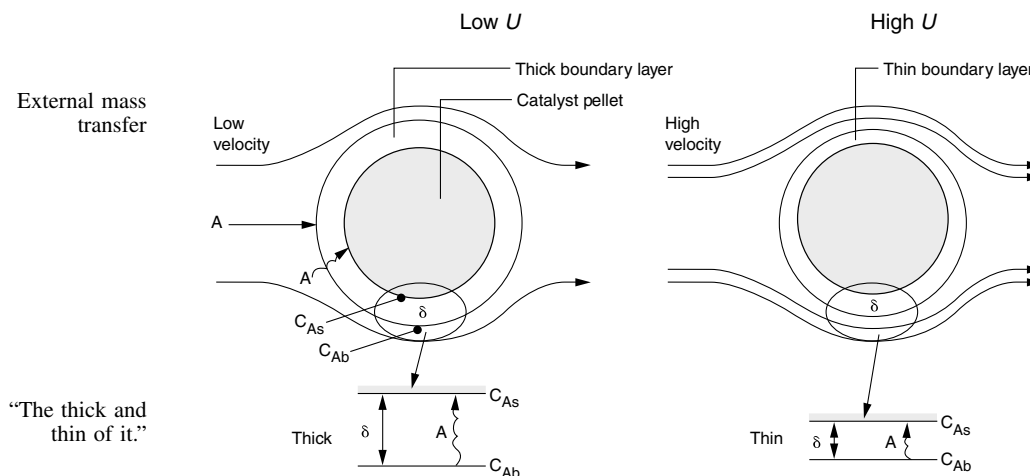


Figure 10-7 Diffusion through the external boundary layer (also see Figure 14-3).

As we see (Chapter 14), the mass-transfer coefficient is inversely proportional to the boundary layer thickness, δ , and directly proportional to the diffusion coefficient (i.e., the diffusivity D_{AB}).

$$k_C = \frac{D_{AB}}{\delta}$$

At low velocities of fluid flow over the pellet, the boundary layer across which A and B must diffuse is thick, and it takes a long time for A to travel to the surface, resulting in a small mass-transfer coefficient k_C . As a result, mass transfer across the boundary layer is slow and limits the rate of the overall reaction. As the velocity over the pellet is increased, the boundary layer becomes thinner and the mass-transfer rate is increased. At very high velocities, the boundary layer thickness, δ , is so small it no longer offers any resistance to the diffusion across

the boundary layer. As a result, external mass transfer no longer limits the rate of reaction. This external resistance also decreases as the particle size is decreased. As the fluid velocity increases and/or the particle diameter decreases, the mass-transfer coefficient increases until a plateau is reached, as shown in Figure 10-8. On this plateau, $C_{Ab} \approx C_{As}$, and one of the other steps in the sequence, i.e., Steps 2 through 6, is the slowest step and limits the overall reaction rate. Further details on external mass transfer are discussed in Chapter 14.

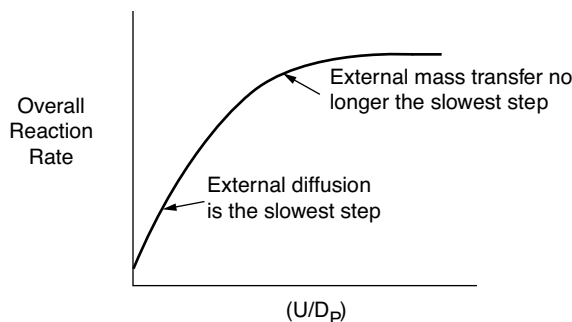


Figure 10-8 Effect of particle size and fluid velocity on the overall reaction rate.

10.2.2 Step 2 Overview: Internal Diffusion

Now consider that we are operating at a fluid velocity where external diffusion is no longer the rate-limiting step and that internal diffusion is the slowest step. In Step 2 the reactant A diffuses from the external pellet surface at a concentration C_{As} into the pellet interior, where the concentration is C_A . As A diffuses into the interior of the pellet, it reacts with catalyst deposited on the sides of the catalyst pellet's pore walls.

For large pellets, it takes a long time for the reactant A to diffuse into the interior, compared to the time that it takes for the reaction to occur on the interior pore surface. Under these circumstances, the reactant is only consumed near the exterior surface of the pellet and the catalyst near the center of the pellet is wasted catalyst. On the other hand, for very small pellets it takes very little time to diffuse into and out of the pellet interior and, as a result, internal diffusion no longer limits the rate of reaction. When internal mass transfer no longer limits the rate of reaction, the rate law can be expressed as

$$\text{Rate} = k_r C_{As}$$

where C_{As} is the concentration at the external pellet surface and k_r is an overall rate constant, which is a function of particle size. The overall rate constant, k_r , increases as the pellet diameter decreases. In Chapter 15, we show that Figure 15-5 can be combined with Equation (15-34) to arrive at the plot of k_r as a function of D_p , shown in Figure 10-9(b).

We see in Figure 10-9 that at small particle sizes, internal diffusion is no longer the slow step and that the surface reaction sequence of adsorption, surface reaction, and desorption (Steps 3, 4, and 5 in Figure 10-5) limit the overall rate

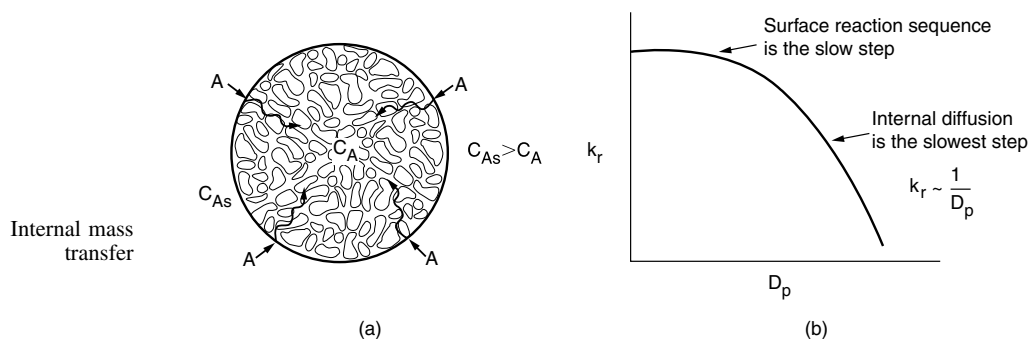


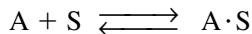
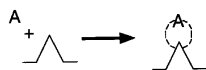
Figure 10-9 Effect of particle size on the overall reaction-rate constant. (a) Branching of a single pore with deposited metal; (b) decrease in rate constant with increasing particle diameter. (See the CRE Web site, Chapter 12.)

of reaction. Consider now one more important point about internal diffusion and surface reaction. These steps (2 through 6) are **not at all** affected by flow conditions external to the pellet.

In the material that follows, we are going to choose our pellet size and external fluid velocity such that neither external diffusion nor internal diffusion is limiting as discussed in Chapters 14 and 15. Instead, we assume that either Step 3 (adsorption), Step 4 (surface reaction), or Step 5 (desorption), or a combination of these steps, limits the overall rate of reaction.

10.2.3 Adsorption Isotherms

Because chemisorption is usually a necessary part of a catalytic process, we shall discuss it before treating catalytic reaction rates. The letter S will represent an active site; alone, it will denote a vacant site, with no atom, molecule, or complex adsorbed on it. The combination of S with another letter (e.g., $A \cdot S$) will mean that one unit of species A will be chemically adsorbed on the site S . Species A can be an atom, molecule, or some other atomic combination, depending on the circumstances. Consequently, the adsorption of A on a site S is represented by



The total molar concentration of active sites per unit mass of catalyst is equal to the number of active sites per unit mass divided by Avogadro's number and will be labeled C_t (mol/g-cat). The molar concentration of vacant sites, C_v (mol/g-cat), is the number of vacant sites per unit mass of catalyst divided by Avogadro's number. In the absence of catalyst deactivation, we assume that the total concentration of active sites, C_t , remains constant. Some further definitions include

P_i = partial pressure of species i in the gas phase, (atm or kPa)

$C_{i \cdot S}$ = surface concentration of sites occupied by species i , (mol/g-cat)

A conceptual model depicting species A and B adsorbed on two different sites is shown in Figure 10-10.



Figure 10-10 Vacant and occupied sites.

For the system shown in Figure 10-10, the total concentration of sites is

Site balance

$$C_t = C_v + C_{A \cdot S} + C_{B \cdot S} \quad (10-1)$$

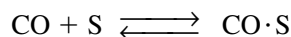
This equation is referred to as a *site balance*. A typical value for the total concentration of sites could be the order of 10^{22} sites/g-cat.

Now consider the adsorption of a nonreacting gas onto the surface of a catalyst. Adsorption data are frequently reported in the form of adsorption *isotherms*. Isotherms portray the amount of a gas adsorbed on a solid at different pressures at a given temperature.

Postulate models; then see which one(s) fit(s) the data.

First, an adsorption mechanism is proposed, and then the isotherm (see Figure 10-11, page 413) obtained from the mechanism is compared with the experimental data. If the isotherm predicted by the model agrees with the experimental data, the model may reasonably describe what is occurring physically in the real system. If the predicted curve does not agree with the experimental data, the model fails to match the physical situation in at least one important characteristic and perhaps more.

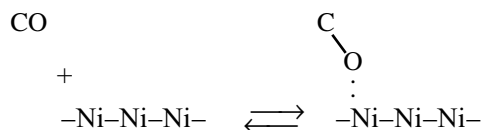
We will consider two types of adsorption: molecular adsorption and dissociative adsorption. To illustrate the difference between molecular adsorption and dissociative adsorption, we will postulate two models for the adsorption of carbon monoxide on metal surfaces. In the molecular adsorption model, CO is adsorbed as molecules, CO,



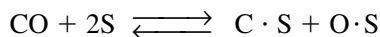
as is the case on nickel

Two models:

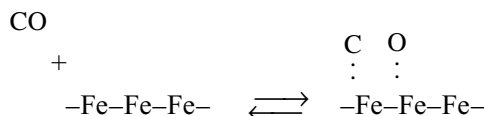
1. Adsorption as CO
2. Adsorption as C and O



In the dissociative adsorption model, carbon monoxide is adsorbed as oxygen and carbon atoms instead of molecular CO



as is the case on iron⁵



⁵ R. I. Masel, *Principles of Adsorption and Reaction on Solid Surfaces* (New York: Wiley, 1996).

The former is called *molecular* or *nondissociated adsorption* (e.g., CO) and the latter is called *dissociative adsorption* (e.g., C and O). Whether a molecule adsorbs nondissociatively or dissociatively depends on the surface.

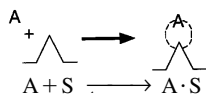
The adsorption of carbon monoxide molecules will be considered first. Because the carbon monoxide does not react further after being adsorbed, we need only to consider the adsorption process:

Molecular Adsorption



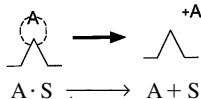
In obtaining a rate law for the rate of adsorption, the reaction in Equation (10-2) can be treated as an *elementary reaction*. The rate of attachment of the carbon monoxide molecules to the active site on the surface is proportional to the number of collisions that these molecules make with a surface active site per second. In other words, a specific fraction of the molecules that strike the surface become adsorbed. The collision rate is, in turn, directly proportional to the carbon monoxide partial pressure, P_{CO} . Because carbon monoxide molecules adsorb only on vacant sites and not on sites already occupied by other carbon monoxide molecules, the rate of attachment is also directly proportional to the concentration of vacant sites, C_v . Combining these two facts means that the rate of attachment of carbon monoxide molecules to the surface is directly proportional to the product of the partial pressure of CO and the concentration of vacant sites; that is,

$$P_{\text{CO}} = C_{\text{CO}}RT$$



$$\text{Rate of attachment} = k_A P_{\text{CO}} C_v$$

The rate of detachment of molecules from the surface can be a first-order process; that is, the detachment of carbon monoxide molecules from the surface is usually directly proportional to the concentration of sites occupied by the adsorbed molecules (e.g., $C_{\text{CO} \cdot \text{S}}$):



$$\text{Rate of detachment} = k_{-A} C_{\text{CO} \cdot \text{S}}$$

The net rate of adsorption is equal to the rate of molecular attachment to the surface minus the rate of detachment from the surface. If k_A and k_{-A} are the constants of proportionality for the attachment and detachment processes, then

$$r_{\text{AD}} = k_A P_{\text{CO}} C_v - k_{-A} C_{\text{CO} \cdot \text{S}} \quad (10-3)$$

The ratio $K_A = k_A/k_{-A}$ is the *adsorption equilibrium constant*. Using K_A to rearrange Equation (10-3) gives

$$\begin{aligned} \text{A} + \text{S} &\xrightleftharpoons{\text{Adsorption}} \text{A} \cdot \text{S} \\ r_{\text{AD}} &= k_A \left(P_A C_v - \frac{C_{\text{A} \cdot \text{S}}}{K_A} \right) \end{aligned}$$

$$r_{\text{AD}} = k_A \left(P_{\text{CO}} C_v - \frac{C_{\text{CO} \cdot \text{S}}}{K_A} \right) \quad (10-4)$$

The adsorption rate constant, k_A , for molecular adsorption is virtually independent of temperature, while the desorption constant, k_{-A} , increases exponentially with increasing temperature. Consequently, the *equilibrium adsorption constant* K_A *decreases exponentially with increasing temperature*.

Because carbon monoxide is the only material adsorbed on the catalyst, the site balance gives

$$C_t = C_v + C_{\text{CO} \cdot \text{S}} \quad (10-5)$$

$$-r'_A = r_{AD} = \left(\frac{\text{mol}}{\text{g-cat} \cdot \text{s}} \right)$$

$$k_A = \left(\frac{1}{\text{atm} \cdot \text{s}} \right)$$

$$P_A = (\text{atm})$$

$$C_v = \left(\frac{\text{mol}}{\text{g-cat}} \right)$$

$$K_A = \left(\frac{1}{\text{atm}} \right)$$

$$C_{A \cdot S} = \left(\frac{\text{mol}}{\text{g-cat}} \right)$$

At equilibrium, the net rate of adsorption equals zero, i.e., $r_{AD} \equiv 0$. Setting the left-hand side of Equation (10-4) equal to zero and solving for the concentration of CO adsorbed on the surface, we get

$$C_{\text{CO} \cdot S} = K_A C_v P_{\text{CO}} \quad (10-6)$$

Using Equation (10-5) to give C_v in terms of $C_{\text{CO} \cdot S}$ and the total number of sites C_t , we can solve for the equilibrium value of $C_{\text{CO} \cdot S}$ in terms of constants and the pressure of carbon monoxide

$$C_{\text{CO} \cdot S} = K_A C_v P_{\text{CO}} = K_A P_{\text{CO}} (C_t - C_{\text{CO} \cdot S})$$

Rearranging gives us

$$C_{\text{CO} \cdot S} = \frac{K_A P_{\text{CO}} C_t}{1 + K_A P_{\text{CO}}} \quad (10-7)$$

This equation thus gives the equilibrium concentration of carbon monoxide adsorbed on the surface, $C_{\text{CO} \cdot S}$, as a function of the partial pressure of carbon monoxide, and is an equation for the adsorption isotherm. This particular type of isotherm equation is called a *Langmuir isotherm*.⁶ **Figure 10-11(a)** shows the Langmuir isotherm for the amount of CO adsorbed per unit mass of catalyst as a function of the partial pressure of CO. For the case of dissociative adsorption, Equation (10-11), **Figure 10-11(b)**, shows the concentration of the atoms C and O adsorbed per unit mass of catalyst.

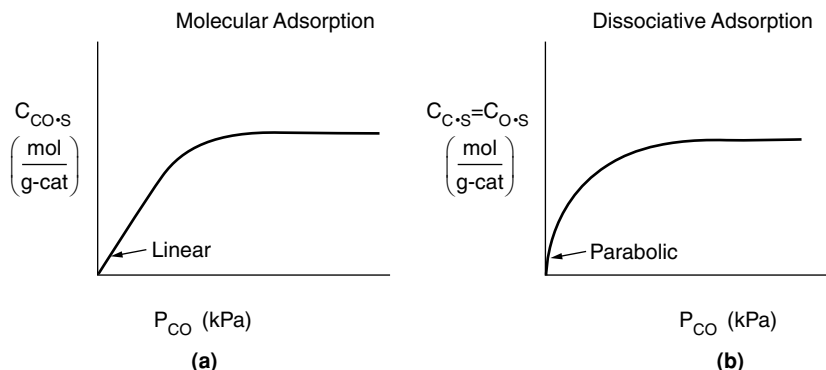


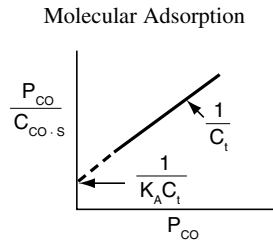
Figure 10-11 Langmuir isotherms for (a) molecular adsorption (b) dissociative adsorption of CO.

One method of checking whether a model (e.g., molecular adsorption versus dissociative adsorption) predicts the behavior of the experimental data is to linearize the model's equation and then plot the indicated variables against one another. For example, the molecular adsorption isotherm, Equation (10-7), may be arranged in the form

$$\frac{P_{\text{CO}}}{C_{\text{CO} \cdot S}} = \frac{1}{K_A C_t} + \frac{P_{\text{CO}}}{C_t} \quad (10-8)$$

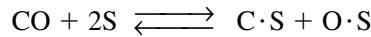
⁶ Named after Irving Langmuir (1881–1957), who first proposed it. He received the Nobel Prize in 1932 for his discoveries in surface chemistry.

and the linearity of a plot of $P_{\text{CO}}/C_{\text{CO}\cdot\text{S}}$ as a function of P_{CO} will determine if the data conform to molecular adsorption, i.e., a Langmuir single-site isotherm.



Next, we derive the isotherm for carbon monoxide disassociating into separate atoms as it adsorbs on the surface, i.e.,

Dissociative
adsorption



When the carbon monoxide molecule dissociates upon adsorption, it is referred to as the *dissociative adsorption* of carbon monoxide. As in the case of molecular adsorption, the rate of adsorption is proportional to the pressure of carbon monoxide in the system because this rate is governed by the number of gaseous collisions with the surface. For a molecule to dissociate as it adsorbs, however, two adjacent vacant active sites are required, rather than the single site needed when a substance adsorbs in its molecular form. The probability of two vacant sites occurring adjacent to one another is proportional to the square of the concentration of vacant sites. These two observations mean that the rate of adsorption is proportional to the product of the carbon monoxide partial pressure and the square of the vacant-site concentration, $P_{\text{CO}}C_v^2$.

For desorption to occur, two occupied sites must be adjacent, meaning that the rate of desorption is proportional to the product of the occupied-site concentration, $(\text{C}\cdot\text{S}) \times (\text{O}\cdot\text{S})$. The net rate of adsorption can then be expressed as

$$r_{\text{AD}} = k_{\text{A}}P_{\text{CO}}C_v^2 - k_{-\text{A}}C_{\text{O}\cdot\text{S}}C_{\text{C}\cdot\text{S}} \quad (10-9)$$

Factoring out k_{A} , the equation for *dissociative adsorption* is

Rate of dissociative
adsorption

$$r_{\text{AD}} = k_{\text{A}} \left(P_{\text{CO}}C_v^2 - \frac{C_{\text{C}\cdot\text{S}}C_{\text{O}\cdot\text{S}}}{K_{\text{A}}} \right)$$

where

$$K_{\text{A}} = \frac{k_{\text{A}}}{k_{-\text{A}}}$$

For dissociative adsorption, both k_{A} and $k_{-\text{A}}$ increase exponentially with increasing temperature, while the *adsorption equilibrium constant* K_{A} decreases with increasing temperature.

At equilibrium, $r_{\text{AD}} \equiv 0$, and

$$k_{\text{A}}P_{\text{CO}}C_v^2 = k_{-\text{A}}C_{\text{C}\cdot\text{S}}C_{\text{O}\cdot\text{S}}$$

For $C_{\text{C}\cdot\text{S}} = C_{\text{O}\cdot\text{S}}$

$$(K_{\text{A}}P_{\text{CO}})^{1/2}C_v = C_{\text{O}\cdot\text{S}} \quad (10-10)$$

Substituting for $C_{\text{C}\cdot\text{S}}$ and $C_{\text{O}\cdot\text{S}}$ in a site balance equation (10-1),

$$\begin{aligned}
 C_t &= C_v + C_{O \cdot S} + C_{C \cdot S} \\
 \text{Site balance:} \quad &= C_v + (K_{CO}P_{CO})^{1/2}C_v + (K_{CO}P_{CO})^{1/2}C_v = C_v(1 + 2(K_{CO}P_{CO})^{1/2})
 \end{aligned}$$

Solving for C_v

$$C_v = C_t / (1 + 2(K_{CO}P_{CO})^{1/2})$$

This value may be substituted into Equation (10-10) to give an expression that can be solved for the equilibrium value of $C_{O \cdot S}$. The resulting equation for the isotherm shown in Figure 10-11(b) is

Langmuir isotherm
for carbon monoxide
dissociative
adsorption as
atomic carbon and
oxygen

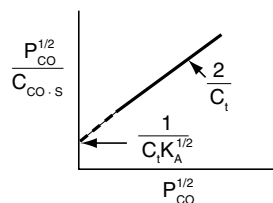
$$C_{O \cdot S} = \frac{(K_A P_{CO})^{1/2} C_t}{1 + 2(K_A P_{CO})^{1/2}} \quad (10-11)$$

Taking the inverse of both sides of the equation, then multiplying through by $(P_{CO})^{1/2}$, yields

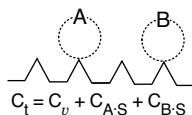
$$\frac{(P_{CO})^{1/2}}{C_{O \cdot S}} = \frac{1}{C_t(K_A)^{1/2}} + \frac{2(P_{CO})^{1/2}}{C_t} \quad (10-12)$$

If dissociative adsorption is the correct model, a plot of $(P_{CO}^{1/2}/C_{O \cdot S})$ versus $P_{CO}^{1/2}$ should be linear with slope $(2/C_t)$.

Dissociative Adsorption



When more than one substance is present, the adsorption isotherm equations are somewhat more complex. The principles are the same, though, and the isotherm equations are easily derived. It is left as an exercise to show that the adsorption isotherm of A in the presence of another adsorbate B is given by the relationship



$$C_{A \cdot S} = \frac{K_A P_A C_t}{1 + K_A P_A + K_B P_B} \quad (10-13)$$

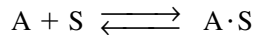
When the adsorption of both A and B are first-order processes, the desorptions are also first order, and both A and B are adsorbed as molecules. The derivations of other Langmuir isotherms are relatively easy.

In obtaining the Langmuir isotherm equations, several aspects of the adsorption system were presupposed in the derivations. The most important of these, and the one that has been subject to the greatest doubt, is that a *uniform* surface is assumed. In other words, any active site has the same attraction for an impinging molecule as does any other active site. Isotherms different from the Langmuir isotherm, such as the Freundlich isotherm, may be derived based on various assumptions concerning the adsorption system, including different types of nonuniform surfaces.

Note assumptions in
the model
and check their
validity.

10.2.4 Surface Reaction

The rate of adsorption of species A onto a solid surface



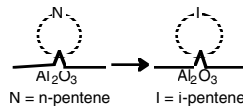
is given by

$$r_{AD} = k_A \left(P_A C_v - \frac{C_{A \cdot S}}{K_A} \right) \quad (10-14)$$

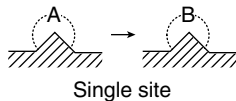
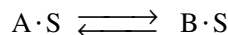
Surface reaction models

After a reactant has been adsorbed onto the surface, i.e., $A \cdot S$, it is capable of reacting in a number of ways to form the reaction product. Three of these ways are:

1. **Single site.** The surface reaction may be a single-site mechanism in which only the site on which the reactant is adsorbed is involved in the reaction. For example, an adsorbed molecule of A may isomerize (or perhaps decompose) directly on the site to which it is attached, such as



The pentane isomerization can be written in generic form as



Each step in the reaction mechanism is elementary, so the surface reaction rate law is

Single Site

$$k_s = \left(\frac{1}{s} \right)$$

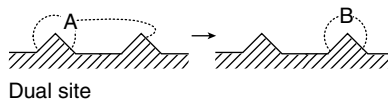
K_s = (dimensionless)

$$r_s = k_s C_{A \cdot S} - k_{-s} C_{B \cdot S} = k_s \left(C_{A \cdot S} - \frac{C_{B \cdot S}}{K_s} \right) \quad (10-15)$$

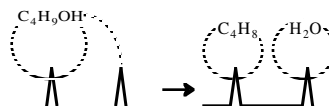
where K_s is the surface-reaction equilibrium constant $K_s = k_s/k_{-s}$

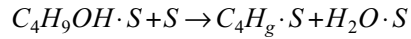
2. **Dual site.** The surface reaction may be a dual-site mechanism in which the adsorbed reactant interacts with another site (either unoccupied or occupied) to form the product.

First type of dual-site mechanism



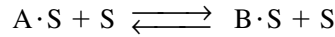
For example, adsorbed A may react with an adjacent vacant site to yield a vacant site and a site on which the product is adsorbed, or as in the case of the dehydration of butanol, the products may adsorb on two adjacent sites.





$$r_s = k_s \left[(C_{C_4H_9OH \cdot S})(C_v) - \frac{(C_{C_4H_g \cdot S})(C_{H_2O \cdot S})}{K_s} \right]$$

For the generic reaction



Dual Site

$$r_s = \left(\frac{\text{mol}}{\text{g-cat} \cdot \text{s}} \right)$$

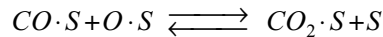
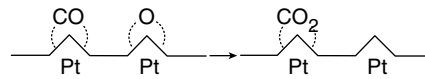
$$k_s = \left(\frac{\text{g-cat}}{\text{mol} \cdot \text{s}} \right)$$

$K_s = (\text{dimensionless})$

the corresponding surface-reaction rate law is

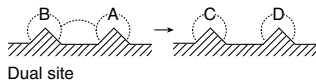
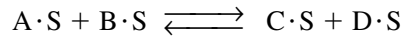
$$r_s = k_s \left(C_{A \cdot S} C_v - \frac{C_{B \cdot S} C_v}{K_s} \right) \quad (10-16)$$

A second dual-site mechanism is the reaction between two adsorbed species, such as the reaction of CO with O.



$$r_s = k_s \left[(C_{CO \cdot S})(C_{O \cdot S}) - \frac{(C_{CO_2 \cdot S})(C_v)}{K_s} \right]$$

For the generic reaction

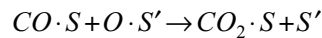
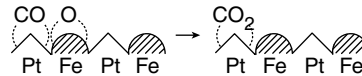


Dual site

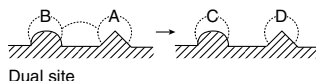
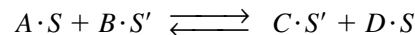
the corresponding surface-reaction rate law is

$$r_s = k_s \left(C_{A \cdot S} C_{B \cdot S} - \frac{C_{C \cdot S} C_{D \cdot S}}{K_s} \right) \quad (10-17)$$

A third dual-site mechanism is the reaction of two species adsorbed on different types of sites S and S', such as the reaction of CO with O.



For the generic reaction



Dual site

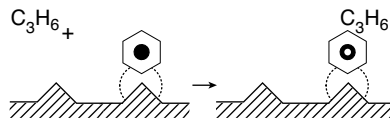
the corresponding surface-reaction rate law is

$$r_s = k_s \left(C_{A \cdot S} C_{B \cdot S'} - \frac{C_{C \cdot S'} C_{D \cdot S}}{K_s} \right) \quad (10-18)$$

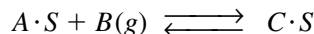
Langmuir–
Hinshelwood
kinetics

Reactions involving either single- or dual-site mechanisms, which were described earlier, are sometimes referred to as following *Langmuir–Hinshelwood kinetics*.

3. **Eley–Rideal.** A third mechanism is the reaction between an adsorbed molecule and a molecule in the gas phase, such as the reaction of propylene and benzene (cf. the reverse reaction in Figure 10-13)



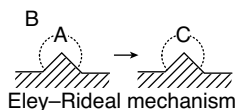
For the generic reaction



the corresponding surface-reaction rate law is

$$r_s = k_s \left(C_{A \cdot S} P_B - \frac{C_{C \cdot S}}{K_s} \right) \quad (10-19)$$

This type of mechanism is referred to as an *Eley–Rideal mechanism*.



$$k_s = \left(\frac{1}{\text{atm} \cdot \text{s}} \right)$$

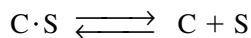
$$K_s = \left(\frac{1}{\text{atm}} \right)$$

10.2.5 Desorption

In each of the preceding cases, the products of the surface reaction adsorbed on the surface are subsequently desorbed into the gas phase. For the desorption of a species (e.g., C)

$$K_{DC} = (\text{atm})$$

$$k_D = \left(\frac{1}{\text{s}} \right)$$



the rate of desorption of C is

$$r_{DC} = k_D \left(C_{C \cdot S} - \frac{P_C C_v}{K_{DC}} \right) \quad (10-20)$$

where K_{DC} is the desorption equilibrium constant with units of atm. Now let's look at the above adsorption from right to left. We note that the desorption step for C is just the reverse of the adsorption step. Consequently, the rate of desorption of C, r_{DC} , is just opposite in sign to the rate of adsorption of C, r_{ADC}

$$r_{DC} = -r_{ADC}$$

In addition, we see that the desorption equilibrium constant K_{DC} is just the reciprocal of the adsorption equilibrium constant for C, K_C

$$K_{DC} = (\text{atm})$$

$$K_C = \left(\frac{1}{\text{atm}} \right)$$

$$K_{DC} = \frac{1}{K_C}$$

in which case the rate of desorption of C can be written

$$r_{DC} = k_D (C_{C \cdot S} - K_C P_C C_v) \quad (10-21)$$

In the material that follows, the form of the equation for the desorption step that we will use to develop our rate laws will be similar to Equation (10-21).

10.2.6 The Rate-Limiting Step

When heterogeneous reactions are carried out at steady state, the rates of each of the three reaction steps in series (adsorption, surface reaction, and desorption) are equal to one another

$$-r'_A = r_{AD} = r_S = r_D$$

However, one particular step in the series is usually found to be *rate-limiting* or *rate-controlling*. That is, if we could make that particular step go faster, the entire reaction would proceed at an accelerated rate. Consider the analogy to the electrical circuit shown in Figure 10-12. A given concentration of reactants is analogous to a given driving force or electromotive force (EMF). The current I (with units of Coulombs/s) is analogous to the rate of

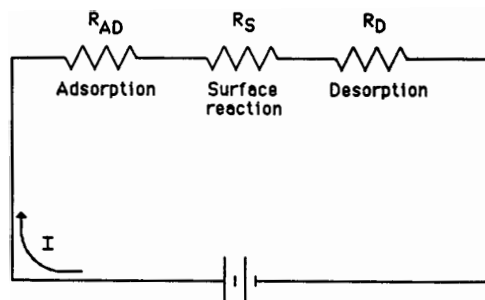


Figure 10-12 Electrical analog to heterogeneous reactions.

reaction, $-r'_A$ (mol/s·g-cat), and a resistance R_i is associated with each step in the series. Because the resistances are in series, the total resistance R_{tot} is just the sum of the individual resistances, for adsorption (R_{AD}), surface reaction (R_S), and desorption (R_D). The current, I , for a given voltage, E , is

$$I = \frac{E}{R_{\text{tot}}} = \frac{E}{R_{AD} + R_S + R_D}$$

The concept of a
rate-limiting step
Who is slowing us
down?

Because we observe only the total resistance, R_{tot} , it is our task to find which resistance is much larger (say, 100 Ω) than the other two resistances (say, 0.1 Ω). Thus, if we could lower the largest resistance, the current I (i.e., $-r'_A$), would be larger for a given voltage, E . Analogously, we want to know which step in the adsorption–reaction–desorption series is limiting the overall rate of reaction.

The approach in determining catalytic and heterogeneous mechanisms is usually termed the *Langmuir–Hinshelwood approach*, since it is derived from ideas proposed by Hinshelwood based on Langmuir's principles for adsorption.⁷ The Langmuir–Hinshelwood approach was popularized by Hougen and Watson and occasionally includes their names.⁸ It consists of first assuming a sequence of steps in the reaction. In writing this sequence, one must choose among such mechanisms as molecular or atomic adsorption, and single- or dual-site reaction. Next, rate laws are written for the individual steps as shown in the

⁷ C. N. Hinshelwood, *The Kinetics of Chemical Change* (Oxford: Clarendon Press, 1940).

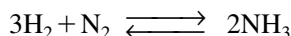
⁸ O. A. Hougen and K. M. Watson, *Ind. Eng. Chem.*, 35, 529 (1943).



preceding section, assuming that all steps are reversible. Finally, a rate-limiting step is postulated, and steps that are not rate-limiting are used to eliminate all coverage-dependent terms. The most questionable assumption in using this technique to obtain a rate law is the hypothesis that the activity of the surface is essentially uniform as far as the various steps in the reaction are concerned.

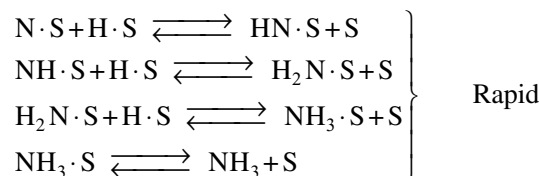
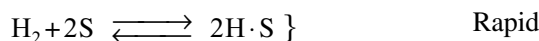
Industrial Example of Adsorption-Limited Reaction

An example of an adsorption-limited reaction is the synthesis of ammonia from hydrogen and nitrogen



over an *iron catalyst* that proceeds by the following mechanism:⁹

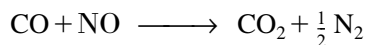
Dissociative
adsorption of
N₂ is rate-limiting



The rate-limiting step is believed to be the adsorption of the N₂ molecule as an N atom.

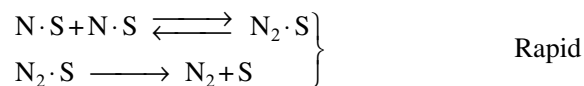
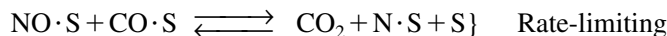
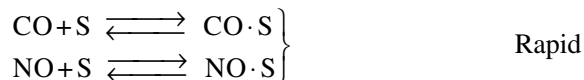
Industrial Example of Surface-Limited Reaction

An example of a surface-limited reaction is the reaction of two noxious automobile exhaust products, CO and NO



carried out over a *copper catalyst* to form environmentally acceptable products, N₂ and CO₂

Surface reaction is
rate-limiting



Analysis of the rate law suggests that CO₂ and N₂ are weakly adsorbed, i.e., have infinitesimally small adsorption constants (see Problem P10-9_B).

⁹ From the literature cited in G. A. Somorjai, *Introduction to Surface Chemistry and Catalysis* (New York: Wiley, 1994), p. 482.

10.3 Synthesizing a Rate Law, Mechanism, and Rate-Limiting Step

We now wish to develop rate laws for catalytic reactions that are not diffusion-limited. In developing the procedure to obtain a mechanism, a rate-limiting step, and a rate law consistent with experimental observation, we shall discuss a particular catalytic reaction, the decomposition of cumene to form benzene and propylene. The overall reaction is



A conceptual model depicting the sequence of steps in this platinum-catalyzed reaction is shown in Figure 10-13. Figure 10-13 is only a schematic representation of the adsorption of cumene; a more realistic model is the formation of a complex of the π orbitals of benzene with the catalytic surface, as shown in Figure 10-14.

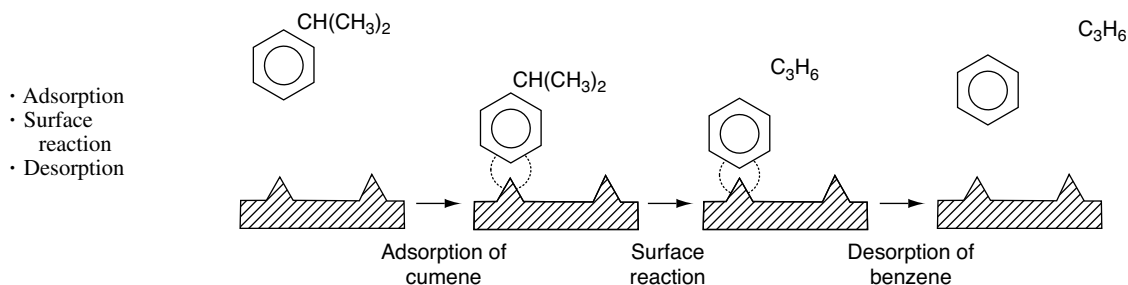


Figure 10-13 Sequence of steps in a reaction-limited catalytic reaction.

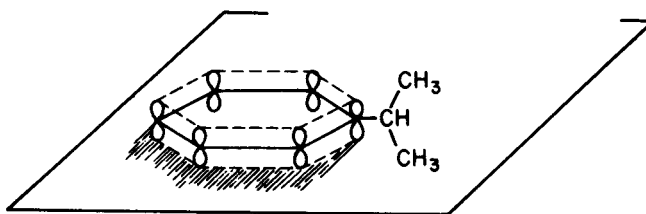
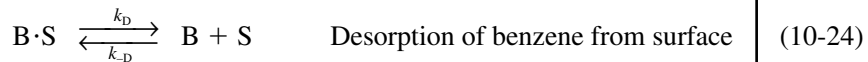
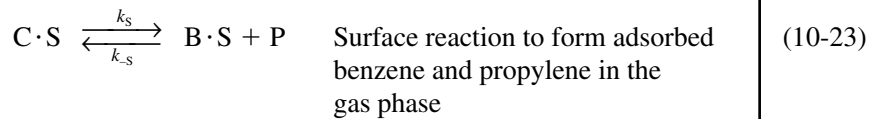
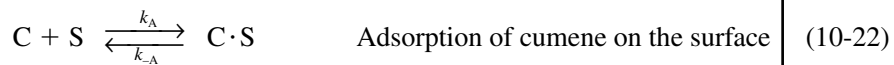


Figure 10-14 π -orbital complex on surface.

The nomenclature in Table 10-3 will be used to denote the various species in this reaction: C = cumene, B = benzene, and P = propylene. The reaction sequence for this decomposition is shown in Table 10-3.

TABLE 10-3 STEPS IN A LANGMUIR–HINSHELWOOD KINETIC MECHANISM

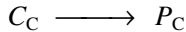
These three steps represent the mechanism for cumene decomposition.



Equations (10-22) through (10-24) represent the mechanism proposed for this reaction.

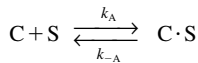
When writing rate laws for these steps, we treat each step as an elementary reaction; the only difference is that the species concentrations in the gas phase are replaced by their respective partial pressures

Ideal gas law
 $P_C = C_C RT$

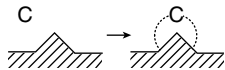


There is no theoretical reason for this replacement of the concentration, C_C , with the partial pressure, P_C ; it is just the convention initiated in the 1930s and used ever since. Fortunately, P_C can be calculated easily and directly from C_C using the ideal gas law (i.e., $P_C = C_C RT$).

The rate expression for the adsorption of cumene as given in Equation (10-22) is



$$r_{AD} = k_A P_C C_v - k_{-A} C_{C \cdot S}$$



Adsorption: $r_{AD} = k_A \left(P_C C_v - \frac{C_{C \cdot S}}{K_C} \right)$

(10-25)

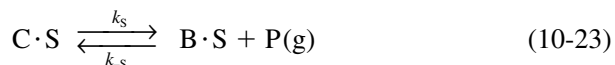
If r_{AD} has units of (mol/g-cat·s) and $C_{C \cdot S}$ has units of (mol cumene adsorbed/g-cat), then typical units of k_A , k_{-A} , and K_C would be

$$[k_A] \equiv (\text{kPa} \cdot \text{s})^{-1} \text{ or } (\text{atm} \cdot \text{h})^{-1}$$

$$[k_{-A}] \equiv \text{h}^{-1} \text{ or } \text{s}^{-1}$$

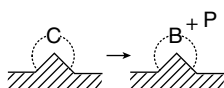
$$[K_C] \equiv \left[\frac{k_A}{k_{-A}} \right] \equiv \text{kPa}^{-1}$$

The rate law for the surface-reaction step producing adsorbed benzene and propylene in the gas phase



is

$$r_S = k_S C_{C \cdot S} - k_{-S} P_P C_{B \cdot S}$$



$$\text{Surface reaction:} \quad r_s = k_s \left(C_{C \cdot S} - \frac{P_P C_{B \cdot S}}{K_S} \right) \quad (10-26)$$

with the *surface reaction equilibrium constant* being

$$K_S = \frac{k_s}{k_{-s}}$$

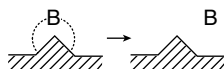
Typical units for k_s and K_S are s^{-1} and kPa, respectively.

Propylene is not adsorbed on the surface. Consequently, its concentration on the surface is zero.

$$C_{P \cdot S} = 0$$

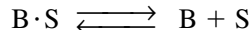
The rate of benzene desorption [see Equation (10-24)] is

$$r_D = k_D C_{B \cdot S} - k_{-D} P_B C_v \quad (10-27)$$



$$\text{Desorption:} \quad r_D = k_D \left(C_{B \cdot S} - \frac{P_B C_v}{K_{DB}} \right) \quad (10-28)$$

Typical units of k_D and K_{DB} are s^{-1} and kPa, respectively. By viewing the desorption of benzene



from right to left, we see that desorption is just the reverse of the adsorption of benzene. Consequently, as mentioned earlier, it is easily shown that the benzene adsorption equilibrium constant K_B is just the reciprocal of the benzene desorption constant K_{DB}

$$K_B = \frac{1}{K_{DB}}$$

and Equation (10-28) can be written as

$$\text{Desorption:} \quad r_D = k_D (C_{B \cdot S} - K_B P_B C_v) \quad (10-29)$$

Because there is no accumulation of reacting species on the surface, the rates of each step in the sequence are all equal as discussed in Figure 10-12:

$$-r'_C = r_{AD} = r_s = r_D \quad (10-30)$$

For the mechanism postulated in the sequence given by Equations (10-22) through (10-24), we wish to determine which step is rate-limiting. We first assume one of the steps to be rate-limiting (rate-controlling) and then formulate the reaction-rate law in terms of the partial pressures of the species present. From this expression we can determine the variation of the initial reaction rate with the initial partial pressures and the initial total pressure. If the predicted rate varies with pressure in the same manner as the rate observed experimentally, the implication is that the assumed mechanism and rate-limiting step are correct.

We will first start our development of the rate laws with the assumption that the adsorption step is rate-limiting and derive the rate law, and then

proceed to assume that each of the other two steps' surface reaction and desorption limit the overall rate and then derive the rate law for each of these other two limiting cases.

10.3.1 Is the Adsorption of Cumene Rate-Limiting?

To answer this question we shall assume that the adsorption of cumene is indeed rate-limiting, derive the corresponding rate law, and then check to see if it is consistent with experimental observation. By postulating that this (or any other) step is rate-limiting, we are assuming that the reaction-rate constant of this step (in this case k_A) is small with respect to the specific rates of the other steps (in this case k_S and k_D).¹⁰ The rate of adsorption is

Need to express C_v
and $C_{C \cdot S}$ in terms
of P_C , P_B , and P_P

$$-r'_C = r_{AD} = k_A \left(P_C C_v - \frac{C_{C \cdot S}}{K_C} \right) \quad (10-25)$$

Because we cannot measure either C_v or $C_{C \cdot S}$, we must replace these variables in the rate law with measurable quantities for the equation to be meaningful.

For steady-state operation we have

$$-r'_C = r_{AD} = r_S = r_D \quad (10-30)$$

For adsorption-limited reactions, k_A is very small and k_S and k_D are very, very large by comparison. Consequently, the ratios r_S/k_S and r_D/k_D are very small (approximately zero), whereas the ratio r_{AD}/k_A is relatively large.

The surface reaction-rate law is

$$r_S = k_S \left(C_{C \cdot S} - \frac{C_{B \cdot S} P_P}{K_S} \right) \quad (10-31)$$

Again, for adsorption-limited reactions, the surface-specific reaction rate k_S is large by comparison, and we can set

$$\frac{r_S}{k_S} \approx 0 \quad (10-32)$$

and solve Equation (10-31) for $C_{C \cdot S}$

$$C_{C \cdot S} = \frac{C_{B \cdot S} P_P}{K_S} \quad (10-33)$$

¹⁰ *Strictly speaking*, one should compare the product $k_A P_C$ with k_S and k_D .

$$r_{AD} = k_A P_C \left[C_v - \frac{C_{C \cdot S}}{K_C P_C} \right]$$

$$\frac{\text{mol}}{\text{s} \cdot \text{kg-cat}} = \left(\frac{1}{\text{s} \cdot \text{atm}} \right) \cdot (\text{atm}) \cdot \left[\frac{\text{mol}}{\text{kg-cat}} \right] = \left[\frac{1}{\text{s}} \right] \frac{\text{mol}}{\text{kg-cat}}$$

Dividing r_{AD} by $k_A P_C$, we note $\frac{r_{AD}}{k_A P_C} = \frac{\text{mol}}{\text{kg-cat}}$. The reason we do this is that in order

to compare terms, the ratios $\left(-\frac{r_{AD}}{k_A P_C} \right)$, $\left(\frac{r_S}{k_S} \right)$, and $\left(\frac{r_D}{k_D} \right)$ must all have the same units

$\left[\frac{\text{mol}}{\text{kg-cat}} \right]$. Luckily for us, the end result is the same, however.

To be able to express $C_{C \cdot S}$ solely in terms of the partial pressures of the species present, we must evaluate $C_{B \cdot S}$. The rate of desorption of benzene is

$$r_D = k_D (C_{B \cdot S} - K_B P_B C_v) \quad (10-29)$$

Using
 $\frac{r_S}{k_S} \approx 0 \approx \frac{r_D}{k_D}$
 to find $C_{B \cdot S}$ and
 $C_{C \cdot S}$ in terms of
 partial pressures

However, for adsorption-limited reactions, k_D is large by comparison, and we can set

$$\frac{r_D}{k_D} \approx 0 \quad (10-34)$$

and then solve Equation (10-29) for $C_{B \cdot S}$

$$C_{B \cdot S} = K_B P_B C_v \quad (10-35)$$

After combining Equations (10-33) and (10-35), we have

$$C_{C \cdot S} = K_B \frac{P_B P_P}{K_S} C_v \quad (10-36)$$

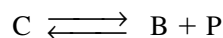
Replacing $C_{C \cdot S}$ in the rate equation by Equation (10-36) and then factoring C_v , we obtain

$$r_{AD} = k_A \left(P_C - \frac{K_B P_B P_P}{K_S K_C} \right) C_v = k_A \left(P_C - \frac{P_B P_P}{K_P} \right) C_v \quad (10-37)$$

Let's look at how the thermodynamic constant pressure equilibrium constant, K_P , found its way into Equation (10-37) and how we can find its value for any reaction. First we observe that at equilibrium $r_{AD} = 0$, Equation (10-37) rearranges to

$$\frac{P_{Be} P_{Pe}}{P_{Ce}} = \frac{K_C K_S}{K_B}$$

We also know from thermodynamics (Appendix C) that for the reaction



also at equilibrium ($-r'_C = 0$), we have the following relationship for partial pressure equilibrium constant K_P

$$K_P = \frac{P_{Be} P_{Pe}}{P_{Ce}}$$

Consequently, the following relationship must hold

$$\boxed{\frac{K_S K_C}{K_B} = K_P} \quad (10-38)$$

The equilibrium constant can be determined from *thermodynamic data* and is related to the change in the Gibbs free energy, ΔG° , by the equation (see Appendix C)

$$\boxed{RT \ln K = -\Delta G^\circ} \quad (10-39)$$

where R is the ideal gas constant and T is the absolute temperature.

The concentration of vacant sites, C_v , can now be eliminated from Equation (10-37) by utilizing the *site balance* to give the total concentration of sites, C_t , which is assumed constant¹¹

$$\text{Total sites} = \text{Vacant sites} + \text{Occupied sites}$$

Because cumene and benzene are adsorbed on the surface, the concentration of occupied sites is $(C_{C \cdot S} + C_{B \cdot S})$, and the total concentration of sites is

$$\text{Site balance} \quad C_t = C_v + C_{C \cdot S} + C_{B \cdot S} \quad (10-40)$$

Substituting Equations (10-35) and (10-36) into Equation (10-40), we have

$$C_t = C_v + \frac{K_B}{K_S} P_B P_C C_v + K_B P_B C_v$$

Solving for C_v , we have

$$C_v = \frac{C_t}{1 + P_B P_C K_B / K_S + K_B P_B} \quad (10-41)$$

Combining Equations (10-41) and (10-37), we find that the rate law for the catalytic decomposition of cumene, assuming that the adsorption of cumene is the rate-limiting step, is

Cumene reaction
rate law if
adsorption were
the limiting step

$$-r'_C = r_{AD} = \frac{C_t k_A (P_C - P_B P_C / K_P)}{1 + K_B P_B P_C / K_S + K_B P_B} \quad (10-42)$$

We now wish to sketch a plot of the initial rate of reaction as a function of the partial pressure of cumene, P_{C0} . Initially, no products are present; consequently, $P_P = P_B = 0$. The initial rate is given by

$$-r'_{C0} = C_t k_A P_{C0} = k P_{C0} \quad (10-43)$$

If the cumene decomposition is adsorption rate limited, then from Equation (10-43) we see that the initial rate will be linear with the initial partial pressure of cumene, as shown in Figure 10-15.

Before checking to see if Figure 10-15 is consistent with experimental observation, we shall derive the corresponding rate laws for the other possible rate-limiting steps and then develop corresponding initial rate plots for the case when the surface reaction is rate-limiting and then for the case when the desorption of benzene is rate-limiting.

¹¹ Some (I won't mention any names) prefer to write the surface reaction rate in terms of the fraction of the surface of sites covered (i.e., f_A) rather than the number of sites $C_{A \cdot S}$ covered, the difference being the multiplication factor of the total site concentration, C_t . In any event, the final form of the rate law is the same because C_t , K_A , k_S , and so on, are all lumped into the reaction-rate constant, k .

If adsorption were rate-limiting, the data should show $-r'_{\text{CO}}$ increasing linearly with P_{CO} .

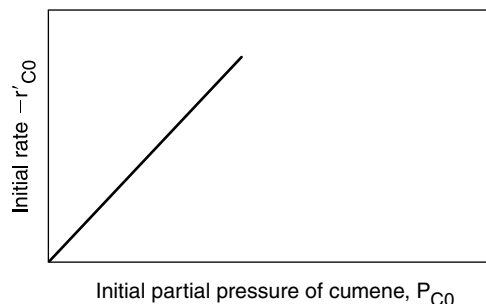


Figure 10-15 Adsorption-limited reaction.

10.3.2 Is the Surface Reaction Rate-Limiting?

The rate of surface reaction is

Single-site
mechanism

$$r_s = k_s \left(C_{\text{C}\cdot\text{S}} - \frac{P_{\text{P}} C_{\text{B}\cdot\text{S}}}{K_{\text{S}}} \right) \quad (10-26)$$

Since we cannot readily measure the concentrations of the adsorbed species, we must utilize the adsorption and desorption steps to eliminate $C_{\text{C}\cdot\text{S}}$ and $C_{\text{B}\cdot\text{S}}$ from this equation.

From the adsorption rate expression in Equation (10-25) and the condition that k_{A} and k_{D} are very large by comparison with k_{S} when the surface reaction is limiting (i.e., $r_{\text{AD}}/k_{\text{A}} \approx 0$),¹² we obtain a relationship for the surface concentration for adsorbed cumene

$$C_{\text{C}\cdot\text{S}} = K_{\text{C}} P_{\text{C}} C_{\text{v}}$$

In a similar manner, the surface concentration of adsorbed benzene can be evaluated from the desorption rate expression, Equation (10-29), together with the approximation

Using
 $\frac{r_{\text{AD}}}{k_{\text{A}}} \approx 0 \approx \frac{r_{\text{D}}}{k_{\text{D}}}$
to find $C_{\text{B}\cdot\text{S}}$ and
 $C_{\text{C}\cdot\text{S}}$ in terms of
partial pressures

$$\text{when } \frac{r_{\text{D}}}{k_{\text{D}}} \approx 0$$

then we get the same result for $C_{\text{B}\cdot\text{S}}$ as before when we had adsorption limitation, i.e.,

$$C_{\text{B}\cdot\text{S}} = K_{\text{B}} P_{\text{B}} C_{\text{v}}$$

Substituting for $C_{\text{B}\cdot\text{S}}$ and $C_{\text{C}\cdot\text{S}}$ in Equation (10-26) gives us

$$r_s = k_s \left(P_{\text{C}} K_{\text{C}} - \frac{K_{\text{B}} P_{\text{B}} P_{\text{P}}}{K_{\text{S}}} \right) C_{\text{v}} = k_s K_{\text{C}} \left(P_{\text{C}} - \frac{P_{\text{B}} P_{\text{P}}}{K_{\text{P}}} \right) C_{\text{v}} \quad (10-26a)$$

where the thermodynamic equilibrium constant was used to replace the ratio of surface reaction and adsorption constants, i.e.,

$$K_{\text{P}} = \frac{K_{\text{C}} K_{\text{S}}}{K_{\text{B}}} \quad (10-26b)$$

¹² See footnote 10 on page 424.

The only variable left to eliminate is C_v and we use a site balance to accomplish this, i.e.,

Site balance
$$C_t = C_v + C_{B \cdot S} + C_{C \cdot S} \quad (10-40)$$

Substituting for concentrations of the adsorbed species, $C_{B \cdot S}$, and $C_{C \cdot S}$, factoring out C_v , and rearranging yields

$$C_v = \frac{C_t}{1 + K_B P_B + K_C P_C}$$

Substituting for C_v in Equation (10-26a)

Cumene rate law
for surface-
reaction-limiting

$$-r'_C = r_S = \frac{\overbrace{k_s C_t K_C}^k (P_C - P_P P_B / K_P)}{1 + P_B K_B + K_C P_C} \quad (10-44)$$

The initial rate of reaction is

$$-r'_{C0} = \frac{\overbrace{k_s C_t K_C}^k P_{C0}}{1 + K_C P_{C0}} = \frac{k P_{C0}}{1 + K_C P_{C0}} \quad (10-45)$$

Figure 10-16 shows the initial rate of reaction as a function of the initial partial pressure of cumene for the case of surface-reaction-limiting.

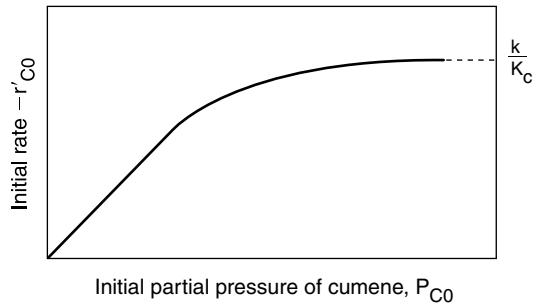


Figure 10-16 Surface-reaction-limited.

At low partial pressures of cumene

$$1 \gg K_C P_{C0}$$

and we observe that the initial rate will increase linearly with the initial partial pressure of cumene:

$$-r'_{C0} \approx k P_{C0}$$

At high partial pressures

$$K_C P_{C0} \gg 1$$

and Equation (10-45) becomes

$$-r'_{C0} \approx \frac{k P_{C0}}{K_C P_{C0}} = \frac{k}{K_C}$$

If surface
reaction were
rate-limiting, the
data would show
this behavior.

and the initial rate is independent of the initial partial pressure of cumene.

10.3.3 Is the Desorption of Benzene Rate-Limiting?

The rate expression for the desorption of benzene is

$$r_D = k_D (C_{B \cdot S} - K_B P_B C_v) \quad (10-29)$$

For desorption-limited reactions, both k_A and k_S are very large compared with k_D , which is small.

From the rate expression for surface reaction, Equation (10-26), we set

$$\frac{r_S}{k_S} \approx 0$$

to obtain

$$C_{B \cdot S} = K_S \left(\frac{C_{C \cdot S}}{P_P} \right) \quad (10-46)$$

Similarly, for the adsorption step, Equation (10-25), we set

$$\frac{r_{AD}}{k_A} \approx 0$$

to obtain

$$C_{C \cdot S} = K_C P_C C_v$$

then substitute for $C_{C \cdot S}$ in Equation (10-46) to obtain

$$C_{B \cdot S} = \frac{K_C K_S P_C C_v}{P_P} \quad (10-47)$$

Combining Equations (10-26b), (10-29), and (10-47) gives us

$$r_D = k_D K_C K_S \left(\frac{P_C}{P_P} - \frac{P_B}{K_P} \right) C_v \quad (10-48)$$

where K_C is the cumene adsorption constant, K_S is the surface-reaction equilibrium constant, and K_P is the thermodynamic gas-phase equilibrium constant, Equation (10-38), for the reaction. The expression for C_v is obtained from a site balance:

$$\text{Site balance: } C_t = C_{C \cdot S} + C_{B \cdot S} + C_v \quad (10-49)$$

After substituting for the respective surface concentrations, we solve the site balance for C_v

$$C_v = \frac{C_t}{1 + K_C K_S P_C / P_P + K_C P_C} \quad (10-49)$$

Replacing C_v in Equation (10-48) by Equation (10-49) and multiplying the numerator and denominator by P_P , we obtain the rate expression for desorption control

Cumene decomposition rate law if desorption were limiting

$$-r'_C = r_D = \frac{k_D C_t K_S K_C (P_C - P_B P_P / K_P)}{P_P + P_C K_C K_S + K_C P_P P_C} \quad (10-50)$$

If desorption limits, the initial rate is independent of the initial partial pressure of cumene.

To determine the dependence of the initial rate of reaction on the initial partial pressure of cumene, we again set $P_P = P_B = 0$, and the rate law reduces to

$$-r'_{C_0} = k_D C_t$$

with the corresponding plot of $-r'_{C_0}$ shown in Figure 10-17. If desorption were rate limiting, we would see that the initial rate of reaction would be independent of the initial partial pressure of cumene.

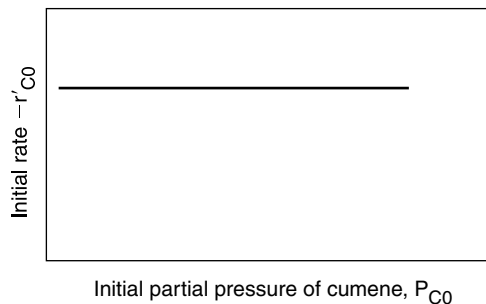


Figure 10-17 Desorption-limited reaction.

10.3.4 Summary of the Cumene Decomposition

Cumene decomposition is surface-reaction-limited

Surface-reaction-limited mechanism is consistent with experimental data.

The experimental observations of $-r'_{C_0}$ as a function of P_{C_0} are shown in Figure 10-18. From the plot in Figure 10-18, we can clearly see that neither adsorption nor desorption is rate-limiting. For the reaction and mechanism given by



the rate law derived by assuming that the surface reaction is rate-limiting agrees with the data.

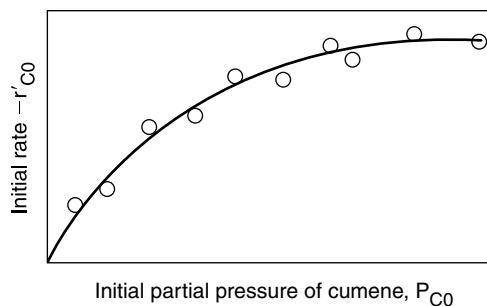


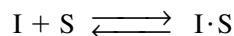
Figure 10-18 Actual initial rate as a function of partial pressure of cumene.

The rate law for the case of no inerts adsorbing on the surface is

$$-r'_C = \frac{k(P_C - P_B P_P / K_P)}{1 + K_B P_B + K_C P_C} \quad (10-44)$$

The forward cumene decomposition reaction is a single-site mechanism involving only adsorbed cumene, while the reverse reaction of propylene in the gas phase reacting with adsorbed benzene is an Eley–Rideal mechanism.

If we were to have an adsorbing inert in the feed, the inert would not participate in the reaction but would occupy active sites on the catalyst surface:



Our site balance is now

$$C_t = C_v + C_{C \cdot S} + C_{B \cdot S} + C_{I \cdot S} \quad (10-51)$$

Because the adsorption of the inert is at equilibrium, the concentration of sites occupied by the inert is

$$C_{I \cdot S} = K_I P_I C_v \quad (10-52)$$

Substituting for the inert sites in the site balance, the rate law for surface reaction control when an adsorbing inert is present is

$$\text{Adsorbing inerts} \quad -r'_C = \frac{k(P_C - P_B P_P / K_P)}{1 + K_C P_C + K_B P_B + K_I P_I} \quad (10-53)$$

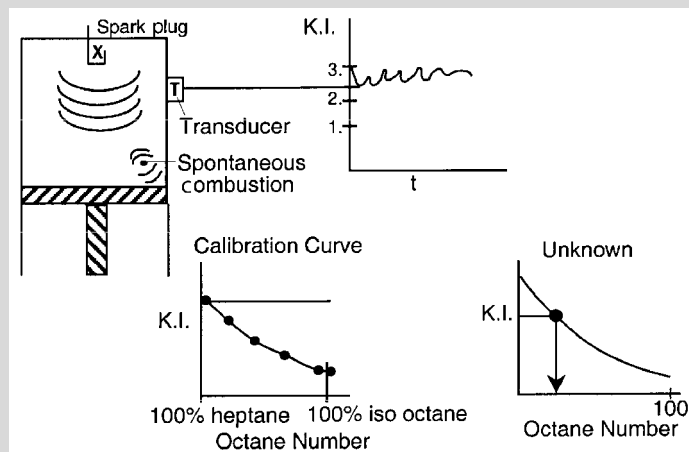
One observes that the rate decreases as the partial pressure of adsorbing inerts increases.

10.3.5 Reforming Catalysts

We now consider a dual-site mechanism, which is a reforming reaction found in petroleum refining to upgrade the octane number of gasoline.

Side Note: Octane Number. Fuels with low octane numbers can produce spontaneous combustion in the car cylinder before the air/fuel mixture is compressed to its desired volume and ignited by the spark plug. The following figure shows the desired combustion wave front moving down from the spark plug and the unwanted spontaneous combustion wave in the lower right-hand corner. This spontaneous combustion produces detonation waves, which constitute engine knock. The lower the octane number, the greater the chance of spontaneous combustion and engine knock.

The octane number of a gasoline is determined from a calibration curve relating knock intensity to the % iso-octane in a mixture of iso-octane and heptane. One way to calibrate the octane number is to place a transducer on the side of the cylinder to measure the knock intensity (K.I.) (pressure pulse) for various mixtures of heptane and iso-octane. The octane number is the percentage of iso-octane in this mixture. That is, pure iso-octane has an octane number of 100, 80% iso-octane/20% heptane has an octane number of 80, and so on. The knock intensity is measured for this



80/20 mixture and recorded. The relative percentages of iso-octane and heptane are changed (e.g., 90/10), and the test is repeated. After a series of experiments, a calibration curve is constructed. The gasoline to be calibrated is then used in the test engine, where the standard knock intensity is measured. Knowing the knock intensity, the octane rating of the fuel is read off the calibration curve. A gasoline with an octane rating of 92 means that it matches the performance of a mixture of 92% iso-octane and 8% heptane. Another way to calibrate the octane number is to set the knock intensity and increase the compression ratio. A fixed percentage of iso-octane and heptane is placed in a test engine and the compression ratio (CR) is increased continually until spontaneous combustion occurs, producing an engine knock. The compression ratio and the corresponding composition of the mixture are then recorded, and the test is repeated to obtain the calibration curve of compression ratio as a function of % iso-octane. After the calibration curve is obtained, the unknown is placed in the cylinder, and the compression ratio (CR) is increased until the set knock intensity is exceeded. The CR is then matched to the calibration curve to find the octane number.

The more compact the hydrocarbon molecule, the less likely it is to cause spontaneous combustion and engine knock. Consequently, it is desired to isomerize straight-chain hydrocarbon molecules into more compact molecules through a catalytic process called *reforming*.

In the U.S., the higher the octane number the greater the cost of a gallon (liter) of gasoline. However, because of government regulations, the cost of a gallon (liter) of gasoline in Jofostan is the same for all octane numbers.

The more compact the molecule, the greater the octane number.

Catalyst manufacture

One common reforming catalyst is platinum on alumina. Platinum on alumina (Al_2O_3) (see SEM photo in Figure 10-19) is a bifunctional catalyst that can be prepared by exposing alumina pellets to a chloroplatinic acid solution, drying, and then heating in air at 775 K to 875 K for several hours. Next, the material is exposed to hydrogen at temperatures around 725 K to 775 K to produce very small clusters of Pt on alumina. These clusters have sizes on the

order of 10 Å, while the alumina pore sizes on which the Pt is deposited are on the order of 100 Å to 10,000 Å (i.e., 10 nm to 1000 nm).

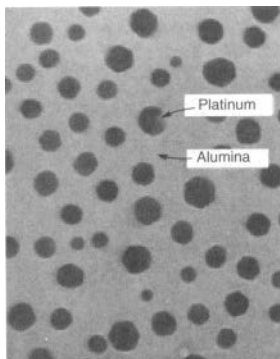
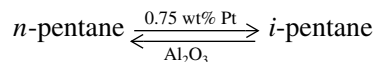


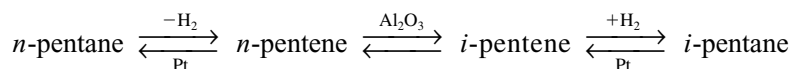
Figure 10-19 Platinum on alumina. (Masel, Richard. *Chemical Kinetics and Catalysis*, p. 700. © John Wiley & Sons, Inc. Reprinted by permission of John Wiley & Sons, Inc. All rights reserved.)

As an example of catalytic reforming, we shall consider the isomerization of *n*-pentane to *i*-pentane

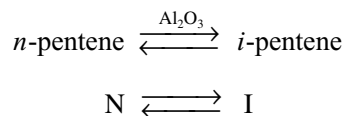


Gasoline	
C ₅	10%
C ₆	10%
C ₇	20%
C ₈	25%
C ₉	20%
C ₁₀	10%
C ₁₁₋₁₂	5%

Normal pentane has an octane number of 62, while *iso*-pentane, which is more compact, has an octane number of 90! The *n*-pentane adsorbs onto the platinum, where it is dehydrogenated to form *n*-pentene. The *n*-pentene desorbs from the platinum and adsorbs onto the alumina, where it is isomerized to *i*-pentene, which then desorbs and subsequently adsorbs onto platinum, where it is hydrogenated to form *i*-pentane. That is

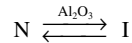


We shall focus on the isomerization step to develop the mechanism and the rate law



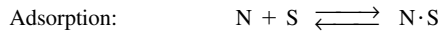
The procedure for formulating a mechanism, rate-limiting step, and corresponding rate law is given in Table 10-4.

TABLE 10-4 ALGORITHM FOR DETERMINING THE REACTION MECHANISM AND RATE-LIMITING STEP

Isomerization of *n*-pentene (N) to *i*-pentene (I) over alumina

Reforming
reaction to increase
octane number
of gasoline

Step 1. *Select a mechanism.* (Let's choose a Dual Site Mechanism)



Treat each reaction step as an elementary reaction when writing rate laws.

Step 2. *Assume a rate-limiting step.* We choose the surface reaction first, because *more than 75% of all heterogeneous reactions that are not diffusion-limited are surface-reaction-limited.* We note that the PSSH must be used when more than one step is limiting (see Section 10.3.6). The rate law for the surface reaction step is

$$-r'_N = r_S = k_S \left(C_v C_{\text{N} \cdot \text{S}} - \frac{C_{\text{I} \cdot \text{S}} C_v}{K_S} \right)$$

Step 3. *Find the expression for concentration of the adsorbed species $C_{i \cdot \text{S}}$.* Use the other steps that are not limiting to solve for $C_{i \cdot \text{S}}$ (e.g., $C_{\text{N} \cdot \text{S}}$ and $C_{\text{I} \cdot \text{S}}$). For this reaction

$$\text{From } \frac{r_{\text{AD}}}{k_A} \approx 0: \quad C_{\text{N} \cdot \text{S}} = P_N K_N C_v$$

$$\text{From } \frac{r_D}{k_D} \approx 0: \quad C_{\text{I} \cdot \text{S}} = \frac{P_I C_v}{K_D} = K_I P_I C_v$$

Step 4. *Write a site balance.*

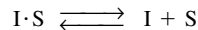
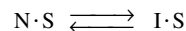
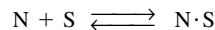
$$C_t = C_v + C_{\text{N} \cdot \text{S}} + C_{\text{I} \cdot \text{S}}$$

Step 5. *Derive the rate law.* Combine Steps 2, 3, and 4 to arrive at the rate law

$$-r'_N = r_S = \frac{\overbrace{k_S C_t^2 K_N}^k (P_N - P_I / K_P)}{(1 + K_N P_N + K_I P_I)^2}$$

Step 6. *Compare with data.* Compare the rate law derived in Step 5 with experimental data. If they agree, there is a good chance that you have found the correct mechanism and rate-limiting step. If your derived rate law (i.e., model) does not agree with the data:

- Assume a different rate-limiting step and repeat Steps 2 through 6.
- If, after assuming that each step is rate-limiting, none of the derived rate laws agrees with the experimental data, select a different mechanism (e.g., a single-site mechanism)



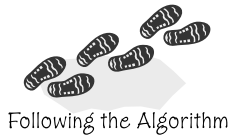
and then proceed through Steps 2 through 6.

The single-site mechanism turns out to be the correct one. For this mechanism, the rate law is

$$-r'_N = \frac{k(P_N - P_I / K_P)}{(1 + K_N P_N + K_I P_I)}$$

- If two or more models agree, the statistical tests discussed in Chapter 7 (e.g., comparison of residuals) should be used to discriminate between them (see the Supplementary Reading).

We note that in Table 10-4 for the dual-site mechanism, the denominator of the rate law for $-r'_A$ is squared (i.e., in Step 5 $[1/(\quad)^2]$), while for a single-site mechanism, it is not squared (i.e., Step 6 $[1/(\quad)]$). This fact is useful when analyzing catalyzed reactor data.



Following the Algorithm

Table 10-5 gives rate laws for different reaction mechanisms that are irreversible and surface-reaction-limited.

TABLE 10-5 IRREVERSIBLE SURFACE-REACTION-LIMITED RATE LAWS

Single site



Dual site



Eley-Rideal



We need a word of caution at this point. Just because the mechanism and rate-limiting step may fit the rate data does not imply that the mechanism is correct.¹³ Usually, spectroscopic measurements are needed to confirm a mechanism absolutely. However, the development of various mechanisms and rate-limiting steps can provide insight into the best way to correlate the data and develop a rate law.

10.3.6 Rate Laws Derived from the Pseudo-Steady-State Hypothesis (PSSH)

In Section 9.1 we discussed the PSSH, where the net rate of formation of *reactive intermediates* was assumed to be zero. An alternative way to derive a catalytic rate law rather than setting

$$\frac{r_{AD}}{k_A} \cong 0$$

is to assume that each species adsorbed on the surface is a *reactive intermediate*. Consequently, the net rate of formation of species *i* adsorbed on the surface will be zero

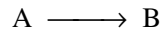
$$r_{i \cdot S}^* = 0 \quad (10-54)$$

The PSSH is primarily used when more than one step is rate-limiting. The isomerization example shown in Table 10-4 is reworked using the PSSH in the Chapter 10 Expanded Material on the CRE Web site.

¹³ R. I. Masel, *Principles of Adsorption and Reaction on Solid Surfaces* (New York: Wiley, 1996), p. 506, www.masel.com. This is a terrific book.

10.3.7 Temperature Dependence of the Rate Law

Consider a surface-reaction-limited irreversible isomerization



in which both A and B are adsorbed on the surface, and the rate law is

$$-r'_A = \frac{kP_A}{1 + K_A P_A + K_B P_B} \quad (10-55)$$

The specific reaction rate, k , will usually follow an Arrhenius temperature dependence and increase exponentially with temperature. However, the adsorption of all species on the surface is exothermic. Consequently, the higher the temperature, the smaller the adsorption equilibrium constant. That is, as the temperature increases, K_A and K_B decrease resulting in less coverage of the surface by A and B. Therefore, at high temperatures, the denominator of catalytic rate laws approaches 1. That is, at high temperatures (low coverage)

$$1 \gg (P_A K_A + P_B K_B)$$

The rate law could then be approximated as

$$-r'_A \simeq kP_A \quad (10-56)$$

or for a reversible isomerization we would have

$$-r'_A \simeq k \left(P_A - \frac{P_B}{K_P} \right) \quad (10-57)$$

Neglecting the
adsorbed species at
high temperatures

Algorithm

Deduce

Rate law

Find

Mechanism

Evaluate

Rate-law

parameters

Design

PBR

CSTR

The algorithm we can use as a start in postulating a reaction mechanism and rate-limiting step is shown in Table 10-4. Again, we can never really prove a mechanism to be correct by comparing the derived rate law with experimental data. Independent spectroscopic experiments are usually needed to confirm the mechanism. We can, however, prove that a proposed mechanism is *inconsistent* with the experimental data by following the algorithm in Table 10-4. Rather than taking all the experimental data and then trying to build a model from the data, Box et al. describe techniques of sequential data collection and model building.¹⁴

10.4 Heterogeneous Data Analysis for Reactor Design

In this section we focus on four operations that chemical reaction engineers need to be able to accomplish:

- (1) Developing an algebraic rate law consistent with experimental observations,
- (2) Analyzing the rate law in such a manner that the rate-law parameters (e.g., k , K_A) can readily be determined from the experimental data,

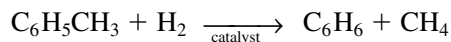


¹⁴ G. E. P. Box, W. G. Hunter, and J. S. Hunter, *Statistics for Engineers* (New York: Wiley, 1978).

- (3) Finding a mechanism and rate-limiting step consistent with the experimental data
- (4) Designing a catalytic reactor to achieve a specified conversion

We shall use the hydrodemethylation of toluene to illustrate these four operations.

Hydrogen and toluene are reacted over a solid mineral catalyst containing clinoptilolite (a crystalline silica-alumina) to form methane and benzene¹⁵



We wish to design a packed-bed reactor and a fluidized CSTR to process a feed consisting of 30% toluene, 45% hydrogen, and 25% inerts. Toluene is fed at a rate of 50 mol/min at a temperature of 640°C and a pressure of 40 atm (4052 kPa). To design the PBR, we must first determine the rate law from the differential reactor data presented in Table 10-6. In this table, we are given the rate of reaction of toluene as a function of the partial pressures of hydrogen (H_2), toluene (T), benzene (B), and methane (M). In the first two runs, methane was introduced into the feed together with hydrogen and toluene, while the other product, benzene, was fed to the reactor together with the reactants only in runs 3, 4, and 6. In runs 5 and 16, both methane and benzene were introduced in the feed. In the remaining runs, none of the products was present in the feedstream. Because the conversion was less than 1% in the differential reactor, the partial pressures of the products, methane and benzene, formed in these runs were essentially zero, and the reaction rates were equivalent to initial rates of reaction.

TABLE 10-6 DATA FROM A DIFFERENTIAL REACTOR

Run	$-r_1' \times 10^{10}$ $\left(\frac{\text{g mol toluene}}{\text{g-cat} \cdot \text{s}}\right)$	Partial Pressure (atm)			
		Toluene (T), P_{T}	Hydrogen (H ₂), P_{H_2}	Methane (M), P_{M}	Benzene (B), P_{B}
Set A					
1	71.0	1	1	1	0
2	71.3	1	1	4	0
Set B					
3	41.6	1	1	0	1
4	19.7	1	1	0	4
5	42.0	1	1	1	1
6	17.1	1	1	0	5
Set C					
7	71.8	1	1	0	0
8	142.0	1	2	0	0
9	284.0	1	4	0	0
Set D					
10	47.0	0.5	1	0	0
11	71.3	1	1	0	0
12	117.0	5	1	0	0
13	127.0	10	1	0	0
14	131.0	15	1	0	0
15	133.0	20	1	0	0
16	41.8	1	1	1	1

Unscramble the data
to find the
rate law

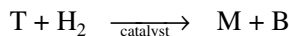
¹⁵ J. Papp, D. Kallo, and G. Schay, *J. Catal.*, 23, 168.

10.4.1 Deducing a Rate Law from the Experimental Data

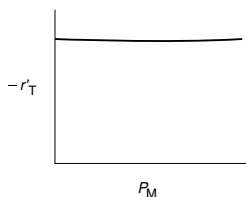


Following the Algorithm

First, let's look at run 3. In run 3, there is no possibility of the reverse reaction taking place because the concentration of methane is zero, i.e., $P_M = 0$, whereas in run 5 the reverse reaction could take place because all products are present. Comparing runs 3 and 5, we see that the initial rate is essentially the same for both runs, and we can assume that the reaction is essentially irreversible.



We now ask what qualitative conclusions can be drawn from the data about the dependence of the rate of disappearance of toluene, $-r'_T$, on the partial pressures of toluene, hydrogen, methane, and benzene.

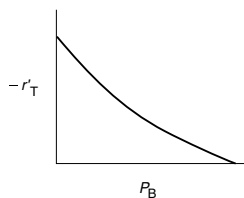


1. *Dependence on the product methane.* If methane were adsorbed on the surface, the partial pressure of methane would appear in the denominator of the rate expression and the rate would vary inversely with methane concentration

$$-r'_T \sim \frac{[\cdot]}{1 + K_M P_M + \dots} \quad (10-67)$$

However, comparing runs 1 and 2 we observe that a fourfold increase in the pressure of methane has little effect on $-r'_T$. Consequently, we assume that methane is either very weakly adsorbed (i.e., $K_M P_M \ll 1$) or goes directly into the gas phase in a manner similar to propylene in the cumene decomposition previously discussed.

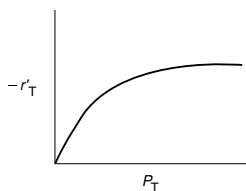
If it is in the denominator, it is probably on the surface.



2. *Dependence on the product benzene.* In runs 3 and 4, we observe that, for fixed concentrations (partial pressures) of hydrogen and toluene, the rate decreases with increasing concentration of benzene. A rate expression in which the benzene partial pressure appears in the denominator could explain this dependency

$$-r'_T \sim \frac{1}{1 + K_B P_B + \dots} \quad (10-68)$$

The type of dependence of $-r'_T$ on P_B given by Equation (10-68) suggests that benzene is adsorbed on the clinoptilolite surface.

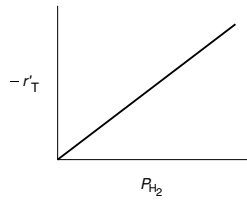


3. *Dependence on toluene.* At low concentrations of toluene (runs 10 and 11), the rate increases with increasing partial pressure of toluene, while at high toluene concentrations (runs 14 and 15), the rate is virtually independent of the toluene partial pressure. A form of the rate expression that would describe this behavior is

$$-r'_T \sim \frac{P_T}{1 + K_T P_T + \dots} \quad (10-69)$$

A combination of Equations (10-68) and (10-69) suggests that the rate law may be of the form

$$-r'_T \sim \frac{P_T}{1 + K_T P_T + K_B P_B + \dots} \quad (10-70)$$



4. *Dependence on hydrogen.* When we compare runs 7, 8, and 9 in Table 10-6, we see that the rate increases linearly with increasing hydrogen concentration, and we conclude that the reaction is first order in H_2 . In light of this fact, hydrogen is either not adsorbed on the surface or its coverage of the surface is extremely low ($1 \gg K_{H_2}P_{H_2}$) for the pressures used. If H_2 were adsorbed, $-r'_T$ would have a dependence on P_{H_2} analogous to the dependence of $-r'_T$ on the partial pressure of toluene, P_T [see Equation (10-69)]. For first-order dependence on H_2 ,

$$-r'_T \sim P_{H_2} \quad (10-71)$$

Combining Equations (10-67) through (10-71), we find that the rate law

$$-r'_T = \frac{kP_{H_2}P_T}{1 + K_B P_B + K_T P_T}$$

is in qualitative agreement with the data shown in Table 10-6.

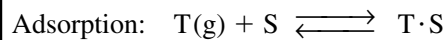
10.4.2 Finding a Mechanism Consistent with Experimental Observations

Approximately
75% of all
heterogeneous
reaction
mechanisms are
surface-reaction-
limited.

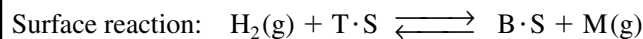
We now propose a mechanism for the hydrodemethylation of toluene. We assume that the reaction follows an *Eley-Rideal mechanism* where toluene is adsorbed on the surface and then reacts with hydrogen in the gas phase to produce benzene adsorbed on the surface and methane in the gas phase. Benzene is then desorbed from the surface. Because approximately 75% to 80% of all heterogeneous reaction mechanisms are surface-reaction-limited rather than adsorption- or desorption-limited, we begin by assuming the reaction between adsorbed toluene and gaseous hydrogen to be reaction-rate-limited. Symbolically, this mechanism and associated rate laws for each elementary step are

Proposed Mechanism

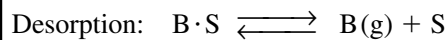
Eley-Rideal
mechanism



$$r_{AD} = k_A \left(C_v P_T - \frac{C_{T \cdot S}}{K_T} \right) \quad (10-72)$$



$$r_S = k_S \left(P_{H_2} C_{T \cdot S} - \frac{C_{B \cdot S} P_M}{K_S} \right) \quad (10-73)$$



$$r_D = k_D (C_{B \cdot S} - K_B P_B C_v) \quad (10-74)$$

For surface-reaction-limited mechanisms

$$r_S = k_S \left(P_{H_2} C_{T \cdot S} - \frac{C_{B \cdot S} P_M}{K_S} \right) \quad (10-73)$$

we see that we need to replace $C_{T \cdot S}$ and $C_{B \cdot S}$ in Equation (10-73) by quantities that we can measure.

For surface-reaction-limited mechanisms, we use the adsorption rate Equation (10-72) for toluene to obtain $C_{T \cdot S}$ ¹⁶, i.e.,

$$\frac{r_{AD}}{k_A} \approx 0$$

Then

$$\boxed{C_{T \cdot S} = K_T P_T C_v} \quad (10-75)$$

and we use the desorption rate Equation (10-74) for benzene to obtain $C_{B \cdot S}$:

$$\frac{r_D}{k_D} \approx 0$$

Then

$$\boxed{C_{B \cdot S} = K_B P_B C_v} \quad (10-76)$$

The total concentration of sites is

Perform a site
balance to obtain C_v .

$$\boxed{C_t = C_v + C_{T \cdot S} + C_{B \cdot S}} \quad (10-77)$$

Substituting Equations (10-75) and (10-76) into Equation (10-77) and rearranging, we obtain

$$C_v = \frac{C_t}{1 + K_T P_T + K_B P_B} \quad (10-78)$$

Next, substitute for $C_{T \cdot S}$ and $C_{B \cdot S}$, and then substitute for C_v in Equation (10-73) to obtain the rate law for the case when the reaction is surface-reaction-rate-limited

$$-r'_T = \frac{\overbrace{C_t k_S K_T}^k (P_{H_2} P_T - P_B P_M / K_P)}{1 + K_T P_T + K_B P_B} \quad (10-79)$$

We have shown by comparing runs 3 and 5 that we can neglect the reverse reaction, i.e., the thermodynamic equilibrium constant K_P is very, very large. Consequently, we obtain

Rate law for
Eley-Rideal
surface-reaction-
limited mechanism

$$\boxed{-r'_T = \frac{k P_{H_2} P_T}{1 + K_B P_B + K_T P_T}} \quad (10-80)$$

Again we note that the adsorption equilibrium constant of a given species is exactly the reciprocal of the desorption equilibrium constant of that species.

10.4.3 Evaluation of the Rate-Law Parameters

In the original work on this reaction by Papp et al.,¹⁷ over 25 models were tested against experimental data, and it was concluded that the preceding

¹⁶ See footnote 10 on page 424.

¹⁷ Ibid.

mechanism and rate-limiting step (i.e., the surface reaction between adsorbed toluene and H_2 gas) is the correct one. Assuming that the reaction is essentially irreversible, the rate law for the reaction on clinoptilolite is

$$-r'_T = k \frac{P_{H_2} P_T}{1 + K_B P_B + K_T P_T} \quad (10-80)$$

We now wish to determine how best to analyze the data to evaluate the rate-law parameters, k , K_T , and K_B . This analysis is referred to as *parameter estimation*.¹⁸ We now rearrange our rate law to obtain a linear relationship between our measured variables. For the rate law given by Equation (10-80), we see that if both sides of Equation (10-80) are divided by $P_{H_2} P_T$ and the equation is then inverted

Linearize the rate equation to extract the rate-law parameters.

$$\frac{P_{H_2} P_T}{-r'_T} = \frac{1}{k} + \frac{K_B P_B}{k} + \frac{K_T P_T}{k} \quad (10-81)$$

A linear least-squares analysis of the data shown in Table 10-6 is presented on the CRE Web site.

The regression techniques described in Chapter 7 could be used to determine the rate-law parameters by using the equation

$$Y_j = a_0 + a_1 X_{1j} + a_2 X_{2j}$$

One can use the linearized least-squares analysis (PRS 7.3) to obtain initial estimates of the parameters k , K_T , K_B , in order to obtain convergence in nonlinear regression. However, in many cases it is possible to use a nonlinear regression analysis directly, as described in Sections 7.5 and 7.6, and in Example 10-1.

Example 10-1 Nonlinear Regression Analysis to Determine the Model Parameters k , K_B , and K_T

- Use nonlinear regression, as discussed in Chapter 7, along with the data in Table 10-6, to find the best estimates of the rate-law parameters k , K_B , and K_T in Equation (10-80).
- Write the rate law solely as a function of the partial pressures.
- Find the ratio of the sites occupied by toluene, $C_{T,s}$, to those occupied by benzene, $C_{B,s}$, at 40% conversion of toluene.

Solution

The data from Table 10-6 were entered into the Polymath nonlinear least-squares program with the following modification. The rates of reaction in column 1 were multiplied by 10^{10} , so that each of the numbers in column 1 was entered directly (i.e., 71.0, 71.3, ...). The model equation was

$$\text{Rate} = \frac{k P_T P_{H_2}}{1 + K_B P_B + K_T P_T} \quad (\text{E10-1.1})$$



Living Example Problem

¹⁸ See the Supplementary Reading (page 492) for a variety of techniques for estimating the rate-law parameters.

Following the step-by-step regression procedure in Chapter 7 and on the CRE Web site *Summary Notes*, we arrive at the following parameter values shown in Table E10-1.1.

TABLE E10-1.1 PARAMETER VALUES

Source data points and calculated data points							Model: RATE = k*PT*PH2/(1+KB*PB+KT*PT)			
	PT	PH2	PB	RATE	RATE calc	Delta RATE	Variable	Initial guess	Value	95% confidence
1	1	1	0	71	71.0197	-0.0196996	k	144.	144.7673	1.240307
2	1	1	0	71.3	71.0197	0.2803004	KB	1.4	1.390525	0.0457965
3	1	1	1	41.6	42.21931	-0.6193089	KT	1.03	1.038411	0.0131585
4	1	1	4	19.7	19.04705	0.6529537	Nonlinear regression settings Max # iterations = 64			
5	1	1	1	42	42.21931	-0.2193089				
6	1	1	5	17.1	16.10129	0.9987095	Precision			
7	1	1	0	71.8	71.0197	0.7803004				
8	1	2	0	142	142.0394	-0.0393992	R^2		0.9999509	
9	1	4	0	284	284.0788	-0.0787985	R^2adj		0.9999434	
10	0.5	1	0	47	47.64574	-0.6457351	Rmsd		0.1128555	
11	1	1	0	71.3	71.0197	0.2803004	Variance		0.2508084	
12	5	1	0	117	116.8977	0.102331				
13	10	1	0	127	127.1662	-0.1661677				
14	15	1	0	131	131.002	-0.0019833				
15	20	1	0	133	133.008	-0.007997				
16	1	1	1	41.8	42.21931	-0.4193089				

- (a) The best estimates are shown in the upper-right-hand box of Table E10-1.1.
 (b) Converting the rate law to kilograms of catalyst and minutes,

$$-r'_T = \frac{1.45 \times 10^{-8} P_T P_{H_2}}{1 + 1.39 P_B + 1.038 P_T} \frac{\text{mol T}}{\text{g-cat} \cdot \text{s}} \times \frac{1000 \text{ g}}{1 \text{ kg}} \times \frac{60 \text{ s}}{\text{min}} \quad (\text{E10-1.2})$$

we have

$$-r'_T = \frac{8.7 \times 10^{-4} P_T P_{H_2}}{1 + 1.39 P_B + 1.038 P_T} \left(\frac{\text{g mol T}}{\text{kg-cat} \cdot \text{min}} \right) \quad (\text{E10-1.3})$$

Ratio of sites
occupied by toluene
to those occupied
by benzene

- (c) After we have the adsorption constants, K_T and K_B , we can calculate the ratio of sites occupied by the various adsorbed species. For example, taking the ratio of Equation (10-75) to Equation (10-76), the ratio of toluene-occupied sites to benzene-occupied sites at 40% conversion is

$$\begin{aligned} \frac{C_{T \cdot S}}{C_{B \cdot S}} &= \frac{C_v K_T P_T}{C_v K_B P_B} = \frac{K_T P_T}{K_B P_B} = \frac{K_T P_{A0} (1-X)}{K_B P_{A0} X} \\ &= \frac{K_T (1-X)}{K_B X} = \frac{1.038 (1-0.4)}{1.39 (0.4)} = 1.12 \end{aligned}$$

We see that at 40% conversion there are approximately 12% more sites occupied by toluene than by benzene. This fact is *common knowledge* to every chemical engineering student at Jofostan University, Riça, Jofostan.

Analysis: This example shows once again how to determine the values of rate-law parameters from experimental data using Polymath regression. It also shows how to calculate the different fraction of sites, both vacant and occupied, as a function of conversion.

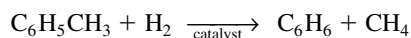
10.4.4 Reactor Design

Our next step is to express the partial pressures P_T , P_B , and P_{H_2} as a function of X , combine the partial pressures with the rate law, $-r'_A$, as a function of conversion, and carry out the integration of the packed-bed design equation

$$\frac{dX}{dW} = \frac{-r'_A}{F_{A0}} \quad (2-17)$$

Example 10-2 Catalytic Reactor Design

The hydrodemethylation of toluene is to be carried out in a PBR catalytic reactor.



The molar feed rate of toluene to the reactor is 50 mol/min, and the reactor inlet is at 40 atm and 640°C. The feed consists of 30% toluene, 45% hydrogen, and 25% inerts. Hydrogen is used in excess to help prevent coking. The pressure-drop parameter, α , is $9.8 \times 10^{-5} \text{ kg}^{-1}$.

- Plot and analyze the conversion, the pressure ratio, p , and the partial pressures of toluene, hydrogen, and benzene as a function of PBR catalyst weight.
- Determine the catalyst weight in a fluidized CSTR with a bulk density of 400 kg/m^3 (0.4 g/cm^3) to achieve 65% conversion.

Solution**(a) PBR with pressure drop****1. Mole Balance:**

Balance on
toluene (T)

$$\frac{dF_T}{dW} = r'_T$$

$$\frac{dX}{dW} = \frac{-r'_T}{F_{T0}} \quad (\text{E10-2.1})$$

2. Rate Law: From Equation (E10-1.1) we have

$$-r'_T = \frac{kP_{H_2}P_T}{1 + K_BP_B + K_TP_T} \quad (\text{E10-2.2})$$

with $k = 0.00087 \text{ mol/atm}^2/\text{kg-cat/min}$, $K_B = 1.39 \text{ atm}^{-1}$, and $K_T = 1.038 \text{ atm}^{-1}$.

3. Stoichiometry:

$$P_T = C_T RT = C_{T0} RT_0 \left(\frac{1-X}{1+\varepsilon X} \right) p = P_{T0} \left(\frac{1-X}{1+\varepsilon X} \right) p$$

$$\varepsilon = y_{T0}\delta = 0.3(0) = 0$$

$$p = \frac{P}{P_0}$$

$$P_T = P_{T0}(1-X)p \quad (\text{E10-2.3})$$

$$P_{H_2} = P_{T0}(\Theta_{H_2} - X)p$$

$$\Theta_{H_2} = \frac{0.45}{0.30} = 1.5$$



Relating
Toluene (T)
Benzene (B)
Hydrogen (H_2)

P_0 = total pressure
at the entrance

Pressure drop in
PBRs is discussed in
Section 5.5.

$$P_{H_2} = P_{T0}(1.5 - X)p \quad (\text{E10-2.4})$$

$$P_B = P_{T0}Xp \quad (\text{E10-2.5})$$

Because $\varepsilon = 0$, we can use the integrated form of the pressure-drop term.

$$p = \frac{P}{P_0} = (1 - \alpha W)^{1/2} \quad (5-33)$$

$$\alpha = 9.8 \times 10^{-5} \text{ kg}^{-1}$$

Note that P_{T0} designates the inlet partial pressure of toluene. In this example, the inlet total pressure is designated P_0 to avoid any confusion. The inlet mole fraction of toluene is 0.3 (i.e., $y_{T0} = 0.3$), so that the inlet partial pressure of toluene is

$$P_{T0} = (0.3)(40) = 12 \text{ atm}$$

We now calculate the maximum catalyst weight we can have, such that the exiting pressure will not fall below atmospheric pressure (i.e., 1.0 atm) for the specified feed rate. This weight is calculated by substituting the entering pressure of 40 atm and the exiting pressure of 1 atm into Equation (5-33), i.e.,

$$p = \frac{P}{P_0} = (1 - \alpha W)^{1/2}$$

$$\frac{1}{40} = (1 - 9.8 \times 10^{-5} W)^{1/2}$$

$$W = 10,197 \text{ kg}$$

4. Evaluate: Consequently, we will set our final weight at 10,000 kg and determine the conversion profile as a function of catalyst weight up to this value. Equations (E10-2.1) through (E10-2.5) are shown in the Polymath program in Table E10-2.1. The conversion is shown as a function of catalyst weight in Figure E10-2.1, and profiles of the partial pressures of toluene, hydrogen, and benzene are shown in Figure E10-2.2. We note that the pressure drop causes (cf. Equation E10-2.5) the partial pressure of benzene to go through a maximum as one traverses the reactor.

TABLE E10-2.1 POLYMATH PROGRAM AND OUTPUT

Differential equations		Calculated values of DEQ variables		
1	$d(X)/d(w) = -rt/FT_0$	Variable	Initial value	Final value
Explicit equations		1	alpha	9.8E-05
1	$FT_0 = 50$	2	FT_0	50.
2	$k = .00087$	3	k	0.00087
3	$KT = 1.038$	4	KB	1.39
4	$KB = 1.39$	5	KT	1.038
5	$\alpha = 0.000098$	6	P	40.
6	$P_0 = 40$	7	PB	0
7	$PT_0 = 0.3 \cdot P_0$	8	PH2	18.
8	$p = (1 - \alpha w)^{0.5}$	9	Po	40.
9	$P = y \cdot P_0$	10	PT	12.
10	$PH_2 = PT_0 \cdot (1.5 - X) \cdot y$	11	PTo	12.
11	$PB = PT_0 \cdot X \cdot y$	12	RATE	0.0139655
12	$PT = PT_0 \cdot (1 - X) \cdot y$	13	rt	-0.0139655
13	$rt = -k \cdot PT \cdot PH_2 / (1 + KB \cdot PB + KT \cdot PT)$	14	w	0
14	$RATE = -rt$	15	X	0
		16	P	1.
				0.1414214



Living Example Problem

Conversion profile
down the
packed bed

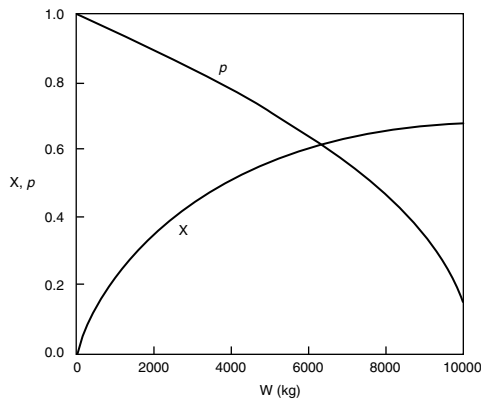


Figure E10-2.1 Conversion and pressure ratio profiles.

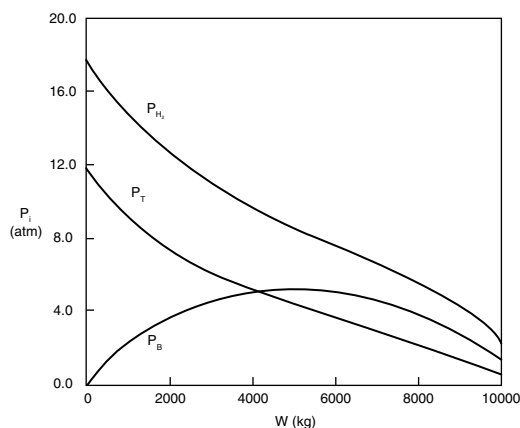


Figure E10-2.2 Partial pressure profiles.

Note the partial pressure of benzene goes through a maximum. Why?

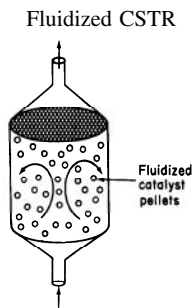
If one had neglected ΔP it could have been very embarrassing.

See YouTube video, "Reaction Engineering Gone Wrong," accessible from the CRE Web site home page.

PBR Analysis: For the case of no pressure drop, the conversion that would have been achieved with 10,000 kg of catalyst would have been 79%, compared with 68.2% when there is pressure drop in the reactor. To carry out this calculation, use the Living Example Program (LEP) on the CRE Web site and simply multiply the pressure-drop parameter by zero, i.e., line (5) would read $\alpha = 0.00098 \times 0$. For the feed rate given, eliminating or minimizing pressure drop would increase the production of benzene by up to 61 million pounds per year! Finally, we note in Figure E10-2.2 that the partial pressure of benzene (P_B) goes through a maximum. This maximum can be explained by recalling that P_B is just the product of the benzene mole fraction (y_B) times the total pressure (P) [i.e., $P_B = y_B P_T$]. Near the middle to end of the bed, benzene is no longer being formed so that y_B stops increasing. However, because of the pressure drop, the total pressure decreases and, as a result, so does P_B .

(b) Fluidized CSTR

We will now calculate the fluidized CSTR catalyst weight necessary to achieve the same (ca.) conversion as in the packed-bed reactor at the same operating conditions. The bulk density in the fluidized reactor is 0.4 g/cm^3 . The design equation is

**1. Mole Balance:**

In	–	Out	+	Gen	=	Accum
F_{T0}	–	F_T	+	$r'_T W$	=	0

Rearranging

$$W = \frac{F_{T0} - F_T}{-r'_T} = \frac{F_{T0} X}{-r'_T} \quad (\text{E10-2.6})$$

2. Rate Law and 3. Stoichiometry same as in part (a) PBR calculation**4. Combine and Evaluate:** Writing Equation (E10-2.2) in terms of conversion (E10-2.3) and then substituting $X = 0.65$ and $P_{T0} = 12$ atm, we have

$$-r'_T = \frac{8.7 \times 10^{-4} P_T P_{H_2}}{1 + 1.39 P_B + 1.038 P_T} = \frac{8.7 \times 10^{-4} P_{T0}^2 (1-X)(1.5-X)}{1 + 1.39 P_{T0} X + 1.038 P_{T0} (1-X)} = 2.3 \times 10^{-3} \frac{\text{mol}}{\text{kg-cat} \cdot \text{min}}$$

$$W = \frac{F_{T0} X}{-r'_T} = \frac{(50 \text{ mol T/min})(0.65)}{2.3 \times 10^{-3} \text{ mol T/kg cat} \cdot \text{min}}$$

$$W = 1.41 \times 10^4 \text{ kg of catalyst}$$

The corresponding reactor volume is

$$V = \frac{W}{\rho_b} = \frac{1.41 \times 10^4 \text{ kg}}{400 \text{ kg/m}^3} = 35.25 \text{ m}^3$$

Analysis: This example used real data and the CRE algorithm to design a PBR and CSTR. An important point is that it showed how one could be embarrassed by not including pressure drop in the design of a packed-bed reactor. We also note that for both the PBR and fluidized CSTR, the values of the catalyst weight and reactor volume are quite high, especially for the low feed rates given. *Consequently, the temperature of the reacting mixture should be increased to reduce the catalyst weight, provided that side reactions and catalyst decay do not become a problem at higher temperatures.*

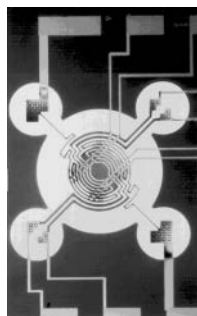
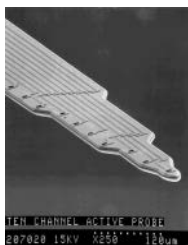
How can the weight of catalyst be reduced?
Raise the temperature?

Example 10-2 illustrated the major activities pertinent to catalytic reactor design described earlier in Figure 10-6. In this example, the rate law was extracted directly from the data and then a mechanism was found that was consistent with experimental observation. Conversely, developing a feasible mechanism may guide one in the synthesis of the rate law.

10.5 Reaction Engineering in Microelectronic Fabrication

10.5.1 Overview

We now extend the principles of the preceding sections to one of the emerging technologies in chemical engineering. Chemical engineers are now playing an



Chemical engineering principles are involved in virtually every step!

important role in the electronics industry. Specifically, they are becoming more involved in the manufacture of electronic and photonic devices, recording materials, and especially medical lab-on-a-chip devices.

Surface reactions play an important role in the manufacture of microelectronic devices. One of the single most important developments of the twentieth century was the invention of the integrated circuit. Advances in the development of integrated circuitry have led to the production of circuits that can be placed on a single semiconductor chip the size of a pinhead and perform a wide variety of tasks by controlling the electron flow through a vast network of channels. These channels, which are made from semiconductors such as silicon, gallium arsenide, indium phosphide, and germanium, have led to the development of a multitude of novel microelectronic devices. Examples of microelectronic sensing devices manufactured using chemical reaction engineering principles are shown in the left-hand margin.

The manufacture of an integrated circuit requires the fabrication of a network of pathways for electrons. The principal reaction engineering steps of the fabrication process include depositing material on the surface of a material called a *substrate* (e.g., by chemical vapor deposition, abbreviated as CVD), changing the conductivity of regions of the surface (e.g., by boron doping or ion implantation), and removing unwanted material (e.g., by etching). By applying these steps systematically, miniature electronic circuits can be fabricated on very small semiconductor chips. The fabrication of microelectronic devices may include as few as 30 or as many as 200 individual steps to produce chips with up to 10^9 elements per chip.

An abbreviated schematic of the steps involved in producing a typical metal-oxide, semiconductor, field-effect transistor (MOSFET) device is shown in Figure 10-20. Starting from the upper left, we see that single-crystal silicon ingots are grown in a Czochralski crystallizer, sliced into wafers, and chemically and physically polished. These polished wafers serve as starting materials for a variety of microelectronic devices. A typical fabrication sequence is shown for processing the wafer, beginning with the formation of an SiO_2 layer on top of the silicon. The SiO_2 layer may be formed either by oxidizing a silicon layer or by laying down a SiO_2 layer by chemical vapor deposition (CVD). Next, the wafer is masked with a polymer photoresist, a template with the pattern to be etched onto the SiO_2 layer is placed over the photoresist, and the wafer is exposed to ultraviolet irradiation. If the mask is a positive photoresist, the light will cause the exposed areas of the polymer to dissolve when the wafer is placed in the developer. On the other hand, when a negative photoresist mask is exposed to ultraviolet irradiation, cross-linking of the polymer chains occurs, and the *unexposed* areas dissolve in the developer. The undeveloped portion of the photoresist (in either case) will protect the covered areas from etching.

After the exposed areas of SiO_2 are etched to form trenches (either by wet etching or by plasma etching), the remaining photoresist is removed. Next, the wafer is placed in a furnace containing gas molecules of the desired dopant, which then diffuse into the exposed silicon. After diffusion of dopant to the desired depth in the wafer, the wafer is removed and then SiO_2 is removed by etching. The sequence of masking, etching, CVD, and metallization continues until the desired device is formed. A schematic of a final chip is

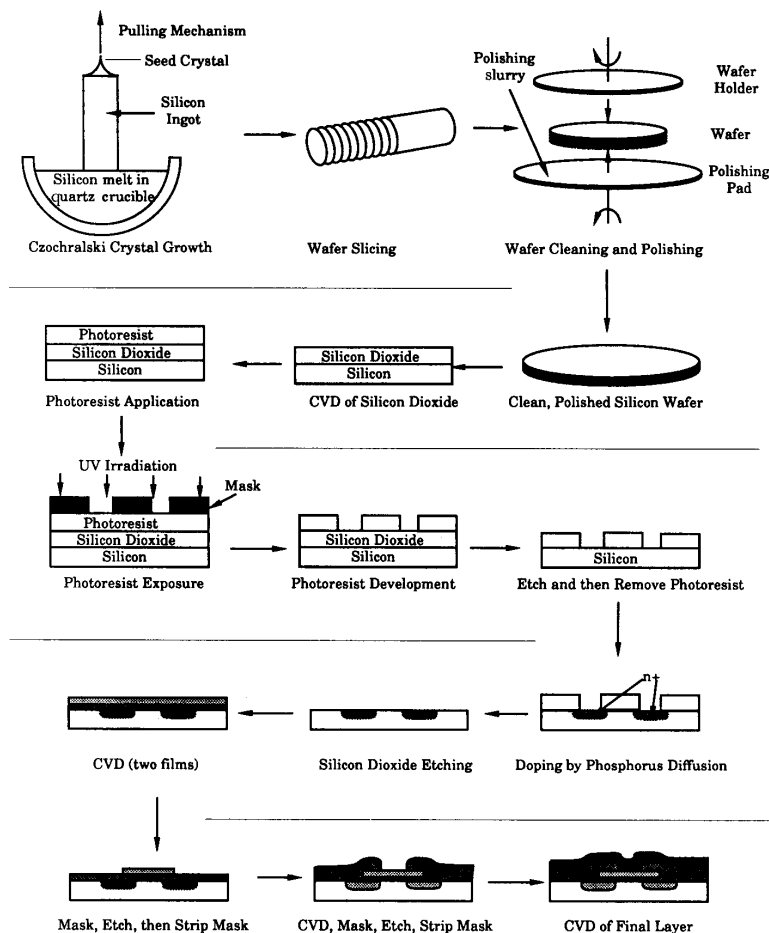


Figure 10-20 Microelectronic fabrication steps.

shown in the lower-right-hand corner of Figure 10-20. In Section 10.5.2, we discuss one of the key processing steps, CVD.

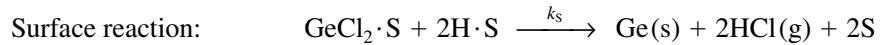
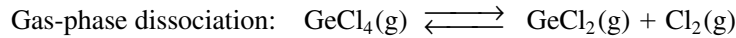
10.5.2 Chemical Vapor Deposition

The mechanisms by which CVD occurs are very similar to those of heterogeneous catalysis discussed earlier in this chapter. The reactant(s) adsorbs on the surface and then reacts on the surface to form a new surface. This process may be followed by a desorption step, depending on the particular reaction.

The growth of a germanium epitaxial film as an interlayer between a gallium arsenide layer and a silicon layer has received attention in the microelectronics industry.¹⁹ Epitaxial germanium is also an important material in the fabrication of tandem solar cells. The growth of germanium films can be accomplished by CVD. A proposed mechanism is

Ge used in
solar cells

¹⁹ H. Ishii and Y. Takahashi, *J. Electrochem. Soc.*, 135, p. 1539.



At first it may appear that a site has been lost when comparing the right- and left-hand sides of the surface reaction step. However, the newly formed germanium atom on the right-hand side is a site for the future adsorption of $\text{H}_2(\text{g})$ or $\text{GeCl}_2(\text{g})$, and there are three sites on both the right- and left-hand sides of the surface reaction step. These sites are shown schematically in Figure 10-21.

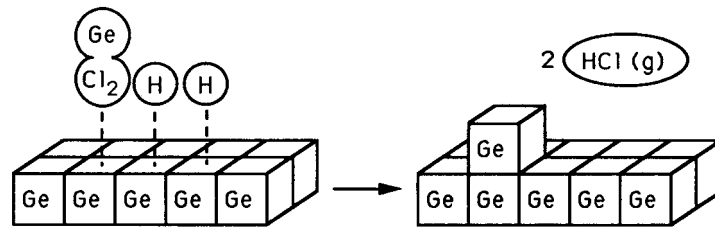


Figure 10-21 CVD surface reaction step for germanium.

The surface reaction between adsorbed molecular hydrogen and germanium dichloride is believed to be rate-limiting. The reaction follows an elementary rate law with the rate being proportional to the fraction of the surface covered by GeCl_2 times the square of the fraction of the surface covered by molecular hydrogen.

Rate law for rate-limiting step

$$r''_{\text{Dep}} = k_s f_{\text{GeCl}_2} f_{\text{H}}^2 \quad (10-82)$$

where r''_{Dep} = deposition rate per unit surface area, nm/s

k_s = surface specific reaction rate, nm/s

f_{GeCl_2} = fraction of the surface occupied by germanium dichloride

f_{H} = fraction of the surface covered by molecular hydrogen

The deposition rate (film growth rate) is usually expressed in nanometers per second and is easily converted to a molar rate ($\text{mol}/\text{m}^2 \cdot \text{s}$) by multiplying by the molar density of solid germanium (mol/m^3).

The difference between developing CVD rate laws and rate laws for catalysis is that the site concentration (e.g., C_v) is replaced by the fractional surface area coverage (e.g., the fraction of the surface that is vacant, f_v). The total fraction of surface available for adsorption should, of course, add up to 1.0.

Area balance

Fractional area balance:

$$f_v + f_{\text{GeCl}_2} + f_{\text{H}} = 1 \quad (10-83)$$

We will first focus our attention on the adsorption of GeCl_2 . The rate of jumping on to the surface is proportional to the partial pressure of GeCl_2 , P_{GeCl_2} , and the fraction of the surface that is vacant, f_v . The net rate of GeCl_2 adsorption is

$$r_{\text{AD}} = k_A \left(f_v P_{\text{GeCl}_2} - \frac{f_{\text{GeCl}_2}}{K_A} \right) \quad (10-84)$$

Since the surface reaction is rate-limiting, in a manner analogous to catalysis reactions, we have for the adsorption of GeCl_2

**Adsorption of
 GeCl_2 not
rate-limiting**

$$\frac{r_{\text{AD}}}{k_A} \approx 0$$

Solving Equation (10-84) for the fractional surface coverage of GeCl_2 gives

$$f_{\text{GeCl}_2} = f_v K_A P_{\text{GeCl}_2} \quad (10-85)$$

For the dissociative adsorption of hydrogen on the Ge surface, the equation analogous to Equation (10-84) is

$$r_{\text{ADH}_2} = k_H \left(P_{\text{H}_2} f_v^2 - \frac{f_{\text{H}}^2}{K_H} \right) \quad (10-86)$$

Since the surface reaction is rate-limiting

**Adsorption of H_2 is
not rate-limiting**

$$\frac{r_{\text{ADH}_2}}{k_H} \approx 0$$

Then

$$f_{\text{H}} = f_v \sqrt{K_H P_{\text{H}_2}} \quad (10-87)$$

Recalling the rate of deposition of germanium, we substitute for f_{GeCl_2} and f_{H} in Equation (10-82) to obtain

$$r''_{\text{Dep}} = f_v^3 k_S K_A P_{\text{GeCl}_2} K_H P_{\text{H}_2} \quad (10-88)$$

We solve for f_v in an identical manner to that for C_v in heterogeneous catalysis. Substituting Equations (10-85) and (10-87) into Equation (10-83) gives

$$f_v + f_v \sqrt{K_H P_{\text{H}_2}} + f_v K_A P_{\text{GeCl}_2} = 1$$

Rearranging yields

$$f_v = \frac{1}{1 + K_A P_{\text{GeCl}_2} + \sqrt{K_H P_{\text{H}_2}}} \quad (10-89)$$

Finally, substituting for f_v in Equation (10-88), we find that

$$r''_{\text{Dep}} = \frac{k_s K_H K_A P_{\text{GeCl}_2} P_{\text{H}_2}}{(1 + K_A P_{\text{GeCl}_2} + \sqrt{K_H P_{\text{H}_2}})^3}$$

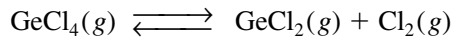
and lumping K_A , K_H , and k_s into a specific reaction rate k' yields

**Rate of deposition
of Ge**

$$r''_{\text{Dep}} = \frac{k' P_{\text{GeCl}_2} P_{\text{H}_2}}{(1 + K_A P_{\text{GeCl}_2} + \sqrt{K_H P_{\text{H}_2}})^3} \quad (10-90)$$

We now need to relate the partial pressure of GeCl_2 to the partial pressure of GeCl_4 in order to calculate the conversion of GeCl_4 . If we assume that the gas-phase reaction

**Equilibrium in
gas phase**



is in equilibrium, we have

$$K_p = \frac{P_{\text{GeCl}_2} P_{\text{Cl}_2}}{P_{\text{GeCl}_4}}$$

$$P_{\text{GeCl}_2} = \frac{P_{\text{GeCl}_4}}{P_{\text{Cl}_2}} \cdot K_p$$

and if hydrogen is weakly adsorbed ($\sqrt{K_H P_{\text{H}_2}} < 1$), we obtain the rate of deposition as

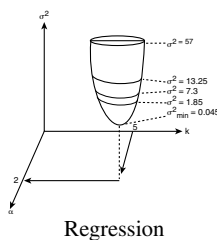
**Rate of deposition
of Ge when H_2 is
weakly adsorbed**

$$r''_{\text{Dep}} = \frac{k P_{\text{GeCl}_4} P_{\text{H}_2} P_{\text{Cl}_2}^2}{(P_{\text{Cl}_2} + K_p P_{\text{GeCl}_4})^3} \quad (10-91)$$

We now can use stoichiometry to express each of the species' partial pressures in terms of conversion and the entering partial pressure of GeCl_4 , $P_{\text{GeCl}_4,0}$, and then proceed to calculate the conversion.

It should also be noted that it is possible that GeCl_2 may also be formed by the reaction of GeCl_4 and a Ge atom on the surface, in which case a different rate law would result.

10.6 Model Discrimination



We have seen that for each mechanism and each rate-limiting step we can derive a rate law. Consequently, if we had three possible mechanisms and three rate-limiting steps for each mechanism, we would have nine possible rate laws to compare with the experimental data. We will use the regression techniques discussed in Chapter 7 to identify which model equation best fits the data by choosing the one with the smaller sums of squares and/or carrying out an F-test. We could also compare the residual plots for each model, which not only show the error associated with each data point but also show if the error

is randomly distributed or if there is a trend in the error. If the error is randomly distributed, this result is an additional indication that the correct rate law has been chosen.



We need to raise a caution here about choosing the model with the smallest sums of squares. The caution is that the model parameter values that give the smallest sum must be realistic. In the case of heterogeneous catalysis, *all values of the adsorption equilibrium constants must be positive*. In addition, if the temperature dependence is given, because adsorption is exothermic, the adsorption equilibrium constant must decrease with increasing temperature. To illustrate these principles, let's look at the following example.

Example 10-3 Hydrogenation of Ethylene to Ethane

The hydrogenation (H) of ethylene (E) of form ethane (EA),



is carried out over a cobalt molybdenum catalyst [*Collect. Czech. Chem. Commun.*, 51, 2760 (1988)]. Carry out a nonlinear regression analysis on the data given in Table E10-3.1, and determine which rate law best describes the data.

TABLE E10-3.1 DIFFERENTIAL REACTOR DATA

Run Number	Reaction Rate (mol/kg-cat. · s)	P_E (atm)	P_{EA} (atm)	P_H (atm)
1	1.04	1	1	1
2	3.13	1	1	3
3	5.21	1	1	5
4	3.82	3	1	3
5	4.19	5	1	3
6	2.391	0.5	1	3
7	3.867	0.5	0.5	5
8	2.199	0.5	3	3
9	0.75	0.5	5	1

Procedure

- Enter data
- Enter model
- Make initial estimates of parameters
- Run regression
- Examine parameters and variance
- Observe error distribution
- Choose model

Determine which of the following rate laws best describes the data in Table E10-3.1.

$$(a) \quad -r'_E = \frac{kP_E P_H}{1 + K_{EA}P_{EA} + K_E P_E}$$

$$(c) \quad -r'_E = \frac{kP_E P_H}{(1 + K_E P_E)^2}$$

$$(b) \quad -r'_E = \frac{kP_E P_H}{1 + K_E P_E}$$

$$(d) \quad -r'_E = kP_E^a P_H^b$$

Solution

Polymath was chosen as the software package to solve this problem. The data in Table E10-3.1 were entered into the program. A screen-shot-by-screen-shot set of instructions on how to carry out the regression is given on the CRE Web site, at the end of the *Summary Notes* for Chapter 7. After entering the data and following the step-by-step procedures, the results shown in Table E10-3.2 were obtained.

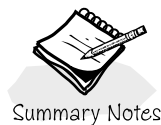




TABLE E10-3.2 RESULTS OF THE POLYMATH NONLINEAR REGRESSION

Model (a)			Model (b)		
Model: RATE = $k \cdot P_E \cdot P_H^2 / (1 + K_{EA} \cdot P_{EA} + K_E \cdot P_E)$			Model: RATE = $k \cdot P_E \cdot P_H^2 / (1 + K_E \cdot P_E)$		
<u>Variable</u>	<u>Value</u>	<u>95% confidence</u>	<u>Variable</u>	<u>Value</u>	<u>95% confidence</u>
k	3.3478805	0.2922517	k	3.1867851	0.287998
K _{EA}	0.0428419	0.0636262	K _E	2.1013363	0.2638835
K _E	2.2110797	0.2392585			
Nonlinear regression settings Max # iterations = 64			Nonlinear regression settings Max # iterations = 64		
Precision R ² = 0.998321 R ² adj = 0.9977614 Rmsd = 0.0191217 Variance = 0.0049361			Precision R ² = 0.9975978 R ² adj = 0.9972547 Rmsd = 0.022872 Variance = 0.0060534		
Model (c)			Model (d)		
Model: RATE = $k \cdot P_E \cdot P_H^2 / (1 + K_E \cdot P_E)^2$			Model: RATE = $k \cdot P_E \cdot a \cdot P_H^2 \cdot b$		
<u>Variable</u>	<u>Value</u>	<u>95% confidence</u>	<u>Variable</u>	<u>Value</u>	<u>95% confidence</u>
k	2.0087761	0.2661838	k	0.8940237	0.2505474
K _E	0.3616652	0.0623045	a	0.2584412	0.0704628
			b	1.0615542	0.2041339
Nonlinear regression settings Max # iterations = 64			Nonlinear regression settings Max # iterations = 64		
Precision R ² = 0.9752762 R ² adj = 0.9717442 Rmsd = 0.0733772 Variance = 0.0623031			Precision R ² = 0.9831504 R ² adj = 0.9775338 Rmsd = 0.0605757 Variance = 0.0495372		

Model (a) Single site, surface-reaction, rate-limited with hydrogen weakly adsorbed

From Table E10-3.2 data, we can obtain

$$-r'_E = \frac{3.348 P_E P_H}{1 + 0.043 P_{EA} + 2.21 P_E} \quad (\text{E10-3.1})$$

We now examine the sums of squares (variance) and range of variables themselves. The sums of squares is reasonable and, in fact, the smallest of all the models at 0.0049. *However*, let's look at K_{EA} . We note that the value for the 95% confidence limit of ± 0.0636 is greater than the nominal value of $K_{EA} = 0.043 \text{ atm}^{-1}$ itself (i.e., $K_{EA} = 0.043 \pm 0.0636$). The 95% confidence limit means that if the experiment were run 100 times and then 95 times it would fall within the range $(-0.021) < K_{EA} < (0.1066)$. Because K_{EA} can never be negative, we are going to reject this model. Consequently, we set $K_{EA} = 0$ and proceed to Model (b).

Model (b) Single site, surface-reaction, rate-limited with ethane and hydrogen weakly adsorbed

From Table E10-3.2 we can obtain

$$-r'_E = \frac{3.187 P_E P_H}{1 + 2.1 P_E} \quad (\text{E10-3.2})$$

The value of the adsorption constant $K_E = 2.1 \text{ atm}^{-1}$ is reasonable and is not negative within the 95% confidence limit. Also, the variance is small at $\sigma_B^2 = 0.0061$.

Model (c) Dual site, surface-reaction, rate-limited with hydrogen and ethane weakly adsorbed

From Table E10-3.2 we can obtain

$$-r'_E = \frac{2.0 P_E P_{H_2}}{(1 + 0.036 P_E)^2} \quad (\text{E10-3.3})$$

While K_E is small, it never goes negative within the 95% confidence interval. The variance of this model at $\sigma_C^2 = 0.0623$ is much larger than the other models. Comparing the variance of model (c) with model (b)

$$\frac{\sigma_C^2}{\sigma_B^2} = \frac{0.0623}{0.0061} = 10.2$$

We see that the σ_C^2 is an order of magnitude greater than σ_B^2 , and therefore we eliminate model (c).²⁰

Model (d) Empirical

Similarly for the power law model, we obtain from Table E10-3.2

$$-r'_E = 0.894 P_E^{0.26} P_{H_2}^{1.06} \quad (\text{E10-3.4})$$

As with model (c), the variance is quite large compared to model (b)

$$\frac{\sigma_D^2}{\sigma_B^2} = \frac{0.049}{0.0061} = 8.03$$

So we also eliminate model (d). For heterogeneous reactions, Langmuir-Hinshelwood rate laws are preferred over power-law models.



Analysis: Choose the Best Model. In this example, we were presented with four rate laws and were asked which law best fits the data. Because all the parameter values are realistic for **model (b)** and the sums of squares are significantly smaller for **model (b)** than for the other models, we choose **model (b)**. We note again that there is a caution we need to point out regarding the use of regression! One cannot simply carry out a regression and then choose the model with the lowest value of the sums of squares. If this were the case, we would have chosen **model (a)**, which had the smallest sums of squares of all the models with $\sigma^2 = 0.0049$. However, one must consider the physical realism of the parameters in the model. In **model (a)** the 95% confidence interval was greater than the parameter itself, thereby yielding negative values of the parameter, K_{AE} , which is physically impossible.

10.7 Catalyst Deactivation

In designing fixed and ideal fluidized-bed catalytic reactors, we have assumed up to now that the activity of the catalyst remains constant throughout the catalyst's life. That is, the total concentration of active sites, C_t , accessible to the reaction does not change with time. Unfortunately, Mother Nature is not so kind as to allow this behavior to be the case in most industrially significant

²⁰ See G. F. Froment and K. B. Bishoff, *Chemical Reaction Analysis and Design*, 2nd ed. (New York: Wiley, 1990), p. 96.

catalytic reactions. One of the most insidious problems in catalysis is the loss of catalytic activity that occurs as the reaction takes place on the catalyst. A wide variety of mechanisms have been proposed by Butt and Petersen, to explain and model catalyst deactivation.²¹

Catalytic deactivation adds another level of complexity to sorting out the reaction rate law parameters and pathways. In addition, we need to make adjustments for the decay of the catalysts in the design of catalytic reactors. *However, please don't worry*, this adjustment is usually made by a quantitative specification of the catalyst's activity, $a(t)$. In analyzing reactions over decaying catalysts, we divide the reactions into two categories: *separable kinetics* and *nonseparable kinetics*. In separable kinetics, we separate the rate law and activity

Separable kinetics: $-r'_A = a(\text{Past history}) \times -r'_A$ (Fresh catalyst)

When the kinetics and activity are separable, it is possible to study catalyst decay and reaction kinetics independently. However, nonseparability

Nonseparable kinetics: $-r'_A = -r'_A$ (Past history, fresh catalyst)

must be accounted for by assuming the existence of a nonideal surface or by describing deactivation by a mechanism composed of several elementary steps.²²

We shall consider only *separable kinetics* and define the activity of the catalyst at time t , $a(t)$, as the ratio of the rate of reaction on a catalyst that has been used for a time t to the rate of reaction on a fresh catalyst ($t = 0$):

$a(t)$: catalyst
activity

$$a(t) = \frac{-r'_A(t)}{-r'_A(t=0)} \quad (10-92)$$

Because of the catalyst decay, the activity decreases with time and a typical curve of the activity as a function of time is shown in Figure 10-22.

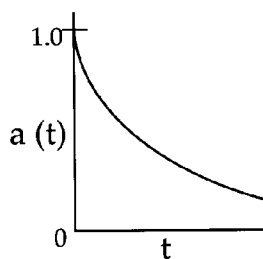


Figure 10-22 Activity as a function of time.

Combining Equations (10-92) and (3-2), the rate of disappearance of reactant A on a catalyst that has been utilized for a time t is

Reaction rate law
accounting for
catalyst activity

$$-r'_A = a(t) k(T) \text{fn}(C_A, C_B, \dots, C_P) \quad (10-93)$$

²¹ J. B. Butt and E. E. Petersen, *Activation, Deactivation and Poisoning of Catalysts* (New York: Academic Press, 1988).

²² D. T. Lynch and G. Emig, *Chem. Eng. Sci.*, 44(6), 1275–1280 (1989).

where $a(t)$ = catalytic activity, time-dependent

$k(T)$ = specific reaction rate, temperature-dependent

C_i = gas-phase concentration of reactants, products, or contaminant

The rate of catalyst decay, r_d , can be expressed in a rate law analogous to Equation (10-93)

Catalyst decay
rate law

$$r_d = -\frac{da}{dt} = p[a(t)]k_d(T)h(C_A, C_B, \dots, C_P) \quad (10-94)$$

e.g., $p[a(t)] = [a(t)]^2$

where $p[a(t)]$ is some function of the catalyst activity, k_d is the specific decay constant, and $h(C_i)$ is the functionality of rate of decay, r_d , on the reacting species concentrations. For the cases presented in this chapter, this functionality either will be independent of concentration (i.e., $h = 1$) or will be a linear function of species concentration (i.e., $h = C_i$).

The functionality of the activity term, $p[a(t)]$, in the decay law can take a variety of forms. For example, for a first-order decay

$$p(a) = a \quad (10-95)$$

and for a second-order decay

$$p(a) = a^2 \quad (10-96)$$

The particular function, $p(a)$, will vary with the gas catalytic system being used and the reason or mechanism for catalytic decay.

10.7.1 Types of Catalyst Deactivation

- Sintering
- Fouling
- Poisoning

There are three categories into which the loss of catalytic activity can traditionally be divided: sintering or aging, fouling or coking, and poisoning.

Deactivation by Sintering (Aging).²³ Sintering, also referred to as aging, is the loss of catalytic activity due to a loss of active surface area resulting from the prolonged exposure to high gas-phase temperatures. The active surface area may be lost either by crystal agglomeration and growth of the metals deposited on the support or by narrowing or closing of the pores inside the catalyst pellet. A change in the surface structure may also result from either surface recrystallization or the formation or elimination of surface defects (active sites). The reforming of heptane over platinum on alumina is an example of catalyst deactivation as a result of sintering.

Figure 10-23 shows the loss of surface area resulting from the flow of the solid porous catalyst support at high temperatures to cause pore closure. Figure 10-24 shows the loss of surface area by atomic migration and agglomeration of small metal sites deposited on the surface into a larger site where the interior

The catalyst support becomes soft and flows, resulting in pore closure.

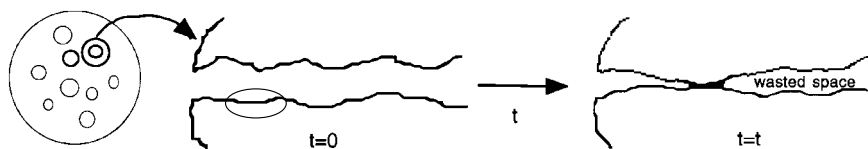


Figure 10-23 Decay by sintering: pore closure; loss of reactive surface area.

²³ See G. C. Kuczynski, ed., *Sintering and Catalysis*, vol. 10 of *Materials Science Research* (New York: Plenum Press, 1975).

The atoms move along the surface and agglomerate.

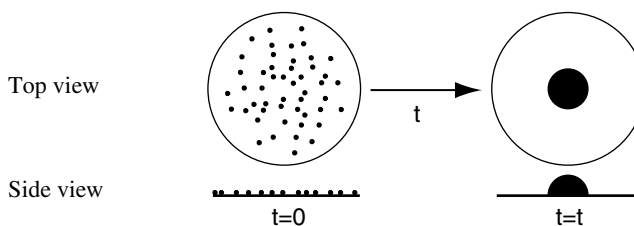


Figure 10-24 Decay by sintering: agglomeration of deposited metal sites; loss of reactive surface area.

atoms are not accessible to the reaction. Sintering is usually negligible at temperatures below 40% of the melting temperature of the solid.²⁴

Deactivation by sintering may in some cases be a function of the main-stream gas concentration. Although other forms of the sintering decay rate laws exist, one of the most commonly used decay laws is second order with respect to the present activity

$$r_d = k_d a^2 = - \frac{da}{dt} \quad (10-97)$$

Integrating, with $a = 1$ at time $t = 0$, yields

Sintering: second-order decay

$$a(t) = \frac{1}{1 + k_d t} \quad (10-98)$$

The amount of sintering is usually measured in terms of the active surface area of the catalyst S_a

$$S_a = \frac{S_{a0}}{1 + k_d t} \quad (10-99)$$

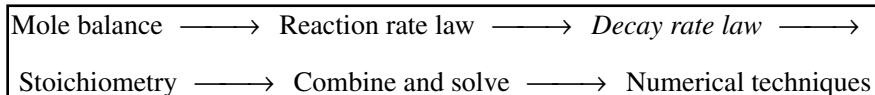
The sintering decay constant, k_d , follows the Arrhenius equation

$$k_d = k_d(T_0) \exp \left[\frac{E_d}{R} \left(\frac{1}{T_0} - \frac{1}{T} \right) \right] \quad (10-100)$$

The decay activation energy, E_d , for the reforming of heptane on Pt/Al₂O₃ is on the order of 70 kcal/mol, which is rather high. As mentioned earlier, sintering can be reduced by keeping the temperature below 0.3 to 0.4 times the metal's melting point.

We will now stop and consider reactor design for a fluid-solid system with decaying catalyst. To analyze these reactors, we only add one step to our algorithm; that is, determine the catalyst decay law. The sequence is shown here.

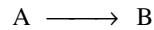
The algorithm



²⁴R. Hughes, *Deactivation of Catalysts* (San Diego: Academic Press, 1984).

Example 10-4 Calculating Conversion with Catalyst Decay in Batch Reactors

The first-order isomerization



is being carried out isothermally in a batch reactor on a catalyst that is decaying as a result of aging. Derive an equation for conversion as a function of time.

Solution

1. Mol Balance:

$$N_{A0} = \frac{dX_d}{dt} = -r'_A W \quad (\text{E10-4.1})$$

where X_d is the conversion of A when the catalyst is decaying.

2. Reaction-Rate Law:

$$-r'_A = k' a(t) C_A \quad (\text{E10-4.2})$$

3. Decay Law: For second-order decay by sintering:

$$a(t) = \frac{1}{1 + k_d t} \quad (10-98)$$

4. Stoichiometry:

$$C_A = C_{A0}(1 - X_d) = \frac{N_{A0}}{V}(1 - X_d) \quad (\text{E10-4.3})$$

5. Combining gives us

$$\frac{dX_d}{dt} = k'(1 - X_d)a \frac{W}{V} \quad (\text{E10-4.4})$$

Let $k = k'W/V$. Substituting for catalyst activity a , we have

$$\frac{dX_d}{dt} = k(1 - X_d)a = k \frac{(1 - X_d)}{(1 + k_d t)} \quad (\text{E10-4.5})$$

where X_d is the conversion when there is decay. We want to compare the conversion with and without catalyst decay.

For no decay $k_d = 0$

$$\frac{dX}{dt} = k(1 - X)$$

The Polymath program and a comparison of the conversion with decay X_d and without decay X are shown below.

One extra step
(number 3) is added
to the algorithm.



Following the Algorithm

POLYMATH Report

Ordinary Differential Equations

Calculated values of DEQ variables

Variable	Initial value	Final value
1 k	0.01	0.01
2 kd	0.1	0.1
3 t	0	500.
4 X	0	0.9932621
5 Xd	0	0.2013514

Differential equations

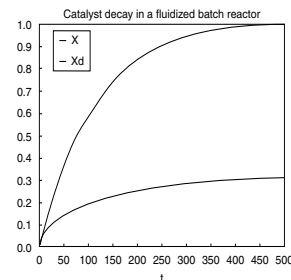
1 $d(x)/d(t) = k^*(1-X)$

2 $d(Xd)/d(t) = k^*(1-X)/(1+kd^*t)$

Explicit equations

1 $k = .01$

2 $kd = .1$



One can also obtain an analytical solution for this reaction order and low decay. Separating variables and integrating yields

$$\int_0^X \frac{dX_d}{1-X_d} = k \int_0^t \frac{dt}{1+k_d t} \quad (\text{E10-4.6})$$

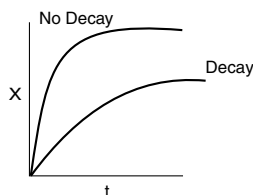
$$\ln \frac{1}{1-X_d} = \frac{k}{k_d} \ln(1+k_d t) \quad (\text{E10-4.7})$$

6. **Solving** for the conversion X_d at any time t , we find that

$$X_d = 1 - \frac{1}{(1+k_d t)^{k/k_d}} \quad (\text{E10-4.8})$$

the analytical solution **without decay**

$$\ln \frac{1}{1-X} = kt \quad (\text{E10-4.9})$$



Analysis: One observes that for long times the conversion with decay approaches a rather flat plateau and reaches conversion of about 30%. This is the conversion that will be achieved in a batch reactor for a first-order reaction when the catalyst decay law is second order. By comparison, we obtain virtually complete conversion in 500 minutes when there is no decay. The purpose of this example was to demonstrate the algorithm for isothermal catalytic reactor design for a decaying catalyst. In problem P10-1B(d) you are asked to sketch the temperature–time trajectories for various values of k and k_d .

Deactivation by Coking or Fouling. This mechanism of decay (see Figures 10-25 and 10-26) is common to reactions involving hydrocarbons. It results from a carbonaceous (coke) material being deposited on the surface of a catalyst.

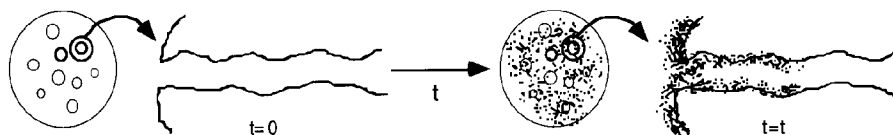


Figure 10-25 Schematic of decay by coking; loss of reactive surface area.

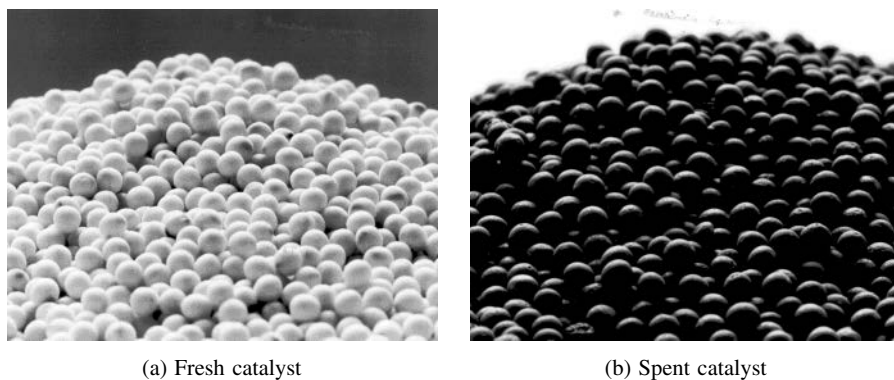


Figure 10-26 Decay by coking. (Photos courtesy of Engelhard catalyst, copyright by Michael Gaffney Photographer, Mendham, NJ.)

The amount of coke on the surface after a time t has been found to obey the following empirical relationship

$$C_C = At^n \quad (10-101)$$

where C_C is the concentration of carbon on the surface (g/m^2) and n and A are fouling parameters, which can be functions of the feed rate. This expression was originally developed by Voorhies and has been found to hold for a wide variety of catalysts and feed streams.²⁵ Representative values of A and n for the cracking of East Texas light gas oil yield²⁶

$$\% \text{ coke} = 0.47 \sqrt{t(\text{min})}$$

Different functionalities between the activity and amount of coke on the surface have been observed. One commonly used form is

$$a = \frac{1}{k_{Ck} C_C^p + 1} \quad (10-102)$$

or, in terms of time, we combine Equations (10-101) and (10-102)

$$a = \frac{1}{k_{Ck} A^p t^{np} + 1} = \frac{1}{1 + k' t^m} \quad (10-103)$$

For light Texas gas oil being cracked at 750°F over a synthetic catalyst for short times, the decay law is

$$a = \frac{1}{1 + 7.6 t^{1/2}} \quad (10-104)$$

where t is in seconds.

Activity for
deactivation by
coking

Other commonly used forms are

$$a = e^{-\alpha_1 C_C} \quad (10-105)$$

²⁵ A. Voorhies, *Ind. Eng. Chem.*, 37, 318 (1945).

²⁶ C. O. Prater and R. M. Lago, *Adv. Catal.*, 8, 293 (1956).

and

$$a = \frac{1}{1 + \alpha_2 C_C} \quad (10-106)$$

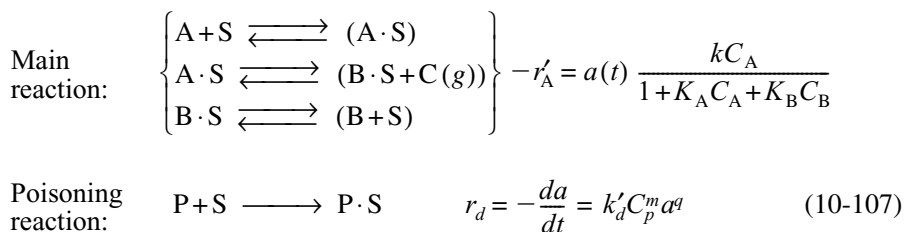
A dimensionless fouling correlation has been developed by Pacheco and Petersen.²⁷

When possible, coking can be reduced by running at elevated pressures (2000 to 3000 kPa) and hydrogen-rich streams. A number of other techniques for minimizing fouling are discussed by Bartholomew.²⁸ Catalysts deactivated by coking can usually be regenerated by burning off the carbon.

Deactivation by Poisoning. Deactivation by this mechanism occurs when the poisoning molecules become irreversibly chemisorbed to active sites, thereby reducing the number of sites available for the main reaction. The poisoning molecule, P, may be a reactant and/or a product in the main reaction, or it may be an impurity in the feed stream.

Side Note. One of the most significant examples of catalyst poisoning occurred at the gasoline pump. Oil companies found that adding lead to the gasoline increased the octane number. The television commercials said “We are going to enhance your gasoline, *but it’s going to cost you* for the added tetra-ethyl lead.” So for many years they used lead as an antiknock component. As awareness grew about NO, HC, and CO emission from the engine, it was decided to add a catalytic afterburner in the exhaust system to reduce these emissions. Unfortunately, it was found that the lead in the gasoline poisoned the reactive catalytic sites. So, the television commercials now said “We are going to take the lead out of gasoline but to receive the same level of performance as without lead, *but it’s going to cost you* because of the added refining costs to raise the octane number.” Do you think that, financially, consumers would have been better off if they never put the lead in the gasoline in the first place?

Poison in the Feed. Many petroleum feed stocks contain trace impurities such as sulfur, lead, and other components that are too costly to remove, yet poison the catalyst slowly over time. For the case of an impurity, P, in the feed stream, such as sulfur, for example, in the reaction sequence



²⁷ M. A. Pacheco and E. E. Petersen, *J. Catal.*, 86, 75 (1984).

²⁸ R. J. Farrauto and C. H. Bartholomew, *Fundamentals of Industrial Catalytic Processes*, 2nd edition (New York: Blackie Academic and Professional, 2006). This book is one of the most definitive resources on catalyst decay.

the surface sites would change with time as shown in Figure 10-27.

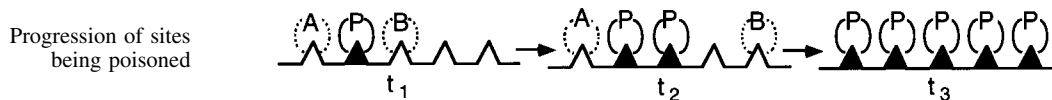


Figure 10-27 Decay by poisoning.

If we assume the rate of removal of the poison, $r_{P,S}$, from the reactant gas stream onto the catalyst sites is proportional to the number of sites that are unpoisoned ($C_{i0} - C_{P,S}$) and the concentration of poison in the gas phase is C_P then

$$r_{P,S} = k_d(C_{i0} - C_{P,S})C_P$$

where $C_{P,S}$ is the concentration of poisoned sites and C_{i0} is the total number of sites initially available. Because every molecule that is adsorbed from the gas phase onto a site is assumed to poison the site, this rate is also equal to the rate of removal of total active sites (C_i) from the surface

$$-\frac{dC_t}{dt} = \frac{dC_{P,S}}{dt} = r_{P,S} = k_d(C_{i0} - C_{P,S})C_P$$

Dividing through by C_{i0} and letting f be the fraction of the total number of sites that have been poisoned yields

$$\frac{df}{dt} = k_d(1 - f)C_P \quad (10-108)$$

The fraction of sites available for adsorption ($1 - f$) is essentially the activity $a(t)$. Consequently, Equation (10-108) becomes

$$\boxed{-\frac{da}{dt} = a(t)k_dC_P} \quad (10-109)$$

A number of examples of catalysts with their corresponding catalyst poisons are given by Farrauto and Bartholomew.²⁹

Packed-Bed Reactors. In packed-bed reactors where the poison is removed from the gas phase by being adsorbed on the specific catalytic sites, the deactivation process can move through the packed bed as a wave front. Here, at the start of the operation, only those sites near the entrance to the reactor will be deactivated because the poison (which is usually present in trace amounts) is removed from the gas phase by the adsorption; consequently, the catalyst sites farther down the reactor will not be affected. However, as time continues, the sites near the entrance of the reactor become saturated, and the poison must travel farther downstream before being adsorbed (removed) from the gas phase and attaching to a site to deactivate it. Figure 10-28 shows the corresponding activity profile for this type of poisoning process. We see in Figure 10-28 that by time t_4 the entire bed has become deactivated. The corresponding overall conversion at the exit of the reactor might vary with time as shown in Figure 10-29.

²⁹ Ibid.

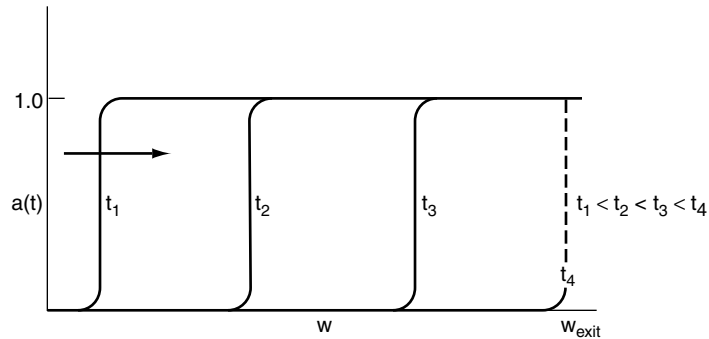


Figure 10-28 Movement of activity front in a packed bed.

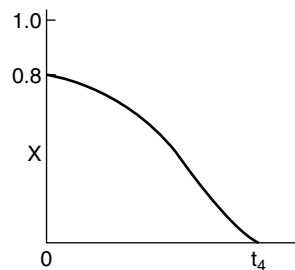
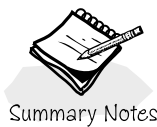


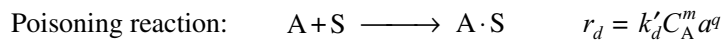
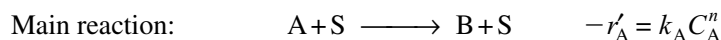
Figure 10-29 Exit conversion as a function of time.



Summary Notes

The partial differential equations that describe the movement of the reaction front shown in Figure 10-28 are derived and solved in an example on the CRE Web site, at the very end of the *Summary Notes* for Chapter 10.

Poisoning by Either Reactants or Products. For the case where the main reactant also acts as a poison, the rate laws are



An example where one of the reactants acts as a poison is in the reaction of CO and H₂ over ruthenium to form methane, with

$$-r_{\text{CO}} = ka(t)C_{\text{CO}}$$

$$-\frac{da}{dt} = r_d = k'_d a(t) C_{\text{CO}}$$

Similar rate laws can be written for the case when the product B acts as a poison.

For *separable deactivation kinetics* resulting from contacting a poison at a constant concentration C_{p_0} and no spatial variation

Separable
deactivation
kinetics

$$-\frac{da}{dt} = r_d = k'_d C_{p_0}^n a^n(t) = k_d a^n \quad (10-110)$$

The solution to this equation for the case of first-order decay, $n = 1$

$$-\frac{da}{dt} = k_d a \quad (10-111)$$

is

$$a = e^{-k_d t} \quad (10-112)$$

Empirical Decay Laws. Table 10-7 gives a number of empirical decay laws along with the reaction systems to which they apply.

TABLE 10-7 DECAY RATE LAWS

<i>Functional Form of Activity</i>	<i>Decay Reaction Order</i>	<i>Differential Form</i>	<i>Integral Form</i>	<i>Examples</i>
Linear	0	$-\frac{da}{dt} = \beta_0$	$a = 1 - \beta_0 t$	Conversion of <i>para</i> -hydrogen on tungsten when poisoned with oxygen ^{a*}
Exponential	1	$-\frac{da}{dt} = \beta_1 a$	$a = e^{-\beta_1 t}$	Ethylene hydrogenation on Cu poisoned with CO ^b Paraffin dehydrogenation on Cr·Al ₂ O ₃ ^c Cracking of gas oil ^d Vinyl chloride monomer formation ^e
Hyperbolic	2	$-\frac{da}{dt} = \beta_2 a^2$	$\frac{1}{a} = 1 + \beta_2 t$	Vinyl chloride monomer formation ^f Cyclohexane dehydrogenation on Pt/Al ₂ O ₃ ^g Isobutylene hydrogenation on Ni ^h
Reciprocal power	$\frac{\beta_3 + 1}{\beta_3} = \gamma$	$-\frac{da}{dt} = \beta_3 a^\gamma A_0^{1/5}$	$a = A_0 t^{-\beta_3}$	Cracking of gas oil and gasoline on clay ⁱ
	$\frac{\beta_4 + 1}{\beta_4} = n$	$-\frac{da}{dt} = \beta_4 a^n A_0^{1/5}$	$a = A_0 t^{-\beta_4}$	Cyclohexane aromatization on NiAl ^j

^{a*}D. D. Eley and E. J. Rideal, *Proc. R. Soc. London*, A178, 429 (1941).

^bR. N. Pease and L. Y. Steward, *J. Am. Chem. Soc.*, 47, 1235 (1925).

^cE. F. K. Herington and E. J. Rideal, *Proc. R. Soc. London*, A184, 434 (1945).

^dV. W. Weekman, *Ind. Eng. Chem. Process Des. Dev.*, 7, 90 (1968).

^eA. F. Ogonye and W. H. Ray, *Ind. Eng. Chem. Process Des. Dev.*, 9, 619 (1970).

^fA. F. Ogonye and W. H. Ray, *Ind. Eng. Chem. Process Des. Dev.*, 10, 410 (1971).

^gH. V. Maat and L. Moscou, *Proc. 3rd Int. Congr. Catal.* (Amsterdam: North-Holland, 1965), p. 1277.

^hA. L. Pozzi and H. F. Rase, *Ind. Eng. Chem.*, 50, 1075 (1958).

ⁱA. Voorhies, Jr., *Ind. Eng. Chem.*, 37, 318 (1945); E. B. Maxted, *Adv. Catal.*, 3, 129 (1951).

^jC. G. Ruderhausen and C. C. Watson, *Chem. Eng. Sci.*, 3, 110 (1954).

Source: J. B. Butt, *Chemical Reactor Engineering—Washington, Advances in Chemistry Series 109* (Washington, D.C.: American Chemical Society, 1972), p. 259. Also see CES 23, 881(1968).

Examples of
reactions with
decaying catalysts
and their decay laws

Key resource for catalyst deactivation

One should also see *Fundamentals of Industrial Catalytic Processes*, by Farrauto and Bartholomew, which contains rate laws similar to those in Table 10-7, and also gives a comprehensive treatment of catalyst deactivation.³⁰

10.7.2 Reactors That Can Be Used to Help Offset Catalyst Decay

We will now consider three reaction systems that can be used to handle systems with decaying catalyst. We will classify these systems as those having slow, moderate, and rapid losses of catalytic activity. To offset the decline in chemical reactivity of decaying catalysts in continuous-flow reactors, the following three methods are commonly used:

Matching the reactor type with speed of catalyst decay

- Slow decay – *Temperature–Time Trajectories* (10.7.3)
- Moderate decay – *Moving-Bed Reactors* (10.7.4)
- Rapid decay – *Straight-Through Transport Reactors* (10.7.5)

10.7.3 Temperature–Time Trajectories

In many large-scale reactors, such as those used for hydrotreating, and reaction systems where deactivation by poisoning occurs, the catalyst decay is relatively slow. In these continuous flow systems, constant conversion is usually necessary in order that subsequent processing steps (e.g., separation) are not upset. One way to maintain a constant conversion with a decaying catalyst in a packed or fluidized bed is to increase the reaction rate by steadily increasing the feed temperature to the reactor. Operation of a “fluidized” bed in this manner is shown in Figure 10-30.

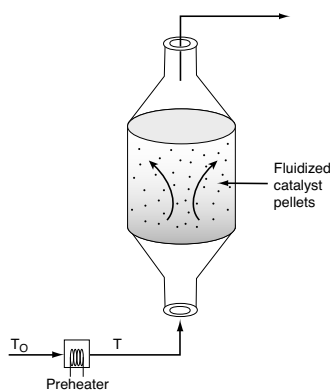


Figure 10-30 Reactor with preheater to continually increase feed temperature.

We are going to increase the feed temperature in such a manner that the reaction rate remains constant with time:

$$-r'_A(t=0, T_0) = -r'_A(t, T) = a(t, T)[-r'_A(t=0, T)]$$

For a first-order reaction we have

$$k(T_0)C_A = a(t, T)k(T)C_A$$

³⁰Ibid.

Slow rate of catalyst decay

We will neglect any variations in concentration so that the product of the activity (a) and specific reaction rate (k) is constant and equal to the specific reaction rate, k_0 at time $t = 0$ and temperature T_0 ; that is

$$k(T)a(t, T) = k_0 \quad (10-113)$$

The goal is to find *how* the temperature should be increased with time (i.e., the temperature–time trajectory) to maintain constant conversion. Using the Arrhenius equation to substitute for k in terms of the activation energy, E_A , gives

$$k_0 e^{(E_A/R)(1/T_0 - 1/T)} a = k_0 \quad (10-114)$$

Solving for $1/T$ yields

Gradually raising the temperature can help offset effects of catalyst decay.

$$\boxed{\frac{1}{T} = \frac{R}{E_A} \ln a + \frac{1}{T_0}} \quad (10-115)$$

The decay law also follows an Arrhenius-type temperature dependence

$$-\frac{da}{dt} = k_{d0} e^{(E_d/R)(1/T_0 - 1/T)} a^n \quad (10-116)$$

where k_{d0} = decay constant at temperature T_0 , s^{-1}

E_A = activation energy for the main reaction (e.g., $A \rightarrow B$), kJ/mol

E_d = activation energy for catalyst decay, kJ/mol

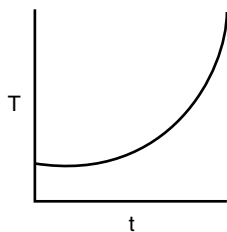
Substituting Equation (10-115) into (10-116) and rearranging yields

$$-\frac{da}{dt} = k_{d0} \exp\left(-\frac{E_d}{E_A} \ln a\right) a^n = k_{d0} a^{(n - E_d/E_A)} \quad (10-117)$$

Integrating with $a = 1$ at $t = 0$ for the case $n \neq (1 + E_d/E_A)$, we obtain

$$t = \frac{1 - a^{1 - n + E_d/E_A}}{k_{d0}(1 - n + E_d/E_A)} \quad (10-118)$$

Solving Equation (10-114) for a and substituting in (10-118) gives

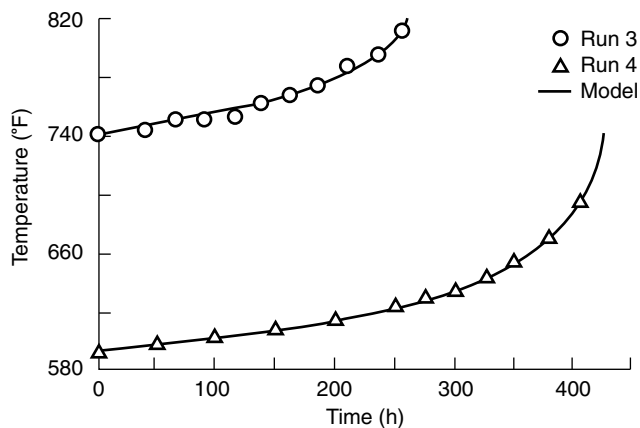


$$\boxed{t = \frac{1 - \exp\left[\frac{E_A - nE_A + E_d}{R} \left(\frac{1}{T} - \frac{1}{T_0}\right)\right]}{k_{d0}(1 - n + E_d/E_A)}} \quad (10-119)$$

Equation (10-119) tells us how the temperature of the catalytic reactor should be increased with time in order for the reaction rate to remain constant.

In many industrial reactions, the decay rate law changes as temperature increases. In hydrocracking, the temperature–time trajectories are divided into three regimes. Initially, there is fouling of the acidic sites of the catalyst followed by a linear regime due to slow coking and, finally, accelerated coking characterized by an exponential increase in temperature. The temperature–time trajectory for a deactivating hydrocracking catalyst is shown in Figure 10-31.

Comparing theory
and experiment



Run	E_A (cal/mol)	A (h^{-1})	E_d (cal/mol)	A_d (h^{-1})
3	30.0	0.52×10^{12}	42.145	1.54×10^{11}
4	30.0	0.52×10^{12}	37.581	2.08×10^{11}

Figure 10-31 Temperature–time trajectories for deactivating hydrocracking catalyst, runs 3 and 4. (Krishnaswamy, S., and J. R. Kittrell. Analysis of Temperature–Time Data for Deactivating Catalysts. *Industry and Engineering Chemistry Process Design and Development*, 1979, 18(3), 399–403. Copyright © 1979, American Chemical Society. Reprinted by permission.)

For a first-order decay, Krishnaswamy and Kittrell's expression, Equation (10-119), for the temperature–time trajectory reduces to

$$t = \frac{E_A}{k_{d0}E_d} [1 - e^{(E_d/R)(1/T - 1/T_0)}] \quad (10-120)$$

10.7.4 Moving-Bed Reactors

Reaction systems with significant catalyst decay require the continual regeneration and/or replacement of the catalyst. Two types of reactors currently in commercial use that accommodate production with decaying catalysts are the moving-bed and straight-through transport reactor. A schematic diagram of a moving-bed reactor (used for catalytic cracking) is shown in Figure 10-32.

The freshly regenerated catalyst enters the top of the reactor and then moves through the reactor as a compact packed bed. The catalyst is coked continually as it moves through the reactor until it exits the reactor into the kiln, where air is used to burn off the carbon. The regenerated catalyst is lifted from the kiln by an airstream and then fed into a separator before it is returned to the reactor. The catalyst pellets are typically between one-quarter and one-eighth in. in diameter.

The reactant feed stream enters at the top of the reactor and flows rapidly through the reactor relative to the flow of the catalyst through the reactor (Figure 10-33). If the feed rates of the catalyst and the reactants do not vary

Moving bed reactor,
used for reactions
with moderate rate
of catalyst decay

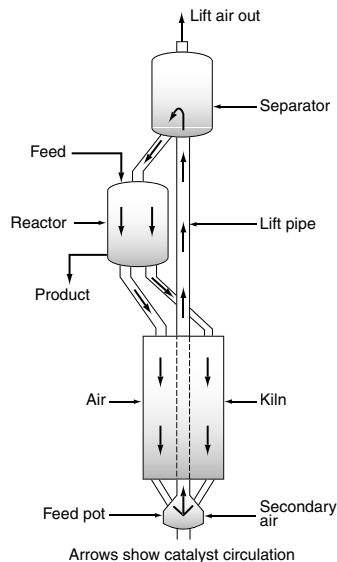


Figure 10-32 Thermoform catalytic cracking (TCC) unit. (Weekman, V. *AIChE Monogr. Ser.*, 75(11), 4 (1979). With permission of the American Institute of Chemical Engineers. Copyright © 1979 AIChE. All rights reserved.)

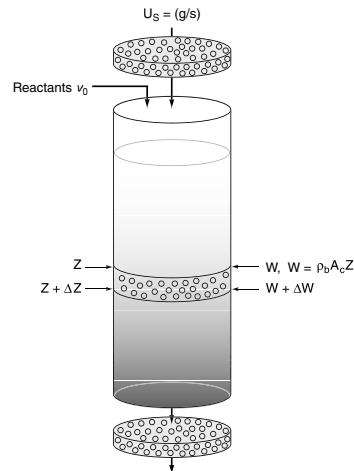


Figure 10-33 Moving-bed reactor schematic. The value of the catalyst contained in a reactor of this type is approximately \$1 million.

with time, the reactor operates at steady state; that is, conditions at any point in the reactor do not change with time. The mole balance on reactant A over ΔW is

$$\left[\begin{array}{c} \text{Molar} \\ \text{flow} \\ \text{rate of A in} \end{array} \right] - \left[\begin{array}{c} \text{Molar} \\ \text{flow} \\ \text{rate of A out} \end{array} \right] + \left[\begin{array}{c} \text{Molar} \\ \text{rate of} \\ \text{generation of A} \end{array} \right] = \left[\begin{array}{c} \text{Molar} \\ \text{rate of} \\ \text{accumulation of A} \end{array} \right]$$

$$F_A(W) - F_A(W+\Delta W) + r'_A \Delta W = 0 \quad (10-121)$$

Dividing by ΔW , letting ΔW approach zero, and expressing the flow rate in terms of conversion gives

Mole Balance
$$F_{A0} \frac{dX}{dW} = -r'_A \quad (2-17)$$

The rate of reaction at any time t is

$$-r'_A = a(t)[-r'_A(t=0)] = a(t)[k \text{ fn}(C_A, C_B, \dots, C_P)] \quad (10-93)$$

The activity, as before, is a function of the time the catalyst has been in contact with the reacting gas stream. The decay rate law is

Decay Law
$$-\frac{da}{dt} = k_d a^n \quad (10-110)$$

We now need to relate the contact time to the weight of the catalyst. Consider a point z in the reactor, where the reactant gas has passed cocurrently through a catalyst weight W . Because the solid catalyst is moving through the bed at a rate U_s (mass per unit time), the time t that the catalyst has been in contact with the gas when the catalyst reaches a point z is

$$t = \frac{W}{U_s} \quad (10-122)$$

If we now differentiate Equation (10-122)

$$dt = \frac{dW}{U_s} \quad (10-123)$$

and combine it with the decay rate law, we obtain

$$-\frac{da}{dW} = \frac{k_d}{U_s} a^n \quad (10-124)$$

The activity equation is combined with the mole balance

The design equation
for moving-bed
reactors

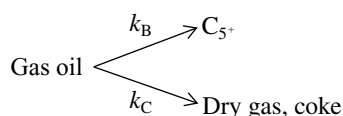
$$\frac{dX}{dW} = \frac{a[-r'_A(t=0)]}{F_{A0}} \quad (10-125)$$

these two coupled differential equations (i.e., Equations (10-124) and (10-125)) are numerically solved simultaneously with an ODE solver, e.g., Polymath.

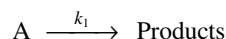


Example 10-5 Catalytic Cracking in a Moving-Bed Reactor

The catalytic cracking of a gas-oil charge, A, to form C_5+ (B) and to form coke and dry gas (C) is to be carried out in a screw-type conveyor, moving-bed reactor at 900°F :



This cracking of gas oil can also be written as



While pure hydrocarbons are known to crack according to a first-order rate law, the fact that the gas-oil exhibits a wide spectrum of cracking rates gives rise to the fact that the lumped cracking rate is well represented by a second-order rate law (see Web Problem CDP5-H_B) with the following specific reaction rate:³¹

$$-r'_A = 600 \frac{(\text{dm})^6}{(\text{kg cat})(\text{mol})(\text{min})} C_A^2$$

The catalytic deactivation is independent of gas-phase concentration and follows a first-order decay rate law, with a decay constant, k_d of 0.72 reciprocal minutes. The feed stream is diluted with nitrogen so that as a first approximation,

³¹Estimated from V. W. Weekman and D. M. Nace, *AIChE J.*, 16, 397 (1970).

volume changes can be neglected with reaction. The reactor contains 22 kg of catalyst that moves through the reactor at a rate of 10 kg/min. The gas-oil is fed at a rate of 30 mol/min at a concentration of 0.075 mol/dm³. Determine the conversion that can be achieved in this reactor.

Solution

1. Mole Balance:

$$F_{A0} \frac{dX}{dW} = a (-r'_A) \quad (\text{E10-5.1})$$

2. Rate Law:

$$-r'_A = kC_A^2 \quad (\text{E10-5.2})$$

3. Decay Law. First-order decay

$$-\frac{da}{dt} = k_d a$$

Using Equation (10-124), we obtain

$$-\frac{da}{dW} = \frac{k_d}{U_s} a \quad (\text{E10-5.3})$$

Integrating

$$a = e^{-(k_d/U_s)W} \quad (\text{E10-5.4})$$

4. Stoichiometry. If $v \approx v_0$, then

$$C_A = C_{A0}(1 - X) \quad (\text{E10-5.5})$$

5. Combining we have

$$\frac{dX}{dW} = a \frac{kC_{A0}^2(1-X)^2}{F_{A0}} \quad (\text{E10-5.6})$$

$$a = e^{-(k_d/U_s)W}$$

The polymath program is shown below along with a conversion profile.

POLYMATH Report

Ordinary Differential Equations

Calculated values of DEQ variables

	Variable	Initial value	Final value
1	a	1.	0.2051528
2	Ca	0.075	0.033453
3	Cao	0.075	0.075
4	Fao	30.	30.
5	k	600.	600.
6	kd	0.72	0.72
7	raprime	-3.375	-0.6714636
8	Us	10.	10.
9	W	0	22.
10	X	0	0.5539595

Differential equations

$$1 \quad d(a)/d(W) = -kd*a/Us$$

$$2 \quad d(X)/d(W) = a*(-raprime)/Fao$$

Explicit equations

$$1 \quad Us = 10$$

$$2 \quad kd = 0.72$$

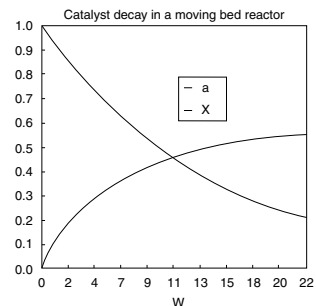
$$3 \quad Fao = 30$$

$$4 \quad Cao = 0.075$$

$$5 \quad Ca = Cao*(1-X)$$

$$6 \quad k = 600$$

$$7 \quad raprime = -k*Ca^2$$



Moving beds:
moderate rate of
catalyst decay

For the simple rate law and activity law given here, we also could have solved this problem analytically.

6. Separating and integrating yields

$$F_{A0} \frac{dX}{dW} = e^{-(k_d/U_s)W} kC_{A0}^2 (1-X)^2$$

$$\frac{F_{A0}}{kC_{A0}^2} \int_0^X \frac{dX}{(1-X)^2} = \int_0^W e^{-(k_d/U_s)W} dW \quad (\text{E10-5.7})$$

$$\frac{X}{1-X} = \frac{kC_{A0}^2 U_s}{F_{A0} k_d} (1 - e^{-k_d W/U_s}) \quad (\text{E10-5.8})$$

7. Numerical evaluation:

$$\begin{aligned} \frac{X}{1-X} &= \frac{0.6 \text{ dm}^6}{\text{mol} \cdot \text{g cat} \cdot \text{min}} \times \frac{(0.075 \text{ mol/dm}^3)^2}{30 \text{ mol/min}} \frac{10,000 \text{ g cat/min}}{0.72 \text{ min}^{-1}} \\ &\quad \times \left(1 - \exp \left[\frac{(-0.72 \text{ min}^{-1})(22 \text{ kg})}{10 \text{ kg/min}} \right] \right) \end{aligned}$$

$$\frac{X}{1-X} = 1.24$$

$$X = 55\%$$

If there were no catalyst decay, the conversion would be

$$\begin{aligned} \frac{X}{1-X} &= \frac{kC_{A0}^2}{F_{A0}} W \\ &= 600 \left(\frac{\text{dm}^6}{(\text{kg cat})(\text{mol})(\text{min})} \times \frac{(0.075 \text{ mol/dm}^3)^2}{(30 \text{ mol/min})} (22 \text{ ckg cat}) \right) \\ &= 7.48 \\ X &= 0.71 \end{aligned}$$

Analysis: The purpose of this example was to show step-by-step how to apply the algorithm to a moving-bed reactor that has been used to reduce the effects of catalyst decay that would occur in a PBR.

We will now rearrange Equation (E10-5.8) to a form more commonly found in the literature. Let λ be a dimensionless decay time

$$\lambda = k_d t = \frac{k_d W}{U_s} \quad (10-126)$$

and Da_2 be the Damköhler number for a second-order reaction (*a reaction rate divided by a transport rate*) for a packed-bed reactor

$$\text{Da}_2 = \frac{(kC_{A0}^2)(W)}{F_{A0}} = \frac{kC_{A0}W}{v_0} \quad (10-127)$$

Through a series of manipulations we arrive at the equation for the conversion in a *moving bed* where a second-order reaction is taking place³²

Second-order
reaction in a
moving-bed reactor

$$X = \frac{Da_2(1-e^{-\lambda})}{\lambda + Da_2(1-e^{-\lambda})} \quad (10-128)$$

Similar equations are given or can easily be obtained for other reaction orders or decay laws.

10.7.5 Straight-Through Transport Reactors (STTR)

This reactor is used for reaction systems in which the catalyst deactivates very rapidly. Commercially, the STTR is used in the production of gasoline from the cracking of heavier petroleum fractions where coking of the catalyst pellets occurs very rapidly. In the STTR, the catalyst pellets and the reactant feed enter together and are transported very rapidly through the reactor. The bulk density of the catalyst particle in the STTR is significantly smaller than in moving-bed reactors, and often the particles are carried through at the same velocity as the gas velocity. In some places the STTR is also referred to as a circulating fluidized bed (CFB). A schematic diagram is shown in Figure 10-34.

A mole balance on the reactant A over the differential reactor volume

$$\Delta V = A_C \Delta z$$

is

$$F_A|_z - F_A|_{z+\Delta z} + r_A A_C \Delta z = 0$$

STTR: Used when
catalyst decay
(usually coking) is
very rapid

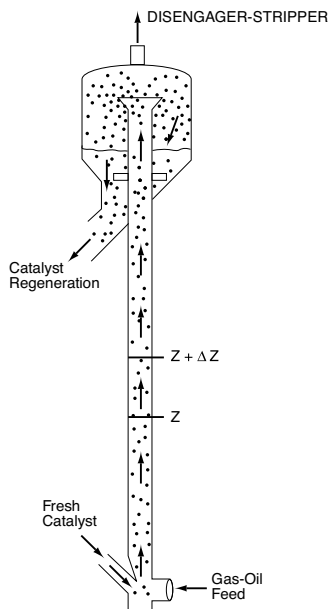


Figure 10-34 Straight-through transport reactor.

³²Ibid.

Dividing by Δz and taking the limit as $\Delta z \rightarrow 0$ and recalling that $r_A = \rho_B r'_A$, we obtain

$$\frac{dF_A}{dz} = r_A A_C = r'_A \rho_B A_C \quad (10-129)$$

In terms of conversion and catalyst activity

$$\frac{dX}{dz} = \left(\frac{\rho_B A_C}{F_{A0}} \right) [-r'_A(t=0)] a(t) \quad (10-130)$$

For a catalyst particle traveling through the reactor with a particle velocity U_p , the time the catalyst pellet has been in the reactor when it reaches a height z is just

$$t = \frac{z}{U_p} \quad (10-131)$$

Substituting for time t in terms of distance z [i.e., $a(t) = a(z/U_p)$], the mole balance now becomes

$$\frac{dX}{dz} = \frac{\rho_B A_C [-r'_A(t=0)] a(z/U_p)}{F_{A0}}$$

The entering molar flow rate, F_{A0} , can be expressed in terms of the gas velocity U_g , C_{A0} , and A_C

$$F_{A0} = U_o A_C C_{A0}$$

Substituting for F_{A0} , we have

$$\boxed{\frac{dX}{dz} = \frac{\rho_B a(z/U_p) [-r'_A(t=0)]}{C_{A0} U_o}} \quad (10-132)$$



Living Example Problem

A typical cost of the catalyst in the reactor system is \$1 million.

Example 10-6 Decay in a Straight-Through Transport Reactor

The vapor-phase cracking of a gas-oil is to be carried out in a straight-through transport reactor (STTR) that is 10-m high and 1.5 m in diameter. Gas-oil is a mixture of normal and branched paraffins (C_{12} – C_{40}), naphthenes, and aromatics, all of which will be lumped as a single species, A. We shall lump the primary hydrocarbon products according to distillate temperature into two respective groups, dry gas (C – C_4) B and gasoline (C_5 – C_{14}) C. The reaction



can be written symbolically as



Both B and C are adsorbed on the surface. The rate law for a gas-oil cracking reaction on fresh catalyst can be approximated by

$$-r'_A = \frac{k'P_A}{1 + K_A P_A + K_B P_B + K_C P_C}$$

with $k' = 0.0014 \text{ kmol/kg-cat} \cdot \text{s} \cdot \text{atm}$, $K_A = 0.05 \text{ atm}^{-1}$, $K_B = 0.15 \text{ atm}^{-1}$, and $K_C = 0.1 \text{ atm}^{-1}$. The catalyst decays by the deposition of coke, which is produced in most cracking reactions along with the reaction products. The decay law is

$$a = \frac{1}{1 + At^{1/2}} \quad \text{with } A = 7.6 \text{ s}^{-1/2}$$

Pure gas-oil enters at a pressure of 12 atm and a temperature of 400°C. The bulk density of catalyst in the STTR is 80 kg-cat/m³. Plot the activity and conversion of gas-oil up the reactor for entering gas velocity $U_0 = 2.5 \text{ m/s}$.

Solution

Mole Balance:

$$F_{A0} \frac{dX}{dz} = -r_A A_c$$

$$\boxed{\frac{dX}{dz} = \frac{-r_A}{U_0 C_{A0}}} \quad (\text{E10-6.1})$$

The height of the catalyst particle at time “t” after entering the STTR is

$$z = \int_0^t U dt$$

Differentiating, we can find a relation between the time the catalyst particle has been in the STTR and reached a height z , which we can use to find the activity a .

$$\frac{dt}{dz} = \frac{1}{U}$$

Rate Law:

$$-r_A = \rho_B (-r'_A) \quad (\text{E10-6.2})$$

$$-r'_A = a [-r'_A(t=0)] \quad (\text{E10-6.3})$$

On fresh catalyst

$$-r'_A(t=0) = k' \frac{P_A}{1 + K_A P_A + K_B P_B + K_C P_C} \quad (\text{E10-6.4})$$

Combining Equations (E10-6.2) through (E10-6.4) gives

$$\boxed{-r_A = a \left(\rho_B k' \frac{P_A}{1 + K_A P_A + K_B P_B + K_C P_C} \right)} \quad (\text{E10-6.5})$$

Decay Law. Assuming that the catalyst particle and gas travel up the reactor at the velocity $U_p = U_g = U$, we obtain

$$t = \frac{z}{U} \quad (\text{E10-6.6})$$



Following the Algorithm

$$a = \frac{1}{1 + A (z/U)^{1/2}} \quad (\text{E10-6.7})$$

where $U = v/A_C = v_0(1 + \varepsilon X)/A_C$ and $A_C = \pi D^2/4$.

Stoichiometry (gas-phase isothermal and no pressure drop):

$$P_A = P_{A0} \frac{1-X}{1+\varepsilon X} \quad (\text{E10-6.8})$$

$$P_B = \frac{P_{A0}X}{1+\varepsilon X} \quad (\text{E10-6.9})$$

$$P_C = P_B \quad (\text{E10-6.10})$$

Parameter Evaluation:

$$\varepsilon = y_{A0}\delta = (1 + 1 - 1) = 1$$

$$U = U_0(1 + \varepsilon X)$$

$$C_{A0} = \frac{P_{A0}}{RT_0} = \frac{12 \text{ atm}}{(0.082 \text{ m}^3 \cdot \text{atm/kmol} \cdot \text{K})(673 \text{ K})} = 0.22 \frac{\text{kmol}}{\text{m}^3}$$

Equations (E10-6.1), (E10-6.5), (E10-6.7), and (E10-6.8) through (E10-6.10) are now combined and solved using an ODE solver. The Polymath program is shown in Table E10-6.1, and the computer output is shown in Figure E10-6.1.

TABLE E10-6.1 EQUATIONS FOR THE STTR: LANGMUIR–HINSHELWOOD KINETICS

POLYMATH Report

Ordinary Differential Equations

Calculated values of DEQ variables

	Variable	Initial value	Final value
1	A	7.6	7.6
2	a	1.	0.0760585
3	Cao	0.2174465	0.2174465
4	D	1.5	1.5
5	eps	1.	1.
6	Ka	0.05	0.05
7	Kb	0.15	0.15
8	Kc	0.1	0.1
9	Kca	2.7593	2.7593
10	kprime	0.0014	0.0014
11	Pa	12.	3.329127
12	Pao	12.	12.
13	Pb	0	4.335437
14	Pc	0	4.335437
15	R	0.082	0.082
16	ra	-0.84	-0.0126024
17	raprime	-0.0105	-0.0001575
18	rho	80.	80.
19	T	673.	673.
20	U	2.5	3.914117
21	Uo	2.5	2.5
22	vo	4.417875	4.417875
23	X	0	0.5656469
24	x	0	10.

Differential equations

$$1 \quad d(X)/d(z) = -ra/U/Cao$$

Explicit equations

- 1 Ka = 0.05
- 2 kb = .15
- 3 Pao = 12
- 4 eps = 1
- 5 A = 7.6
- 6 R = 0.082
- 7 T = 400+273
- 8 rho = 80
- 9 kprime = 0.0014
- 10 D = 1.5
- 11 Uo = 2.5
- 12 Kc = 0.1
- 13 U = Uo*(1+eps*X)
- 14 Pa = Pao*(1-X)/(1+eps*X)
- 15 Pb = Pao*X/(1+eps*X)
- 16 vo = Uo*3.1416*D*D/4
- 17 Cao = Pao/R/T
- 18 Kca = Ka*R/T
- 19 Pc = Pb
- 20 a = 1/(1+A*(z/U)^0.5)
- 21 raprime = a*(-kprime*Pa/(1+Ka*Pa+Kb*Pb+Kc*Pc))
- 22 ra = rho*raprime



Living Example Problem

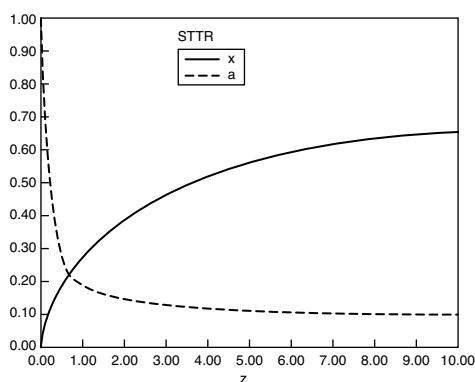


Figure E10-6.1 Activity and conversion profiles.

Analysis: In this example we applied the algorithm to a STTR in which the gas velocity and, hence, particle velocity increases as we move through the reactor. The decay is quite rapid and the activity is only 15% of its initial value at $z = 3$ m into the reactor and the conversion begins to plateau at $z = 6$ m at 60% conversion. If there were no catalyst decay ($a = 0$) the conversion would have been 97%.

Closure. After reading this chapter, the reader should be able to discuss the steps in a heterogeneous reaction (adsorption, surface reaction, and desorption) and describe what is meant by a rate-limiting step. The differences between molecular adsorption and dissociated adsorption should be explained by the reader, as should the different types of surface reactions (single site, dual site, and Eley–Rideal). Given heterogeneous reaction-rate data, the reader should be able to analyze the data and develop a rate law for Langmuir–Hinshelwood kinetics. The reader should also be able to discriminate between rate laws to find which rate law best fits the data. After evaluating the rate-law parameters, the reader can proceed to the design of PBRs and fluidized CSTRs.

Applications of CRE in the electronics industry were discussed and readers should be able to describe the analogy between Langmuir–Hinshelwood kinetics and chemical vapor deposition (CVD), and to derive a rate law for CVD mechanisms.

Because of the high temperatures and harsh environment, catalysts do not maintain their original activity and the catalysts decay during the course of the reaction. The reader should be able to discuss the three basic types of catalyst decay (sintering, coking or fouling, and poisoning). In addition, the reader should be able to suggest reactors (e.g., moving bed) to use to offset catalyst decay and should be able to carry out calculations to predict the conversion.

SUMMARY

1. Types of adsorption:
 - a. Chemisorption
 - b. Physical adsorption
2. The **Langmuir isotherm** relating the concentration of species A on the surface to the partial pressure of A in the gas phase is

$$C_{A \cdot S} = \frac{K_A C_t P_A}{1 + K_A P_A} \quad (\text{S10-1})$$

3. The sequence of steps for the solid-catalyzed isomerization



is:

- a. **Mass transfer of A** from the bulk fluid to the external surface of the pellet
 - b. **Diffusion of A** into the interior of the pellet
 - c. **Adsorption of A** onto the catalytic surface
 - d. **Surface reaction of A** to form **B**
 - e. **Desorption of B** from the surface
 - f. **Diffusion of B** from the pellet interior to the external surface
 - g. **Mass transfer of B** away from the solid surface to the bulk fluid
4. Assuming that mass transfer is not rate-limiting, the rate of adsorption is

$$r_{AD} = k_A \left(C_v P_A - \frac{C_{A \cdot S}}{K_A} \right) \quad (\text{S10-3})$$

The rate of surface reaction is

$$r_S = k_S \left(C_{A \cdot S} - \frac{C_{B \cdot S}}{K_S} \right) \quad (\text{S10-4})$$

The rate of desorption is

$$r_D = k_D (C_{B \cdot S} - K_B P_B C_v) \quad (\text{S10-5})$$

At steady state

$$-r'_A = r_{AD} = r_S = r_D \quad (\text{S10-6})$$

If there are no inhibitors present, the total concentration of sites is

$$C_t = C_v + C_{A \cdot S} + C_{B \cdot S} \quad (\text{S10-7})$$

5. If we assume that the surface reaction is rate-limiting, we set

$$\frac{r_{AD}}{k_A} \simeq 0 \quad \frac{r_D}{k_D} \simeq 0$$

and solve for $C_{A \cdot S}$ and $C_{B \cdot S}$ in terms of P_A and P_B . After substitution of these quantities in Equation (S10-4), the concentration of vacant sites is eliminated with the aid of Equation (S10-7)

$$-r'_A = r_S = \frac{\overbrace{C_1 k_S K_A}^k (P_A - P_B / K_P)}{1 + K_A P_A + K_B P_B} \quad (\text{S10-8})$$

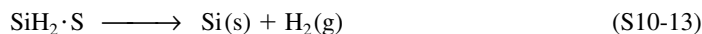
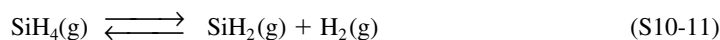
Recall that the equilibrium constant for desorption of species B is the reciprocal of the equilibrium constant for the adsorption of species B

$$K_B = \frac{1}{K_{DB}} \quad (\text{S10-9})$$

and the thermodynamic equilibrium constant, K_P , is

$$K_P = K_A K_S / K_B \quad (\text{S10-10})$$

6. Chemical vapor deposition



$$r_{\text{Dep}} = \frac{kP_{\text{SiH}_4}}{P_{\text{H}_2} + KP_{\text{SiH}_4}} \quad (\text{S10-14})$$

7. Catalyst deactivation. The catalyst activity is defined as

$$a(t) = \frac{-r'_A(t)}{-r'_A(t=0)} \quad (\text{S10-15})$$

The rate of reaction at any time t is

$$-r'_A = a(t)k(T) \text{fn}(C_A, C_B, \dots, C_P) \quad (\text{S10-16})$$

The rate of catalyst decay is

$$r_d = -\frac{da}{dt} = p[a(t)]k_d(T)g(C_A, C_B, \dots, C_P) \quad (\text{S10-17})$$

For first-order decay

$$p(a) = a \quad (\text{S10-18})$$

For second-order decay

$$p(a) = a^2 \quad (\text{S10-19})$$

8. For slow catalyst decay, the idea of a **temperature–time trajectory** is to increase the temperature in such a way that the rate of reaction remains constant.

9. The coupled differential equations to be solved for a **moving-bed reactor** are

$$F_{A0} \frac{dX}{dW} = a(-r'_A) \quad (\text{S10-20})$$

For n th-order activity decay and m order in a gas-phase concentration of species i

$$-\frac{da}{dW} = \frac{k_d a^n C_i^m}{U_s} \quad (\text{S10-21})$$

$$t = \frac{W}{U_s} \quad (\text{S10-22})$$

10. The coupled differential equations to be solved in a **straight-through transport reactor** for the case when the particle and gas velocities, U , are identical are

$$\frac{dX}{dz} = \frac{a(t)[-r'_A(t=0)]}{U_g} \left(\frac{\rho_b A_c}{C_{A0}} \right) \quad (\text{S10-23})$$

$$t = \frac{z}{U_g} \quad (\text{S10-24})$$

For coking

$$a(t) = \frac{1}{1 + At^{1/2}} \quad (\text{S10-25})$$

ODE SOLVER ALGORITHM

The isomerization $A \rightarrow B$ is carried out over a decaying catalyst in a *moving-bed reactor*. Pure A enters the reactor and the catalyst flows through the reactor at a rate of 2.0 kg/s.

$$\frac{dX}{dW} = \frac{-r'_A}{F_{A0}}$$

$$k = 0.1 \text{ mol}/(\text{kg-cat} \cdot \text{s} \cdot \text{atm})$$

$$r'_A = \frac{-akP_A}{1 + K_A P_A}$$

$$K_A = 1.5 \text{ atm}^{-1}$$

$$\frac{da}{dW} = \frac{-k_d a^2 P_B}{U_s}$$

$$k_d = \frac{0.75}{\text{s} \cdot \text{atm}}$$

$$P_A = P_{A0}(1 - X)p$$

$$F_{A0} = 10 \text{ mol/s}$$

$$P_B = P_{A0} X p$$

$$P_{A0} = 20 \text{ atm}$$

$$\frac{dp}{dW} = -\frac{\alpha}{2p}$$

$$U_s = 2.0 \text{ kg-cat/s}$$

$$\alpha = 0.0019 \text{ kg}^{-1}$$

$$W_f = 500 \text{ kg-cat}$$

CRE WEB SITE MATERIALS

- Expanded Material

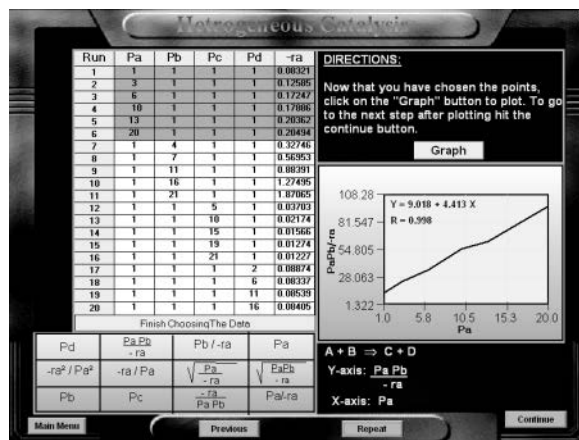
1. Puzzle Problem "What's Wrong with this Solution?"
2. Turn Over Frequency (TOF)
3. Algorithm When More than One of the Three Steps in a Catalytic Reaction Mechanism (Adsorption, Surface Reaction, Desorption) Is Rate Limiting
4. Additional Homework Problems

- Learning Resources

1. Summary Notes for Chapter 10
2. Interactive Computer Modules
 - A. Heterogeneous Catalysis



Summary Notes



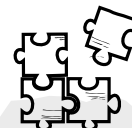
3. Solved Problems

Example Chapter 10 CRE Web Site: 10-1 Analysis of a Heterogeneous Reaction [Class Problem University of Michigan]

Example Chapter 10 CRE Web Site: 10-2 Least Squares Analysis to Determine the Rate Law Parameters k , k_T , and k_B

Example Chapter 10 CRE Web Site: 10-3 Decay in a Straight-Through Reactor

Example Chapter 10 CRE Web Site: 10-4 Catalyst Poisoning in a Batch Reactor



Solved Problems

- Living Example Problems

1. Example 10-2 Regression Analysis to Determine Model Parameters
2. Example 10-3 Fixed-Bed Reactor Design
3. Example 10-4 Model Discrimination
4. Example 10-5 Catalyst Decay in a Batch Reactor
5. Example 10-6 Catalyst Decay in a Fluidized Bed Modeled as a CSTR
6. Example 10-8 Decay in a Straight-Through Transport Reactor



Living Example Problem

- Professional Reference Shelf

R10.1. Classification of Catalysts

R10.2. Hydrogen Adsorption

A. Molecular Adsorption

B. Dissociative Adsorption

R10.2. Analysis of Catalyst Decay Laws

A. Integral Method

B. Differential Method

R10.3. Etching of Semiconductors

A. Dry Etching

B. Wet Etching

C. Dissolution Catalysis



Reference Shelf

R10.4. *Catalyst Deactivation*

- A. Type of Catalyst Deactivation
- B. Temperature–Time Trajectories
- C. Moving-Bed Reactors
- D. Straight-Through Transport Reactors

**After Reading Each Page in
This Book, Ask Yourself a Question
About What You Read**

QUESTIONS AND PROBLEMS

The subscript to each of the problem numbers indicates the level of difficulty: A, least difficult; D, most difficult.

A = ● B = ■ C = ◆ D = ◆◆

Questions

Q10-1_A Read over the problems at the end of this chapter. Make up an original problem that uses the concepts presented in this chapter. See Problem P5-1_A for guidelines. To obtain a solution:

- (a) Create your data and reaction.
- (b) Use a real reaction and real data.

The journals listed at the end of Chapter 1 may be useful for part (b).

- (c) Choose an FAQ from Chapter 10 and explain why it was most helpful.
- (d) Listen to the audios  on the *Summary Notes*, pick one, and explain why it was most helpful.

Problems

- P10-1_B** (a) **Example 10-1.** Plot and analyze (1) the ratio of toluene-occupied sites to benzene-occupied sites, (2) the fraction of vacant sites, and (3) the fraction of benzene-occupied sites as a function of conversion at 1 atm.
- (b) **Example 10-2.** (1) What if the entering pressure were increased to 80 atm or reduced 1 atm, how would your answers change? (2) What if the molar flow rate were reduced by 50%, how would X and y change? (3) What catalyst weight would be required for 60% conversion?
- (c) **Example 10-3.** (1) How would your answers change if the following data for run 10 were incorporated in your regression table?

$$-r'_E = 0.8 \text{ mol/kg-cat} \cdot \text{s}, P_E = 0.5 \text{ atm}, P_{EA} = 15 \text{ atm}, P_H = 2 \text{ atm}.$$

- (2) How do the rate laws (e) and (f)

$$(e) \quad -r'_E = \frac{k P_E P_H}{(1 + K_A P_{EA} + K_E P_E)^2} \quad (f) \quad -r'_E = \frac{k P_H P_E}{1 + K_A P_{EA}}$$

compare with the other rate laws used to model the data?

- (3) Write a question for this problem that involves critical thinking and explain why it involves critical thinking.
- (d) **Example 10-4.** Vary k and k_d , and describe what you find.
- (e) **Example 10-5.** (1) Sketch X vs. t for various values of k_d and k . Pay particular attention to the ratio k/k_d . (2) Repeat (1) for this example (i.e., the plotting of X vs. t) for a second-order reaction with $(C_{A0} = 1 \text{ mol/dm}^3)$ and first-order decay. (3) Repeat (2) for this example for a first-order reaction and first-order decay. (4) Repeat (1) for this example for a second-order reaction $(C_{A0} = 1 \text{ mol/dm}^3)$ and a second-order decay.

- (f) **Example 10-7.** (1) What if the solids and reactants entered from opposite ends of the reactor? How would your answers change? (2) What if the decay in the moving bed were second order? By how much must the catalyst charge, U_s , be increased to obtain the same conversion? (3) What if $\varepsilon = 2$ (e.g., $A \rightarrow 3B$) instead of zero, how would the results be affected?
- (g) **Example 10-8.** (1) What if you varied the parameters P_{A0} , U_g , A , and k' in the STTR? What parameter has the greatest effect on either increasing or decreasing the conversion? Ask questions such as: What is the effect of varying the ratio of k to U_g or of k to A on the conversion? Make a plot of conversion versus distance as U_g is varied between 0.5 and 50 m/s. Sketch the activity and conversion profiles for $U_g = 0.025, 0.25, 2.5$, and 25 m/s. What generalizations can you make? Plot the exit conversion and activity as a function of gas velocity between velocities of 0.02 and 50 m/s. What gas velocity do you suggest operating at? What is the corresponding entering volumetric flow rate? What concerns do you have operating at the velocity you selected? Would you like to choose another velocity? If so, what is it?
- (h) What if you were asked to sketch the temperature–time trajectories and to find the catalyst lifetimes for first- and for second-order decay when $E_A = 35$ kcal/mol, $E_d = 10$ kcal/mol, $k_{d0} = 0.01$ day⁻¹, and $T_0 = 400$ K? How would the trajectory of the catalyst lifetime change if $E_A = 10$ kcal/mol and $E_d = 35$ kcal/mol? At what values of k_{d0} and ratios of E_d to E_A would temperature–time trajectories not be effective? What would your temperature–time trajectory look like if $n = 1 + E_d/E_A$?
- (i) Write a question for this problem that involves critical thinking and explain why it involves critical thinking.

P10-2_A Download the Interactive Computer Games (**ICG**) from the CRE Web site. Play the game and then record your performance number for the game, which indicates your mastery of the material. Your professor has the key to decode your performance number. (This ICG is a little longer than the other ICGs.) ICG Heterogeneous Catalysis Performance # _____.

P10-3_A t-Butyl alcohol (TBA) is an important octane enhancer that is used to replace lead additives in gasoline [*Ind. Eng. Chem. Res.*, 27, 2224 (1988)]. TBA was produced by the liquid-phase hydration (W) of isobutene (I) over an Amberlyst-15 catalyst. The system is normally a multiphase mixture of hydrocarbon, water, and solid catalysts. However, the use of cosolvents or excess TBA can achieve reasonable miscibility.

The reaction mechanism is believed to be



Derive a rate law assuming:

- (a) The surface reaction is rate-limiting.
 (b) The adsorption of isobutene is limiting.
 (c) The reaction follows Eley–Rideal kinetics



and the surface reaction is limiting.

- (d) Isobutene (I) and water (W) are adsorbed on different sites.

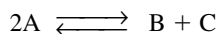


TBA is *not* on the surface, and the surface reaction is rate-limiting.

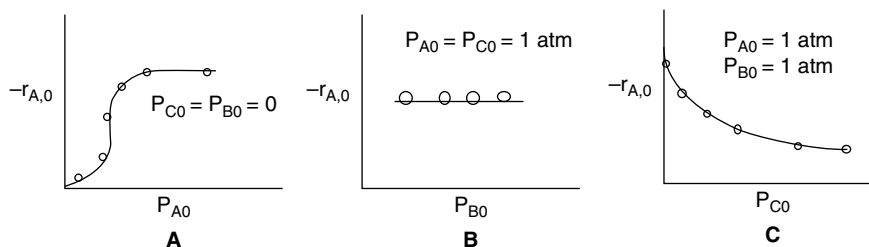
$$\left[\text{Ans.: } r'_{TBA} = -r'_1 = \frac{k [C_I C_W - C_{TBA} / K_c]}{(1 + K_W C_W)(1 + K_I C_I)} \right]$$

- (e) What generalization can you make by comparing the rate laws derived in parts (a) through (d)?

P10-4_B Consider the catalytic reaction as a function of the initial partial pressures



The rate of disappearance of species A was obtained in a differential reactor and is shown below.



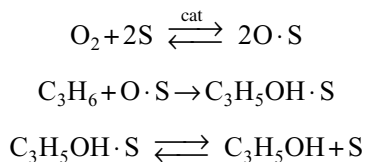
- What species are on the surface?
- What does Figure B tell you about the reversibility and what's adsorbed on the surface?
- Derive the rate law and suggest a rate-limiting step consistent with the above figures.
- How would you plot your data to linearize the initial rate data in Figure A?
- Assuming pure A is fed, and the adsorption constants for A and C are $K_A = 0.5 \text{ atm}^{-1}$ and $K_C = 0.25 \text{ atm}^{-1}$ respectively, at what conversion are the number of sites with A adsorbed on the surface and C adsorbed on the surface equal? (Ans.: $X = 0.66$)

P10-5_A The rate law for the hydrogenation (H) of ethylene (E) to form ethane (A) over a cobalt-molybdenum catalyst [Collection Czech. Chem. Commun., 51, 2760 (1988)] is

$$-r'_E = \frac{kP_E P_H}{1 + K_E P_E}$$

- Suggest a mechanism and rate-limiting step consistent with the above rate law.
- What was the most difficult part in finding the mechanism?

P10-6_B The formation of propanol on a catalytic surface is believed to proceed by the following mechanism



Suggest a rate-limiting step and derive a rate law.

P10-7_B The dehydration of *n*-butyl alcohol (butanol) over an alumina-silica catalyst was investigated by J. F. Maurer (Ph.D. thesis, University of Michigan). The data in Figure P10-7_B were obtained at 750°F in a modified differential reactor. The feed consisted of pure butanol.

- Suggest a mechanism and rate-limiting step that are consistent with the experimental data.
- Evaluate the rate-law parameters.
- At the point where the initial rate is a maximum, what is the fraction of vacant sites? What is the fraction of occupied sites by both A and B? % vacant = 0.41
- What generalizations can you make from studying this problem?
- Write a question that requires critical thinking and then explain why your question requires critical thinking. *Hint:* See Preface Section I.
- Apply one or more of the six ideas in Preface Table P-4, page xxviii, to this problem.

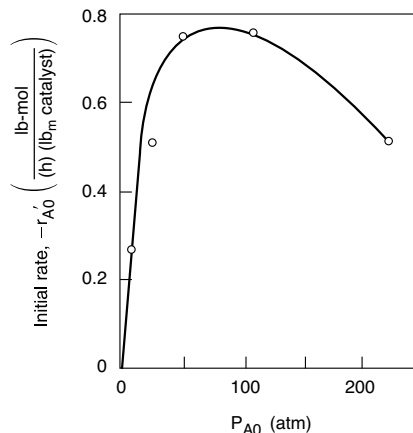
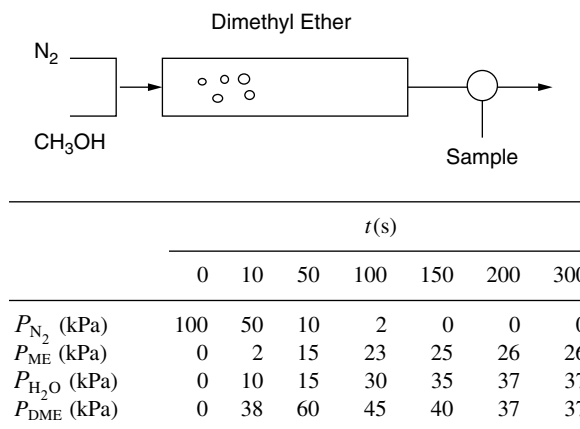
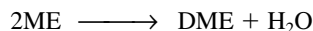
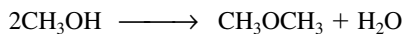


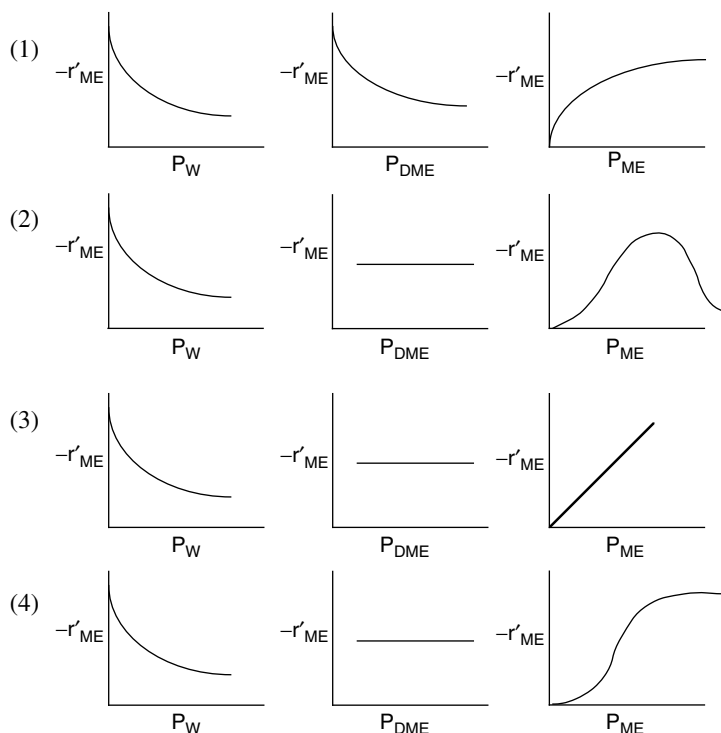
Figure P10-7_B Initial reaction rate as function of initial partial pressure of butanol.

P10-8_B The catalytic dehydration of methanol (ME) to form dimethyl ether (DME) and water was carried out over an ion exchange catalyst [K. Klusacek, *Collection Czech. Chem. Commun.*, 49, 170 (1984)]. The packed bed was initially filled with nitrogen, and at $t = 0$. The N_2 feed is switched to pure methanol vapor entering the reactor at 413 K, 100 kPa, and 0.2 cm³/s. The following partial pressures were recorded at the exit to the differential reactor containing 1.0 g of catalyst in 4.5 cm³ of reactor volume.



Use parts (a) through (f) to lead you to suggest a mechanism, rate-limiting step, and rate law consistent with this data.

- Using the data above, sketch the exit concentrations as a function of time.
- Which species took longer than others to exit the reactor in the gas phase? What could have caused this difference in exit times?
- What species are adsorbed on the surface?
- Are any species not adsorbed on the surface? If so, which ones?
- Which set of figures, (1)-(4) below, correctly describes the functionality of the chemical reaction rate with the partial pressures P_W , P_{DME} , and P_{ME} ?
- Derive a rate law for the catalytic dehydration of methanol. Dimethyl Ether $2ME \rightarrow DME + W$.

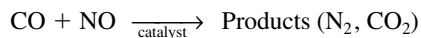


P10-9_B In 1981, the U.S. government put forth the following plan for automobile manufacturers to reduce emissions from automobiles over the next few years.

	Year		
	1981	1993	2010
Hydrocarbons	0.41	0.25	0.125
CO	3.4	3.4	1.7
NO	1.0	0.4	0.2

All values are in grams per mile. An automobile emitting 3.74 lb_m of CO and 0.37 lb_m of NO on a journey of 1000 miles would meet the current government requirements.

To remove oxides of nitrogen (assumed to be NO) from automobile exhaust, a scheme has been proposed that uses unburned carbon monoxide (CO) in the exhaust to reduce the NO over a solid catalyst, according to the reaction



Experimental data for a particular solid catalyst indicate that the reaction rate can be well represented over a large range of temperatures by

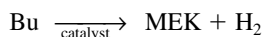
$$-r'_N = \frac{kP_N P_C}{(1 + K_1 P_N + K_2 P_C)^2} \quad (\text{P10-9.1})$$

where P_N = gas-phase partial pressure of NO
 P_C = gas-phase partial pressure of CO
 k, K_1, K_2 = coefficients depending only on temperature



- (a) Propose an adsorption–surface reaction–desorption mechanism and rate-limiting step that are consistent with the experimentally observed rate law. Do you need to assume any species are weakly adsorbed to get agreement with Equation (P10-9.1)?
- (b) A certain engineer thinks that it would be desirable to operate with a very large stoichiometric excess of CO to minimize catalytic reactor volume. Do you agree or disagree? Explain.
- (c) What would be the relevance of the problem if everyone were driving a hybrid by 2018? A driverless car by 2020?

P10-10_B Methyl ethyl ketone (MEK) is an important industrial solvent that can be produced from the dehydrogenation of butan-2-ol (Bu) over a zinc oxide catalyst [*Ind. Eng. Chem. Res.*, 27, 2050 (1988)]:

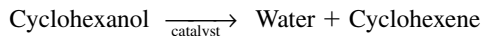


The following data giving the reaction rate for MEK were obtained in a differential reactor at 490°C.

P_{Bu} (atm)	2	0.1	0.5	1	2	1
P_{MEK} (atm)	5	0	2	1	0	0
P_{H_2} (atm)	0	0	1	1	0	10
r'_{MEK} (mol/h·g-cat.)	0.044	0.040	0.069	0.060	0.043	0.059

- (a) Suggest a rate law consistent with the experimental data.
- (b) Suggest a reaction mechanism and rate-limiting step consistent with the rate law. *Hint*: Some species might be weakly adsorbed.
- (c) Apply one or more of the six ideas in Preface Table P-4, page xxviii, to this problem.
- (d) Plot conversion (up to 90%) and reaction rate as a function of catalyst weight for an entering molar flow rate of pure butan-2-ol of 10 mol/min at an entering pressure $P_0 = 10$ atm up to a catalyst weight $W_{\text{max}} = 23$ kg.
- (e) Write a question that requires critical thinking and then explain why your question requires critical thinking. *Hint*: See Preface Section I.
- (f) Repeat part (d), accounting for pressure drop and $\alpha = 0.03$ kg⁻¹. Plot p and X as a function of catalyst weight down the reactor.

P10-11_B Cyclohexanol was passed over a catalyst to form water and cyclohexene:



It is suspected that the reaction may involve a dual-site mechanism, but it is not known for certain. It is believed that the adsorption equilibrium constant for cyclohexanol is around 1.0 and is roughly one or two orders of magnitude greater than the adsorption equilibrium constants for the other compounds. Using these data:

TABLE P10-11_B DATA FOR CATALYTIC FORMATION OF CYCLOHEXENE

Run	Reaction Rate (mol/dm ³ ·s) × 10 ⁵	Partial Pressure of Cyclohexanol (atm)	Partial Pressure of Cyclohexene (atm)	Partial Pressure of Steam (H ₂ O) (atm)
1	3.3	1	1	1
2	1.05	5	1	1
3	0.565	10	1	1
4	1.826	2	5	1
5	1.49	2	10	1
6	1.36	3	0	5
7	1.08	3	0	10
8	0.862	1	10	10
9	0	0	5	8
10	1.37	3	3	3

- (a) Suggest a rate law and mechanism consistent with the data given here.
 (b) Determine the values of the rate-law parameters. (*Ind. Eng. Chem. Res.*, 32, 2626–2632.)
 (c) Why do you think estimates of the rate-law parameters were given?
 (d) For an entering molar flow rate of cyclohexanol of 10 mol/s at a partial pressure of 15 atm, what catalyst weight is necessary to achieve 85% conversion when the bulk density is 1500 gm/dm³?

P10-12_B Experimental data for the gas-phase catalytic reaction



is shown below. The limiting step in the reaction is known to be irreversible, so that the overall reaction is irreversible. The reaction was carried out in a differential reactor to which A, B, and C were all fed.

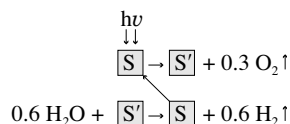
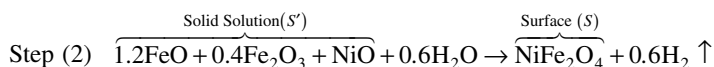
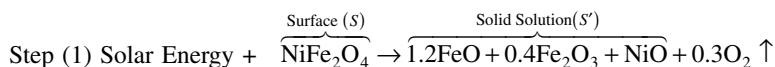
Run Number	P_A (atm)	P_B (atm)	P_C (atm)	Reaction rate (mol)/(g-cat • s)
1	1	1	2	0.114
2	1	10	2	1.140
3	10	1	2	0.180
4	1	20	2	2.273
5	1	20	10	0.926
6	20	1	2	0.186
7	0.1	1	2	0.0243

- (a) Suggest a rate law consistent with the experimental data. *Hint:* Sketch $(-r'_A)$ as a function of P_A , as a function of P_B , and as a function of P_C .
 (b) From your rate expression, which species can you conclude are adsorbed on the surface?
 (c) Suggest a rate law and then show that your mechanism is constant with the rate law in part (a).
 (d) For an entering partial pressure of A of 2 atm in a PBR, what is the ratio of A to C sites at 80% conversion of A?
 (e) At what conversion are the number of A and C sites equal? (*Ans.*: $X = 0.235$)
 (f) What catalyst weight is necessary to achieve 90% conversion of A for a stoichiometric feed and flow of A 2 mol/s? (*Ans.*: $W = 8.9$ g-cat)

If necessary, feel free to use none, any, or all of the following parameter values:

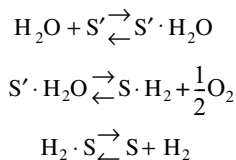
$$k = 2.5 \frac{\text{mol}}{\text{atm}^2 \text{g-cat} \cdot \text{s}}, K_A = 4 \text{ atm}^{-1}, K_C = 13 \text{ atm}^{-1}, K_I = 10 \text{ atm}^{-1}$$

P10-13_B Solar Energy Capture: Water Splitting. Hydrogen and O₂ can be combined in fuel cells to generate electricity. Solar energy can be used to split water to generate the raw reactants H₂ and O₂ for fuel cells. One method of solar thermal reduction is with NiFe₂O₄ in the sequence

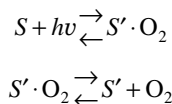


We note NiFe_2O_4 is regenerated in this process.³³

- (a) Derive a rate law for Step (2), assuming that water adsorbs on the solid solution as a single-site mechanism and that the reaction is irreversible.
- (b) Repeat (a) when the reaction is reversible and the solid solution adsorption site for water (S') is different than the NiFe_2O_4 site for the adsorption of H_2 , (S).



- (c) How would your rate law change if we included Step 1?



P10-14_A Vanadium oxides are of interest for various sensor applications, owing to the sharp metal–insulator transitions they undergo as a function of temperature, pressure, or stress. Vanadium triisopropoxide (VTIPO) was used to grow vanadium oxide films by *chemical vapor deposition* [*J. Electrochem. Soc.*, 136, 897 (1989)]. The deposition rate as a function of VTIP pressure for two different temperatures follows.

$T = 120^\circ\text{C}$:

Growth Rate ($\mu\text{m/h}$)	0.004	0.015	0.025	0.04	0.068	0.08	0.095	0.1
VTIPO Pressure (torr)	0.1	0.2	0.3	0.5	0.8	1.0	1.5	2.0

$T = 200^\circ\text{C}$:

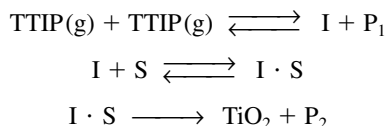
Growth Rate ($\mu\text{m/h}$)	0.028	0.45	1.8	2.8	7.2
VTIPO Pressure (torr)	0.05	0.2	0.4	0.5	0.8

In light of the material presented in this chapter, analyze the data and describe your results. Specify where additional data should be taken.

P10-15_A Titanium dioxide is a wide-bandgap semiconductor that is showing promise as an insulating dielectric in VLSI capacitors and for use in solar cells. Thin films of TiO_2 are to be prepared by *chemical vapor deposition* from gaseous titanium tetraisopropoxide (TTIP). The overall reaction is



The reaction mechanism in a CVD reactor is believed to be [K. L. Siefeling and G. L. Griffin, *J. Electrochem. Soc.*, 137, 814 (1990)]



³³ Scheffe, J.R., J. Li, and A. W. Weimer, "A Spinel Ferrite/Hercynite Water-Splitting Redox Cycle," *International Journal of Hydrogen Energy*, 35, 3333–3340 (2010).

where I is an active intermediate and P_1 is one set of reaction products (e.g., H_2O , C_3H_6), and P_2 is another set. Assuming the homogeneous gas-phase reaction for TTIP is in equilibrium, derive a rate law for the deposition of TiO_2 . The experimental results show that at $200^\circ C$ the reaction is second order at low partial pressures of TTIP and zero order at high partial pressures, while at $300^\circ C$ the reaction is second order in TTIP over the entire pressure range. Discuss these results in light of the rate law you derived.

P10-16_B The dehydrogenation of methylcyclohexane (M) to produce toluene (T) was carried out over a 0.3% Pt/ Al_2O_3 catalyst in a differential catalytic reactor. The reaction is carried out in the presence of hydrogen (H_2) to avoid coking [*J. Phys. Chem.*, 64, 1559 (1960)].

(a) Determine the model parameters for each of the following rate laws.

$$\begin{aligned} (1) \quad -r'_M &= kP_M^\alpha P_{H_2}^\beta & (3) \quad -r'_M &= \frac{kP_M P_{H_2}}{(1 + K_M P_M)^2} \\ (2) \quad -r'_M &= \frac{kP_M}{1 + K_M P_M} & (4) \quad -r'_M &= \frac{kP_M P_{H_2}}{1 + K_M P_M + K_{H_2} P_{H_2}} \end{aligned}$$

Use the data in Table P10-16_B below.

(b) Which rate law best describes the data? *Hint*: Neither K_{H_2} or K_M can take on negative values.

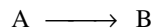
(c) Where would you place additional data points?

(d) Suggest a mechanism and rate-limiting step consistent with the rate law you have chosen.

TABLE P10-16_B DEHYDROGENATION OF METHYLCYCLOHEXANE

P_{H_2} (atm)	P_M (atm)	$r'_T \left(\frac{\text{mol toluene}}{\text{s} \cdot \text{kg-cat}} \right)$
1	1	1.2
1.5	1	1.25
0.5	1	1.30
0.5	0.5	1.1
1	0.25	0.92
0.5	0.1	0.64
3	3	1.27
1	4	1.28
3	2	1.25
4	1	1.30
0.5	0.25	0.94
2	0.05	0.41

P10-17_A Sketch *qualitatively* the reactant, product, and activity profiles as a function of length at various times for a *packed-bed reactor* for each of the following cases. In addition, sketch the effluent concentration of A as a function of time. The reaction is a simple isomerization:



(a) Rate law: $-r'_A = kaC_A$

Decay law: $r_d = k_d a C_A$

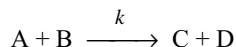
Case I: $k_d \ll k$, Case II: $k_d = k$, Case III: $k_d \gg k$

(b) $-r'_A = kaC_A$ and $r_d = k_d a^2$

(c) $-r'_A = kaC_A$ and $r_d = k_d a C_B$

- (d) Sketch similar profiles for the rate laws in parts (a) and (c) in a *moving-bed reactor* with the solids entering at the same end of the reactor as the reactant.
- (e) Repeat part (d) for the case where the solids and the reactant enter at opposite ends.

P10-18_B The elementary irreversible gas-phase catalytic reaction



is to be carried out in a moving-bed reactor at constant temperature. The reactor contains 5 kg of catalyst. The feed is stoichiometric in A and B. The entering concentration of A is 0.2 mol/dm³. The catalyst decay law is zero order with $k_D = 0.2 \text{ s}^{-1}$ and $k = 1.0 \text{ dm}^6/(\text{mol} \cdot \text{kg-cat} \cdot \text{s})$ and the volumetric flow rate is $v_0 = 1 \text{ dm}^3/\text{s}$.

- (a) What conversion will be achieved for a catalyst feed rate of 0.5 kg/s? (**Ans.:** $X = 0.2$)
- (b) Sketch the catalyst activity as a function of catalyst weight (i.e., distance) down the reactor length for a catalyst feed rate of 0.5 kg/s.
- (c) What is the maximum conversion that could be achieved (i.e., at an infinite catalyst loading rate)?
- (d) What catalyst loading rate is necessary to achieve 40% conversion? (**Ans.:** $U_s = 1.5 \text{ kg/s}$)
- (e) At what catalyst loading rate (kg/s) will the catalyst activity be exactly zero at the exit of the reactor?
- (f) What does an activity of zero mean? Can catalyst activity be less than zero?
- (g) How would your answer in part (a) change if the catalyst and reactant were fed at opposite ends? Compare with part (a).
- (h) Now consider the reaction to be zero order with $k = 0.2 \text{ mol/kg-cat} \cdot \text{min}$.

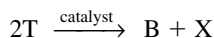
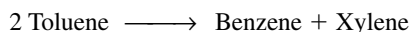
The economics:

- The product sells for \$160 per gram mole.
- The cost of operating the bed is \$10 per kilogram of catalyst exiting the bed.

What is the feed rate of solids (kg/min) that will give the maximum profit? (**Ans.:** $U_s = 4 \text{ kg/min.}$)

(Note: For the purpose of this calculation, ignore all other costs, such as the cost of the reactant, the cost to the company of providing free lunches to workers, etc.)

P10-19_B With the increasing demand for xylene in the petrochemical industry, the production of xylene from toluene disproportionation has gained attention in recent years [*Ind. Eng. Chem. Res.*, 26, 1854 (1987)]. This reaction,



was studied over a hydrogen mordenite catalyst that decays with time. As a first approximation, assume that the catalyst follows second-order decay

$$r_d = k_d a^2$$

and the rate law for low conversions is

$$-r'_T = k_T P_T a$$

with $k_T = 20 \text{ g mol/h} \cdot \text{kg-cat} \cdot \text{atm}$ and $k_d = 1.6 \text{ h}^{-1}$ at 735 K.

- (a) Compare the conversion-time curves in a batch reactor containing 5 kg-cat at different initial partial pressures (1 atm, 10 atm, etc.). The reaction volume containing pure toluene initially is 1 dm³ and the temperature is 735 K.
- (b) What conversion can be achieved in a *moving-bed reactor* containing 50 kg of catalyst with a catalyst feed rate of 2 kg/h? Toluene is fed at a pressure of 2 atm and a rate of 10 mol/min.
- (c) Explore the effect of catalyst feed rate on conversion.
- (d) Suppose that $E_T = 25 \text{ kcal/mol}$ and $E_d = 10 \text{ kcal/mol}$. What would the temperature-time trajectory look like for a CSTR? What if $E_T = 10 \text{ kcal/mol}$ and $E_d = 25 \text{ kcal/mol}$?

- (e) The decay law more closely follows the equation

$$r_d = k_d P_T^2 a^2$$

with $k_d = 0.2 \text{ atm}^{-2} \text{ h}^{-1}$. Redo parts (b) and (c) for these conditions.

P10-20_A The vapor-phase cracking of gas-oil in Example 10-6 is carried out over a different catalyst, for which the rate law is

$$-r'_A = k' P_A^2 \quad \text{with } k' = 5 \times 10^{-5} \frac{\text{kmol}}{\text{kg-cat} \cdot \text{s} \cdot \text{atm}^2}$$

- (a) Assuming that you can vary the entering pressure and gas velocity, what operating conditions would you recommend?

- (b) What could go wrong with the conditions you chose?

Now assume the decay law is

$$-\frac{da}{dt} = k_D a C_{\text{coke}} \quad \text{with } k_D = 100 \frac{\text{dm}^3}{\text{mol} \cdot \text{s}} \text{ at } 400^\circ\text{C}$$

where the concentration, C_{coke} , in mol/dm^3 , can be determined from a stoichiometric table.

- (c) For a temperature of 400°C and a reactor height of 15 m, what gas velocity do you recommend? Explain. What is the corresponding conversion?
- (d) The reaction is now to be carried in an STTR 15 m high and 1.5 m in diameter. The gas velocity is 2.5 m/s. You can operate in the temperature range between 100 and 500°C . What temperature do you choose, and what is the corresponding conversion?
- (e) What would the temperature–time trajectory look like for a CSTR?

Additional information:

$$E_R = 3000 \text{ cal/mol}$$

$$E_D = 15,000 \text{ cal/mol}$$

P10-21_C When the impurity cumene hydroperoxide is present in trace amounts in a cumene feed stream, it can deactivate the silica-alumina catalyst over which cumene is being cracked to form benzene and propylene. The following data were taken at 1 atm and 420°C in a differential reactor. The feed consists of cumene and a trace (0.08 mol %) of cumene hydroperoxide (CHP).

Benzene in Exit Stream (mol %)	2	1.62	1.31	1.06	0.85	0.56	0.37	0.24
t (s)	0	50	100	150	200	300	400	500

- (a) Determine the order of decay and the decay constant. (**Ans.:** $k_d = 4.27 \times 10^{-3} \text{ s}^{-1}$.)
- (b) As a first approximation (actually a rather good one), we shall neglect the denominator of the catalytic rate law and consider the reaction to be first order in cumene. Given that the specific reaction rate with respect to cumene is $k = 3.8 \times 10^3 \text{ mol/kg fresh cat} \cdot \text{s} \cdot \text{atm}$, the molar flow rate of cumene (99.92% cumene, 0.08% CHP) is 200 mol/min, the entering concentration is 0.06 kmol/ m^3 , the catalyst weight is 100 kg, and the velocity of solids is 1.0 kg/min, what conversion of cumene will be achieved in a *moving-bed reactor*?

P10-22_C The decomposition of spartanol to wulfrene and CO_2 is often carried out at high temperatures [J. Theor. Exp., 15, 15 (2014)]. Consequently, the denominator of the catalytic rate law is easily approximated as unity, and the reaction is first order with an activation energy of 150 kJ/mol.

Fortunately, the reaction is irreversible. Unfortunately, the catalyst over which the reaction occurs decays with time on stream. The following conversion-time data were obtained in a differential reactor:

For $T = 500$ K:

t (days)	0	20	40	60	80	120
X (%)	1	0.7	0.56	0.45	0.38	0.29

For $T = 550$ K:

t (days)	0	5	10	15	20	30	40
X (%)	2	1.2	0.89	0.69	0.57	0.42	0.33

(a) If the initial temperature of the catalyst is 480 K, determine the *temperature-time trajectory* to maintain a constant conversion.

(b) What is the catalyst lifetime?

P10-23_B The hydrogenation of ethylbenzene to ethylcyclohexane over a nickel-mordenite catalyst is zero order in both reactants up to an ethylbenzene conversion of 75% [*Ind. Eng. Chem. Res.*, 28 (3), 260 (1989)]. At 553 K, $k = 5.8$ mol ethylbenzene/(dm³ of catalyst · h). When a 100-ppm thiophene concentration entered the system, the ethylbenzene conversion began to drop.

Time (h)	0	1	2	4	6	8	12
Conversion	0.92	0.82	0.75	0.50	0.30	0.21	0.10

The reaction was carried out at 3 MPa and a molar ratio of $H_2/ETB = 10$. Discuss the catalyst decay. Be quantitative where possible.

SUPPLEMENTARY READING

1. A terrific discussion of heterogeneous catalytic mechanisms and rate-controlling steps may or may not be found in

BURGESS, THORNTON W., *The Adventures of Grandfather Frog*. New York: Dover Publications, Inc., 1915.

MASEL, R. I., *Principles of Adsorption and Reaction on Solid Surfaces*. New York: Wiley, 1996. A great reference.

SOMORJAI, G. A., *Introduction to Surface Chemistry and Catalysis*. New York: Wiley, 1994.

2. A truly excellent discussion of the types and rates of adsorption together with techniques used in measuring catalytic surface areas is presented in

MASEL, R. I., *Principles of Adsorption and Reaction on Solid Surfaces*. New York: Wiley, 1996.

3. Techniques for discriminating between mechanisms and models can be found in

BOX, G. E. P., W. G. HUNTER, and J. S. HUNTER, *Statistics for Experimenters*. New York: Wiley, 1978.

4. Examples of applications of catalytic principles to microelectronic manufacturing can be found in

BUTT, JOHN B., *Reaction Kinetics and Reactor Design. Second Edition, Revised and Expanded*. New York: Marcel Dekker, Inc., 1999.

DOBKIN, D. M., and M. K. ZURAW. *Principles of Chemical Vapor Deposition*. The Netherlands: Kluwer Academic Publishers, 2003.

Mass Transfer **14**

Limitations in Reacting Systems

Giving up is the ultimate tragedy.

—Robert J. Donovan

or

It ain't over 'til it's over.

—Yogi Berra
NY Yankees

Overview. Many industrial reactions are carried out at high temperatures where the overall rate of reaction is limited by the rate of mass transfer of reactants between the bulk fluid and the catalytic surface. By mass transfer, we mean any process in which diffusion plays a role. Under these circumstances our generation term becomes a little more complicated as we cannot directly use the rate laws discussed in Chapter 3. Now we have to consider the fluid velocity and the fluid properties when writing the mole balance. In the rate laws and catalytic reaction steps described in Chapter 10 (diffusion, adsorption, surface reaction, desorption, and diffusion), we neglected the diffusion steps.

In this chapter we discuss how to determine the rate of reaction and how to size reactors when the reactions are limited by mass transfer. To do this we

- Present the fundamentals of diffusion and molar flux, and then write the mole balance in terms of the mole fluxes for rectangular and for cylindrical coordinates (Section 14.1).
- Incorporate Fick's first law into our mole balance in order to describe flow, diffusion, and reaction (Section 14.2).
- Model diffusion through a stagnant film to a reacting surface (Section 14.3).

- Introduce the mass transfer coefficient, k_c , and describe how it is used to design mass transfer limited reactions (Section 14.4).
- Focus on one of the engineer's most important skills, i.e., to answer "What if..." questions, as *Robert the Worrier* does (Section 14.5).

14.1 Diffusion Fundamentals

The Algorithm

1. Mole balance
2. Rate law
3. Stoichiometry
4. Combine
5. Evaluate

The first step in our CRE algorithm is the mole balance, which we now need to extend to include the molar flux, W_{Az} , and diffusional effects. The molar flow rate of A in a given direction, such as the z direction down the length of a tubular reactor, is just the product of the flux, W_{Az} (mol/m² • s), and the cross-sectional area, A_c (m²); that is,

$$F_{Az} = A_c W_{Az}$$

In the previous chapters, we have only considered plug flow with no diffusion superimposed, in which case

$$W_{Az} = \frac{C_A v}{A_C}$$

We now drop the plug-flow assumption and extend our discussion of mass transfer in catalytic and other mass-transfer limited reactions. In Chapter 10 we focused on the middle three steps (3, 4, and 5) in a catalytic reaction and neglected steps (1), (2), (6), and (7) by assuming the reaction was surface-reaction limited. In this chapter we describe the first and last steps (1) and (7), as well as showing other applications in which mass transfer plays a role.

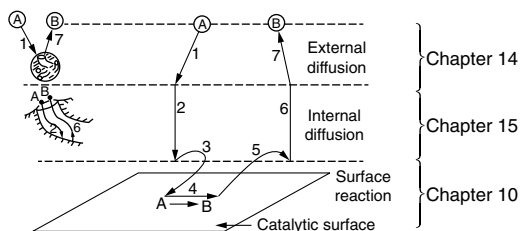


Figure 14-1 Steps in a heterogeneous catalytic reaction.

Where are we going ??:[†]

We want to arrive at the mole balance that incorporates both diffusion and reaction effects, such as Equation (14-16) on page 685. I.e.,

$$D_{AB} \frac{d^2 C_A}{dz^2} - U_z \frac{dC_A}{dz} + r_A = 0$$

[†] "If you don't know where you are going, you'll probably wind up some place else."
Yogi Berra, NY Yankees

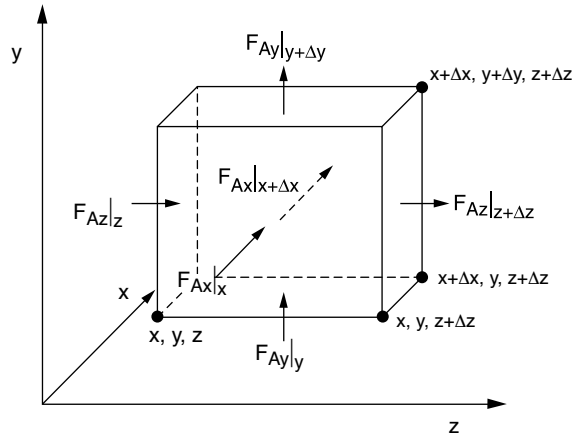
We begin with Section 14.1.1 where we write the mole balance on Species A in three dimensions in terms of the molar flux, $\mathbf{W}_A$. In Section 14.1.2 we write $\mathbf{W}_A$ in terms of the bulk flow of A in the fluid, $\mathbf{B}_A$ and the diffusion flux $\mathbf{J}_A$ of A that is superimposed on bulk flow. In Section 14.1.3 we use the previous two subsections as a basis to finally write the molar flux, $\mathbf{W}_A$, in terms of concentration using Fick's first law, $\mathbf{J}_A$, and the bulk flow, $\mathbf{B}_A$. Next, in Section 14.2 we combine diffusion convective transport and reaction in our mole balance.

14.1.1 Definitions

Diffusion is the spontaneous intermingling or mixing of atoms or molecules by random thermal motion. It gives rise to motion of the species *relative* to motion of the mixture. In the absence of other gradients (such as temperature, electric potential, or gravitational potential), molecules of a given species within a single phase will always diffuse from regions of higher concentrations to regions of lower concentrations. This gradient results in a molar flux of the species (e.g., A), $\mathbf{W}_A$ (moles/area·time), in the direction of the concentration gradient. The flux of A, $\mathbf{W}_A$, is relative to a fixed coordinate (e.g., the lab bench) and is a vector quantity with typical units of mol/m²·s. In rectangular coordinates

$$\mathbf{W}_A = iW_{Ax} + jW_{Ay} + kW_{Az} \quad (14-1)$$

We now apply the mole balance to species A, which flows and reacts in an element of volume $\Delta V = \Delta x \Delta y \Delta z$ to obtain the variation of the molar fluxes in three dimensions.



$$F_{Az} = W_{Az} \Delta x \Delta y$$

$$F_{Ay} = W_{Ay} \Delta x \Delta z$$

$$F_{Ax} = W_{Ax} \Delta y \Delta z$$

Mole Balance

$$\begin{aligned}
 & \left[\begin{array}{c} \text{Molar} \\ \text{flow rate} \\ \text{in} \end{array} \right]_{-z} - \left[\begin{array}{c} \text{Molar} \\ \text{flow rate} \\ \text{out} \end{array} \right]_{-z+\Delta z} + \left[\begin{array}{c} \text{Molar} \\ \text{flow rate} \\ \text{in} \end{array} \right]_{-y} - \left[\begin{array}{c} \text{Molar} \\ \text{flow rate} \\ \text{out} \end{array} \right]_{-y+\Delta y} + \\
 & \Delta x \Delta y W_{Az}|_z - \Delta x \Delta y W_{Az}|_{z+\Delta z} + \Delta x \Delta z W_{Ay}|_y - \Delta x \Delta z W_{Ay}|_{y+\Delta y} + \\
 & \left[\begin{array}{c} \text{Molar} \\ \text{flow rate} \\ \text{in} \end{array} \right]_{-x} - \left[\begin{array}{c} \text{Molar} \\ \text{flow rate} \\ \text{out} \end{array} \right]_{-x+\Delta x} + \left[\begin{array}{c} \text{Rate of} \\ \text{generation} \end{array} \right] = \left[\begin{array}{c} \text{Rate of} \\ \text{accumulation} \end{array} \right] \\
 & \Delta z \Delta y W_{Ax}|_x - \Delta z \Delta y W_{Ax}|_{x+\Delta x} + r_A \Delta x \Delta y \Delta z = \Delta x \Delta y \Delta z \frac{\partial C_A}{\partial t}
 \end{aligned}$$

where r_A is the rate of generation of A by reaction per unit volume (e.g., mol/m³/h).

Dividing by $\Delta x \Delta y \Delta z$ and taking the limit as they go to zero, we obtain the molar flux balance in rectangular coordinates

$$\boxed{-\frac{\partial W_{Ax}}{\partial x} - \frac{\partial W_{Ay}}{\partial y} - \frac{\partial W_{Az}}{\partial z} + r_A = \frac{\partial C_A}{\partial t}} \quad (14-2)$$

The corresponding balance in cylindrical coordinates with no variation in the rotation about the **z**-axis is

COMSOL

$$\boxed{-\frac{1}{r} \frac{\partial}{\partial r}(r W_{Ar}) - \frac{\partial W_{Az}}{\partial z} + r_A = \frac{\partial C_A}{\partial t}} \quad (14-3)$$

We will now evaluate the flux terms $\mathbf{W}_A$. We have taken the time to derive the molar flux equations in this form because they are now in a form that is consistent with the partial differential equation (PDE) solver COMSOL, which is accessible from the CRE Web site.

14.1.2 Molar Flux

The molar flux of A, $\mathbf{W}_A$, is the result of two contributions: $\mathbf{J}_A$, the molecular diffusion flux relative to the bulk motion of the fluid produced by a concentration gradient, and $\mathbf{B}_A$, the flux resulting from the bulk motion of the fluid:

Total flux =
diffusion + bulk
motion

$$\boxed{\mathbf{W}_A = \mathbf{J}_A + \mathbf{B}_A} \quad (14-4)$$

The bulk-flow term for species A is the total flux of all molecules relative to a fixed coordinate times the mole fraction of A, y_A ; i.e., $\mathbf{B}_A = y_A \sum \mathbf{W}_i$.

For a two-component system of A diffusing in B, the flux of A is

$$\mathbf{W}_A = \mathbf{J}_A + y_A(\mathbf{W}_A + \mathbf{W}_B) \quad (4-5)$$

The diffusional flux, $\mathbf{J}_A$, is the flux of A molecules that is superimposed on the bulk flow. It tells how fast A is moving ahead of the bulk flow velocity, i.e., the molar average velocity.

The flux of species A, $\mathbf{W}_A$, is wrt a *fixed coordinate system* (e.g., the lab bench) and is just the concentration of A, C_A , times the particle velocity of species A, $\mathbf{U}_A$, at that point

$$\mathbf{W}_A = \mathbf{U}_A C_A$$

$$\frac{\text{mol}}{\text{m}^2 \text{ s}} = \left(\frac{\text{m}}{\text{s}} \right) \left(\frac{\text{mol}}{\text{m}^3} \right)$$

By particle velocities, we mean the vector average of millions of molecules of A at a given point. Similarity for species B: $\mathbf{W}_B = \mathbf{U}_B C_B$; substituting into the bulk-flow term

$$\mathbf{B}_A = y_A \sum \mathbf{W}_i = y_A (\mathbf{W}_A + \mathbf{W}_B) = y_A (C_A \mathbf{U}_A + C_B \mathbf{U}_B)$$

Writing the concentration of A and B in the generic form in terms of the mole fraction, y_i , and the total concentration, c , i.e., $C_i = y_i c$, and then factoring out the total concentration, c , the bulk flow, $\mathbf{B}_A$, is

Molar
average
velocity

$$\mathbf{B}_A = (c y_A) (y_A \mathbf{U}_A + y_B \mathbf{U}_B) = C_A \mathbf{U}$$

where $\mathbf{U}$ is the molar average velocity: $\mathbf{U} = \sum y_i \mathbf{U}_i$. The molar flux of A can now be written as

$$\boxed{\mathbf{W}_A = \mathbf{J}_A + C_A \mathbf{U}} \quad (14-6)$$

We now need to determine the equation for the molar flux of A, $\mathbf{J}_A$, that is superimposed on the molar average velocity.

14.1.3 Fick's First Law

Experimentation
with frog legs led to
Fick's first law.

Our discussion on diffusion will be restricted primarily to binary systems containing only species A and B. We now wish to determine *how* the molar diffusive flux of a species (i.e., $\mathbf{J}_A$) is related to its concentration gradient. As an aid in the discussion of the transport law that is ordinarily used to describe diffusion, recall similar laws from other transport processes. For example, in conductive heat transfer the constitutive equation relating the heat flux $\mathbf{q}$ and the temperature gradient is Fourier's law, $\mathbf{q} = -k_t \nabla T$, where k_t is the thermal conductivity.

In rectangular coordinates, the gradient is in the form

Constitutive
equations in heat,
momentum, and
mass transfer

$$\nabla = i \frac{\partial}{\partial x} + j \frac{\partial}{\partial y} + k \frac{\partial}{\partial z}$$

The mass transfer law for the diffusional flux of A resulting from a concentration gradient is analogous to Fourier's law for heat transfer and is given by **Fick's first law**

$$\mathbf{J}_A = -D_{AB} \nabla C_A \quad (14-7)$$

D_{AB} is the diffusivity of A in B $\left(\frac{\text{m}^2}{\text{s}} \right)$. Combining Equations (14-7) and (14-6), we obtain an expression for the molar flux of A in terms of concentration for constant total concentration

Molar flux equation

$$\boxed{\mathbf{W}_A = -D_{AB} \nabla C_A + C_A \mathbf{U}} \quad (14-8)$$

In one dimension, i.e., z , the molar flux term is

$$W_{Az} = -D_{AB} \frac{dC_A}{dz} + C_A U_z \quad (14-8a)$$

14.2 Binary Diffusion

Although many systems involve more than two components, the diffusion of each species can be treated as if it were diffusing through another single species rather than through a mixture by defining an effective diffusivity.

14.2.1 Evaluating the Molar Flux

Now the task is to evaluate the bulk-flow term.

We now consider five typical cases in Table 14-1 of A diffusing in B. Substituting Equation (14-7) into Equation (14-6) we obtain

$$W_A = -D_{AB} \nabla C_A + y_A (W_A + W_B) \quad (14-9)$$

TABLE 14-1 EVALUATING W_A FOR SPECIES A DIFFUSING IN SPECIES B

- (1) Equal molar counter diffusion (EMCD) of species A and B. For every molecule of A that diffuses in the forward direction, one molecule of B diffuses in the reverse direction

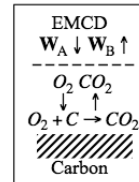
$$W_A = -W_B$$

$$W_A = J_A = -D_{AB} \nabla C_A \quad (14-10)$$

An example of EMCD is the oxidation of solid carbon; for every mole of oxygen that diffuses to the surface to react with the carbon surface, one mole of carbon dioxide diffuses away from the surface. $W_{O_2} = -W_{CO_2}$

- (2) Species A diffusing through stagnant species B ($W_B = 0$). This situation usually occurs when a solid boundary is involved and there is a stagnant fluid layer next to the boundary through which A is diffusing

$$W_A = J_A + y_A W_A$$



$$W_A = \frac{J_A}{1 - y_A} = -\frac{D_{AB} \nabla C_A}{1 - y_A} = +c D_{AB} \nabla \ln(1 - y_A) \quad (14-11)$$

- (3) Bulk flow of A is much greater than molecular diffusion of A, i.e., $B_A \gg J_A$

$$W_A = B_A = y_A (W_A + W_B) = C_A U \quad (14-12)$$

This case is the plug-flow model we have been using in the previous chapters in this book

$$F_A = W_A A_c = C_A \overbrace{U A_c}^v = v C_A$$

- (4) For small bulk flow $J_A \gg B_A$, we get the same result as EMCD, i.e., Equation (14-10)

$$W_A = J_A = -D_{AB} \nabla C_A \quad (14-10)$$

- (5) Knudsen Diffusion: Occurs in porous catalysts where the diffusing molecules collide more often with the pore walls than with each other

$$W_A = J_A = -D_K \nabla C_A \quad (14-13)$$

and D_K is the Knudsen diffusion.¹

¹ C. N. Satterfield, *Mass Transfer in Heterogeneous Catalysis* (Cambridge: MIT Press, 1970), pp. 41–42, discusses Knudsen flow in catalysis and gives the expression for calculating D_K .

14.2.2 Diffusion and Convective Transport

When accounting for diffusional effects, the molar flow rate of species A, F_A , in a specific direction z , is the product of molar flux in that direction, W_{Az} , and the cross-sectional area normal to the direction of flow, A_c

$$F_{Az} = A_c W_{Az}$$

In terms of concentration, the flux is

$$W_{Az} = -D_{AB} \frac{dC_A}{dz} + C_A U_z$$

The molar flow rate is

$$F_{Az} = W_{Az} A_c = \left[-D_{AB} \frac{dC_A}{dz} + C_A U_z \right] A_c \quad (14-14)$$

Similar expressions follow for W_{Ax} and W_{Ay} . Substituting for the flux W_{Ax} , W_{Ay} , and W_{Az} into Equation (14-2), we obtain

Flow, diffusion, and
reaction
This form is
used in COMSOL
Multiphysics.

$$D_{AB} \left[\frac{\partial^2 C_A}{\partial x^2} + \frac{\partial^2 C_A}{\partial y^2} + \frac{\partial^2 C_A}{\partial z^2} \right] - U_x \frac{\partial C_A}{\partial x} - U_y \frac{\partial C_A}{\partial y} - U_z \frac{\partial C_A}{\partial z} + r_A = \frac{\partial C_A}{\partial t} \quad (14-15)$$

Equation (14-15) is in a user-friendly form to apply to the PDE solver, COMSOL. For one dimension at steady state, Equation (14-15) reduces to

$$D_{AB} \frac{d^2 C_A}{dz^2} - U_z \frac{dC_A}{dz} + r_A = 0 \quad (14-16)$$

In order to solve Equation (14-16) we need to specify the boundary conditions. In this chapter we will consider some of the simple boundary conditions, and in Chapter 18 we will consider the more complicated boundary conditions, such as the Danckwerts' boundary conditions.

We will now use this form of the molar flow rate in our mole balance in the z direction of a tubular flow reactor

$$\frac{dF_A}{dV} = \frac{d(A_c W_{Az})}{d(A_c z)} = \frac{dW_{Az}}{dz} = r_A \quad (14-17)$$

However, we first have to discuss the boundary conditions in solving this equation.

14.2.3 Boundary Conditions

The most common boundary conditions are presented in Table 14-2.

TABLE 14-2 TYPES OF BOUNDARY CONDITIONS

1. Specify a concentration at a boundary (e.g., $z = 0$, $C_A = C_{A0}$).
For an instantaneous reaction at a boundary, the concentration of the reactants at the boundary is taken to be zero (e.g., $C_{As} = 0$). See Chapter 18 for the more exact and complicated Danckwerts' boundary conditions at $z = 0$ and $z = L$.

2. Specify a flux at a boundary.

- a. No mass transfer to a boundary

$$W_A = 0 \quad (14-18)$$

for example, at the wall of a nonreacting pipe. Species A cannot diffuse into the solid pipe wall so $W_A = 0$ and then

$$\frac{dC_A}{dr} = 0 \quad \text{at } r = R \quad (14-19)$$

That is, because the diffusivity is finite, the only way the flux can be zero is if the concentration gradient is zero.

- b. Set the molar flux to the surface equal to the rate of reaction on the surface

$$W_A(\text{surface}) = -r_A''(\text{surface}) \quad (14-20)$$

- c. Set the molar flux to the boundary equal to convective transport across a boundary layer

$$W_A(\text{boundary}) = k_c(C_{As} - C_{Ab}) \quad (14-21)$$

where k_c is the mass transfer coefficient and C_{As} and C_{Ab} are the surface and bulk concentrations, respectively.

3. Planes of symmetry. When the concentration profile is symmetrical about a plane, the concentration gradient is zero in that plane of symmetry. For example, in the case of radial diffusion in a pipe, at the center of the pipe

$$\frac{dC_A}{dr} = 0 \quad \text{at } r = 0 \quad (14-22)$$

14.2.4 Temperature and Pressure Dependence of D_{AB}

Before closing this brief discussion on mass-transfer fundamentals, further mention should be made of the diffusion coefficient.² Equations for predicting gas diffusivities are given by Fuller and are also given in Perry's *Handbook*.^{3,4} The orders of magnitude of the diffusivities for gases, liquids, and solids and the manner in which they vary with temperature and pressure are given in Table 14-3.⁵ We note that the Knudsen, liquid, and solid diffusivities are independent of total pressure.

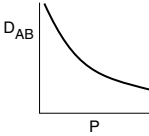
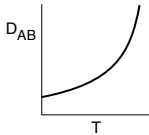
² For further discussion of mass-transfer fundamentals, see R. B. Bird, W. E. Stewart, and E. N. Lightfoot, *Transport Phenomena*, 2nd ed. (New York: Wiley, 2002).

³ E. N. Fuller, P. D. Schettler, and J. C. Giddings, *Ind. Eng. Chem.*, 58(5), 19 (1966). Several other equations for predicting diffusion coefficients can be found in B. E. Poling, J. M. Prausnitz, and J. P. O'Connell, *The Properties of Gases and Liquids*, 5th ed. (New York: McGraw-Hill, 2001).

⁴ R. H. Perry and D. W. Green, *Chemical Engineer's Handbook*, 7th ed. (New York: McGraw-Hill, 1999).

⁵ To estimate liquid diffusivities for binary systems, see K. A. Reddy and L. K. Doraiswamy, *Ind. Eng. Chem. Fund.*, 6, 77 (1967).

TABLE 14-3 DIFFUSIVITY RELATIONSHIPS FOR GASES, LIQUIDS, AND SOLIDS

	Phase	Order of Magnitude		Temperature and Pressure Dependences ^a
		cm ² /s	m ² /s	
It is important to know the magnitude and the T and P dependence of the diffusivity. Gas:  Liquid: 	Gas			
	Bulk	10 ⁻¹	10 ⁻⁵	$D_{AB}(T_2, P_2) = D_{AB}(T_1, P_1) \frac{P_1}{P_2} \left(\frac{T_2}{T_1} \right)^{1.75}$
	Knudsen	10 ⁻²	10 ⁻⁶	$D_A(T_2) = D_A(T_1) \left(\frac{T_2}{T_1} \right)^{1/2}$
	Liquid	10 ⁻⁵	10 ⁻⁹	$D_{AB}(T_2) = D_{AB}(T_1) \frac{\mu_1}{\mu_2} \left(\frac{T_2}{T_1} \right)$
	Solid	10 ⁻⁹	10 ⁻¹³	$D_{AB}(T_2) = D_{AB}(T_1) \exp \left[\frac{E_D}{R} \left(\frac{T_2 - T_1}{T_1 T_2} \right) \right]$

^a μ_1 , μ_2 , liquid viscosities at temperatures T_1 and T_2 , respectively; E_D , diffusion activation energy.

14.2.5 Steps in Modeling Diffusion to a Reacting Surface

We first consider the diffusion of species A through a stagnant film in which no reaction takes place to a catalytic surface where Species A reacts instantaneously by upon reaching the surface, i.e., $C_{As} \equiv 0$. Consequently, the rate of diffusion through the stagnant film equals the rate of reaction on the surface. The first steps in modeling are:

Steps in modeling mass transfer

- Step 1:* Perform a differential mole balance on a particular species A or use the general mole balance, to obtain an equation for W_{Az} , e.g., Equation (14-2).
- Step 2:* Replace W_{Az} by the appropriate expression for the concentration gradient.
- Step 3:* State the boundary conditions.
- Step 4:* Solve for the concentration profile.
- Step 5:* Solve for the molar flux.

In Section 14.3, we are going to apply this algorithm to one of the most important cases, diffusion through a boundary layer. Here, we consider the boundary layer to be a hypothetical “stagnant film” in which all the resistance to mass transfer is lumped.

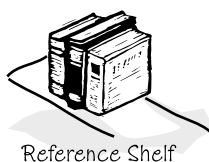
14.2.6 Modeling Diffusion with Chemical Reaction

Next, we consider the situation where species A reacts as it diffuses through the stagnant film.⁶ This table will provide the foundation for problems with diffusion and reaction in both Chapters 14 and 15.

The purpose of presenting algorithms (e.g., Table 14-4) to solve reaction engineering problems is to give the readers a starting point or framework with which to work if they were to get stuck. It is expected that once readers are

Use Table 14-4 to
Move
In $\longleftrightarrow$ Out
of the algorithm
(Steps 1 $\rightarrow$ 6)
to generate
creative solutions.

⁶ E. L. Cussler, *Diffusion Mass Transfer in Fluid Systems*, 2nd ed. (New York: Cambridge University Press, 1997).



Expanding the previous six modeling steps just a bit

TABLE 14-4 STEPS IN MODELING CHEMICAL SYSTEMS WITH DIFFUSION AND REACTION

1. Define the problem and state the assumptions.
2. Define the system on which the balances are to be made.
3. Perform a differential mole balance on a particular species.
4. Obtain a differential equation in W_A by rearranging your balance equation properly and taking the limit as the volume of the element goes to zero.
5. Substitute the appropriate expression involving the concentration gradient for W_A from Section 14.2 to obtain a second-order differential equation for the concentration of A.^a
6. Express the reaction rate r_A (if any) in terms of concentration and substitute into the differential equation.
7. State the appropriate boundary and initial conditions.
8. Put the differential equations and boundary conditions in dimensionless form.
9. Solve the resulting differential equation for the concentration profile.
10. Differentiate this concentration profile to obtain an expression for the molar flux of A.
11. Substitute numerical values for symbols.

^aIn some instances it may be easier to integrate the resulting differential equation in Step 4 before substituting for W_A .

familiar and comfortable using the algorithm/framework, they will be able to move in and out of the framework as they develop creative solutions to non-standard chemical reaction engineering problems.

14.3 Diffusion Through a Stagnant Film

To begin our discussion on the diffusion of reactants from the bulk fluid to the external surface of a catalyst, we shall focus attention on the flow past a single catalyst pellet. Reaction takes place only on the external catalyst surface and not in the fluid surrounding it. The fluid velocity in the vicinity of the spherical pellet will vary with position around the sphere. The hydrodynamic boundary layer is usually defined as the distance from a solid object to where the fluid velocity is 99% of the bulk velocity, U_0 . Similarly, the mass transfer boundary-layer thickness, δ , is defined as the distance from a solid object to where the concentration of the diffusing species reaches 99% of the bulk concentration.

A reasonable representation of the concentration profile for a reactant A diffusing to the external surface is shown in Figure 14-2. As illustrated, the change in concentration of A from C_{Ab} to C_{As} takes place in a very narrow fluid layer next to the surface of the sphere. Nearly all of the resistance to mass transfer is found in this layer.

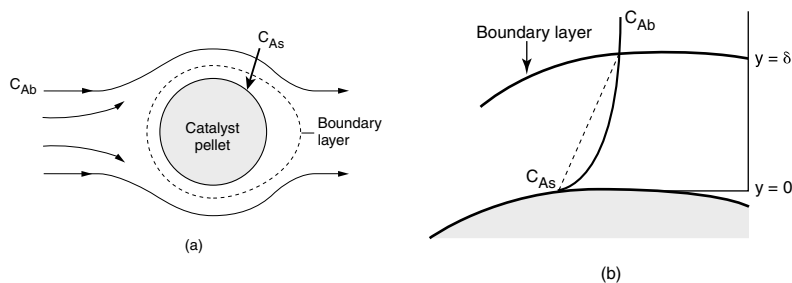


Figure 14-2 Boundary layer around the surface of a catalyst pellet.

The concept of a hypothetical stagnant film within which all the resistance to external mass transfer exists

A useful way of modeling diffusive transport is to treat the fluid layer next to a solid boundary as a hypothetical stagnant film of thickness δ , which we cannot measure. We say that *all* the resistance to mass transfer is found (i.e., lumped) within this hypothetical stagnant film of thickness δ , and the properties (i.e., concentration, temperature) of the fluid at the outer edge of the film are identical to those of the bulk fluid. This model can readily be used to solve the differential equation for diffusion through a stagnant film. The dashed line in Figure 14-2b represents the concentration profile predicted by the hypothetical stagnant film model, while the solid line gives the actual profile. If the film thickness is much smaller than the radius of the pellet (which is usually the case), curvature effects can be neglected. As a result, only the one-dimensional diffusion equation must be solved, as was shown in Figure 14-3.

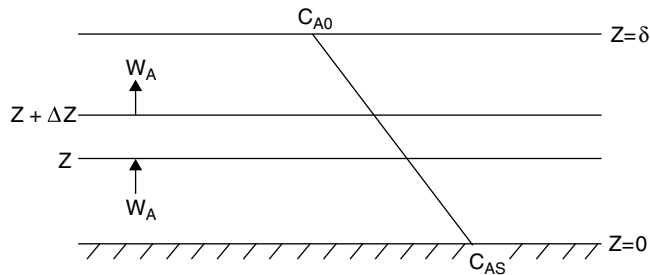


Figure 14-3 Concentration profile for dilute concentration in stagnant film mode

We are going to carry out a mole balance on species A diffusing through the fluid between $z = z$ and $z = z + \Delta z$ at steady state for the unit cross-sectional area, A_c

$$\begin{array}{ccccccc} \text{In} & - & \text{Out} & + & \text{Generation} & = & \text{Accumulation} \\ W_{Az}|_z & - & W_{Az}|_{z+\Delta z} & + & 0 & = & 0 \end{array}$$

dividing by Δz and taking the limit as $\Delta z \rightarrow 0$

$$\frac{dW_{Az}}{dz} = 0$$

For diffusion through a stagnant film at dilute concentrations

$$J_A \gg y_A (W_A + W_B) \quad (14-23)$$

or for EMCD, we have using Fick's first law

$$W_{Az} = -D_{AB} \frac{dC_A}{dz} \quad (14-24)$$

Substituting for W_{Az} and dividing by D_{AB} we have

$$\frac{d^2 C_A}{dz^2} = 0$$

Integrating twice to get $C_A = K_1 z + K_2$, using the boundary conditions at

$$z = 0 \quad C_A = C_{As}$$

$$z = \delta \quad C_A = C_{A0}$$

we obtain the concentration profile

$$C_A = C_{As} + (C_{Ab} - C_{As}) \frac{z}{\delta} \quad (14-25)$$

To find the flux to the surface we substitute Equation (14-25) into Equation (14-24) to obtain

$$W_{Az} = \frac{D_{AB}}{\delta} [C_{Ab} - C_{As}] \quad (14-26)$$

At steady state the flux of A to the surface will be equal to the rate of reaction of A on the surface. We also note that another example of diffusion through a stagnant film as applied to transdermal drug delivery is given in the Chapter 14 *Expanded Material* on the CRE Web site.

14.4 The Mass Transfer Coefficient

We now interpret the ratio (D_{AB}/δ) in Equation (14-26).

While the boundary-layer thickness will vary around the sphere, we will take it to have a mean film thickness δ . The ratio of the diffusivity D_{AB} to the film thickness δ is the mass transfer coefficient, k_c , that is,

The mass transfer
coefficient

$$k_c = \frac{D_{AB}}{\delta} \quad (14-27)$$

Combining Equations (14-26) and (14-27), we obtain the average molar flux from the bulk fluid to the surface

Molar flux of A to
the surface

$$W_{Az} = k_c (C_{Ab} - C_{As}) \quad (14-28)$$

In this stagnant film model, we consider all the resistance to mass transfer to be lumped into the thickness δ . The reciprocal of the mass transfer coefficient can be thought of as this resistance

$$W_{Az} = \text{Flux} = \frac{\text{Driving force}}{\text{Resistance}} = \frac{C_{Ab} - C_{As}}{(1/k_c)}$$

14.4.1 Correlations for the Mass Transfer Coefficient

The mass transfer coefficient k_c is analogous to the heat transfer coefficient h . The heat flux q from the bulk fluid at a temperature T_0 to a solid surface at T_s is

$$q_z = h(T_0 - T_s) \quad (14-29)$$

For forced convection, the heat transfer coefficient is normally correlated in terms of three dimensionless groups: the Nusselt number, Nu; the Reynolds number, Re; and the Prandtl number, Pr. For the single spherical pellets discussed here, Nu and Re take the following forms

$$\text{Nu} = \frac{hd_p}{k_t} \quad (14-30)$$

$$\text{Re} = \frac{U\rho d_p}{\mu} \quad (14-31)$$

The Prandtl number is not dependent on the geometry of the system

$$\text{Pr} = \frac{\mu C_p}{k_t} = \frac{\mu}{\rho} \left(\frac{\rho C_p}{k_t} \right) = \frac{\nu}{\alpha_t} \quad (14-32)$$

The Nusselt, Prandtl, and Reynolds numbers are used in forced convection heat transfer correlations.

where $\alpha_t = k_t/\rho C_p$ = thermal diffusivity, m^2/s

μ = viscosity, $\text{kg}/\text{m}\cdot\text{s}$

ρ = fluid density, kg/m^3

$\nu = \frac{\mu}{\rho}$ = kinematic viscosity (momentum diffusivity), m^2/s

d_p = diameter of pellet, m

U = free-stream velocity, m/s

k_t = thermal conductivity, $\text{J}/\text{K}\cdot\text{m}\cdot\text{s}$

h = heat transfer coefficient, $\text{J}/\text{m}^2\cdot\text{s}\cdot\text{K}$ or $\text{Watts}/\text{m}^2\text{ K}$

C_p = heat capacity, $\text{J}/\text{kg}/\text{K}$

The other symbols are as defined previously.

The heat transfer correlation relating the Nusselt number to the Prandtl and Reynolds numbers for flow around a sphere is⁷

$$\text{Nu} = 2 + 0.6\text{Re}^{1/2}\text{Pr}^{1/3} \quad (14-33)$$

Although this correlation can be used over a wide range of Reynolds numbers, it can be shown theoretically that if a sphere is immersed in a stagnant fluid ($\text{Re} = 0$), then

$$\text{Nu} = 2 \quad (14-34)$$

and that at higher Reynolds numbers in which the boundary layer remains laminar, we can neglect the 2 in Equation (14-34), in which case the Nusselt number becomes

$$\text{Nu} \approx 0.6\text{Re}^{1/2}\text{Pr}^{1/3} \quad (14-35)$$

Although further discussion of heat transfer correlations is no doubt worthwhile, it will not help us to determine the mass transfer coefficient and the mass flux from the bulk fluid to the external pellet surface. However, the preceding discussion on heat transfer was not entirely futile because, for similar geometries, *the heat and mass transfer correlations are analogous*. If a heat transfer correlation for the Nusselt number exists, the mass transfer coefficient

Converting a heat transfer correlation to a mass transfer correlation

⁷ W. E. Ranz and W. R. Marshall, Jr., *Chem. Eng. Prog.*, 48, 141–146, 173–180 (1952).

[†] Strictly speaking, replacing the Nusselt number by the Sherwood number is only valid for situations where the Lewis number, Le, is close to 1. $\text{Le} = \frac{\text{Sc}}{\text{Pe}} = \frac{\alpha_t}{D_{AB}}$

can be estimated by replacing the Nusselt and Prandtl numbers in this correlation by the Sherwood and Schmidt numbers, respectively:

$$\text{Sh} \longrightarrow \text{Nu}$$

$$\text{Sc} \longrightarrow \text{Pr}$$

The heat and mass transfer coefficients are analogous.[†] The corresponding fluxes are

$$q_z = h(T - T_s) \quad (14-36)$$

$$W_{Az} = k_c(C_A - C_{As})$$

The one-dimensional differential forms of the mass flux for EMCD and the heat flux are, respectively,

For EMCD the
heat and molar flux
equations are
analogous.

$$W_{Az} = -D_{AB} \frac{dC_A}{dz} \quad (\text{E14-1.3})$$

$$q_z = -k_t \frac{dT}{dz} \quad (14-37)$$

If we replace h by k_c and k_t by D_{AB} in Equation (14-30), i.e.,

$$\left. \begin{array}{l} h \longrightarrow k_c \\ k_t \longrightarrow D_{AB} \end{array} \right\} \text{Nu} \longrightarrow \text{Sh}$$

we obtain the mass transfer Nusselt number (i.e., the Sherwood number)

$$\text{Sherwood number} \quad \text{Sh} = \frac{k_c d_p}{D_{AB}} = \frac{(\text{m/s})(\text{m})}{\text{m}^2/\text{s}} \text{ dimensionless} \quad (14-38)$$

The Prandtl number is the ratio of the kinematic viscosity (i.e., the momentum diffusivity) to the thermal diffusivity. Because the Schmidt number is analogous to the Prandtl number, one would expect that Sc is the ratio of the momentum diffusivity (i.e., the kinematic viscosity), ν , to the mass diffusivity D_{AB} . Indeed, this is true

$$\alpha_t \longrightarrow D_{AB}$$

The Schmidt number is

$$\text{Schmidt number} \quad \text{Sc} = \frac{\nu}{D_{AB}} = \frac{\text{m}^2/\text{s}}{\text{m}^2/\text{s}} \text{ dimensionless} \quad (14-39)$$

Consequently, the correlation for mass transfer for flow around a spherical pellet is analogous to that given for heat transfer, Equation (14-33); that is,

$$\text{Sh} = 2 + 0.6\text{Re}^{1/2}\text{Sc}^{1/3} \quad (14-40)$$

This relationship is often referred to as the *Frössling correlation*.⁸

A few additional correlations for the Sherwood number from which one can determine the mass transfer coefficient are given in Table 14-5.

⁸ N. Frössling, *Gerlands Beitr. Geophys.*, 52, 170 (1938).

The Sherwood, Reynolds, and Schmidt numbers are used in forced convection mass transfer correlations.

TABLE 14-5 MASS TRANSFER CORRELATIONS

Turbulent flow, mass transfer to pipe wall	$Sh = .332 (Re)^{1/2} (Sc)^{1/3}$
Mass transfer to a single sphere	$Sh = 2 + 0.6 Re^{1/2} Sc^{1/3}$
Mass transfer in fluidized beds	$\phi J_D = \frac{0.765}{Re^{.82}} + \frac{0.365}{Re^{0.386}}$
Mass transfer to packed beds	$\phi J_D = 0.453 Re^{0.453}$
	$J_D = \frac{Sh}{ReSc^{1/3}}$

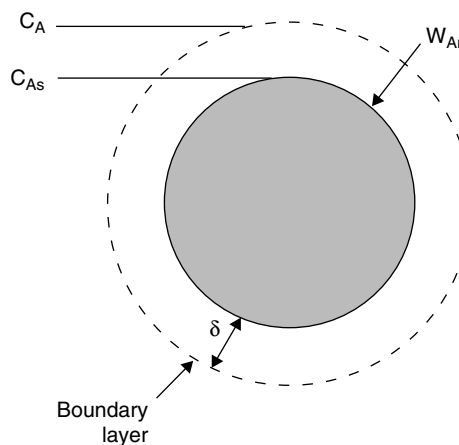
14.4.2 Mass Transfer to a Single Particle

In this section we consider two limiting cases of diffusion and reaction on a catalyst particle.⁹ In the first case, the reaction is so rapid that the rate of diffusion of the reactant to the surface limits the reaction rate. In the second case, the reaction is so slow that virtually no concentration gradient exists in the gas phase (i.e., rapid diffusion with respect to surface reaction).

Example 14-1 Rapid Reaction on the Surface of a Catalyst

If the surface reaction is rapid, then diffusion limits the overall rate.

Calculate the molar flux, W_{Ar} , of reactant A to a single catalyst pellet 1 cm in diameter suspended in a large body of liquid B. The reactant is present in dilute concentrations, and the reaction is considered to take place instantaneously at the external pellet surface (i.e., $C_{As} \approx 0$). The bulk concentration of the reactant A is 1.0 M, and the free-stream liquid velocity past the sphere is 0.1 m/s. The kinematic viscosity (i.e., $\frac{\mu}{\rho}$) is 0.5 centistoke (cS; 1 centistoke = 10^{-6} m²/s), and the liquid diffusivity of A in B is $D_{AB} = 10^{-10}$ m²/s, at 300 K.



⁹ A comprehensive list of correlations for mass transfer to particles is given by G. A. Hughmark, *Ind. Eng. Chem. Fund.*, 19(2), 198 (1980).

Solution

For dilute concentrations of the solute, the radial flux is

$$W_{Ar} = k_c(C_{Ab} - C_{As}) \quad (14-28)$$

Because reaction is assumed to occur instantaneously on the external surface of the pellet, $C_{As} = 0$. Also, C_{Ab} is given as 1 mol/dm³. The mass transfer coefficient for single spheres is calculated from the Frössling correlation

$$\text{Sh} = \frac{k_c d_p}{D_{AB}} = 2 + 0.6\text{Re}^{1/2}\text{Sc}^{1/3} \quad (14-41)$$

$$\text{Re} = \frac{\rho d_p U}{\mu} = \frac{d_p U}{\nu} = \frac{(0.01 \text{ m})(0.1 \text{ m/s})}{0.5 \times 10^{-6} \text{ m}^2/\text{s}} = 2000$$

$$\text{Sc} = \frac{\nu}{D_{AB}} = \frac{5 \times 10^{-7} \text{ m}^2/\text{s}}{10^{-10} \text{ m}^2/\text{s}} = 5000$$

Substituting these values into Equation (14-40) gives us

$$\text{Sh} = 2 + 0.6(2000)^{0.5}(5000)^{1/3} = 460.7 \quad (\text{E14-1.1})$$

$$k_c = \frac{D_{AB}}{d_p} \text{Sh} = \frac{10^{-10} \text{ m}^2/\text{s}}{0.01 \text{ m}} \times 460.7 = 4.61 \times 10^{-6} \text{ m/s} \quad (\text{E14-1.2})$$

$$C_{Ab} = 1.0 \text{ mol/dm}^3 = 10^3 \text{ mol/m}^3$$

Substituting for k_c and C_{Ab} in Equation (14-26), the molar flux to the surface is

$$W_{Ar} = (4.61 \times 10^{-6} \text{ m/s})(10^3 - 0) \text{ mol/m}^3 = 4.61 \times 10^{-3} \text{ mol/m}^2 \cdot \text{s}$$

Because $W_{Ar} = -r''_{As}$, this rate is also the rate of reaction per unit surface area of catalyst.

$$-r''_{As} = 0.0046 \text{ mol/m}^2 \cdot \text{s} = 0.46 \text{ mol/dm}^2 \cdot \text{s}$$

Analysis: In this example we calculated the rate of reaction on the external surface of a catalyst pellet when external mass transfer was limiting the reaction rate. To determine the rate of reaction, we used correlations to calculate the mass transfer coefficient and then used k_c to calculate the flux to the surface, which in turn was equal to the rate of surface reaction.

In Example 14-1, the surface reaction was extremely rapid and the rate of mass transfer to the surface dictated the overall rate of reaction. We now consider a more general case. The isomerization



is taking place on the surface of a solid sphere (Figure 14-4). The surface reaction follows a Langmuir–Hinshelwood single-site mechanism for which the rate law is

$$-r''_{As} = \frac{k_r C_{As}}{1 + K_A C_{As} + K_B C_{Bs}} \quad (14-41)$$

Liquid Phase

$$\text{Re} = 2000$$

$$\text{Sc} = 5000$$

$$\text{Sh} = 460$$

$$k_c = 4.6 \times 10^{-6} \text{ m/s}$$

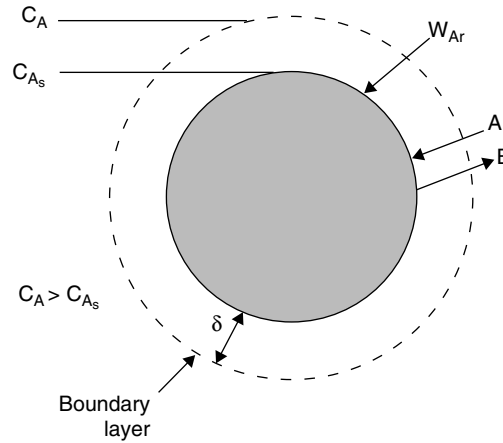


Figure 14-4 Diffusion to, and reaction on, external surface of pellet.

The temperature is sufficiently high that we only need to consider the case of very weak adsorption (i.e., low surface coverage) of A and B; thus

$$(K_B C_{Bs} + K_A C_{As}) \ll 1$$

Therefore, the rate law becomes apparent first order

$$-r''_{As} = k_r C_{As} \quad (14-42)$$

Using boundary conditions 2b and 2c in Table 14-1, we obtain

$$W_A|_{\text{surface}} = -r''_{As} \quad (14-43)$$

$$W_A = k_c(C_A - C_{As}) = k_r C_{As} \quad (14-44)$$

The concentration C_{As} is not as easily measured as the bulk concentration. Consequently, we need to eliminate C_{As} from the equation for the flux and rate of reaction. Solving Equation (14-44) for C_{As} yields

$$C_{As} = \frac{k_c C_A}{k_r + k_c} \quad (14-45)$$

and the rate of reaction on the surface becomes

Molar flux of A to the surface is equal to the rate of consumption of A on the surface.

$$W_A = -r''_{As} = \frac{k_c k_r C_A}{k_r + k_c} \quad (14-46)$$

One will often find the flux to or from the surface written in terms of an *effective* transport coefficient k_{eff}

$$W_A = -r''_{As} = k_{\text{eff}} C_A \quad (14-47)$$

where

$$k_{\text{eff}} = \frac{k_c k_r}{k_c + k_r}$$

Rapid Reaction. We first consider how the overall rate of reaction may be increased when the rate of mass transfer to the surface limits the overall rate of reaction. Under these circumstances, the specific reaction rate constant is much greater than the mass transfer coefficient

$$k_r \gg k_c$$

and

$$k_{\text{eff}} = k_c$$

$$W_A = -r''_{As} = \frac{k_c C_A}{1 + k_c/k_r} \approx k_c C_A \quad (14-48)$$

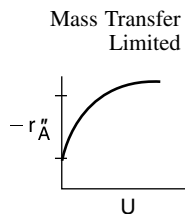
To increase the rate of reaction per unit surface area of a solid sphere, one must increase C_A and/or k_c . In this gas-phase catalytic reaction example, and for most liquids, the Schmidt number is sufficiently large that the number 2 in Equation (14-40) is negligible with respect to the second term when the Reynolds number is greater than 25. As a result, Equation (14-40) gives

$$\begin{aligned} k_c &= 0.6 \left(\frac{D_{AB}}{d_p} \right) \text{Re}^{1/2} \text{Sc}^{1/3} \\ &= 0.6 \left(\frac{D_{AB}}{d_p} \right) \left(\frac{U d_p}{\nu} \right)^{1/2} \left(\frac{\nu}{D_{AB}} \right)^{1/3} \end{aligned}$$

$$k_c = 0.6 \times \frac{D_{AB}^{2/3}}{\nu^{1/6}} \times \frac{U^{1/2}}{d_p^{1/2}} \quad (14-49)$$

$$k_c = 0.6 \times (\text{Term 1}) \times (\text{Term 2})$$

It is important to know how the mass transfer coefficient varies with fluid velocity, particle size, and physical properties.



Term 1 is a function of the physical properties D_{AB} and ν , which depend on temperature and pressure only. The diffusivity always increases with increasing temperature for both gas and liquid systems. However, the kinematic viscosity ν increases with temperature ($\nu \propto T^{3/2}$) for gases and decreases exponentially with temperature for liquids. Term 2 is a function of flow conditions and particle size. Consequently, to increase k_c and thus the overall rate of reaction per unit surface area, one may either decrease the particle size or increase the velocity of the fluid flowing past the particle. For this particular case of flow past a single sphere, we see that if the velocity is doubled, the mass transfer coefficient and consequently the rate of reaction is increased by a factor of

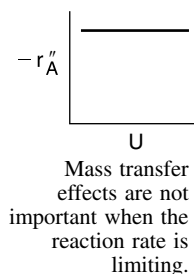
$$(U_2/U_1)^{0.5} = 2^{0.5} = 1.41 \text{ or } 41\%$$

Reaction Rate Limited

Slow Reaction. Here, the specific reaction rate constant is small with respect to the mass transfer coefficient

$$k_r \ll k_c$$

$$W_A = -r''_{As} = \frac{k_r C_A}{1 + k_r/k_c} \approx k_r C_A \quad (14-50)$$



The specific reaction rate is independent of the velocity of fluid and for the solid sphere considered here, independent of particle size. *However*, for porous catalyst pellets, k_r may depend on particle size for certain situations, as shown in Chapter 15. We will continue this discussion in Section 14.5.

Figure 14-5 shows the variation in reaction rate with Term 2 in Equation (14-49), the ratio of velocity to particle size. At low velocities, the mass transfer boundary-layer thickness is large and diffusion limits the reaction. As the velocity past the sphere is increased, the boundary-layer thickness decreases, and the mass transfer across the boundary layer no longer limits the rate of reaction. One also notes that for a given (i.e., fixed) velocity, reaction-limiting conditions can be achieved by using very small particles. However, the smaller the particle size, the greater the pressure drop in a packed bed. When one is obtaining reaction-rate data in the laboratory, one must operate at sufficiently high velocities or sufficiently small particle sizes to ensure that the reaction is not mass transfer-limited when collecting data.

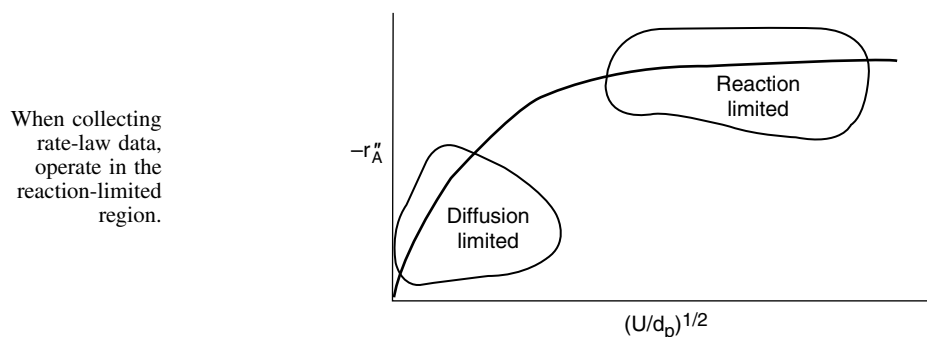
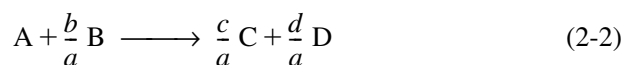


Figure 14-5 Regions of mass transfer-limited and reaction-limited reactions.

14.4.3 Mass Transfer-Limited Reactions in Packed Beds

A number of industrial reactions are potentially mass transfer-limited because they may be carried out at high temperatures without the occurrence of undesirable side reactions. In mass transfer-dominated reactions, the surface reaction is so rapid that the rate of transfer of reactant from the bulk gas or liquid phase to the surface limits the overall rate of reaction. Consequently, mass transfer-limited reactions respond quite differently to changes in temperature and flow conditions than do the rate-limited reactions discussed in previous chapters. In this section the basic equations describing the variation of conversion with the various reactor design parameters (catalyst weight, flow conditions) will be developed. To achieve this goal, we begin by carrying out a mole balance on the following generic mass transfer-limited reaction



carried out in a packed-bed reactor (Figure 14-6). A steady-state mole balance on reactant A in the reactor segment between z and $z + \Delta z$ is

$$\begin{aligned} \left[\begin{array}{c} \text{Molar} \\ \text{rate in} \end{array} \right] - \left[\begin{array}{c} \text{Molar} \\ \text{rate out} \end{array} \right] + \left[\begin{array}{c} \text{Molar rate of} \\ \text{generation} \end{array} \right] &= \left[\begin{array}{c} \text{Molar rate of} \\ \text{accumulation} \end{array} \right] \\ F_{Az}|_z - F_{Az}|_{z+\Delta z} + r_A'' a_c (A_c \Delta z) &= 0 \end{aligned} \quad (14-51)$$

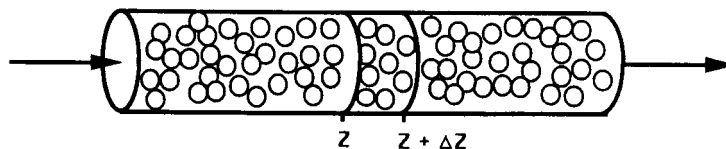


Figure 14-6 Packed-bed reactor.

where r_A'' = rate of generation of A per unit of catalytic surface area, $\text{mol/s} \cdot \text{m}^2$

a_c = external surface area of catalyst per volume of catalytic bed, m^2/m^3

$$a_c = \frac{\text{Volume of solid}}{\text{Volume of bed}} \times \frac{\text{Surface area}}{\text{Volume of solid}} = (1 - \phi) [\pi d_p^2 / (\pi d_p^3 / 6)]$$

$$= 6(1 - \phi) / d_p \text{ for packed beds, } \text{m}^2/\text{m}^3$$

ϕ = porosity of the bed (i.e., void fraction)¹⁰

d_p = particle diameter, m

A_c = cross-sectional area of tube containing the catalyst, m^2

Dividing Equation (14-51) by $A_c \Delta z$ and taking the limit as $\Delta z \rightarrow 0$, we have

$$-\frac{1}{A_c} \left(\frac{dF_{Az}}{dz} \right) + r_A'' a_c = 0 \quad (14-52)$$

We now need to express F_{Az} and r_A'' in terms of concentration.

The molar flow rate of A in the axial direction is

$$F_{Az} = A_c W_{Az} = (J_{Az} + B_{Az}) A_c \quad (14-53)$$

In almost all situations involving flow in packed-bed reactors, the amount of material transported by diffusion or dispersion in the axial direction is negligible compared with that transported by convection (i.e., bulk flow)

$$J_{Az} \ll B_{Az}$$

(In Chapter 18 we consider the case when dispersive effects (e.g., diffusion) must be taken into account.) Neglecting dispersion, Equation (14-14) becomes

$$F_{Az} = A_c W_{Az} = A_c B_{Az} = U C_A A_c \quad (14-54)$$

where U is the superficial molar average velocity through the bed (m/s). Substituting for F_{Az} in Equation (14-52) gives us

$$-\frac{d(C_A U)}{dz} + r_A'' a_c = 0 \quad (14-55)$$

¹⁰In the nomenclature for Chapter 4, for the Ergun equation for pressure drop.

For the case of constant superficial velocity U

Differential equation describing flow and reaction in a packed bed

$$-U \frac{dC_A}{dz} + r_A'' a_c = 0 \quad (14-56)$$

For reactions at steady state, the molar flux of A to the particle surface, W_{Ar} (mol/m²·s) (see Figure 14-7), is equal to the rate of disappearance of A on the surface $-r_A''$ (mol/m²·s); that is

$$-r_A'' = W_{Ar} \quad (14-57)$$

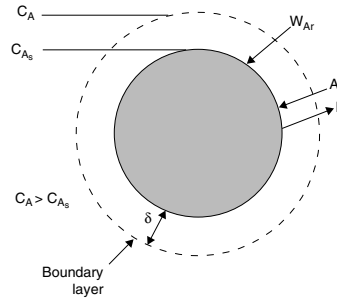
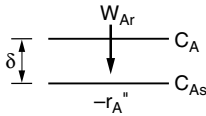


Figure 14-7 Diffusion across stagnant film surrounding catalyst pellet.



From Section 14.4, the boundary condition at the external surface is

$$-r_A'' = W_{Ar} = k_c(C_A - C_{As}) \quad (14-58)$$

where k_c = mass transfer coefficient = (D_{AB}/δ) , (m/s)

C_A = bulk concentration of A (mol/m³)

C_{As} = concentration of A at the catalytic surface (mol/m³)

Substituting for r_A'' in Equation (14-56), we have

$$-U \frac{dC_A}{dz} - k_c a_c (C_A - C_{As}) = 0 \quad (14-59)$$

In reactions that are completely mass transfer-limited, it is not necessary to know the rate law.

In most mass transfer-limited reactions, the surface concentration is negligible with respect to the bulk concentration (i.e., $C_A \gg C_{As}$)

$$-U \frac{dC_A}{dz} = k_c a_c C_A \quad (14-60)$$

Integrating with the limit, at $z = 0$, $C_A = C_{A0}$

$$\frac{C_A}{C_{A0}} = \exp\left(-\frac{k_c a_c}{U} z\right) \quad (14-61)$$

The corresponding variation of reaction rate along the length of the reactor is

$$-r_A'' = k_c C_{A0} \exp\left(-\frac{k_c a_c}{U} z\right) \quad (14-62)$$

The concentration and conversion profiles down a reactor of length L are shown in Figure 14-8.

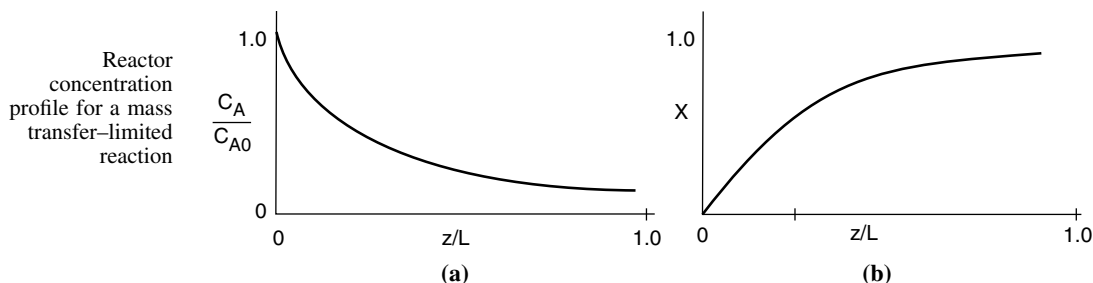


Figure 14-8 Axial concentration (a) and conversion (b) profiles in a packed bed.

To determine the reactor length L necessary to achieve a conversion X , we combine the definition of conversion

$$X = \frac{C_{A0} - C_{AL}}{C_{A0}} \quad (14-63)$$

with the evaluation of Equation (14-61) at $z = L$ to obtain

$$\ln \frac{1}{1-X} = \frac{k_c a_c}{U} L \quad (14-64)$$

14.4.4 Robert the Worrier



Robert is an engineer who is always worried (which is a Jofostanian trait). He thinks something bad will happen if we change an operating condition such as flow rate or temperature or an equipment parameter such as particle size. Robert's motto is "If it ain't broke, don't fix it." We can help Robert be a little more adventuresome by analyzing how the important parameters vary as we change operating conditions in order to predict the outcome of such a change. We first look at Equation (14-64) and see that conversion depends upon the parameters k_c , a_c , U , and L . We now examine how each of these parameters will change as we change operating conditions. We first consider the effects of temperature and flow rate on conversion.

To learn the effect of flow rate on conversion, we need to know how flow rate affects the mass transfer coefficient. That is, we must determine the correlation for the mass transfer coefficient for the particular geometry and flow field. For flow through a packed bed, the correlation given by Thoenes and Kramers for $0.25 < \phi < 0.5$, $40 < \text{Re}' < 4000$, and $1 < \text{Sc} < 4000$ is¹¹

$$\text{Sh}' = 1.0(\text{Re}')^{1/2} \text{Sc}^{1/3} \quad (14-65)$$

Thoenes–Kramers
correlation for flow
through packed
beds

$$\left[\frac{k_c d_p}{D_{AB}} \left(\frac{\phi}{1-\phi} \right) \frac{1}{\gamma} \right] = \left[\frac{U d_p \rho}{\mu (1-\phi) \gamma} \right]^{1/2} \left(\frac{\mu}{\rho D_{AB}} \right)^{1/3} \quad (14-66)$$

¹¹D. Thoenes, Jr. and H. Kramers, *Chem. Eng. Sci.*, 8, 271 (1958).

where $Re' = \frac{Re}{(1 - \phi)\gamma}$

$$Sh' = \frac{Sh \phi}{(1 - \phi)\gamma}$$

d_p = particle diameter (equivalent diameter of a sphere of the same volume), m

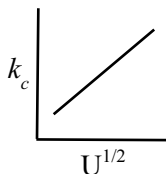
$$= [(6/\pi) (\text{volume of pellet})]^{1/3}, \text{ m}$$

ϕ = void fraction (porosity) of packed bed

γ = shape factor (external surface area divided by πd_p^2)

and

U, ρ, μ, ν and D_{AB} are as previously defined.



For constant fluid properties and particle diameter

$$k_c \propto U^{1/2} \quad (14-67)$$

We see that the mass transfer coefficient increases with the square root of the superficial velocity through the bed. Therefore, *for a fixed concentration, C_A* , such as that found in a differential reactor, the rate of reaction should vary with $U^{1/2}$

$$-r_A'' \propto k_c C_A \propto U^{1/2}$$

For diffusion-limited reactions, reaction rate depends on particle size and fluid velocity.

However, if the gas velocity is continually increased, a point is reached where the reaction becomes reaction rate-limited and, consequently, is independent of the superficial gas velocity, as shown in Figure 14-5.

Most mass transfer correlations in the literature are reported in terms of the Colburn J factor (i.e., J_D) as a function of the Reynolds number. The relationship between J_D and the numbers we have been discussing is

Colburn J factor

$$J_D = \frac{Sh}{Sc^{1/3} Re} \quad (14-68)$$

Figure 14-9 shows data from a number of investigations for the J factor as a function of the Reynolds number for a wide range of particle shapes and gas-flow conditions. *Note:* There are serious deviations from the Colburn analogy when the concentration gradient and temperature gradient are coupled, as shown by Venkatesan and Fogler.¹²

Dwidevi and Upadhyay review a number of mass transfer correlations for both fixed and fluidized beds and arrive at the following correlation, which is valid for both gases ($Re > 10$) and liquids ($Re > 0.01$) in either fixed or fluidized beds:¹³

A correlation for flow through packed beds in terms of the Colburn J factor

$$\phi J_D = \frac{0.765}{Re^{0.82}} + \frac{0.365}{Re^{0.386}} \quad (14-69)$$

¹²R. Venkatesan and H. S. Fogler, *AIChE J.*, 50, 1623 (July 2004).

¹³P. N. Dwidevi and S. N. Upadhyay, *Ind. Eng. Chem. Process Des. Dev.*, 16, 157 (1977).

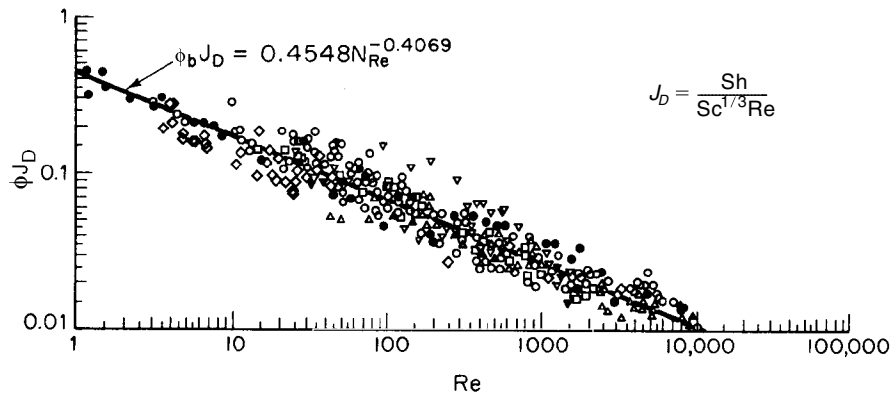


Figure 14-9 Mass transfer correlation for packed beds. $\phi_b \equiv \phi$ [Reprinted by permission. Copyright © 1977, American Chemical Society. Dwivedi, P. N. and S. N. Upadhyay, "Particle-Fluid Mass Transfer in Fixed and Fluidized Beds," *Industrial & Engineering Chemistry Process Design and Development*, 1977, 16 (2), 157–165.]

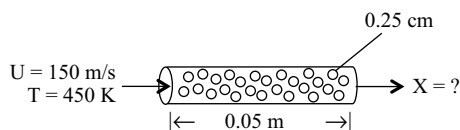
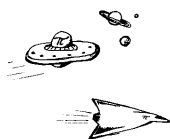
For nonspherical particles, the equivalent diameter used in the Reynolds and Sherwood numbers is $d_p = \sqrt{A_p/\pi} = 0.564 \sqrt{A_p}$, where A_p is the external surface area of the pellet.

To obtain correlations for mass transfer coefficients for a variety of systems and geometries, see either D. Kunii and O. Levenspiel, *Fluidization Engineering*, 2nd ed. (Butterworth-Heinemann, 1991), Chap. 7, or W. L. McCabe, J. C. Smith, and P. Harriott, *Unit Operations in Chemical Engineering*, 6th ed. (New York: McGraw-Hill, 2000). For other correlations for packed beds with different packing arrangements, see I. Colquhoun-Lee and J. Stepanek, *Chemical Engineer*, 108 (Feb. 1974).

Example 14-2 Mass Transfer Effects in Maneuvering a Space Satellite

Hydrazine has been studied extensively for use in monopropellant thrusters for space flights of long duration. Thrusters are used for altitude control of communication satellites. Here, the decomposition of hydrazine over a packed bed of alumina-supported iridium catalyst is of interest.¹⁴ In a proposed study, a 2% hydrazine in 98% helium mixture is to be passed over a packed bed of cylindrical particles 0.25 cm in diameter and 0.5 cm in length at a gas-phase velocity of 150 m/s and a temperature of 450 K. The kinematic viscosity of helium at this temperature is $4.94 \times 10^{-5} \text{ m}^2/\text{s}$. The hydrazine decomposition reaction is believed to be externally mass transfer-limited under these conditions. If the packed bed is 0.05 m in length, what conversion can be expected? Assume isothermal operation.

Actual case history
and current
application



¹⁴O. I. Smith and W. C. Solomon, *Ind. Eng. Chem. Fund.*, 21, 374.



Tedious reading and calculations, but we gotta know how to do the nitty-gritty.

Representative values

Gas Phase $Re' = 15173$ $Sc = 0.35$ $Sh' = 86.66$ $k_c = 6.15 \text{ m/s}$

Additional information:

$$D_{AB} = 0.69 \times 10^{-4} \text{ m}^2/\text{s at } 298 \text{ K}$$

Bed porosity: 40%

Bed fluidicity: 95.7%

Solution

The following solution is detailed and a bit tedious, but it is important to know the details of how a mass transfer coefficient is calculated.

Rearranging Equation (14-64) gives us

$$X = 1 - e^{-(k_c a_c / U)L} \quad (\text{E14-2.1})$$

(a) Using the Thoenes–Kramers correlation to calculate the mass transfer coefficient, k_c

1. First we find the volume-average particle diameter

$$d_p = \left(\frac{6V}{\pi} \right)^{1/3} = \left(6 \frac{\pi D^2 L}{\pi} \right)^{1/3} \quad (\text{E14-2.2})$$

$$= [1.5(0.0025 \text{ m})^2 (0.005 \text{ m})]^{1/3} = 3.61 \times 10^{-3} \text{ m}$$

2. Surface area per volume of bed

$$a_c = 6 \left(\frac{1-0.4}{d_p} \right) = 6 \left(\frac{1-0.4}{3.61 \times 10^{-3} \text{ m}} \right) = 998 \text{ m}^2/\text{m}^3 \quad (\text{E14-2.3})$$

3. Mass transfer coefficient

$$Re = \frac{d_p U}{\nu} = \frac{(3.61 \times 10^{-3} \text{ m})(150 \text{ m/s})}{4.94 \times 10^{-4} \text{ m}^2/\text{s}} = 10942$$

For cylindrical pellets

$$\gamma = \frac{2\pi r L_p + 2\pi r^2}{\pi d_p^2} = \frac{(2)(0.0025/2)(0.005) + (2)(0.0025/2)^2}{(3.61 \times 10^{-3})^2} = 1.20 \quad (\text{E14-2.4})$$

$$Re' = \frac{Re}{(1-\phi)\gamma} = \frac{10942}{(0.6)(1.2)} = 15173$$

Correcting the diffusivity to 450 K using Table 14-2 gives us

$$D_{AB}(450 \text{ K}) = D_{AB}(298 \text{ K}) \times \left(\frac{450}{298} \right)^{1.75} = (0.69 \times 10^{-4} \text{ m}^2/\text{s})(2.06)$$

$$D_{AB}(450 \text{ K}) = 1.42 \times 10^{-4} \text{ m}^2/\text{s} \quad (\text{E14-2.5})$$

$$Sc = \frac{\nu}{D_{AB}} = \frac{4.94 \times 10^{-5} \text{ m}^2/\text{s}}{1.42 \times 10^{-4} \text{ m}^2/\text{s}} = 0.35$$

Substituting Re' and Sc into Equation (14-65) yields

$$Sh' = (15173.92)^{1/2} (0.35)^{1/3} = (123.18)(0.70) = 86.66 \quad (\text{E14-2.6})$$

$$k_c = \frac{D_{AB}(1-\phi)}{d_p \phi} \gamma (Sh') = \left(\frac{1.42 \times 10^{-4} \text{ m}^2/\text{s}}{3.61 \times 10^{-3} \text{ m}} \right) \left(\frac{1-0.4}{0.4} \right) \times (1.2)(86.66)$$

$$k_c = 6.15 \text{ m/s} \quad (\text{E14-2.7})$$



Once again the
nitty-gritty

Typical
values

Gas Phase

$$\text{Re} = 11996$$

$$J_D = 0.025$$

$$\text{Sc} = 0.35$$

$$\text{Sh} = 212$$

$$k_c = 7.58 \text{ m/s}$$

Fluidicity??
Red herring!

The conversion is

$$\begin{aligned} X &= 1 - \exp \left[-(6.15 \text{ m/s}) \left(\frac{998 \text{ m}^2/\text{m}^3}{150 \text{ m/s}} \right) (0.05 \text{ m}) \right] \\ &= 1 - 0.13 \approx 0.87 \end{aligned} \quad (\text{E14-2.8})$$

We find 87% conversion.

(b) Colburn J_D factor to calculate k_c . To find k_c , we first calculate the surface-area-average particle diameter.

For cylindrical pellets, the external surface area is

$$A = \pi d L_p + 2\pi \left(\frac{d^2}{4} \right) \quad (\text{E14-2.9})$$

$$d_p = \sqrt{\frac{A}{\pi}} = \sqrt{\frac{\pi d L_p + 2\pi (d^2/4)}{\pi}} \quad (\text{E14-2.10})$$

$$= \sqrt{(0.0025)(0.005) + \frac{(0.0025)^2}{2}} = 3.95 \times 10^{-3} \text{ m}$$

$$a_c = \frac{6(1-\phi)}{d_p} = 910.74 \text{ m}^2/\text{m}^3$$

$$\begin{aligned} \text{Re} &= \frac{d_p U}{\nu} = \frac{(3.95 \times 10^{-3} \text{ m})(150 \text{ m/s})}{4.94 \times 10^{-5} \text{ m}^2/\text{s}} \\ &= 11996.04 \end{aligned}$$

$$\phi J_D = \frac{0.765}{\text{Re}^{0.82}} + \frac{0.365}{\text{Re}^{0.386}} \quad (14-69)$$

$$\begin{aligned} &= \frac{0.765}{(11996)^{0.82}} + \frac{0.365}{(11996)^{0.386}} = 3.5 \times 10^{-4} + 9.7 \times 10^{-3} \quad (\text{E14-2.11}) \\ &= 0.010 \end{aligned}$$

$$J_D = \frac{0.010}{0.4} = 0.25 \quad (\text{E14-2.12})$$

$$\begin{aligned} \text{Sh} &= \text{Sc}^{1/3} \text{Re}(J_D) \\ &= (0.35)^{1/3} (11996)(0.025) = 212 \end{aligned} \quad (\text{E14-2.13})$$

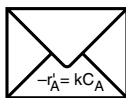
$$k_c = \frac{D_{AB}}{d_p} \text{Sh} = \frac{1.42 \times 10^{-4}}{3.95 \times 10^{-3}} (212) = 7.63 \text{ m/s}$$

$$\begin{aligned} \text{Then } X &= 1 - \exp \left[-(7.63 \text{ m/s}) \left(\frac{910 \text{ m}^2/\text{m}^3}{150 \text{ m/s}} \right) (0.05 \text{ m}) \right] \\ &\approx 0.9 \end{aligned} \quad (\text{E14-2.14})$$

If there were such a thing as the **bed fluidicity**, given in the problem statement, it would be a useless piece of information. Make sure that you know what information you need to solve problems, and go after it. Do not let additional data confuse you or lead you astray with useless information or facts that represent someone else's bias, and which are probably not well founded.

14.5 What If . . . ? (Parameter Sensitivity)

J. D. Goddard's



Back of the
Envelope

Find out how the
mass transfer
coefficient varies
with changes in
physical properties
and *system*
properties.

As we have stressed many times, one of the most important skills of an engineer is to be able to predict the effects of changes of system variables on the operation of a process. The engineer needs to determine these effects quickly through approximate but reasonably close calculations, which are sometimes referred to as “back-of-the-envelope calculations.”¹⁵ This type of calculation is used to answer such questions as “**What** will happen **if** I decrease the particle size?” “**What if** I triple the flow rate through the reactor?”

To help answer these questions, we recall Equation (14-49) and our discussion on page 696. There, we showed the mass transfer coefficient for a packed bed was related to the product of two terms: Term 1 was dependent on the physical properties and Term 2 was dependent on the system properties. Re-writing Equation (14-41) as

$$k_c \propto \left(\frac{D_{AB}^{2/3}}{\nu^{1/6}} \right) \left(\frac{U^{1/2}}{d_p^{1/2}} \right) \quad (14-70)$$

one observes from this equation that the mass transfer coefficient increases as the particle size decreases. The use of sufficiently small particles offers another technique to escape from the mass transfer-limited regime into the reaction-rate-limited regime.



Reactors in series
versus reactors in
parallel

Example 14-3 The Case of Divide and Be Conquered

A mass transfer-limited reaction is being carried out in two reactors of equal volume and packing, connected in series as shown in Figure E14-3.1. Currently, 86.5% conversion is being achieved with this arrangement. It is suggested that the reactors be separated and the flow rate be divided equally among each of the two reactors (Figure E14-3.2) to decrease the pressure drop and hence the pumping requirements. In terms of achieving a higher conversion, Robert is wondering if this is a good idea.

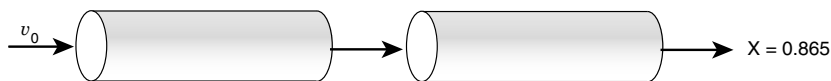


Figure E14-3.1 Series arrangement.

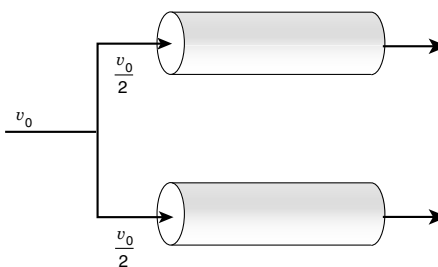


Figure E14-3.2 Parallel arrangement.

¹⁵Prof. J. D. Goddard, University of Michigan, 1963–1976. Currently at University of California, San Diego.

Solution

For the series arrangement we were given, $X_1 = 0.865$, and for the parallel arrangement, the conversion is unknown, i.e., $X_2 = ?$ As a first approximation, we neglect the effects of small changes in temperature and pressure on mass transfer. We recall Equation (14-64), which gives conversion as a function of reactor length. For a mass transfer-limited reaction

$$\ln \frac{1}{1-X} = \frac{k_c a_c}{U} L \quad (14-64)$$

For case 1, the undivided system

$$\left(\ln \frac{1}{1-X_1} \right) = \frac{k_{c1} a_c}{U_1} L_1 \quad (E14-3.1)$$

$$X_1 = 0.865$$

For case 2, the divided system

$$\left(\ln \frac{1}{1-X_2} \right) = \frac{k_{c2} a_c}{U_2} L_2 \quad (E14-3.2)$$

$$X_2 = ?$$

We now take the ratio of case 2 (divided system) to case 1 (undivided system)

$$\frac{\ln \frac{1}{1-X_2}}{\ln \frac{1}{1-X_1}} = \frac{k_{c2}}{k_{c1}} \left(\frac{L_2}{L_1} \right) \frac{U_1}{U_2} \quad (E14-3.3)$$

The surface area per unit volume a_c is the same for both systems.

From the conditions of the problem statement we know that

$$L_2 = \frac{1}{2} L_1, U_2 = \frac{1}{2} U_1, \text{ and } X_1 = 0.865$$

$$X_2 = ?$$

However, we must also consider the effect of the division on the mass transfer coefficient. From Equation (14-70) we know that

$$k_c \propto U^{1/2}$$

Then

$$\frac{k_{c2}}{k_{c1}} = \left(\frac{U_2}{U_1} \right)^{1/2} \quad (E14-3.4)$$

Multiplying by the ratio of superficial velocities yields

$$\frac{U_1}{U_2} \left(\frac{k_{c2}}{k_{c1}} \right) = \left(\frac{U_1}{U_2} \right)^{1/2} \quad (E14-3.5)$$

$$\ln \frac{1}{1-X_2} = \left(\ln \frac{1}{1-X_1} \right) \frac{L_2}{L_1} \left(\frac{U_1}{U_2} \right)^{1/2} \quad (E14-3.6)$$

Bad idea!! Robert was right to worry.



Robert worries if this temperature increase will be worth the trouble.

$$= \left(\ln \frac{1}{1 - 0.865} \right) \left[\frac{\frac{1}{2} L_1}{L_1} \left(\frac{U_1}{\frac{1}{2} U_1} \right)^{1/2} \right]$$

$$= 2.00 \left(\frac{1}{2} \right) \sqrt{2} = 1.414$$

Solving for X_2 gives us

$$X_2 = 0.76$$

Analysis: Consequently, we see that although the divided arrangement will have the advantage of a smaller pressure drop across the bed, it is a bad idea in terms of conversion. Recall that the series arrangement gave $X_1 = 0.865$; therefore ($X_2 < X_1$). Bad idea!! But every chemical engineering student in Jofostan knew that! Recall that if the reaction were reaction rate–limited, both arrangements would give the same conversion.

Example 14–4 The Case of the Overenthusiastic Engineers

The same reaction as that in Example 14–3 is being carried out in the same two reactors in series. A new engineer suggests that the rate of reaction could be increased by a factor of 2^{10} by increasing the reaction temperature from 400°C to 500°C , reasoning that the reaction rate doubles for every 10°C increase in temperature. Another engineer arrives on the scene and berates the new engineer with quotations from Chapter 3 concerning this rule of thumb. She points out that it is valid only for a specific activation energy within a specific temperature range. She then suggests that he go ahead with the proposed temperature increase but should only expect an increase on the order of 2^3 or 2^4 . What do you think? Who is correct?

Solution

Because almost all surface reaction rates increase more rapidly with temperature than do diffusion rates, increasing the temperature will only increase the degree to which the reaction is mass transfer–limited.

We now consider the following two cases:

Case 1: $T = 400^\circ\text{C}$ $X = 0.865$

Case 2: $T = 500^\circ\text{C}$ $X = ?$

Taking the ratio of case 2 to case 1 and noting that the reactor length is the same for both cases ($L_1 = L_2$), we obtain

$$\frac{\ln \frac{1}{1 - X_2}}{\ln \frac{1}{1 - X_1}} = \frac{k_{c2} \left(\frac{L_2}{L_1} \right) \frac{U_1}{U_2}}{k_{c1} \left(\frac{L_2}{L_1} \right) \frac{U_1}{U_2}} = \frac{k_{c2}}{k_{c1}} \left(\frac{U_1}{U_2} \right) \quad (\text{E14-4.1})$$

The molar feed rate F_{T0} remains unchanged

$$F_{T0} = v_{01} \left(\frac{P_{01}}{RT_{01}} \right) = v_{02} \left(\frac{P_{02}}{RT_{02}} \right) \quad (\text{E14-4.2})$$



the pressure remains constant so

$$\frac{v_{01}}{T_1} = \frac{v_{02}}{T_2}$$

Because $v = A_c U$, the superficial velocity at temperature T_2 is

$$U_2 = \frac{T_2}{T_1} U_1 \quad (\text{E14-4.3})$$

We now wish to learn the dependence of the mass transfer coefficient on temperature

$$k_c \propto \left(\frac{U^{1/2}}{d_p^{1/2}} \right) \left(\frac{D_{AB}^{2/3}}{v^{1/6}} \right) \quad (\text{E14-4.4})$$

Taking the ratio of case 2 to case 1 and realizing that the particle diameter is the same for both cases gives us

$$\frac{k_{c2}}{k_{c1}} = \left(\frac{U_2}{U_1} \right)^{1/2} \left(\frac{D_{AB2}}{D_{AB1}} \right)^{2/3} \left(\frac{v_1}{v_2} \right)^{1/6} \quad (\text{E14-4.5})$$

The temperature dependence of the gas-phase diffusivity is (from Table 14-2)

$$D_{AB} \propto T^{1.75} \quad (\text{E14-4.6})$$

For most gases, viscosity increases with increasing temperature according to the relation

$$\mu \propto T^{1/2}$$

From the ideal gas law

$$\rho \propto T^{-1}$$

Then

$$v = \frac{\mu}{\rho} \propto T^{3/2} \quad (\text{E14-4.7})$$

It's really important to know how to do this type of analysis.

$$\frac{\ln \frac{1}{1-X_2}}{\ln \frac{1}{1-X_1}} = \frac{U_1}{U_2} \left(\frac{k_{c2}}{k_{c1}} \right) = \left(\frac{U_1}{U_2} \right)^{1/2} \left(\frac{D_{AB2}}{D_{AB1}} \right)^{2/3} \left(\frac{v_1}{v_2} \right)^{1/6} \quad (\text{E14-4.8})$$

$$= \left(\frac{T_1}{T_2} \right)^{1/2} \left[\left(\frac{T_2}{T_1} \right)^{1.75} \right]^{2/3} \left[\left(\frac{T_1}{T_2} \right)^{3/2} \right]^{1/6}$$

$$\frac{U_1 k_{c2}}{U_2 k_{c1}} = \left(\frac{T_1}{T_2} \right)^{1/2} \left(\frac{T_2}{T_1} \right)^{7/6} \left(\frac{T_1}{T_2} \right)^{1/4} = \left(\frac{T_2}{T_1} \right)^{5/12} \quad (\text{E14-4.9})$$

$$= \left(\frac{773}{673} \right)^{5/12} = 1.059$$

Rearranging Equation (E14-4.1) in the form

$$\ln \frac{1}{1-X_2} = \frac{k_{c2} U_1}{k_{c1} U_2} \ln \frac{1}{1-X_1}$$

Bad idea!! Robert was right to worry.



Important concept

$$\ln \frac{1}{1-X_2} = 1.059 \left(\ln \frac{1}{1-0.865} \right) = 1.059(2) \quad (\text{E14-4.10})$$

$$X_2 = 0.88$$

Analysis: Consequently, we see that increasing the temperature from 400°C to 500°C increases the conversion by only 1.7%, i.e., $X = 0.865$ compared to $X = 0.88$. Bad idea! Bad, bad idea! Both engineers would have benefited from a more thorough study of this chapter.

For a packed catalyst bed, the temperature-dependence part of the mass transfer coefficient for a gas-phase reaction can be written as

$$k_c \propto U^{1/2} (D_{AB}^{2/3} / \nu^{1/6}) \quad (14-71)$$

$$k_c \propto U^{1/2} T^{11/12} \quad (14-72)$$

Depending on how one fixes or changes the molar feed rate, F_{T0} , U may also depend on the feed temperature. **As an engineer, it is extremely important that you reason out the effects of changing conditions**, as illustrated in the preceding two examples.

Closure. After completing this chapter, the reader should be able to define and describe molecular diffusion and how it varies with temperature and pressure, the molar flux, bulk flow, the mass transfer coefficient, the Sherwood and Schmidt numbers, and the correlations for the mass transfer coefficient. The reader should be able to choose the appropriate correlation and calculate the mass transfer coefficient, the molar flux, and the rate of reaction. The reader should also be able to describe the regimes and conditions under which mass transfer-limited reactions occur and when reaction-rate-limited reactions occur, and to make calculations of the rates of reaction and mass transfer for each case. One of the most important areas for the reader to apply the knowledge of this (and other chapters) is in his or her ability to ask and answer “What if . . .” questions.

SUMMARY

1. The molar flux of A in a binary mixture of A and B is

$$\mathbf{W}_A = -D_{AB} \nabla C_A + y_A (\mathbf{W}_A + \mathbf{W}_B) \quad (\text{S14-1})$$

- a. For equimolar counterdiffusion (EMCD) **or** for dilute concentration of the solute

$$\mathbf{W}_A = \mathbf{J}_A = -D_{AB} \nabla C_A \quad (\text{S14-2})$$

- b. For diffusion through a stagnant gas

$$\mathbf{W}_A = c D_{AB} \nabla \ln(1 - y_A) \quad (\text{S14-3})$$

c. For negligible diffusion

$$\mathbf{W}_A = y_A \mathbf{W} = y_A (\mathbf{W}_A + \mathbf{W}_B) = \mathbf{C}_A \mathbf{U} \quad (\text{S14-4})$$

2. The rate of mass transfer from the bulk fluid to a boundary at concentration C_{As} is

$$W_A = k_c (C_{Ab} - C_{As}) \quad (\text{S14-5})$$

where k_c is the mass transfer coefficient.

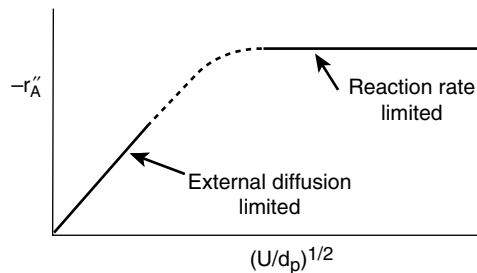
3. The Sherwood and Schmidt numbers are, respectively,

$$\text{Sh} = \frac{k_c d_p}{D_{AB}} \quad (\text{S14-6})$$

$$\text{Sc} = \frac{\nu}{D_{AB}} \quad (\text{S14-7})$$

Representative Values	
Liquid Phase	Gas Phase
$\text{Re} \sim 5000$	$\text{Re} \sim 500$
$\text{Sc} \sim 4000$	$\text{Sc} \sim 1$
$\text{Sh} \sim 500$	$\text{Sh} \sim 10$
$k_c = 10^{-2} \text{ m/s}$	$k_c = 5 \text{ m/s}$

4. If a heat transfer correlation exists for a given system and geometry, the mass transfer correlation may be found by replacing the Nusselt number by the Sherwood number and the Prandtl number by the Schmidt number in the existing heat transfer correlation.
5. Increasing the gas-phase velocity and decreasing the particle size will increase the overall rate of reaction for reactions that are externally mass transfer-limited.



6. The conversion for externally mass transfer-limited reactions can be found from the equation

$$\ln \frac{1}{1-X} = \frac{k_c a_c}{U} L \quad (\text{S14-8})$$

7. Back-of-the-envelope calculations should be carried out to determine the magnitude and direction that changes in process variables will have on conversion. **What if . . . ?**

CRE WEB SITE MATERIALS

- Expanded Material on the Website

- Transdermal Drug Delivery

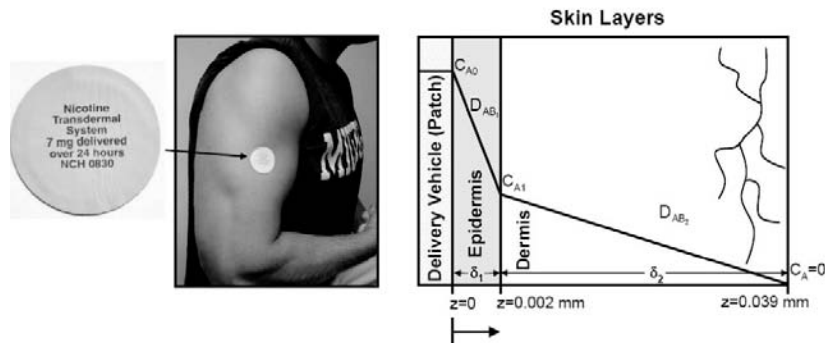


Figure WE14-1.1 Transdermal-drug-delivery schematic.

- Shrinking Core Model

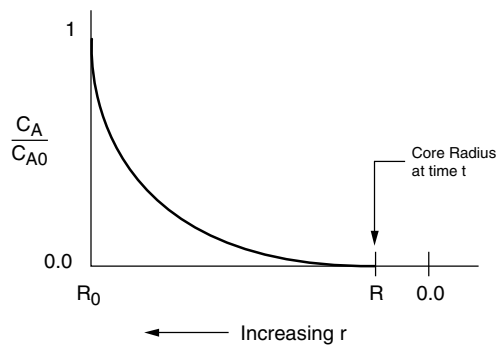


Figure 11-14 Oxygen concentration profile shown from the external radius of the pellet (R_0) to the pellet center. The gas-carbon interface is located at R .

- Diffusion through Film to a Catalyst Particle

- WP14-(a) Revisit Transdermal Drug Delivery

- Additional Homework Problems

- Learning Resources

- Summary Notes

Diffusion through a Stagnant Film

- Solved Problems

Example CD14-1 Calculating Steady State Diffusion

Example CD14-2 Relative Fluxes W_A , B_A , and J_A

Example CD14-3 Diffusion through a Stagnant Gas

Example CD14-4 Measuring Gas-Phase Diffusivities

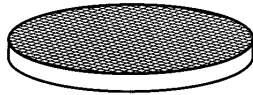
Example CD14-5 Diffusion through a Film to a Catalyst Particle

Example CD14-6 Measuring Liquid-Phase Diffusivities

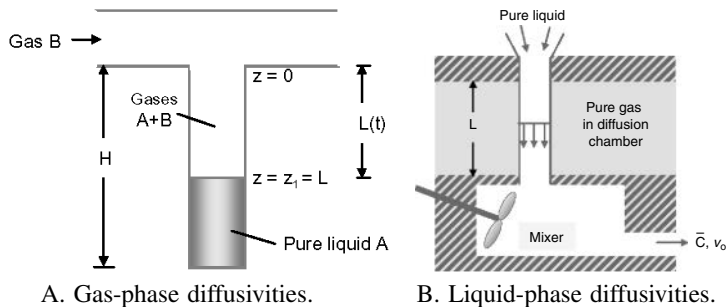
• **Professional Reference Shelf**

R14.1. *Mass Transfer-Limited Reactions on Metallic Surfaces*

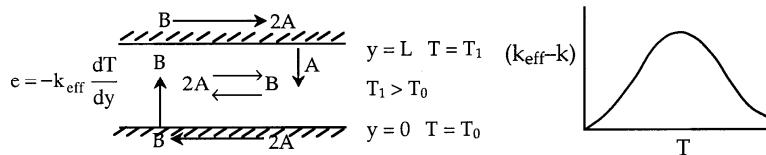
- A. Catalyst Monoliths
- B. Wire Gauze Reactors



R14.2. *Methods to Experimentally Measure Diffusivities*



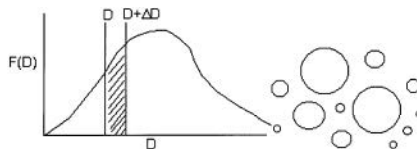
R14.3. *Facilitated Heat Transfer*



R14.4. *Shrinking Core Model*

R14.5. *Dissolution of Monodisperse Particles*

R14.6. *Dissolution of Polydisperse Solids (e.g., pills in the stomach)*



Ibuprofen

QUESTIONS AND PROBLEMS

The subscript to each of the problem numbers indicates the level of difficulty: A, least difficult; D, most difficult.

A = ● B = ■ C = ◆ D = ◆◆

Questions

Q14-1 Read over the problems at the end of this chapter. Make up an original problem that uses the concepts presented in this chapter. See problem P5-1_A for the guidelines. To obtain a solution:

- (a) Make up your data and reaction.
- (b) Use a real reaction and real data.

The journals listed at the end of Chapter 1 may be useful for part (b).



Creative Thinking

- Q14-2** (*Sergeant Ambergromby*). Capt. Apollo is piloting a shuttlecraft on his way to space station Klingon. Just as he is about to maneuver to dock his craft using the hydrazine system discussed in **Example 14-2**, the shuttle craft's thrusters do not respond properly and it crashes into the station, killing Capt. Apollo (*Star Wars 7* (fall 2015)). An investigation reveals that Lt. Darkside prepared the packed beds used to maneuver the shuttle and Lt. Data prepared the hydrazine-helium gas mixture. Foul play is suspected and Sgt. Ambergromby arrives on the scene to investigate.
- What are the first three questions he asks?
 - Make a list of possible explanations for the crash, supporting each one by an equation or reason.

Problems

- P14-1_B** (a) **Example 14-1**. How would your answers change if the temperature was increased by 50°C, the particle diameter was doubled, and fluid velocity was cut in half? Assume properties of water can be used for this system.
- (b) **Example 14-2**. How would your answers change if you had a 50–50 mixture of hydrazine and helium? If you increase d_p by a factor of 5?
- (c) **Example 14-3**. What if you were asked for representative values for Re, Sc, Sh, and k_c for both liquid- and gas-phase systems for a velocity of 10 cm/s and a pipe diameter of 5 cm (or a packed-bed diameter of 0.2 cm)? What numbers would you give?
- (d) **Example 14-4**. How would your answers change if the reaction were carried out in the liquid phase where kinematic viscosity varied as $\nu(T_2) = \nu(T_1)\exp\left[-4000K\left(\frac{1}{T_1} - \frac{1}{T_2}\right)\right]$?

- P14-2_B** Assume the minimum respiration rate of a chipmunk is 1.5 micromoles of O₂/min. The corresponding volumetric rate of gas intake is 0.05 dm³/min at STP.

- What is the deepest a chipmunk can burrow a 3-cm diameter hole beneath the surface in Ann Arbor, Michigan?
 $D_{AB} = 1.8 \times 10^{-5}$ m²/s
- In Boulder, Colorado?
- How would your answers to (a) and (b) change in the dead of winter when $T = 0^\circ\text{F}$?
- Critique and extend this problem (e.g., CO₂ poisoning). Thanks to Professor Robert Kabel at Pennsylvania State University.

Hint: Review derivations and equations for W_A and W_B to see how they can be applied to this problem.

- P14-3_B** Pure oxygen is being absorbed by xylene in a catalyzed reaction in the experimental apparatus sketched in Figure P14-3_B. Under constant conditions of temperature and liquid composition, the following data were obtained:

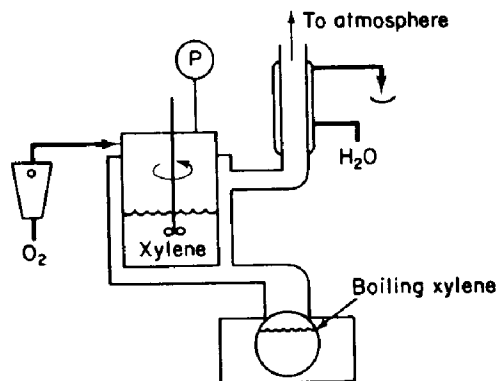


Figure P14-3_B

Stirrer Speed (rpm)	Rate of Uptake of O ₂ (mL/h) for System Pressure (absolute)			
	1.2 atm	1.6 atm	2.0 atm	3.0 atm
400	15	31	75	152
800	20	59	102	205
1200	21	62	105	208
1600	21	61	106	207

No gaseous products were formed by the chemical reaction. What would you conclude about the relative importance of liquid-phase diffusion and about the order of the kinetics of this reaction? (**California Professional Engineers Exam**)

- P14-4_C** In a diving-chamber experiment, a human subject breathed a mixture of O₂ and He while small areas of his skin were exposed to nitrogen gas. After some time, the exposed areas became blotchy, with small blisters forming on the skin. Model the skin as consisting of two adjacent layers, one of thickness δ_1 and the other of thickness δ_2 . If counterdiffusion of He out through the skin occurs at the same time as N₂ diffuses into the skin, at what point in the skin layers is the sum of the partial pressures a maximum? If the saturation partial pressure for the sum of the gases is 101 kPa, can the blisters be a result of the sum of the gas partial pressures exceeding the saturation partial pressure and the gas coming out of the solution (i.e., the skin)?



Before answering any of these questions, derive the concentration profiles for N₂ and He in the skin layers.

Diffusivity of He and N₂ in the inner skin layer = 5×10^{-7} cm²/s and 1.5×10^{-7} cm²/s, respectively

Diffusivity of He and N₂ in the outer skin layer = 10^{-5} cm²/s and 3.3×10^{-4} cm²/s, respectively

	External Skin Boundary Partial Pressure	Internal Skin Boundary Partial Pressure
N ₂	101 kPa	0
He	0	81 kPa
δ_1	20 μ m	Stratum corneum
δ_2	80 μ m	Epidermis

Hint: See Transdermal Drug Delivery in Expanded Material on the CRE Web site.

- P14-5_B** The decomposition of cyclohexane to benzene and hydrogen is mass transfer-limited at high temperatures. The reaction is carried out in a 5-cm-ID pipe 20 m in length packed with cylindrical pellets 0.5 cm in diameter and 0.5 cm in length. The pellets are coated with the catalyst only on the outside. The bed porosity is 40%. The entering volumetric flow rate is 60 dm³/min.

- Calculate the number of pipes necessary to achieve 99.9% conversion of cyclohexane from an entering gas stream of 5% cyclohexane and 95% H₂ at 2 atm and 500°C.
- Plot conversion as a function of pipe length.
- How much would your answer change if the pellet diameter and length were each cut in half?
- How would your answer to part (a) change if the feed were pure cyclohexane?
- What do you believe is the point of this problem? Is the focus really *green* CRE? How so?



Green engineering

P14-6_C Lead titanate, PbTiO_3 , is a material having remarkable ferroelectric, pyroelectric, and piezoelectric properties [*J. Elec. Chem. Soc.*, 135, 3137 (1988)]. A thin film of PbTiO_3 was deposited in a CVD reactor. The deposition rate is given below as a function of a temperature and flow rate over the film.

Flow Rate (SCCM)	Temperature (°C)	Deposition Rate ($\text{mg}/\text{cm}^2 \cdot \text{h}$)	Flow Rate (SCCM)	Temperature (°C)	Deposition Rate ($\text{mg}/\text{cm}^2 \cdot \text{h}$)
500	650	0.2	750	650	0.53
	750	0.8		750	1.45
	800	1.2		800	2.0
600	650	0.35	1000	650	0.55
	750	1.0		750	1.5
	800	1.5		800	2.0

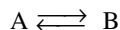
What are all the things, qualitative and quantitative, that you can learn from these data?

P14-7_B A plant is removing a trace of Cl_2 from a waste-gas stream by passing it over a solid granulum absorbent in a tubular packed bed (Figure P14-7). At present, 63.2% removal is being accomplished, but it is believed that greater removal could be achieved if the flow rate were increased by a factor of 4, the particle diameter were decreased by a factor of 3, and the packed tube length increased by 50%. What percentage of chlorine would be removed under the proposed scheme? (The chlorine transferring to the absorbent is removed completely by a virtually instantaneous chemical reaction.) (*Ans.*: 98%)



Figure P14-7

P14-8_B In a certain chemical plant, a reversible fluid-phase isomerization



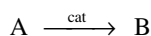
is carried out over a solid catalyst in a tubular packed-bed reactor. If the reaction is so rapid that mass transfer between the catalyst surface and the bulk fluid is rate-limiting, show that the kinetics are described in terms of the bulk concentrations C_A and C_B by

$$-r_A'' = \frac{k_B[C_A - (1/K)C_B]}{1/K + k_B/k_A}$$

where $-r_A'$ = moles of A reacting per unit area catalyst per
 k_A, k_B = transfer coefficients for A and B
 K = equilibrium constant

It is desired to double the capacity of the existing plant by processing twice the feed of reactant A while maintaining the same fractional conversion of A to B in the reactor. How much larger a reactor, in terms of catalyst weight, would be required if all other operating variables are held constant? You may use the Thoenes–Kramers correlation for mass transfer coefficients in a packed bed.

P14-9_B The irreversible gas-phase reaction



is carried out adiabatically over a packed bed of solid catalyst particles. The reaction is first order in the concentration of A on the catalyst surface

$$-r_{As}' = k' C_{As}$$

The feed consists of 50% (mole) A and 50% inerts, and enters the bed at a temperature of 300 K. The entering volumetric flow rate is 10 dm³/s (i.e., 10,000 cm³/s). The relationship between the Sherwood number and the Reynolds number is

$$\text{Sh} = 100 \text{Re}^{1/2}$$

As a first approximation, one may neglect pressure drop. The entering concentration of A is 1.0 M. Calculate the catalyst weight necessary to achieve 60% conversion of A for

- (a) isothermal operation.
- (b) adiabatic operation.

Additional information:

Kinematic viscosity: $\mu/\rho = 0.02 \text{ cm}^2/\text{s}$

Particle diameter: $d_p = 0.1 \text{ cm}$

Superficial velocity: $U = 10 \text{ cm/s}$

Catalyst surface area/mass of catalyst bed: $a = 60 \text{ cm}^2/\text{g-cat}$

Diffusivity of A: $D_e = 10^{-2} \text{ cm}^2/\text{s}$

Heat of reaction: $\Delta H_{\text{Rx}}^\circ = -10,000 \text{ cal/g mol A}$

Heat capacities: $C_{pA} = C_{pB} = 25 \text{ cal/g mol} \cdot \text{K}$, $C_{pS}(\text{solvent}) = 75 \text{ cal/g mol} \cdot \text{K}$

$$k'(300 \text{ K}) = 0.01 \text{ cm}^3/\text{s} \cdot \text{g-cat with } E = 4000 \text{ cal/mol}$$

P14-10_B *Transdermal Drug Delivery.* See photo on page 713. The principles of steady-state diffusion have been used in a number of drug-delivery systems. Specifically, medicated patches are commonly attached to the skin to deliver drugs for nicotine withdrawal, birth control, and motion sickness, to name a few. The U.S. transdermal drug-delivery is a multi-billion dollar market. Equations similar to Equation (14-24) have been used to model the release, diffusion, and absorption of the drug from the patch into the body. The figure shown in the *Expanded Material* on page 713 shows a drug-delivery vehicle (patch) along with the concentration gradient in the epidermis and dermis skin layers.

- (a) Use a shell balance to show

$$\frac{dW_{Az}}{dz} = 0$$

- (b) Show the concentration profile in the epidermis layer

$$\frac{C_{A0} - C_A}{C_{A0} - C_{A1}} = \frac{z}{\delta_1}$$

- (c) Show the concentration profile in the dermis layer

$$\frac{C_A}{C_{A1}} = \frac{\delta_2 - z}{\delta_2 - \delta_1}$$

- (d) Equate the fluxes using $W_{A1} = -D_{A1} \frac{dC_A}{dz}$ and $W_{A2} = -D_{A2} \frac{dC_A}{dz}$ at $z = \delta_1$ to show

$$C_{A1} = \frac{D_{A1} \frac{C_{A0}}{\delta_1}}{\frac{D_{A2}}{\delta_2 - \delta_1} + \frac{D_{A1}}{\delta_1}}$$

- (e) What are the concentration profiles in the dermis and epidermis layers?
 (f) Show the flux in the dermis layer is

$$W_{Az} = \frac{D_{A2}}{\delta_2 - \delta_1} \left[\frac{D_{A1} \frac{C_{A0}}{\delta_1}}{\frac{D_{A2}}{\delta_2 - \delta_1} + \frac{D_{A1}}{\delta_1}} \right]$$

- (g) What is the flux in the epidermis layer?

P14-11_D (*Estimating glacial ages*) The following oxygen-18 data were obtained from soil samples taken at different depths in Ontario, Canada. Assuming that all the ^{18}O was laid down during the last glacial age and that the transport of ^{18}O to the surface takes place by molecular diffusion, estimate the number of years since the last glacial age from the following data. Independent measurements give the diffusivity of ^{18}O in soil as $2.64 \times 10^{-10} \text{ m}^2/\text{s}$.



Figure P14-11 Glaciers.

Depth (m)	(surface)					
	0	3	6	9	12	18
^{18}O Conc. Ratio (C/C_0)	0	0.35	0.65	0.83	0.94	1.0

C_0 is the concentration of ^{18}O at 25 m. *Hint:* A knowledge of error function solutions may or may not be helpful. (*Ans.:* $t = 5,616$ years)

JOURNAL CRITIQUE PROBLEMS

P14C-1 The decomposition of nitric oxide on a heated platinum wire is discussed in *Chem. Eng. Sci.*, 30, 781. After making some assumptions about the density and the temperatures of the wire and atmosphere, and using a correlation for convective heat transfer, determine if mass transfer limitations are a problem in this reaction.

SUPPLEMENTARY READING

1. The fundamentals of diffusional mass transfer may or may not be found in

BIRD, R. B., W. E. STEWART, and E. N. LIGHTFOOT, *Transport Phenomena*, 2nd ed. New York: Wiley, 2002, Chaps. 17 and 18.

Collins, S., *Mockingjay* (The Final Book of the Hunger Games). New York: Scholastic, 2014.

CUSSLER, E. L., *Diffusion Mass Transfer in Fluid Systems*, 3rd ed. New York: Cambridge University Press, 2009.

GEANKOPLIS, C. J., *Transport Processes and Unit Operations*. Upper Saddle River, NJ: Prentice Hall, 2003.

LEVICH, V. G., *Physiochemical Hydrodynamics*. Upper Saddle River, NJ: Prentice Hall, 1962, Chaps. 1 and 4.

2. Experimental values of the diffusivity can be found in a number of sources, two of which are

PERRY, R. H., D. W. GREEN, and J. O. MALONEY, *Chemical Engineers' Handbook*, 8th ed. New York: McGraw-Hill, 2007.

SHERWOOD, T. K., R. L. PIGFORD, and C. R. WILKE, *Mass Transfer*. New York: McGraw-Hill, 1975.

3. A number of correlations for the mass transfer coefficient can be found in

LYDERSEN, A. L., *Mass Transfer in Engineering Practice*. New York: Wiley-Interscience, 1983, Chap. 1.

MCCABE, W. L., J. C. SMITH, and P. HARRIOTT, *Unit Operations of Chemical Engineering*, 6th ed. New York: McGraw-Hill, 2000, Chap. 17.

Diffusion and Reaction 15

Research is to see what everybody else sees, and
to think what nobody else has thought.

—Albert Szent-Gyorgyi

The concentration in
the internal surface
of the pellet is less
than that of the
external surface.

Overview. In this chapter we will develop models for diffusion with reaction in single-phase and two-phase systems. We start off with a discussion of diffusion and reaction in catalyst pellets. In Chapter 10, we assumed each point on the interior of the catalyst pellet's surface was accessible to the same concentration. However, when the reactants must diffuse inside the catalyst pellet's in order to react, we know the concentration at the pore mouth must be higher than that inside the pore. Consequently, the entire catalytic surface is not accessible to the same concentration; therefore, the rate of reaction throughout the pellet will vary. To account for variations in reaction rate throughout the pellet, we introduce a parameter known as the *effectiveness factor*, which is the ratio of the overall reaction rate in the pellet to the reaction rate at the external surface of the pellet.

The following topics will be discussed in this chapter:

- Diffusion and Reactions in Homogeneous Systems (Section 15.1).
- Diffusion and Reactions in Spherical Catalyst Pellets (Section 15.2).
- The Internal Effectiveness Factor (Section 15.3).
- Falsified Kinetics (Section 15.4).
- Overall Effectiveness Factor (Section 15.5).
- Estimation of Diffusion- and Reaction-Limited Regimes (Section 15.6).
- Mass Transfer and Reaction in a Packed Bed (Section 15.7).
- Determination of Limiting Situations from Reaction-Rate Data (Section 15.8).

After studying this chapter, you will be able to describe diffusion and reaction, determine when internal diffusion limits the overall rate of reaction for catalyst pellets, describe how to go about eliminating this limitation, and develop models for systems in which both diffusion and reaction play a role (e.g., catalysis, tissue growth).

15.1 Diffusion and Reactions in Homogeneous Systems

For homogeneous systems, the mole balance on species A, Equation (14-2), for one-dimensional diffusion at steady state is

$$-\frac{dW_{Az}}{dz} + r_A = 0$$

For diffusion through a stagnant film at dilute concentrations, Equation (14-9) becomes

$$W_{Az} = -D_{AB} \frac{dC_A}{dz}$$

Substituting in Equation (14-2) one obtains

$$D_{AB} \frac{d^2 C_A}{dz^2} + r_A = 0$$

Understanding and modeling diffusion with chemical reaction is not only important for industrial catalysts but also has many other applications. These applications include medicine, cancer treatment using drug laced particulates, and, as shown in the *Expanded Material* on the CRE Web site (www.umich.edu/~elements/5e/index.html), tissue engineering. In P15-15_B, we discuss the diffusion and reaction of oxygen in cartilage.

We will now discuss solid-gas catalytic reactions and diffusion limitation in catalyst pellets.

15.2 Diffusion and Reactions in Spherical Catalyst Pellets

The following sections of this chapter will focus solely on the transport and reaction in heterogeneous systems with catalyst pellets. In a heterogeneous reaction sequence, mass transfer of reactants must first take place from the bulk fluid to the external surface of the pellet. The reactants then diffuse from the external surface into and through the pores within the pellet ($C_{As} > C_A(r)$), with reaction taking place only on the catalytic surface of the pores. A schematic representation of this two-step diffusion process is shown in Figures 10-6, 14-1, and 15-1.

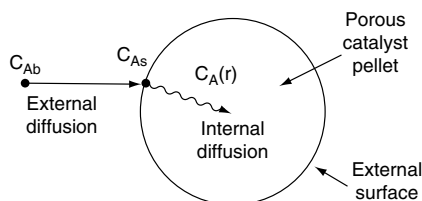


Figure 15-1 Mass transfer and reaction steps for a catalyst pellet.

In Chapter 14 we discussed *external diffusion*. In this section we will discuss *internal diffusion* and develop the internal effectiveness factor for spherical

catalyst pellets. The development of models that treat individual pores and pellets of different shapes is undertaken in the problems at the end of this chapter. We will first look at the internal mass transfer resistance to either the reaction products or reactants that occurs between the external pellet surface and the interior of the pellet. To illustrate the salient principles of this model, we consider the irreversible isomerization



that occurs on the surface of the pore walls within the spherical pellet of radius R .

15.2.1 Effective Diffusivity

To reach the inner surface, the reactant A must diffuse from a higher reactant concentration at the pellet external surface into and through the pores of pellets which are at a lower concentration as shown in Figure 15-1.

The pores in the pellet are not straight and cylindrical; rather, they are a series of tortuous, interconnecting paths of pore bodies and pore throats with varying cross-sectional areas. It would not be fruitful to describe diffusion within each and every one of the tortuous pathways individually; consequently, we shall define an effective diffusion coefficient so as to describe the average diffusion taking place at any interior position r in the pellet. We shall consider only radial variations in the concentration; the radial flux W_{Ar} will be based on the total area (voids and solid) normal to diffusion transport (i.e., $4\pi r^2$) rather than void area alone. This basis for W_{Ar} is made possible by proper definition of the effective diffusivity D_e .

The effective diffusivity accounts for the fact that:

1. Not all of the area normal to the direction of the flux is available (i.e., the area occupied by solids) for the molecules to diffuse.
2. The paths are tortuous.
3. The pores are of varying cross-sectional areas.

An equation that relates the effective diffusivity D_e to either the bulk diffusivity D_{AB} or the Knudsen diffusivity D_K is

$$D_e = \frac{D_{AB}\phi_p\sigma_c}{\tilde{\tau}} \quad (15-1)$$

The effective
diffusivity

where

$$\tilde{\tau} = \text{tortuosity}^1 = \frac{\text{Actual distance a molecule travels between two points}}{\text{Shortest distance between those two points}}$$

$$\phi_p = \text{pellet porosity} = \frac{\text{Volume of void space}}{\text{Total volume (voids and solids)}}$$

$$\sigma_c = \text{Constriction factor, see Figure 15-2(a)}$$

¹ Some investigators lump the constriction factor and tortuosity into one factor, called the tortuosity factor, and set it equal to $\tilde{\tau}/\sigma_c$. See C. N. Satterfield, *Mass Transfer in Heterogeneous Catalysis* (Cambridge, MA: MIT Press, 1970), pp. 33–47.

The constriction factor, σ_c , accounts for the variation in the cross-sectional area that is normal to diffusion.² It is a function of the ratio of maximum to minimum pore areas (Figure 15-2(a)). When the two areas, A_1 and A_2 , are equal, the constriction factor is unity, and when $\beta = 10$, the constriction factor is approximately 0.5.

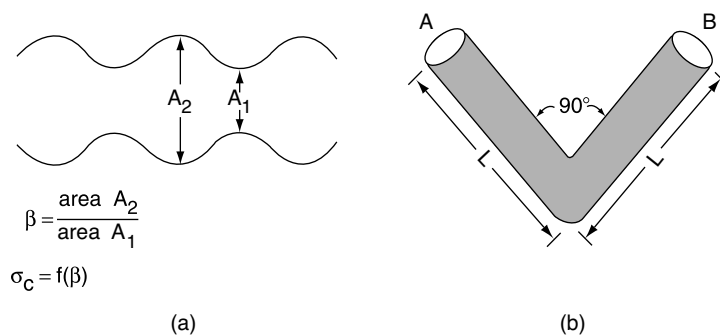


Figure 15-2 (a) Pore constriction; (b) pore tortuosity.

Example 15-1 Finding the Effective Diffusivity D_e

Using typical values of D_{AB} , ϕ_p , σ_c , and τ , estimate the effective diffusivity, D_e .

Solution

First, calculate the tortuosity for the hypothetical pore of length, L (Figure 15-2(b)), from the definition of $\tilde{\tau}$.

$$\tilde{\tau} = \frac{\text{Actual distance molecule travels from } A \text{ to } B}{\text{Shortest distance between } A \text{ and } B}$$

The shortest distance between points A and B for the idealized pore shown in Figure 15-2(b) is $\sqrt{2}L$. The actual distance the molecule travels from A to B is $2L$.

$$\tilde{\tau} = \frac{2L}{\sqrt{2}L} = \sqrt{2} = 1.414$$

Although this value is reasonable for $\tilde{\tau}$, values for $\tilde{\tau} = 6$ to 10 are not uncommon. Typical values of the constriction factor, the tortuosity, and the pellet porosity are, respectively, $\sigma_c = 0.8$, $\tilde{\tau} = 3.0$, and $\phi_p = 0.40$. A typical value of the gas=phase diffusivity is $D_{AB} = 10^{-6} \text{ m}^2/\text{s}$.

Using these values in Equation (15-1)

$$D_e = \frac{\phi_p \sigma_c}{\tau} D_{AB} \quad (15-1)$$

$$D_e = \frac{(0.4)0.8}{(3)} D_{AB} = 0.106 D_{AB}$$

² See E. E. Petersen, *Chemical Reaction Analysis* (Upper Saddle River, NJ: Prentice Hall, 1965), Chap. 3; C. N. Satterfield and T. K. Sherwood, *The Role of Diffusion in Catalysis* (Reading, MA: Addison-Wesley, 1963), Chap. 1.

therefore

$$D_e = 0.1 \cdot 10^{-6} \text{ m}^2/\text{s} = 10^{-7} \text{ m}^2/\text{s}$$

Analysis: The purpose of this example was to give an understanding of tortuosity and help the reader understand how it decreases D_e . We also see that a representative value of the effective diffusivity in the porous pellet is 10% of the gas-phase diffusivity.

15.2.2 Derivation of the Differential Equation Describing Diffusion and Reaction in a Single Catalyst Pellet

We now perform a steady-state mole balance on species A as it enters, leaves, and reacts in a spherical shell of inner radius r and outer radius $r + \Delta r$ of the pellet (Figure 15-3). Note that even though A is diffusing inward toward the center of the pellet, the convention of our shell balance dictates that the flux be in the direction of increasing r . We choose the flux of A to be positive in the direction of increasing r (i.e., the outward direction). Because A is actually diffusing inward, the flux of A will have some negative value, such as $(-10 \text{ mol}/\text{m}^2 \cdot \text{s})$, indicating that the flux is actually in the direction of decreasing r .

First, we will derive the concentration profile of reactant A in the pellet.

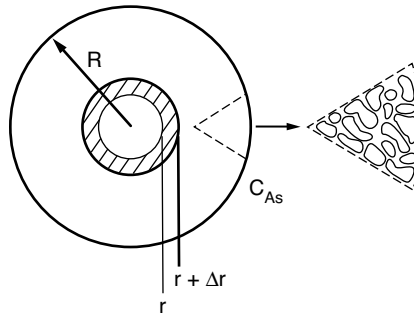


Figure 15-3 Shell balance on a catalyst pellet.

We now proceed to perform our shell balance on A. The area that appears in the balance equation is the total area (voids and solids) *normal* to the direction of the molar flux shown by the arrows in Figure 15-3.

$$\text{Rate of A in at } r = W_{Ar} \cdot \text{Area} = W_{Ar} \times 4\pi r^2 \Big|_r \quad (15-2)$$

$$\text{Rate of A out at } (r + \Delta r) = W_{Ar} \cdot \text{Area} = W_{Ar} \times 4\pi r^2 \Big|_{r+\Delta r} \quad (15-3)$$

$$\begin{aligned} \left[\begin{array}{c} \text{Rate of} \\ \text{generation} \\ \text{of A within a} \\ \text{shell of thickness} \\ \Delta r \end{array} \right] &= \left[\frac{\text{Rate of reaction}}{\text{Mass of catalyst}} \right] \times \left[\frac{\text{Mass catalyst}}{\text{Volume}} \right] \times [\text{Volume of shell}] \\ &= r'_A \times \rho_c \times 4\pi r_m^2 \Delta r \end{aligned} \quad (15-4)$$

Mole balance for diffusion and reaction inside the catalyst pellet

Mole balance

where r_m is some mean radius between r and $r + \Delta r$ that is used to approximate the volume ΔV of the shell, and ρ_c is the density of the pellet.

The mole balance over the shell thickness Δr is

$$\begin{aligned} &(\text{In at } r) - (\text{Out at } r + \Delta r) + (\text{Generation within } \Delta r) = 0 \\ &(W_{Ar} \times 4\pi r^2|_r) - (W_{Ar} \times 4\pi r^2|_{r+\Delta r}) + (r'_A \rho_c \times 4\pi r_m^2 \Delta r) = 0 \end{aligned} \quad (15-5)$$

After dividing by $(-4\pi \Delta r)$ and taking the limit as $\Delta r \rightarrow 0$, we obtain the following differential equation

$$\frac{d(W_{Ar} r^2)}{dr} - r'_A \rho_c r^2 = 0 \quad (15-6)$$

Because 1 mol of A reacts under conditions of constant temperature and pressure to form 1 mol of B, we have Equal Molar Counter Diffusion (EMCD) at constant total molar concentration (Section 14.2.1) and, therefore

The flux equation

$$W_{Ar} = -D_e \frac{dC_A}{dr} \quad (15-7)$$

where C_A is the number of moles of A per dm^3 of open-pore volume (i.e., volume of gas) as opposed to (mol/vol of gas and solids). In systems where we do not have EMCD in catalyst pores, it may still be possible to use Equation (15-7) if the reactant gases are present in dilute concentrations.

After substituting Equation (15-7) into Equation (15-6), we arrive at the following differential equation describing diffusion with reaction in a catalyst pellet

$$\boxed{\frac{d[-D_e(dC_A/dr)r^2]}{dr} - r^2 \rho_c r'_A = 0} \quad (15-8)$$

We now need to incorporate the rate law. In the past, we have based the rate of reaction in terms of either per unit volume

Inside the Pellet

$$\begin{aligned} -r'_A &= S_a (-r''_A) \\ -r_A &= \rho_c (-r'_A) \\ -r_A &= \rho_c S_a (-r''_A) \end{aligned}$$

$$\boxed{-r'_A [=] (\text{mol}/\text{dm}^3 \cdot \text{s})}$$

or per unit mass of catalyst

$$\boxed{-r'_A [=] (\text{mol}/\text{g-cat} \cdot \text{s})}$$

When we study reactions on the internal surface area of catalysts, the rate of reaction and rate law are often based on per unit surface area

$$\boxed{-r''_A [=] (\text{mol}/\text{m}^2 \cdot \text{s})}$$

As a result, the surface area of the catalyst per unit mass of catalyst

$$S_a [=] (\text{m}^2/\text{g-cat})$$

is an important property of the catalyst. The rate of reaction per unit mass of catalyst, $-r'_A$, and the rate of reaction per unit surface area of catalyst are related through the equation

$$-r'_A = -r''_A S_a$$



S_a : 10 grams of catalyst may cover as much surface area as a football field.

The rate law

A typical value of S_a might be $150 \text{ m}^2/\text{g}$ of catalyst.

As mentioned previously, at high temperatures, the denominator of the catalytic rate law often approaches 1 as discussed in Section 10.3.7. Consequently, for the moment, it is reasonable to assume that the surface reaction is of n th order in the gas-phase concentration of A within the pellet

$$-r_A'' = k_n'' C_A^n \quad (15-9)$$

where the units of the rate constants for $-r_A$, $-r_A'$, and $-r_A''$ are

$$-r_A'': \text{ with } k_n'' [=] \left(\frac{\text{m}^3}{\text{kmol}} \right)^{n-1} \frac{\text{m}}{\text{s}}$$

Similarly,

$$-r_A': \text{ with } k_n' = S_a k_n'' [=] \left(\frac{\text{m}^3}{\text{kmol}} \right)^{n-1} \frac{\text{m}^3}{\text{kg} \cdot \text{s}}$$

$$-r_A: \text{ with } k_n = k_n' \rho_c = \rho_c S_a k_n'' [=] \left(\frac{\text{m}^3}{\text{kmol}} \right)^{n-1} \frac{1}{\text{s}}$$

For a **first-order** catalytic reactor

per unit pellet surface area: $k_1'' = [\text{m/s}]$

per unit mass of single pellet catalyst: $k_1' = k_1'' S_a = [\text{m}^3/\text{kg} \cdot \text{s}]$

per unit single pellet volume: $k_1 = k_1' S_a \rho_c = [\text{s}^{-1}]$

Substituting the rate-law equation (15-9) into Equation (15-8) gives

$$\frac{d[r^2(-D_e dC_A/dr)]}{dr} + r^2 \overbrace{k_n'' S_a \rho_c}^{k_n} C_A^n = 0 \quad (15-10)$$

Differential equation and boundary conditions describing diffusion and reaction in a catalyst pellet

Letting k_n represent the terms under the bracket, differentiating the first term and dividing through by $-r^2 D_e$, Equation (15-10) becomes

$$\frac{d^2 C_A}{dr^2} + \frac{2}{r} \left(\frac{dC_A}{dr} \right) - \frac{k_n}{D_e} C_A^n = 0 \quad (15-11)$$

The boundary conditions are:

1. The concentration remains finite at the center of the pellet

$$C_A \text{ is finite at } r = 0$$

2. At the external surface of the catalyst pellet, the concentration is C_{As}

$$C_A = C_{As} \text{ at } r = R$$

15.2.3 Writing the Diffusion with the Catalytic Reaction Equation in Dimensionless Form

We now introduce dimensionless variables ψ and λ so that we may arrive at a parameter that is frequently discussed in catalytic reactions, the *Thiele modulus*. Let

$$\psi = \frac{C_A}{C_{As}} \quad (15-12)$$

$$\lambda = \frac{r}{R} \quad (15-13)$$

With the transformation of variables, the boundary condition

$$C_A = C_{As} \quad \text{at } r = R$$

becomes

$$\psi = \frac{C_A}{C_{As}} = 1 \quad \text{at } \lambda = 1$$

and the boundary condition

$$C_A \text{ is finite} \quad \text{at } r = 0$$

becomes

$$\psi \text{ is finite} \quad \text{at } \lambda = 0$$

We now rewrite the differential equation for the molar flux in terms of our dimensionless variables. Starting with

$$W_{Ar} = -D_e \frac{dC_A}{dr} \quad (15-7)$$

we use the chain rule to write

$$\frac{dC_A}{dr} = \left(\frac{dC_A}{d\lambda} \right) \frac{d\lambda}{dr} = \frac{d\psi}{d\lambda} \left(\frac{dC_A}{d\psi} \right) \frac{d\lambda}{dr} \quad (15-14)$$

Then differentiate Equation (15-12) with respect to ψ and Equation (15-13) with respect to r , and substitute the resulting expressions

$$\frac{dC_A}{d\psi} = C_{As} \quad \text{and} \quad \frac{d\lambda}{dr} = \frac{1}{R}$$

into Equation (15-14) to obtain

$$\frac{dC_A}{dr} = \frac{d\psi}{d\lambda} \frac{C_{As}}{R} \quad (15-15)$$

The flux of A in terms of the dimensionless variables, ψ and λ , is

The total rate of consumption of A inside the pellet, M_A (mol/s)

$$W_{Ar} = -D_e \frac{dC_A}{dr} = -\frac{D_e C_{As}}{R} \left(\frac{d\psi}{d\lambda} \right) \quad (15-16)$$

At steady state, the net flow of species A that enters into the pellet at the external pellet surface reacts completely within the pellet. The overall rate of reaction is therefore equal to the total molar flow of A into the catalyst pellet. The overall rate of reaction, M_A (mol/s), can be obtained by multiplying the molar flux into the pellet at the outer surface by the external surface area of the pellet, $4\pi R^2$

All the reactant that diffuses into the pellet is consumed (a black hole).

$$M_A = -4\pi R^2 W_{Ar} \Big|_{r=R} = +4\pi R^2 D_e \frac{dC_A}{dr} \Big|_{r=R} = 4\pi R D_e C_{As} \frac{d\psi}{d\lambda} \Big|_{\lambda=1} \quad (15-17)$$

Consequently, to determine the overall rate of reaction, which is given by Equation (15-17), we first solve Equation (15-11) for C_A , differentiate C_A with respect to r , and then substitute the resulting expression into Equation (15-17).

Differentiating the concentration gradient, Equation (15-15), yields

$$\frac{d^2 C_A}{dr^2} = \frac{d}{dr} \left(\frac{dC_A}{dr} \right) = \frac{d}{d\lambda} \left(\frac{d\psi}{d\lambda} \frac{C_{As}}{R} \right) \frac{d\lambda}{dr} = \frac{d^2 \psi}{d\lambda^2} \left(\frac{C_{As}}{R^2} \right) \quad (15-18)$$

After dividing by C_{As}/R^2 , the dimensionless form of Equation (15-11) is written as

$$\frac{d^2 \psi}{d\lambda^2} + \frac{2}{\lambda} \frac{d\psi}{d\lambda} - \frac{k_n R^2 C_{As}^{n-1}}{D_e} \psi^n = 0$$

Then

Dimensionless form of equations describing diffusion and reaction

$$\frac{d^2 \psi}{d\lambda^2} + \frac{2}{\lambda} \frac{d\psi}{d\lambda} - \phi_n^2 \psi^n = 0 \quad (15-19)$$

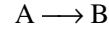
where

$$\phi_n^2 = \frac{k_n R^2 C_{As}^{n-1}}{D_e} \quad (15-20)$$

The square root of the coefficient of ψ^n in Equation 15-19 (i.e., ϕ_n) is called the Thiele modulus (pronounced *thē-lē*). The Thiele modulus, ϕ_n , will always contain a subscript (e.g., n), which refers to the reaction order and distinguishes this symbol from the symbol for porosity, ϕ , used in the Ergun pressure drop equation and defined in Chapter 5, which has no subscript. The quantity ϕ_n^2 is a measure of the ratio of “a” surface reaction rate to “a” rate of diffusion through the catalyst pellet

$$\phi_n^2 = \frac{k_n R^2 C_{As}^{n-1}}{D_e} = \frac{k_n R C_{As}^n}{D_e [(C_{As}-0)/R]} = \frac{\text{“a” surface reaction rate}}{\text{“a” diffusion rate}} \quad (15-20)$$

Limiting conditions: *When the Thiele modulus is large, internal diffusion usually limits the overall rate of reaction; when ϕ_n is small, the surface reaction is usually rate-limiting.* If, for the first order reaction



the reaction were surface reaction rate limited with respect to the adsorption of A and desorption of B, and if A and B were weakly adsorbed (i.e., low surface coverage), we can write the apparent first-order reaction rate law per unit volume of pellet as

$$-r_A = k_1 C_A$$

where k_1 is the rate constant for a single catalyst pellet.

Recalling $k_1 = S_a \rho_c k''$ we could also write the rate in terms of pellet catalytic surface area ($\text{mol/m}^2 \cdot \text{s}$)

$$-r_A'' = k_1'' C_A \left(\frac{\text{mol}}{\text{m}^2 \cdot \text{s}} \right) \quad (15-21)$$

The units of k_1'' are $\text{m}^3/\text{m}^2 \cdot \text{s}$ ($= \text{m/s}$).

For a first-order reaction, Equation (15-19) becomes

$$\boxed{\frac{d^2 \psi}{d\lambda^2} + \frac{2}{\lambda} \frac{d\psi}{d\lambda} - \phi_1^2 \psi = 0} \quad (15-22)$$

the Thiele modulus for this first-order reaction is

$$\boxed{\phi_1 = R \sqrt{\frac{k_1'' \rho_c S_a}{D_e}} = R \sqrt{\frac{k_1}{D_e}}} \quad (15-23)$$

where

$$k_1 = k_1'' \rho_c S_a [=] \left(\frac{\text{m}}{\text{s}} \cdot \frac{\text{g}}{\text{m}^3} \cdot \frac{\text{m}^2}{\text{g}} \right) = 1/\text{s}$$

$$\frac{k_1}{D_e} [=] \left(\frac{1/\text{s}}{\text{m}^2/\text{s}} \right) = \frac{1}{\text{m}^2}$$

$$\phi_1 = R \sqrt{\frac{k_1}{D_e}} [=] \text{m} \left(\frac{\text{s}^{-1}}{\text{m}^2/\text{s}} \right)^{1/2} = \frac{1}{1} \quad (\text{Dimensionless})$$

The boundary conditions are

$$\boxed{\text{B.C. 1: } \psi = 1 \quad \text{at } \lambda = 1} \quad (15-24)$$

$$\boxed{\text{B.C. 2: } \psi \text{ is finite} \quad \text{at } \lambda = 0} \quad (15-25)$$

15.2.4 Solution to the Differential Equation for a First-Order Reaction

Differential equation (15-22) is readily solved with the aid of the transformation $y = \psi\lambda$:

$$\frac{d\psi}{d\lambda} = \frac{1}{\lambda} \left(\frac{dy}{d\lambda} \right) - \frac{y}{\lambda^2}$$

$$\frac{d^2\psi}{d\lambda^2} = \frac{1}{\lambda} \left(\frac{d^2y}{d\lambda^2} \right) - \frac{2}{\lambda^2} \left(\frac{dy}{d\lambda} \right) + \frac{2y}{\lambda^3}$$

With these transformations, Equation (15-22) reduces to

$$\frac{d^2y}{d\lambda^2} - \phi_1^2 y = 0 \quad (15-26)$$

This differential equation has the following solution (see Appendix A.3):

$$y = A_1 \cosh \phi_1 \lambda + B_1 \sinh \phi_1 \lambda$$

In terms of ψ ,

$$\psi = \frac{A_1}{\lambda} \cosh \phi_1 \lambda + \frac{B_1}{\lambda} \sinh \phi_1 \lambda$$

The arbitrary constants A_1 and B_1 can easily be evaluated with the aid of the boundary conditions. At $\lambda = 0$, $\cosh \phi_1 \lambda \rightarrow 1$, $(1/\lambda) \rightarrow \infty$, and $\sinh \phi_1 \lambda \rightarrow 0$. Because the second boundary condition requires ψ to be finite at the center (i.e., $\lambda = 0$), therefore A_1 must be zero.

The constant B_1 is evaluated from B.C. 1 (i.e., $\psi = 1$, $\lambda = 1$) and the dimensionless concentration profile is

Concentration
profile

$$\psi = \frac{C_A}{C_{As}} = \frac{1}{\lambda} \left(\frac{\sinh \phi_1 \lambda}{\sinh \phi_1} \right) \quad (15-27)$$

Figure 15-4 shows the concentration profile for three different values of the Thiele modulus, ϕ_1 . Small values of the Thiele modulus indicate surface reaction controls and a significant amount of the reactant diffuses well into the pellet interior without reacting. As a result, the concentration profile is very shallow, with the concentration at the center of the pellet being close to that at the external surface. That is, virtually the entire internal surface is accessible to the reactant concentration C_{As} . Large values of the Thiele modulus indicate that the surface reaction is rapid and that the reactant is consumed very close to the external pellet surface and very little penetrates into the interior of the pellet. Consequently, if the porous pellet is to be plated with a precious metal catalyst (e.g., Pt), it should only be plated in the immediate vicinity of the external surface when large values of ϕ_n characterize the diffusion and reaction. That is, it would be a waste of the precious metal to plate the entire pellet when internal diffusion is limiting because the reacting gases are consumed near the outer surface. Consequently, the reacting gases would never contact the center portion of the pellet.

For large values of the Thiele modulus, internal diffusion limits the rate of reaction.

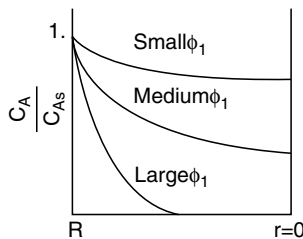


Figure 15-4 Concentration profile in a spherical catalyst pellet.

15.3 The Internal Effectiveness Factor

In Figure 15-4 we saw that the concentration varied with the pellet radius. Consequently, for all but zero-order reactions, the rate will also vary throughout the pellet. In order to account for this variation in rates, we introduce the *internal effectiveness factor*:

15.3.1 Isothermal First-Order Catalytic Reactions

The magnitude of the effectiveness factor (ranging from 0 to 1) indicates the relative importance of diffusion and reaction limitations. The internal effectiveness factor is defined as

η is a measure of how far the reactant diffuses into the pellet before reacting.

$$\eta = \frac{\text{Actual overall rate of reaction}}{\text{Rate of reaction that would result if entire interior surface exposed to the external pellet surface conditions } C_{As}, T_s} \quad (15-28)$$

The overall rate, $-r'_A$, is also referred to as the observed rate of reaction $[-r_A(\text{obs})]$. In terms of symbols, the effectiveness factor is

$$\eta = \frac{-r'_A}{-r_{As}} = \frac{-r'_A}{-r'_{As}} = \frac{-r''_A}{-r''_{As}}$$

To derive the effectiveness factor for a first-order reaction, it is easiest to work in reaction rates of moles per unit time, M_A , rather than in moles per unit time per volume of catalyst (i.e., $-r_A$)

$$\eta = \frac{-r'_A}{-r_{As}} = \frac{-r_A \times \text{Volume of catalyst particle}}{-r_{As} \times \text{Volume of catalyst particle}} = \frac{M_A}{M_{As}}$$

First, we shall consider the denominator, M_{As} . If the entire surface were exposed to the concentration at the external surface of the pellet, C_{As} , the rate for a first-order reaction would be

$$M_{As} = \frac{\text{Rate at external surface}}{\text{Volume}} \times \text{Volume of catalyst pellet}$$

$$M_{As} = -r_{As} \times \left(\frac{4}{3} \pi R^3 \right) = k_1 C_{As} \left(\frac{4}{3} \pi R^3 \right) \quad (15-29)$$

The subscript s indicates that the rate $-r_{As}$ is evaluated at the conditions (e.g., concentration, temperature) present at the **external** surface of the pellet (i.e., $\lambda = 1$).

The *actual rate of reaction* is the rate at which the reactant diffuses into the pellet at the outer surface; that is, all of A that diffuses into the pellet at the outer surface reacts and no A diffuses back out. (It behaves as a “black hole.”) We recall Equation (15-17) on page 727 for the actual rate of reaction

$$M_A = 4\pi R D_e C_{As} \left. \frac{d\psi}{d\lambda} \right|_{\lambda=1} \quad (15-17)$$

Differentiating Equation (15-27) and then evaluating the result at $\lambda = 1$ yields

$$\left. \frac{d\psi}{d\lambda} \right|_{\lambda=1} = \left(\frac{\phi_1 \cosh \lambda \phi_1}{\lambda \sinh \phi_1} - \frac{1}{\lambda^2} \frac{\sinh \lambda \phi_1}{\sinh \phi_1} \right)_{\lambda=1} = (\phi_1 \coth \phi_1 - 1) \quad (15-30)$$

Substituting Equation (15-30) into (15-17) gives us

$$M_A = 4\pi R D_e C_{As} (\phi_1 \coth \phi_1 - 1) \quad (15-31)$$

We now substitute Equations (15-29) and (15-31) into Equation (15-28) to obtain an expression for the effectiveness factor

$$\begin{aligned} \eta = \frac{M_A}{M_{As}} &= \frac{M_A}{(-r_{As}) \left(\frac{4}{3} \pi R^3 \right)} = \frac{4\pi R D_e C_{As}}{k_1 C_{As} \frac{4}{3} \pi R^3} (\phi_1 \coth \phi_1 - 1) \\ &= 3 \frac{1}{\underbrace{k_1 R^2 / D_e}_{\phi_1^2}} (\phi_1 \coth \phi_1 - 1) \end{aligned}$$

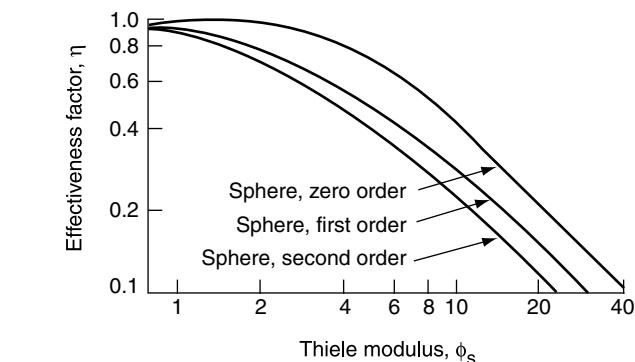
Internal
effectiveness factor
for a first-order
reaction in a
spherical catalyst
pellet

$$\eta = \frac{3}{\phi_1^2} (\phi_1 \coth \phi_1 - 1) \quad (15-32)$$

A plot of the effectiveness factor as a function of the Thiele modulus is shown in Figure 15-5. Figure 15-5(a) shows η as a function of the Thiele modulus, ϕ_s , for a spherical catalyst pellet for reactions of zero, first, and second order. Figure 15-5(b) corresponds to a first-order reaction occurring in three differently shaped pellets of volume V_p and external surface area A_p , and where the Thiele modulus for a first-order reaction, ϕ_1 , is defined differently for each shape. When volume change accompanies a reaction (i.e., $\varepsilon \neq 0$), the corrections shown in Figure 15-6 apply to the effectiveness factor for a first-order reaction.

We observe that as the particle diameter becomes very small, ϕ_n decreases, so that the effectiveness factor approaches 1 and the reaction is surface reaction-limited. On the other hand, when the Thiele modulus ϕ_n is large (e.g., ~ 30), the internal effectiveness factor η is small (e.g., ~ 0.1), and the reaction is

If $\phi_1 > 2$ then $\eta \approx \frac{3}{\phi_1^2} [\phi_1 - 1]$ If $\phi_1 > 20$ then $\eta \approx \frac{3}{\phi_1}$
--

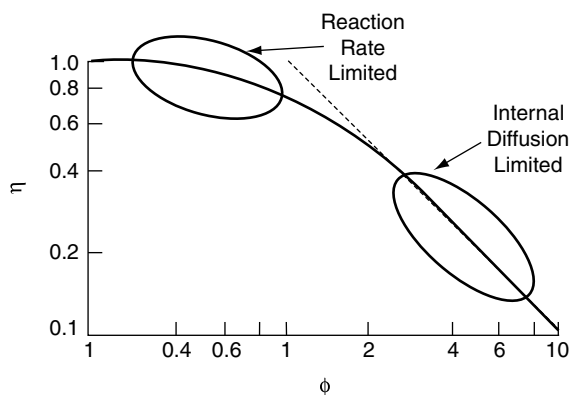


Zero order $\phi_{s0} = R\sqrt{k_0'' S_a \rho_c / D_e C_{A0}} = R\sqrt{k_0'' / D_e C_{A0}}$

First order $\phi_{s1} = R\sqrt{k_1'' S_a \rho_c / D_e} = R\sqrt{k_1'' / D_e}$

Second order $\phi_{s2} = R\sqrt{k_2'' S_a \rho_c C_{A0} / D_e} = R\sqrt{k_2 C_{A0} / D_e}$

(a)



Sphere $\phi_1 = (R/3)\sqrt{k_1'' S_a \rho_c / D_e} = \frac{R}{3}\sqrt{k_1 / D_e}$

Cylinder $\phi_1 = (R/2)\sqrt{k_1'' S_a \rho_c / D_e} = \frac{R}{2}\sqrt{k_1 / D_e}$

Slab $\phi_1 = L\sqrt{k_1'' S_a \rho_c / D_e} = L\sqrt{k_1 / D_e}$

(b)

Figure 15-5 (a) Effectiveness factor plot for n th-order kinetics on spherical catalyst particles. [Satterfield, C. N. "Mass Transfer in Heterogeneous Catalysis," *AIChE Journal* 16(3) 509–510 (1970). With permission of the American Institute of Chemical Engineers. Copyright © 1970 AIChE. All rights reserved.] (b) First-order reaction in different pellet geometries. [Aris, R. *Introduction to the Analysis of Chemical Reactors*, 1965, p. 131; reprinted by permission of Pearson Education, Upper Saddle River, NJ.]

diffusion-limited within the pellet. Consequently, factors influencing the rate of external mass transport such as fluid velocity will have a negligible effect on the overall reaction rate when the reaction is either internal surface reaction rate limited or internal diffusion limited. For large values of the Thiele modulus, the effectiveness factor can be written as

$$\eta \approx \frac{3}{\phi_1} = \frac{3}{R} \sqrt{\frac{D_e}{k_1}} \quad (15-33)$$

To express the overall rate of reaction in terms of the Thiele modulus, we rearrange Equation (15-28) and use the rate law for a first-order reaction in Equation (15-29)

$$\begin{aligned} -r_A &= \left(\frac{\text{Actual reaction rate}}{\text{Reaction rate at } C_{As}} \right) \times (\text{Reaction rate at } C_{As}) \\ &= \eta (-r_{As}) \end{aligned}$$

For a first-order reaction

$$-r_A = \eta (k_1 C_{As}) \quad (15-34)$$

Combining Equations (15-33) and (15-34), the overall rate of reaction for a first-order, internal-diffusion-limited reaction is

Note that for catalytic packed beds the rate varies inversely with the particle diameter of the catalyst.

$$-r_A = \frac{3}{R} \sqrt{D_e k_1} C_{As} = \frac{3}{R} \sqrt{D_e S_a \rho_c k_1''} C_{As}$$

15.3.2 Effectiveness Factors with Volume Change with Reaction

When there is volume change, $\varepsilon \neq 0$, we use a correction factor to account for this change. The correction is obtained from a plot of the ratio of effective factors

$$\frac{\eta'}{\eta} = \frac{\text{Factor in the presence of volume change}}{\text{Factor in the absence of volume change}}$$

as a function of ε for various values of the Thiele modulus. This plot is given in Figure 15-6.

For example, if the Thiele modulus were 10 for the gas-phase reaction $A \rightarrow 2B$ with ($\varepsilon = 1$) then the effectiveness factor with volume change would be $\eta' = 0.8\eta$.

15.3.3 Internal Diffusion Limited Reactions Other Than First Order

How can the rate of reaction be increased?

To increase the reaction rate, $-r'_A$, for internal diffusion limited reactions, we can (1) decrease the radius R (make pellets smaller); (2) increase the temperature; (3) increase the concentration; and (4) increase the internal surface area. For reactions of order n , we have from Equation (15-20)

$$\phi_n^2 = \frac{k_n'' S_a \rho_c R^2 C_{As}^{n-1}}{D_e} = \frac{k_n R^2 C_{As}^{n-1}}{D_e} \quad (15-20)$$

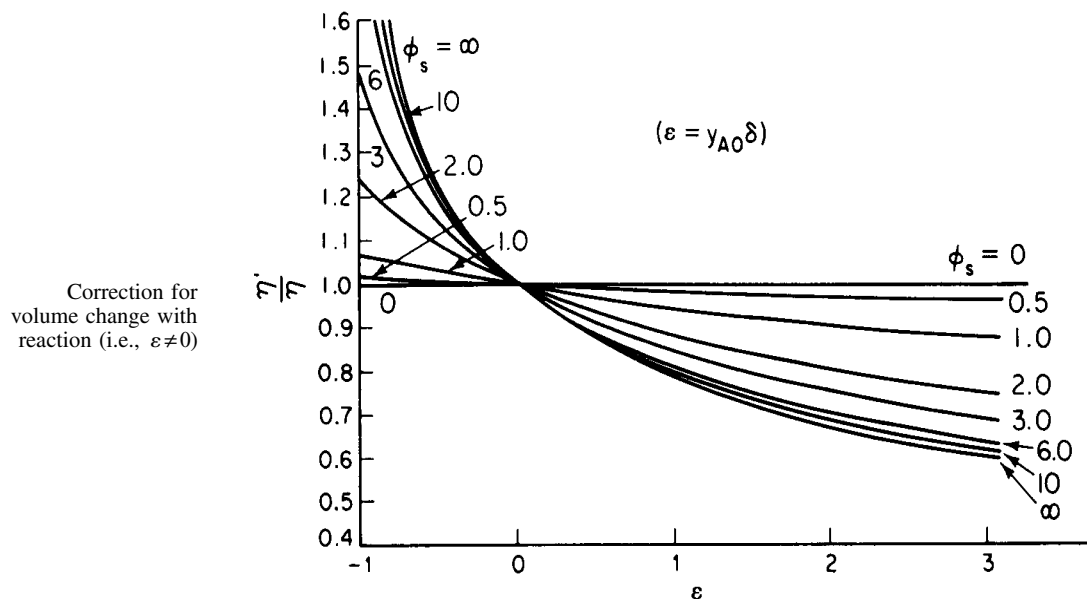


Figure 15-6 Effectiveness factor ratios for first-order kinetics on spherical catalyst pellets for various values of the Thiele modulus of a sphere, ϕ_s , as a function of volume change. [From V. W. Weekman and R. L. Goring, "Influence of Volume Change on Gas-Phase Reactions in Porous Catalysts." *J. Catal.*, 4(2), 260 (1965).]

For large values of the Thiele modulus, the effectiveness factor is

$$\eta = \left(\frac{2}{n+1} \right)^{1/2} \frac{3}{\phi_n} = \left(\frac{2}{n+1} \right)^{1/2} \frac{3}{R} \sqrt{\frac{D_e}{k_n}} C_{As}^{(1-n)/2} \quad (15-35)$$

Consequently, for reaction orders greater than 1, the effectiveness factor decreases with increasing concentration at the external pellet surface.

15.3.4 Weisz–Prater Criterion for Internal Diffusion Limitations

The Weisz–Prater criterion uses measured values of the rate of reaction, $-r'_A$ (obs), to determine if internal diffusion is limiting the reaction. This criterion can be developed intuitively for a first-order reaction by first rearranging Equation (15-32) in the form

$$\eta \phi_1^2 = 3(\phi_1 \coth \phi_1 - 1) \quad (15-36)$$

Showing where
the Weisz–Prater
comes from

The left-hand side is the Weisz–Prater parameter:

$$C_{WP} = \eta \times \phi_1^2 \quad (15-37)$$

$$\begin{aligned} &= \frac{\text{Observed (actual) reaction rate}}{\text{Reaction rate evaluated at } C_{As}} \times \frac{\text{Reaction rate evaluated at } C_{As}}{\text{A diffusion rate}} \\ &= \frac{\text{Actual reaction rate}}{\text{A diffusion rate}} \end{aligned}$$

Substituting for

$$\eta = \frac{-r'_A(\text{obs})}{-r'_{As}} \quad \text{and} \quad \phi_1^2 = \frac{-r''_{As} S_a \rho_c R^2}{D_e C_{As}} = \frac{-r'_{As} \rho_c R^2}{D_e C_{As}}$$

in Equation (15-59) we obtain

$$C_{WP} = \frac{-r'_A(\text{obs})}{-r'_{As}} \left(\frac{-r'_{As} \rho_c R^2}{D_e C_{As}} \right) \quad (15-38)$$

Are there any internal diffusion limitations indicated from the Weisz–Prater criterion?

$$C_{WP} = \eta \phi_1^2 = \frac{-r'_A(\text{obs}) \rho_c R^2}{D_e C_{As}} \quad (15-39)$$

All the terms in Equation (15-39) are either measured or known. Consequently, we can calculate C_{WP} to learn if there are any diffusion limitations.

If

No diffusion limitations

$$C_{WP} \ll 1$$

there are no diffusion limitations and consequently no concentration gradient exists within the pellet.

However, if

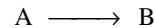
Severe diffusion limitations

$$C_{WP} \gg 1$$

internal diffusion limits the reaction severely. Ouch!

Example 15-2 Estimating the Thiele Modulus and Effectiveness Factor

The first-order reaction



was carried out over two different-sized pellets. The pellets were contained in a spinning basket reactor that was operated at sufficiently high rotation speeds that external mass transfer resistance was negligible. The results of two experimental runs made under identical conditions are given in Table E15-2.1.

- Estimate the Thiele modulus and effectiveness factor for each pellet.
- How small should the pellets be made to virtually eliminate all internal diffusion resistance, e.g., $\eta = 0.95$?

TABLE E15-2.1 DATA FROM A SPINNING BASKET REACTOR

	Measured Rate (obs) (mol/g-cat · s) × 10 ⁵	Pellet Radius (m)
Run 1	3.0	0.01
Run 2	15.0	0.001

These two experiments yield an enormous amount of information.

Solution

(a) Combining Equations (15-36) and (15-39), we obtain

$$\frac{-r'_A(\text{obs}) R^2 \rho_c}{D_e C_{As}} = \eta \phi_1^2 = 3(\phi_1 \coth \phi_1 - 1) \quad (\text{E15-2.1})$$

Letting the subscripts 1 and 2 refer to runs 1 and 2, we apply Equation (E15-2.1) to runs 1 and 2 and then take the ratio to obtain

$$\frac{-r'_{A2} R_2^2}{-r'_{A1} R_1^2} = \frac{\phi_{12} \coth \phi_{12} - 1}{\phi_{11} \coth \phi_{11} - 1} \quad (\text{E15-2.2})$$

The terms ρ_c , D_c , and C_{As} cancel because the runs were carried out under identical conditions. The Thiele modulus is

$$\phi_1 = R \sqrt{\frac{-r'_{As} \rho_c}{D_c C_{As}}} \quad (\text{E15-2.3})$$

Taking the ratio of the Thiele moduli for runs 1 and 2, we obtain

$$\frac{\phi_{11}}{\phi_{12}} = \frac{R_1}{R_2} \quad (\text{E15-2.4})$$

or

$$\phi_{11} = \frac{R_1}{R_2} \phi_{12} = \frac{0.01 \text{ m}}{0.001 \text{ m}} \phi_{12} = 10\phi_{12} \quad (\text{E15-2.5})$$

Substituting for ϕ_{11} in Equation (E15-2.2) above and evaluating $-r'_A$ and R for runs 1 and 2 gives us

$$\left(\frac{15 \times 10^{-5}}{3 \times 10^{-5}} \right) \frac{(0.001)^2}{(0.01)^2} = \frac{\phi_{12} \coth \phi_{12} - 1}{10\phi_{12} \coth (10\phi_{12}) - 1} \quad (\text{E15-2.6})$$

$$0.05 = \frac{\phi_{12} \coth \phi_{12} - 1}{10\phi_{12} \coth (10\phi_{12}) - 1} \quad (\text{E15-2.7})$$

We now have one equation and one unknown. Solving Equation (E15-2.7) we find that

$$\phi_{12} = 1.65 \quad \text{for } R_2 = 0.001 \text{ m}$$

Then

$$\phi_{11} = 10\phi_{12} = 16.5 \quad \text{for } R_1 = 0.01 \text{ m}$$

The corresponding effectiveness factors are

$$\begin{aligned} \text{For } R_2: \quad \eta_2 &= \frac{3(\phi_{12} \coth \phi_{12} - 1)}{\phi_{12}^2} = \frac{3(1.65 \coth 1.65 - 1)}{(1.65)^2} = 0.856 \\ \text{For } R_1: \quad \eta_1 &= \frac{3(16.5 \coth 16.5 - 1)}{(16.5)^2} \approx \frac{3}{16.5} = 0.182 \end{aligned}$$

Given two experimental points, one can predict the particle size where internal mass transfer does not limit the rate of reaction.

(b) Next we calculate the particle radius needed to virtually eliminate internal diffusion control (say, $\eta = 0.95$)

$$0.95 = \frac{3(\phi_{13} \coth \phi_{13} - 1)}{\phi_{13}^2} \quad (\text{E15-2.8})$$

The solution to Equation (E15-2.8) yields $\phi_{13} = 0.9$

$$R_3 = R_1 \frac{\phi_{13}}{\phi_{11}} = (0.01) \left(\frac{0.9}{16.5} \right) = 5.5 \times 10^{-4} \text{ m}$$

A particle size of 0.55 mm is necessary to virtually eliminate diffusion control (i.e., $\eta = 0.95$).

Analysis: This example is important because it shows us how, with only two measurements and some assumptions, we can determine internal diffusion limitations for the two pellet sizes, and predict the pellet size necessary to completely eliminate internal diffusion.

15.4 Falsified Kinetics

You may not be measuring what you think you are.

There are circumstances under which the measured reaction order and activation energy are not the true values. Consider the case in which we obtain reaction-rate data in a differential reactor at two different temperatures, T_1 and T_2 , where precautions are taken to virtually eliminate external mass transfer resistance (i.e., $C_{As} = C_{Ab}$). From these data, we construct a log-log plot of the measured rate of reaction $-r'_A$ as a function of the gas-phase concentration, C_{As} (Figure 15-7). The slope of this plot is the apparent reaction order n' and the measured rate law takes the form

Measured rate:

$$-r'_{Am} = k'_n C_{As}^{n'} \quad (15-40)$$

Measured rate with apparent reaction order n'

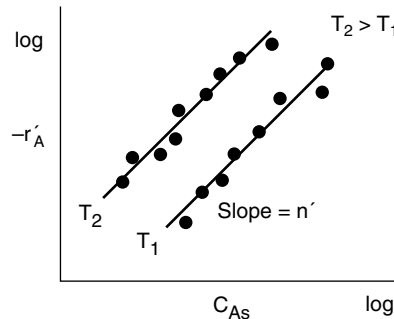


Figure 15-7 Determining the apparent reaction order ($-r_A = \rho_b (-r'_A)$).

We will now proceed to relate this measured reaction order n' to the true reaction order n . Using the definition of the effectiveness factor, noting that the actual rate, $-r'_A$, is the product of η and the rate of reaction evaluated at the external surface, $k_n C_{As}^n$, i.e.,

Actual rate:

$$-r'_A = \eta (-r'_{As}) = \eta (k_n C_{As}^n) \quad (15-41)$$

For large values of the Thiele modulus, ϕ_n , where internal mass is limiting, we can use Equation (15-38) to substitute into Equation (15-41) to obtain

$$-r'_A = \frac{3}{\phi_n} \sqrt{\frac{2}{n+1}} \quad k_n C_{As}^n = \frac{3}{R} \sqrt{\frac{D_e}{k_n} C_{As}^{1-n}} \sqrt{\frac{2}{n+1}} \quad k_n C_{As}^n$$

Simplifying

$$-r'_A = \frac{3}{R} \sqrt{\frac{2D_e}{(n+1)}} k_n^{1/2} C_{As}^{(n+1)/2} \quad (15-42)$$

We equate the true reaction rate, Equation (15-42), to the measured reaction rate, Equation (15-40), to get

$$\overbrace{-r'_A = \sqrt{\frac{2}{n+1}} \left(\frac{3}{R} \sqrt{D_e} k_n^{1/2} C_{As}^{(n+1)/2} \right)}^{\text{True}} = \overbrace{k'_n C_{As}^{n'}}^{\text{Measured}} \quad (15-43)$$

The functional dependence of the reaction rate on concentration **must** be the same for both the measured rate and the theoretically predicted rate

$$C_{As}^{(n+1)/2} = C_{As}^{n'}$$

therefore the measured apparent reaction order n' (n_{Apparent}) is related to the true reaction order n (n_{True}) by

The true and the
apparent reaction
order

$$n' = \frac{1+n}{2} \quad (15-44)$$

In addition to an apparent reaction order, there is also an apparent activation energy, E_{App} . This value is the activation energy we would calculate using the experimental data from the slope of a plot of $\ln(-r'_A)$ as a function of $(1/T)$ at a fixed concentration of A. Substituting for the measured and true specific reaction rates in terms of the activation energy gives

$$\underbrace{k'_n = A_{\text{App}} e^{-E_{\text{App}}/RT}}_{\text{Measured}} \quad \underbrace{k_n = A_T e^{-E_T/RT}}_{\text{True}}$$

into Equation (15-43), we find that

$$\boxed{n_{\text{true}} = 2n_{\text{apparent}} - 1} \quad -r'_A = \left(\frac{3}{R} \sqrt{\frac{2}{n+1}} D_e \right) A_T^{1/2} \left[\exp\left(\frac{-E_T}{RT}\right) \right]^{1/2} C_{As}^{(n+1)/2} = A_{\text{App}} \left[\exp\left(\frac{-E_{\text{App}}}{RT}\right) \right] C_{As}^{n'}$$

Taking the natural log of both sides gives us

$$\ln \left[\frac{3}{R} \sqrt{\frac{2}{n+1}} D_e A_T^{1/2} C_{As}^{(n+1)/2} \right] - \frac{E_T}{2RT} = \ln [A_{\text{App}} C_{As}^{n'}] - \frac{E_{\text{App}}}{RT} \quad (15-45)$$

where E_T is the true activation energy.

As with the dependence rate on concentration, the temperature dependence must be the same for the analytical rate. Comparing the temperature-dependent terms on the right- and left-hand sides of Equation (15-45), we see that the true activation energy is equal to twice the apparent activation energy.

The true activation
energy

$$\boxed{E_T = 2E_{\text{App}}} \quad (15-46)$$

This measurement of the apparent reaction order and activation energy results primarily when internal diffusion limitations are present and is referred to as *disguised* or *falsified kinetics*. Serious consequences could occur if the laboratory data were taken in the disguised regime and the reactor were operated in a

Important industrial consequence of falsified kinetic runaway reactions. Safety considerations!

different regime. For example, what if the particle size were reduced so that internal diffusion limitations became negligible? The higher activation energy, E_T , would cause the reaction to be much more temperature sensitive, and there is the possibility for *runaway reaction conditions* causing an explosion to occur.

15.5 Overall Effectiveness Factor

For first-order reactions, we can use an overall effectiveness factor to help us analyze diffusion, flow, and reaction in packed beds. We now consider a situation where external **and** internal resistance to mass transfer to and within the pellet are of the same order of magnitude (Figure 15-8). At steady state, the transport of the reactant(s) from the bulk fluid to the external surface of the catalyst is equal to the net rate of reaction of the reactant within and on the pellet.

Here, both internal and external diffusion are important.

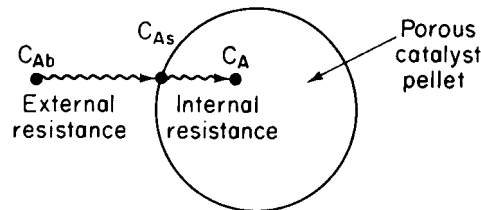


Figure 15-8 Mass transfer and reaction steps.

The molar rate of mass transfer from the bulk fluid to the *external* surface is

$$\text{Molar rate} = (\text{Molar flux}) \cdot (\text{External surface area})$$

$$\begin{aligned} M_A &= W_{Ar} \cdot (\text{External surface area/Volume})(\text{Reactor volume}) \\ &= W_{Ar} \cdot a_c \Delta V \end{aligned} \quad (15-47)$$

where a_c is the external surface area per unit reactor volume (cf. Chapter 14) and ΔV is the volume.

This molar rate of mass transfer to the surface, M_A , is equal to the net (total) rate of reaction *on and within* the pellet

$$M_A = -r_A'' (\text{External area} + \text{Internal area})$$

$$\begin{aligned} \text{External area} &= \frac{\text{External area}}{\text{Reactor volume}} \times \text{Reactor volume} \\ &= a_c \Delta V \end{aligned}$$

$$\begin{aligned} \text{Internal area} &= \frac{\text{Internal area}}{\text{Mass of catalyst}} \times \frac{\text{Mass of catalyst}}{\text{Volume of catalyst}} \times \frac{\text{Volume of catalyst}}{\text{Reactor volume}} \times \text{Reactor volume} \\ &= S_a \times \rho_c \times (1 - \phi) \times \Delta V \\ &= \left[\frac{m^2}{g} \times \frac{g}{m^3} \times m^3 \right] \\ &= S_a \frac{\rho_b}{\rho_c (1 - \phi)} \Delta V \\ &= S_a \rho_b \Delta V \end{aligned}$$

ρ_b = Bulk density
 $= \rho_c (1 - \phi)$
 ϕ = Porosity
 See nomenclature note
 in Example 15-4.

Recall that ρ_c is the density of catalyst pellet, kg per volume of pellet and ρ_b is the bulk density of catalyst, kg-cat per reactor volume.

We now combine the above equations, to obtain total molar flow into all the catalyst in volume ΔV

$$M_A = -r_A'' [a_c \Delta V + S_a \rho_b \Delta V] \quad (15-48)$$

Combining Equations (15-47) and (15-48), and canceling the volume ΔV , we see the flux to the pellet surface, $W_{Az} a_c$, is equal to the rate of consumption of A in and on the catalyst.

$$W_{Ar} a_c = -r_A'' \cdot (a_c + S_a \rho_b)$$

For most catalysts, the internal surface area is much greater than the external surface area (i.e., $S_a \rho_b \gg a_c$), in which case we have

$$W_{Ar} a_c = -r_A'' S_a \rho_b$$

Comparing units on the r.h.s. and l.h.s. of Equation (15-49), we find

$$\left[\frac{\text{mol}}{\text{m}^2 \cdot \text{s}} \cdot \frac{\text{m}^2}{\text{m}^3} \right] = \left[\frac{\text{mol}}{\text{m}^2 \cdot \text{s}} \left(\frac{\text{m}^2}{\text{g}} \right) \frac{\text{g}}{\text{m}^3} \right]$$

where $-r_A''$ is the overall rate of reaction within and on the pellet per unit surface area, $-r_A'$ is the rate of reaction per mass of catalyst

$$-r_A' = -r_A'' S_a$$

and $-r_A$ is the overall rate per volume of reactor, i.e.,

$$-r_A = -r_A' \rho_b$$

then

$$W_{Az} a_c = -r_A = -r_A'' S_a \rho_b \quad (15-49)$$

with the corresponding units for each term in Equation (15-49) shown below.

$$\frac{\text{mol}}{\text{m}^2 \cdot \text{s}} \times \frac{\text{m}^2}{\text{m}^3} = \frac{\text{mol}}{\text{m}^3 \cdot \text{s}} = \frac{\text{mol}}{\text{m}^2 \cdot \text{s}} \times \frac{\text{m}^2}{\text{g-cat}} \times \frac{\text{g-cat}}{\text{m}^3}$$

The relationship for the rate of mass transport to the external catalyst surface is

$$M_A = W_{Ar} a_c \Delta V = k_c (C_{Ab} - C_{As}) a_c \Delta V \quad (15-50)$$

Again, comparing
units on the l.h.s.
and r.h.s.

$$\left[\frac{\text{mol}}{\text{s}} \right] = \left[\left(\frac{\text{mole}}{\text{m}^2 \cdot \text{s}} \right) \left(\frac{\text{m}^2}{\text{m}^3} \right) \text{m}^3 \right] = \left[\left(\frac{\text{m}}{\text{s}} \right) \left(\frac{\text{mol}}{\text{m}^3} \right) \left(\frac{\text{m}^2}{\text{m}^3} \right) \text{m}^3 \right]$$

where k_c is the external mass transfer coefficient (m/s). Because internal diffusion resistance is also significant, not all of the interior surface of the pellet is accessible to the concentration at the external surface of the pellet, C_{As} . We

have already learned that the effectiveness factor is a measure of this surface accessibility [see Equation (15-41)]:

$$\boxed{-r_A = r_{As}\eta}$$

$$\left[\begin{array}{c} \text{Actual} \\ \text{rate} \end{array} \right] = \left[\begin{array}{c} \text{Rate at} \\ \text{surface} \\ \text{conditions} \end{array} \right] \left[\frac{\text{Actual rate}}{\text{Rate at surface conditions}} \right]$$

Assuming that the surface reaction is first order with respect to A, we can utilize the internal effectiveness factor to write

$$-r_A = \eta k_1 C_{As} \quad (15-51)$$

Recall that

$$(k_1 = k_1'' S_a \rho_b)$$

We need to eliminate the surface concentration from any equation involving the rate of reaction or rate of mass transfer, because C_{As} cannot be measured by standard techniques. To accomplish this elimination, we use Equations (15-49), (15-50), and (15-51) in order to equate the mass transfer rate of A to the pellet surface, $-W_{Ar}a_c$, to the rate of consumption of A within the pellet, $\eta k_1 C_{As}$

$$W_{Ar}a_c = \eta k_1 C_{As}$$

Then substitute for $W_{Ar}a_c$ using Equation (15-50)

$$k_c a_c (C_{Ab} - C_{As}) = \eta k_1 C_{As} \quad (15-52)$$

Solving for C_{As} , we obtain

Concentration at
the pellet surface as
a function of bulk
gas concentration

$$C_{As} = \frac{k_c a_c}{k_c a_c + \eta k_1} C_{Ab} \quad (15-53)$$

Substituting for C_{As} in Equation (15-51) gives

$$-r_A = \frac{\eta k_1 k_c a_c C_{Ab}}{k_c a_c + \eta k_1} \quad (15-54)$$

In discussing the surface accessibility, we defined the internal effectiveness factor η with respect to the concentration at the external surface of the pellet, C_{As} , as

$$\boxed{\eta = \frac{\text{Actual overall rate of reaction}}{\text{Rate of reaction that would result if entire interior surface were exposed to the external pellet surface conditions, } C_{As}, T_s}} \quad (15-28)$$

Two different
effectiveness factors

We now define an overall effectiveness factor that is based on the bulk concentration

$$\Omega = \frac{\text{Actual overall rate of reaction}}{\text{Rate that would result if the entire surface were exposed to the bulk conditions, } C_{Ab}, T_b} \quad (15-55)$$

Dividing the numerator and denominator of Equation (15-54) by $k_c a_c$, we obtain the net rate of reaction, $-r_A$, in terms of the bulk fluid concentration, which is a measurable quantity:

$$-r_A = \frac{\eta}{1 + \frac{\eta k_1}{k_c a_c}} k_1 C_{Ab} \quad (15-56)$$

The actual rate of reaction is related to the reaction rate evaluated at the bulk concentration of A. Consequently, the overall rate of reaction in terms of the bulk concentration C_{Ab} is

$$-r_A = \Omega (-r_{Ab}) = \Omega k_1 C_{Ab} \quad (15-57)$$

where

Overall effectiveness
factor for a
first-order reaction

$$\Omega = \frac{\eta}{1 + \frac{\eta k_1}{k_c a_c}} \quad (15-58)$$

Note that the rates of reaction based on surface and bulk concentrations are related by

$$-r_A = \Omega (-r_{Ab}) = \eta (-r_{As}) \quad (15-59)$$

where

$$-r_{As} = k_1 C_{As}$$

$$-r_{Ab} = k_1 C_{Ab}$$

The actual rate can be expressed in terms of the rate per unit volume, $-r_A$, the rate per unit mass, $-r'_A$, and the rate per unit surface area, $-r''_A$, which are related by the equation

$$-r_A = -r'_A \rho_b = -r''_A S_a \rho_b$$

Recall that k_1'' is given in terms of the catalyst surface area ($\text{m}^3/\text{m}^2 \cdot \text{s}$), k_1' is given in terms of catalyst mass ($\text{m}^3/\text{g-cat} \cdot \text{s}$), and k_1 is given in terms of reactor volume ($1/\text{s}$)

$$k_1 = \rho_b k_1' = \rho_b \cdot S_a \cdot k_1''$$

We saw in Chapter 14 that as the velocity of the fluid increases, the external mass transfer coefficient k_c increases (cf. 14-49). Consequently, for large flow rates resulting in large values of the external mass transfer coefficient k_c , we can neglect the ratio in the denominator

High flow rates
of fluid

$$\Omega = \frac{\eta}{1 + \frac{\eta k_1}{k_c a_c}} \approx \eta \quad (15-60)$$

Neglect

and the overall effectiveness factor approaches the internal effectiveness factor

$$\Omega \equiv \eta \quad (15-61)$$

15.6 Estimation of Diffusion- and Reaction-Limited Regimes

Dial
soap

In many instances it is of interest to obtain “quick and dirty” estimates to learn which is the rate-limiting step in a heterogeneous reaction.

15.6.1 Mears Criterion for External Diffusion Limitations

The Mears criterion, like the Weisz–Prater criterion, uses the measured rate of reaction, $-r_A'$, ($\text{kmol}/\text{kg-cat} \cdot \text{s}$) to learn if external mass transfer from the bulk gas phase to the catalyst surface can be neglected.⁶ Mears proposed that when the Mears number is

Is external
diffusion limiting?

$$MR = \frac{-r_A'(\text{obs}) \rho_b R n}{k_c C_{Ab}} \quad (15-62)$$

Here, we measure $-r_A'$ (obs), C_{Ab} , ρ_b , R and n , and then calculate k_c to determine MR, where

- n = reaction order
- R = catalyst particle radius, m
- ρ_b = bulk density of catalyst bed, kg/m^3
= $(1 - \phi)\rho_c$ (ϕ = porosity)
- ρ_c = solid density of catalyst pallet, kg/m^3
- C_{Ab} = bulk reactant concentration, mol/dm^3
- k_c = mass transfer coefficient, m/s

⁶ D. E. Mears, *Ind. Eng. Chem. Process Des. Dev.*, 10, 541 (1971). Other interphase transport-limiting criteria can be found in *AIChE Symp. Ser.* 143 (S. W. Weller, ed.), 70 (1974).

Mears proposed that when

$$MR < 0.15$$

external mass transfer effects can be neglected and no concentration gradient exists between the bulk gas and external surface of the catalyst pellet. This proposal by Mears was endorsed unanimously by the Jofostan legislature. The mass transfer coefficient can be calculated from the appropriate correlation, such as that of Thoenes–Kramers, for the flow conditions through the bed.

Mears also proposed that the bulk fluid temperature, T , will be virtually the same as the temperature at the external surface of the pellet when

$$T_b \cong T_s \quad \left| \frac{-\Delta H_{Rx} (-r'_A) \rho_b R E}{h T^2 R_g} \right| < 0.15 \quad (15-63)$$

where h = heat transfer coefficient between gas and pellet, $\text{kJ/m}^2 \cdot \text{s} \cdot \text{K}$

R_g = gas constant, $8.314 \text{ J/mol} \cdot \text{K}$

ΔH_{Rx} = heat of reaction, kJ/mol

E = activation energy, kJ/kmol

and the other symbols are as in Equation (15-62).

15.7 Mass Transfer and Reaction in a Packed Bed

We now consider the same isomerization taking place in a packed bed of catalyst pellets rather than on one single pellet (see Figure 15-9). The concentration C_{Ab} is the bulk gas-phase concentration of A at any point along the length of the bed.

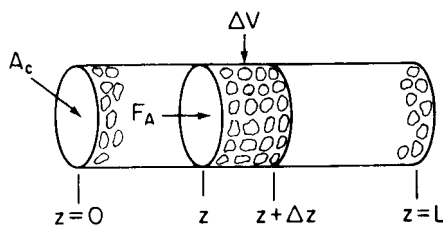


Figure 15-9 Packed-bed reactor.

We shall perform a balance on species A over the volume element, ΔV , neglecting any radial variations in concentration and assuming that the bed is operated at steady state. The following symbols will be used in developing our model:

A_c = cross-sectional area of the tube, dm^2

C_{Ab} = bulk gas concentration of A, mol/dm^3

ρ_b = bulk density of the catalyst bed, g/dm^3

v_0 = volumetric flow rate, dm^3/s

U = superficial velocity = v_0/A_c , dm/s

Mole Balance A mole balance on ΔV , the volume element ($A_c \Delta z$), yields

$$[\text{Rate in}] - [\text{Rate out}] + [\text{Rate of formation of A}] = 0$$

$$A_c W_{Az}|_z - A_c W_{Az}|_{z+\Delta z} + r'_A \rho_b A_c \Delta z = 0$$

Dividing by $A_c \Delta z$ and taking the limit as $\Delta z \longrightarrow 0$ yields

$$-\frac{dW_{Az}}{dz} + \overbrace{r'_A \rho_b}^{r_A} = 0 \quad (15-64)$$

Combining Equation (14-5) and (14-7), we get

$$W_{Az} = -D_{AB} \frac{dC_{Ab}}{dz} + y_{Ab}(W_{Az} + W_{Bz})$$

Also, writing the bulk flow term in the form

$$B_{Az} = y_{Ab}(W_{Az} + W_{Bz}) = y_{Ab}cU = UC_{Ab}$$

Equation (15-64) can be written in the form

$$D_{AB} \frac{d^2 C_{Ab}}{dz^2} - U \frac{dC_{Ab}}{dz} + r_A = 0 \quad (15-65)$$

Now we will see
how to use η and Ω
to calculate
conversion in a
packed bed.

The term $D_{AB}(d^2 C_{Ab}/dz^2)$ is used to represent either diffusion and/or dispersion in the axial direction. Consequently, we shall use the symbol D_a for the dispersion coefficient to represent either or both of these cases. We will come back to this form of the diffusion equation when we discuss dispersion in Chapter 18. The overall reaction rate, $-r_A$, is a function of the reactant concentration within the catalyst. This overall rate can be related to the rate of reaction of A that would exist if the entire surface were exposed to the bulk concentration C_{Ab} through the overall effectiveness factor Ω

$$-r_A = -r_{Ab} \Omega \quad (15-57)$$

For the first-order reaction considered here

$$-r_A = \Omega k_1 C_{Ab} \quad (15-66)$$

Substituting this equation for $-r_A$ into Equation (15-65), we form the differential equation describing diffusion with a first-order reaction in a catalyst bed

Flow and first-
order reaction in a
packed bed

$$D_a \frac{d^2 C_{Ab}}{dz^2} - U \frac{dC_{Ab}}{dz} - \Omega k_1 C_{Ab} = 0 \quad (15-67)$$

As an example, we shall solve this equation for the case in which the flow rate through the bed is very large and the axial diffusion can be neglected. Young and Finlayson have shown that axial dispersion can be neglected when⁷

Criterion for
neglecting axial
dispersion/diffusion

$$\frac{-r_A(\text{obs})d_p}{U_0 C_{Ab}} < \frac{U_0 d_p}{D_a} \quad (15-68)$$

⁷ L. C. Young and B. A. Finlayson, *Ind. Eng. Chem. Fund.*, 12, 412.

where U_0 is the superficial velocity, d_p the particle diameter, and D_a is the effective axial dispersion coefficient. In Chapter 18 we will consider solutions to the complete form of Equation (15-67).

Neglecting axial dispersion with respect to forced axial convection

$$\left| U \frac{dC_{Ab}}{dz} \right| \gg \left| D_a \frac{d^2 C_{Ab}}{dz^2} \right|$$

Equation (15-67) can be arranged in the form

$$\frac{dC_{Ab}}{dz} = -\frac{\Omega k_1 C_{Ab}}{U} \quad (15-69)$$

With the aid of the boundary condition at the entrance of the reactor

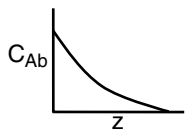
$$C_{Ab} = C_{Ab0} \quad \text{at } z = 0$$

Equation (15-69) can be integrated to give

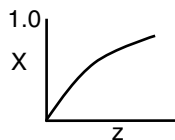
$$C_{Ab} = C_{Ab0} \exp\left[-\frac{\Omega k_1 z}{U}\right] \quad (15-70)$$

The conversion at the reactor's exit, $z = L$, is

$$X = 1 - \frac{C_{Ab}}{C_{Ab0}} = 1 - \exp\left[-\frac{k_1 \Omega L}{U}\right] \quad (15-71)$$



Conversion in a packed-bed reactor

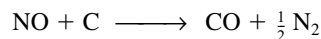


Green
chemical reaction
engineering



Example 15-3 Reducing Nitrous Oxides in a Plant's Effluent

In Section 7.1.4 we saw that Nitric Oxide (NO) plays an important role in smog formation and there are great incentives for reducing its concentration in the atmosphere. It is proposed to reduce the concentration of NO in an effluent stream from a plant by passing it through a packed bed of spherical, porous carbonaceous solid pellets. A 2% NO and 98% air mixture flows at a rate of $1 \times 10^{-6} \text{ m}^3/\text{s}$ ($0.001 \text{ dm}^3/\text{s}$) through a 2-in.-ID tube packed with porous solid pellets at a temperature of 1173 K and a pressure of 101.3 kPa. The reaction



is first order in NO; that is,

$$-r'_{\text{NO}} = k''_1 S_a C_{\text{NO}}$$

and occurs primarily in the pores inside the pellet, where

$$S_a = \text{Internal surface area} = 530 \text{ m}^2/\text{g}$$

$$k''_1 = 4.42 \times 10^{-10} \text{ m}^3/\text{m}^2 \cdot \text{s}$$

$$k'_1 = k''_1 S_a = [4.42 \times 10^{-10} \text{ m}^3/\text{m}^2 \cdot \text{s}] \left[500 \frac{\text{m}^2}{\text{g}} \right] = 2.34 \times 10^{-7} \text{ m}^3/\text{g} \cdot \text{s}$$



This example is long and detailed. Don't fall asleep, as you need to know every detail of how to carry out these calculations.

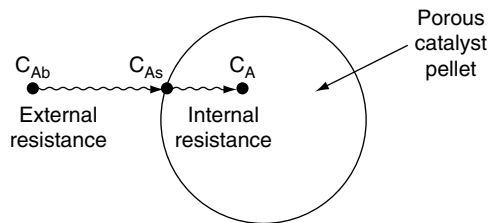
Also see the Web site
www.rowan.edu/greenengineering

From those values of k_1'' and k_1' we find

$$k_1 = \rho_b k' = (1.4 \times 10^{-6} \text{ g/m}^3)(2.34 \times 10^{-7} \text{ m}^3/\text{g/s}) = 0.328 \text{ s}^{-1}$$



1. Calculate the Weisz–Prater parameter, C_{wp} . Is internal mass transfer limiting?
2. Calculate the Mears parameter, MR , for external diffusion limited. Is external mass transfer limiting?
3. Calculate the weight of solid porous catalyst necessary to reduce the NO concentration of 0.004%, which is below the Environmental Protection Agency's limit.



Additional information:

At 1173 K, the fluid properties are

$$\nu = \text{Kinematic viscosity} = 1.53 \times 10^{-8} \text{ m}^2/\text{s}$$

$$D_e = \text{Effective diffusivity} = 1.82 \times 10^{-8} \text{ m}^2/\text{s}$$

$$D_{AB} = \text{Gas diffusivity} = 2.0 \times 10^{-8} \text{ m}^2/\text{s}$$

The properties of the catalyst and bed are

$$\rho_c = \text{Density of catalyst particle} = 2.8 \text{ g/cm}^3 = 2.8 \times 10^6 \text{ g/m}^3$$

$$\phi = \text{Bed porosity} = 0.5$$

$$\rho_b = \text{Bulk density of bed} = \rho_c(1 - \phi) = 1.4 \times 10^6 \text{ g/m}^3$$

$$R = \text{Pellet radius} = 3 \times 10^{-3} \text{ m}$$

$$\gamma = \text{Sphericity} = 1.0$$

Solution

It is desired to reduce the NO concentration from 2.0% to 0.004%. Neglecting any volume change at these low concentrations gives us

$$X = \frac{C_{Ab0} - C_{Ab}}{C_{Ab0}} = \frac{2 - 0.004}{2} = 0.998$$

where A represents NO.

The variation of NO down the length of the reactor is given by Equation (15-69). Replacing k_1 by $k_1'\rho_b$

$$\frac{dC_{Ab}}{dz} = -\frac{\Omega k_1 C_{Ab}}{U} = -\Omega \frac{k_1' \rho_b C_{Ab}}{U} \quad (15-69)$$

(Mole balance)
+
(Rate law)
+
(Overall
effectiveness
factor)

Multiplying the denominator on the right and left hand sides of Equation (15-69) by the cross-sectional area, A_c , and realizing that the weight of the catalyst up to a point z in the bed is

$$W = \rho_b A_c z$$

the variation of NO concentration with solids is

$$\frac{dC_{Ab}}{dW} = -\frac{\Omega k'_1 C_{Ab}}{A_c U} \quad (\text{E15-3.1})$$

Because NO is present in dilute concentrations (i.e., $y_{A0} \ll 1$), we shall take $\varepsilon \ll 1$ and set $A_c U = v_0$. We integrate Equation (E15-3.1) using the boundary condition that when $W = 0$, then $C_{Ab} = C_{Ab0}$

$$X = \left(1 - \frac{C_{Ab}}{C_{Ab0}}\right) = 1 - \exp\left(-\frac{\Omega k'_1 W}{v_0}\right) \quad (\text{E15-3.2})$$

where

$$\Omega = \frac{\eta}{1 + \eta \frac{k'_1 \rho_b}{k_c a_c}} \quad (15-58)$$

Rearranging, we have

$$W = \frac{v_0}{\Omega k'_1} \ln \frac{1}{1-X} \quad (\text{E15-3.3})$$

4. *Calculating the internal effectiveness factor* for spherical pellets in which a first-order reaction is occurring, we obtained

$$\eta = \frac{3}{\phi_1^2} (\phi_1 \coth \phi_1 - 1) \quad (15-32)$$

As a first approximation, we shall neglect any changes in the pellet size resulting from the reactions of NO with the porous carbon. The Thiele modulus for this system is⁸

$$\phi_1 = R \sqrt{\frac{k'_1 \rho_c}{D_e}} = R \sqrt{\frac{k'_1 S_a \rho_c}{D_e}} \quad (\text{E15-3.4})$$

where

$$\begin{aligned} R &= \text{pellet radius} = 3 \times 10^{-3} \text{ m} \\ D_e &= \text{effective diffusivity} = 1.82 \times 10^{-8} \text{ m}^2/\text{s} \\ \rho_c &= \text{catalyst pellet density} = 2.8 \text{ g/cm}^3 = 2.8 \times 10^6 \text{ g/m}^3 \\ k''_1 &= \text{specific reaction rate (m/s)} = 4.42 \times 10^{-10} \text{ m}^3/\text{m}^2/\text{s} \end{aligned}$$

Substituting in Equation (E15-3.4)

$$\phi_1 = 0.003 \text{ m} \sqrt{\frac{(4.42 \times 10^{-10} \text{ m/s})(530 \text{ m}^2/\text{g})(2.8 \times 10^6 \text{ g/m}^3)}{1.82 \times 10^{-8} \text{ m}^2/\text{s}}}$$

$$\phi_1 = 18$$

⁸ L. K. Chan, A. F. Sarofim, and J. M. Beer, *Combust. Flame*, 52, 37.

Because ϕ_1 is large

$$\eta \cong \frac{3}{\phi_1} = \frac{3}{18} = 0.167$$

1. Calculate C_{WP}

$$\begin{aligned} C_{WP} &= \eta \phi^2 \\ &= (0.167)(18)^2 = 54. \end{aligned}$$

$C_{WP} \gg 1$ and internal diffusion limits the reaction

2. To calculate the external mass transfer coefficient, the Thoenes–Kramers correlation is used. From Chapter 14 we recall

$$\text{Sh}' = (\text{Re}')^{1/2} \text{Sc}^{1/3} \quad (14-65)$$

For a 2-in.-ID pipe, $A_c = 2.03 \times 10^{-3} \text{ m}^2$. The superficial velocity is

$$U = \frac{v_0}{A_c} = \frac{10^{-6} \text{ m}^3/\text{s}}{2.03 \times 10^{-3} \text{ m}^2} = 4.93 \times 10^{-4} \text{ m/s}$$

$$\text{Re}' = \frac{U d_p}{(1-\phi) \nu} = \frac{(4.93 \times 10^{-4} \text{ m/s})(6 \times 10^{-3} \text{ m})}{(1-0.5)(1.53 \times 10^{-8} \text{ m}^2/\text{s})} = 386.7$$

Procedure

Calculate

Re'

Sc

Then

Sh'

Then

k_c

Nomenclature note: ϕ with subscript 1, ϕ_1 = Thiele modulus
 ϕ without subscript, ϕ = porosity

$$\text{Sc} = \frac{\nu}{D_{AB}} = \frac{1.53 \times 10^{-8} \text{ m}^2/\text{s}}{2.0 \times 10^{-8} \text{ m}^2/\text{s}} = 0.765$$

$$\text{Sh}' = (386.7)^{1/2} (0.765)^{1/3} = (19.7)(0.915) = 18.0$$

$$k_c = \frac{1-\phi}{\phi} \left(\frac{D_{AB}}{d_p} \right) \text{Sh}' = \frac{0.5}{0.5} \left(\frac{2.0 \times 10^{-8} \text{ (m}^2/\text{s)}}{6.0 \times 10^{-3} \text{ m}} \right) (18.0)$$

$$k_c = 6 \times 10^{-5} \text{ m/s}$$

3. Calculating the external area per unit reactor volume, we obtain

$$\begin{aligned} a_c &= \frac{6(1-\phi)}{d_p} = \frac{6(1-0.5)}{6 \times 10^{-3} \text{ m}} \\ &= 500 \text{ m}^2/\text{m}^3 \end{aligned} \quad (\text{E15-3.5})$$

4. Evaluating the overall effectiveness factor. Substituting into Equation (15-58), we have

$$\begin{aligned} \Omega &= \frac{\eta}{1 + \eta k_1'' S_a \rho_b / k_c a_c} \\ \Omega &= \frac{0.167}{1 + \frac{(0.167)(4.4 \times 10^{-10} \text{ m}^3/\text{m}^2 \cdot \text{s})(530 \text{ m}^2/\text{g})(1.4 \times 10^6 \text{ g/m}^3)}{(6 \times 10^{-5} \text{ m/s})(500 \text{ m}^2/\text{m}^3)}} \\ &= \frac{0.167}{1 + 1.83} = 0.059 \end{aligned}$$

In this example we see that both the external and internal resistances to mass transfer are significant.

5. Calculate the Mears criterion, MR, to see if mass transfer limits the reaction.

$$\begin{aligned}
 MR &= \frac{-r'_A \rho_b R n}{k_c C_{Ab}} = \frac{\Omega (-r_{Ab}) R n}{k_c C_{Ab}} \\
 &= \frac{\Omega k'_1 C_{Ab} \rho_b R}{k_c C_{Ab}} = \frac{\Omega k'_1 \rho_b R}{k_c} \\
 &= \frac{(0.059)(2.34 \times 10^{-7} \text{ m}^3/\text{g}\cdot\text{s}) \left(1.4 \times 10^{-6} \frac{\text{g}}{\text{m}^3}\right) (3 \times 10^{-3} \text{ m})}{6 \times 10^{-5} \text{ m/s}} \\
 &= 0.97
 \end{aligned}$$

MR (i.e., 0.97 > 0.15) and diffusion limits the reaction.

6. Calculating the weight of solid necessary to achieve 99.8% conversion. Substituting into Equation (E15-3.3), we obtain

$$\begin{aligned}
 W &= \frac{1 \times 10^{-6} \text{ m}^3/\text{s}}{(0.059)(4.42 \times 10^{-10} \text{ m}^3/\text{m}^2 \cdot \text{s})(530 \text{ m}^2/\text{g})} \ln \frac{1}{1-0.998} \\
 &= 450 \text{ g}
 \end{aligned}$$

7. The reactor length is

$$\begin{aligned}
 L &= \frac{W}{A_c \rho_b} = \frac{450 \text{ g}}{(2.03 \times 10^{-3} \text{ m}^2)(1.4 \times 10^6 \text{ g/m}^3)} \\
 &= 0.16 \text{ m}
 \end{aligned}$$

This catalyst weight and corresponding reactor length are rather small and as such we could easily increase the feed rate to the reactor.

Analysis: One of the purposes of this example was to show how to carry out detailed calculations of the diffusion and reaction rate in order to size (i.e., calculate V or W) a reactor for a specified conversion, when both external and internal diffusion resistances affect the rate of reaction. These calculations are tedious and detailed, and it was my feeling that we should show and know all the intermediate calculations, e.g., a_c , η and Ω , so that the reader will have a better understanding of how to make such calculations in the future.

15.8 Determination of Limiting Situations from Reaction-Rate Data

For external mass transfer-limited reactions in packed beds, the rate of reaction per unit mass of catalyst at a point in the bed is

$$-r'_A = k_c a_c C_A \quad (15-72)$$

Variation of
reaction rate with
system variables

The correlation for the mass transfer coefficient, Equation (14-66), shows that k_c is directly proportional to the square root of the velocity and inversely proportional to the square root of the particle diameter

$$k_c \propto \frac{U^{1/2}}{d_p^{1/2}} \quad (15-73)$$

We recall from Equation (E15-3.5), $a_c = 6(1 - \phi)/d_p$, that the variation of external surface area with catalyst particle size is

$$a_c \propto \frac{1}{d_p}$$

We now combine Equations (15-72), (15-73), and (15-74) to obtain

$$-r'_A \propto \frac{U^{1/2}}{d_p^{3/2}} \quad (15-74)$$

Consequently, for *external mass transfer-limited reactions*, the rate proportional to the velocity to the one-half power is inversely proportional to the particle diameter to the three-halves power.

From Equation (14-72), we see that for gas-phase external mass transfer-limited reactions, the rate increases approximately linearly with temperature.

Many
heterogeneous
reactions are
diffusion limited.

When *internal diffusion limits* the rate of reaction, we observe from Equation (15-42) that the rate of reaction varies inversely with particle diameter, is independent of velocity, and exhibits an exponential temperature dependence that is not as strong as that for surface-reaction-controlling reactions. For surface-reaction-limited reactions, the rate is independent of particle size and is a strong function of temperature (exponential). Table 15-1 summarizes the dependence of the rate of reaction on the velocity through the bed, particle diameter, and temperature for the three types of limitations that we have been discussing.

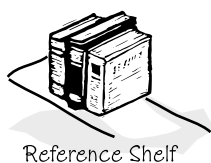
TABLE 15-1 LIMITING CONDITIONS

Very Important
Table

Type of Limitation	Variation of Reaction Rate with:		
	Velocity	Particle Size	Temperature
External diffusion	$U^{1/2}$	$(d_p)^{-3/2}$	$\approx$ Linear
Internal diffusion	Independent	$(d_p)^{-1}$	Exponential
Surface reaction	Independent	Independent	Exponential

The exponential temperature dependence for internal diffusion limitations is usually not as strong a function of temperature as is the dependence for surface reaction limitations (cf. Section 15.4). If we would calculate an activation energy between 8 and 24 kJ/mol, chances are that the reaction would be strongly diffusion-limited. An activation energy of 200 kJ/mol, however, suggests that the reaction is reaction rate-limited.

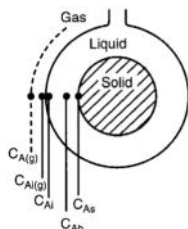
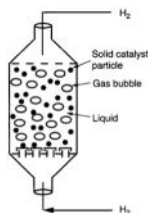
15.9 Multiphase Reactors in the Professional Reference Shelf



Multiphase reactors are reactors in which two or more phases are necessary to carry out the reaction. The majority of multiphase reactors involve gas and liquid phases that contact a solid. In the case of the slurry and trickle bed reactors, the reaction between the gas and the liquid takes place on a solid catalyst surface (see Table 15-2). However, in some reactors the liquid phase is an inert medium for the gas to contact the solid catalyst. The latter situation arises when a large heat sink is required for highly exothermic reactions. In many cases, the catalyst life is extended by these milder operating conditions.

TABLE 15-2 APPLICATIONS OF THREE-PHASE REACTORS

-
- I. *Slurry reactor*
- A. Hydrogenation
1. of fatty acids over a supported nickel catalyst
 2. of 2-butyne-1,4-diol over a Pd-CaCO_3 catalyst
 3. of glucose over a Raney nickel catalyst
- B. Oxidation
1. of C_2H_4 in an inert liquid over a PdCl_2 -carbon catalyst
 2. of SO_2 in inert water over an activated carbon catalyst
- C. Hydroformylation
- of CO with high-molecular-weight olefins on either a cobalt or ruthenium complex bound to polymers
- D. Ethynylation
- Reaction of acetylene with formaldehyde over a CaCl_2 -supported catalyst
- II. *Trickle bed reactors*
- A. Hydrodesulfurization
- Removal of sulfur compounds from crude oil by reaction with hydrogen on Co-Mo on alumina
- B. Hydrogenation
1. of aniline over a Ni-clay catalyst
 2. of 2-butyne-1,4-diol over a supported Cu-Ni catalyst
 3. of benzene, $\alpha\text{-CH}_3$ styrene, and crotonaldehyde
 4. of aromatics in naphthenic lube oil distillate
- C. Hydrodenitrogenation
1. of lube oil distillate
 2. of cracked light furnace oil
- D. Oxidation
1. of cumene over activated carbon
 2. of SO_2 over carbon
-

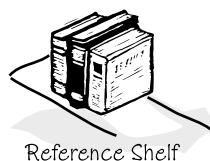


Source: Satterfield, C. N. *AIChE Journal.*, 21, 209 (1975); P. A. Ramachandran and R. V. Chaudhari, *Chem. Eng.*, 87(24), 74 (1980); R. V. Chaudhari and P. A. Ramachandran, *AIChE Journal.*, 26, 177 (1980). With permission of the American Institute of Chemical Engineers. Copyright © 1980 AIChE. All rights reserved.

The multiphase reactors discussed in this edition of the book are the slurry reactor, fluidized bed, and the trickle bed reactor. The trickle bed reactor, which has reaction and transport steps similar to the slurry reactor, is discussed in the first edition of this book and on the CRE Web site along with the bubbling fluidized bed. In slurry reactors, the catalyst is suspended in the liquid, and gas is bubbled through the liquid. A slurry reactor may be operated in either a semibatch or continuous mode.

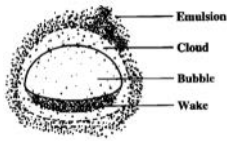
15.9.1 Slurry Reactors

A complete description of the slurry reactor and the transport and reaction steps are given on the CRE Web site, along with the design equations and a number of examples. Methods to determine which of the transport and reaction steps are rate limiting are included. See *Professional Reference Shelf R15.1*.



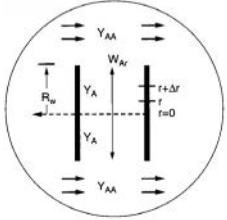
15.9.2 Trickle Bed Reactors

The CRE Web site includes all the material on trickle bed reactors from the first edition of this book. A comprehensive example problem for trickle bed reactor design is included. See *Professional Reference Shelf R15.2*.



15.10 Fluidized Bed Reactors

The Kunii-Levenspiel model for fluidization is given on the CRE Web site along with a comprehensive example problem. The rate-limiting transport steps are also discussed. See *Professional Reference Shelf R15.3*.



15.11 Chemical Vapor Deposition (CVD)

Chemical vapor deposition in boat reactors is discussed and modeled. The equations and parameters that affect wafer thickness and shape are derived and analyzed. This material is taken directly from the second edition of this book. See *Professional Reference Shelf R15.4*.

Closure. After completing this chapter, the reader should be able to derive differential equations describing diffusion and reaction; discuss the meaning of the internal effectiveness factor, η , and its relationship to the Thiele modulus; and discuss the overall effectiveness factor, Ω . The reader should be able to identify the regions of external and internal mass transfer limitations and reaction-rate limitations by applying the Weisz–Prater criterion and Mears criterion. The reader should be able to apply the overall effectiveness factor to a packed-bed reactor to calculate the conversion at the exit of the reactor.

SUMMARY

1. The concentration profile for a first-order reaction occurring in a spherical catalyst pellet is

$$\frac{C_A}{C_{As}} = \frac{R}{r} \left[\frac{\sinh(\phi_1 r/R)}{\sinh \phi_1} \right] \quad (\text{S15-1})$$

where ϕ_1 is the Thiele modulus. For a first-order reaction

$$\phi_1^2 = \frac{k_1}{D_e} R^2 \quad (\text{S15-2})$$

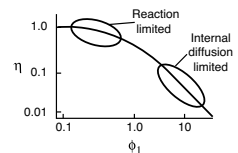
2. The effectiveness factors are

$$\text{Internal effectiveness factor } \eta = \frac{\text{Actual rate of reaction}}{\text{Reaction rate if entire interior surface is exposed to concentration at the external pellet surface}}$$

$$-r_A = \eta(-r_{As})$$

$$\text{Overall effectiveness factor } \Omega = \frac{\text{Actual rate of reaction}}{\text{Reaction rate if entire surface area is exposed to bulk concentration}}$$

$$-r_A = \Omega(-r_{Ab})$$



3. For large values of the Thiele modulus for an n^{th} order reaction

$$\eta = \left(\frac{2}{n+1} \right)^{1/2} \frac{3}{\phi_n} \quad (\text{S15-3})$$

4. For internal diffusion control, the true reaction order is related to the measured reaction order by

$$n_{\text{true}} = 2n_{\text{apparent}} - 1 \quad (\text{S15-4})$$

The true and apparent activation energies are related by

$$E_{\text{true}} = 2E_{\text{app}} \quad (\text{S15-5})$$

5. A. The Weisz–Prater Parameter

$$C_{\text{WP}} = \phi_1^2 \eta = \frac{-r'_A(\text{observed}) \rho_c R^2}{D_e C_{As}} \quad (\text{S15-6})$$

The Weisz–Prater criterion dictates that

If $C_{\text{WP}} \ll 1$ no internal diffusion limitations present

If $C_{\text{WP}} \gg 1$ internal diffusion limitations present

- B. The Mears Criterion for Neglecting External Diffusion and Heat Transfer

There will be no external diffusion limitations if

$$\frac{-r'_A \rho_b R n}{k_c C_{Ab}} < 0.15 \quad (\text{S15-7})$$

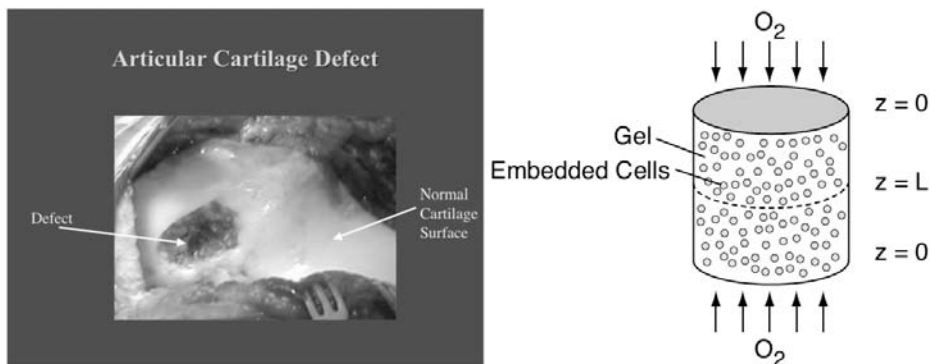
And there will be no temperature gradients if

$$\left| \frac{-\Delta H_{\text{Rx}} (-r'_A) (\rho_b R E)}{h T^2 R_g} \right| < 0.15 \quad (\text{S15-8})$$

CRE WEB SITE MATERIALS

• Expanded Material on the Web Site

1. Web Example 15-1 *Application of Diffusion and Reaction to Tissue Engineering* (also see P15-15_B)



From Prof. Kristi Anseth, University of Colorado.

Figure WE15-1.1 Damaged cartilage. (*Newsweek*, September 3, 2001.) Photo of “damaged cartilage” originally appeared in “21 Million Americans Suffer from Arthritis,” by Jerry Adler.

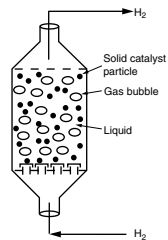
2. 15.3.3 Effectiveness Factor for Nonisothermal First-Order Catalytic Reactions
3. Problem 15-12_C Diffusion and Reaction in Spherical Catalyst Pellets
4. Additional Homework Problems

- **Learning Resources**

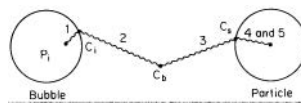
1. Summary Notes
2. Solved Problems

- **Professional Reference Shelf**

R15.1. Slurry Reactors



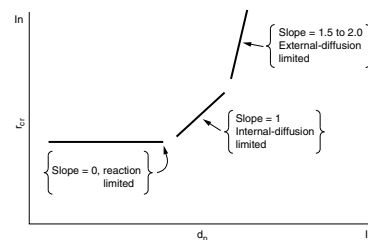
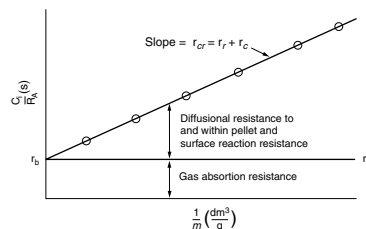
Transport Steps and Resistances



- A. Description of the Use of Slurry Reactors
 - Example R15-1 Industrial Slurry Reactor
- B. Reaction and Transport Steps in a Slurry Reactor

$$\frac{C_i}{R_A} = \frac{1}{k_b a_b} + \frac{1}{m} \left(\frac{r_c}{k_b a_b} + \frac{r_r}{k\eta} \right)$$

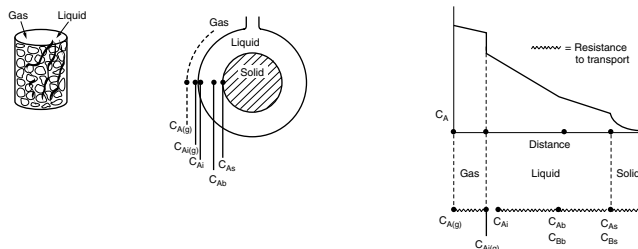
- C. Determining the Rate-Limiting Step
 1. Effect of Loading, Particle Size, and Gas Adsorption
 2. Effect of Shear
 - Example R15-2 Determining the Controlling Resistance
- D. Slurry Reactor Design



Example R15-3 Slurry Reactor Design

R15.2. *Trickle Bed Reactors*

A. Fundamentals



B. Limiting Situations

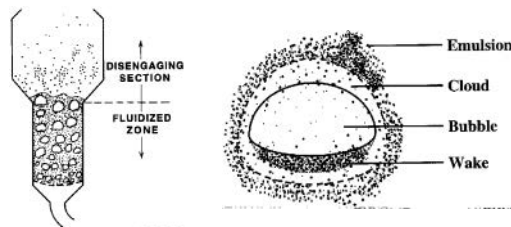
C. Evaluating the Transport Coefficients

$$-r'_A = \frac{1/H}{\underbrace{\frac{(1-\phi)\rho_c}{Hk_g a_i} + \frac{(1-\phi)\rho_c}{k_l a_i} + \frac{1}{k_c a_p} + \frac{1}{\eta k C_{Bs}}}} C_A(g) \quad \frac{\text{mol}}{\text{g-cat} \cdot \text{s}}$$

k_{vg}

R15.3. *Fluidized Bed Reactors*

A. Descriptive Behavior of the Kunii-Levenspiel Bubbling Bed Model



B. Mechanics of Fluidized Beds

Example R15-4 Maximum Solids Hold-Up

C. Mass Transfer in Fluidized Beds

D. Reaction in a Fluidized Bed

E. Solution to the Balance Equations for a First-Order Reaction

$$W = \frac{\rho_c A_c u_b (1 - \epsilon_{mf})(1 - \delta)}{k_{cat} K_R} \ln \frac{1}{1 - X}$$

$$K_R = \gamma_b + \frac{1}{\frac{k_{cat}}{K_{bc}} + \frac{1}{\gamma_c + \frac{1}{\frac{1}{\gamma_e} + \frac{k_{cat}}{K_{ce}}}}}$$

Example R15-5 Catalytic Oxidation of Ammonia

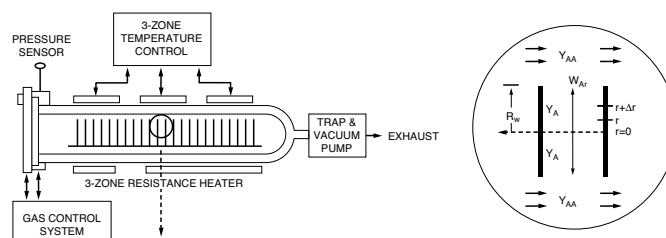
F. Limiting Situations

Example R15-6 Calculation of the Resistances

Example R15-7 Effect of Particle Size on Catalyst Weight for a Slow Reaction

Example R15-8 Effect of Catalyst Weight for a Rapid Reaction

R15.4. Chemical Vapor Deposition Reactors



A. Chemical Reaction Engineering in Microelectronic Processing

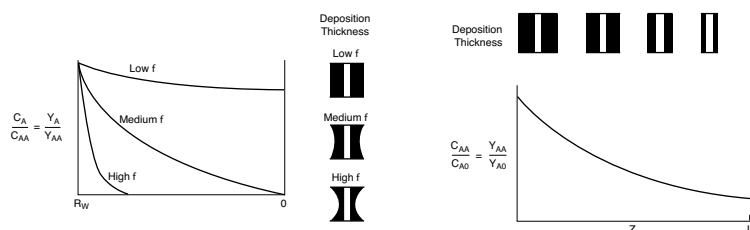
B. Fundamentals of CVD

C. Effectiveness Factors for Boat Reactors

$$\eta = \frac{2I_1(\phi_1)}{\phi_1 I_0(\phi_1)}$$

Example R15-9 Diffusion Between Wafers

Example R15-10 CVD Boat Reactor



QUESTIONS AND PROBLEMS

The subscript to each of the problem numbers indicates the level of difficulty: A, least difficult; D, most difficult.

A = ● B = ■ C = ◆ D = ◆◆

Questions

Q15-1_C Make up an original problem using the concepts presented in Section _____ (your instructor will specify the section). Extra credit will be given if you obtain and use real data from the literature. (See Problem P5-1_A for the guidelines.)

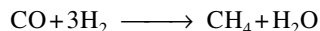
Q15-2 Suppose someone had used the false kinetics (i.e., wrong E , wrong n)? Would the catalyst weight be oversized or undersized? What are other positive or negative effects that occur?

Problems

- P15-1_B** (a) **Example 15-1. Effective Diffusivity.** Make a sketch of a diffusion path for which the tortuosity is 5. How would your effective gas-phase diffusivity change if the absolute pressure were tripled and the temperature were increased by 50%?
- (b) **Example 15-2.** If possible, determine the percent of the total resistance for (1) internal diffusion and (2) for reaction rate for each of the three particles studied. Apply the Weisz-Prater criteria to a particle 0.005 m in diameter.
- (c) **Example 15-3. Overall Effectiveness Factor.** (1) Calculate the percent of the total resistance for the resistance of external diffusion, internal diffusion, and surface reaction. Qualitatively, how would each of your percentages change (2) If the temperature were increased significantly? (3) If the gas velocity were tripled? (4) If the particle size were decreased by a factor of 2? How would the reactor length change in each case? (5) What length would be required to achieve 99.99% conversion of the pollutant NO?

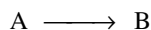
What if...

- (d) you applied the Mears and Weisz-Prater criteria to Examples 15-4 and 15-3? What would you find? What would you learn if $\Delta H_{\text{Rx}} = -25 \text{ kcal/mol}$, $h = 100 \text{ Btu/h} \cdot \text{ft}^2 \cdot ^\circ\text{F}$, and $E = 20 \text{ kcal/mol}$?
- (e) your internal surface area decreased with time because of sintering (see Section 10.7). Describe how your effectiveness factor would change and the rate of reaction change with time if $k_d = 0.01 \text{ h}^{-1}$ and $\eta = 0.01$ at $t = 0$? Explain, being as quantitative as possible when you can.
- (f) you were to assume that the resistance to gas absorption in the CRE Web site *Professional Reference Shelf R15.1* was the same as in *Professional Reference Shelf R15.3* and that the liquid-phase reactor volume in *Professional Reference Shelf R15.3* was 50% of the total? Could you determine the limiting resistance? If so, what is it? What other things could you calculate in *Professional Reference Shelf R15.1* (e.g., selectivity, conversion, molar flow rates in and out)? *Hint:* Some of the other reactions that occur include



- (g) the temperature in the CRE Web site Example R15.2 were increased? How would the relative resistances in the slurry reactor change?
- (h) you were asked for all the things that could go wrong in the operation of a slurry reactor? What would you say?

P15-2_B *Concept problem:* The catalytic reaction



takes place within a fixed bed containing spherical porous catalyst X22. Figure P15-2_B shows the overall rates of reaction at a point in the reactor as a function of temperature for various entering total molar flow rates, F_{T0} .

- (a) Is the reaction limited by external diffusion?
- (b) If your answer to part (a) was “yes,” under what conditions of those shown (i.e., T , F_{T0}) is the reaction limited by external diffusion?
- (c) Is the reaction “reaction-rate-limited”?
- (d) If your answer to part (c) was “yes,” under what conditions of those shown (i.e., T , F_{T0}) is the reaction limited by the rate of the surface reactions?
- (e) Is the reaction limited by internal diffusion?
- (f) If your answer to part (e) was “yes,” under what conditions of those shown (i.e., T , F_{T0}) is the reaction limited by the rate of internal diffusion?
- (g) For a flow rate of 10 g mol/h, determine (if possible) the overall effectiveness factor, Ω , at 360 K.
- (h) Estimate (if possible) the internal effectiveness factor, η , at 367 K. (**Ans.:** $\eta = 0.86$)
- (i) If the concentration at the external catalyst surface is 0.01 mol/dm³, calculate (if possible) the concentration at $r = R/2$ inside the porous catalyst at 367 K. (Assume a first-order reaction.)

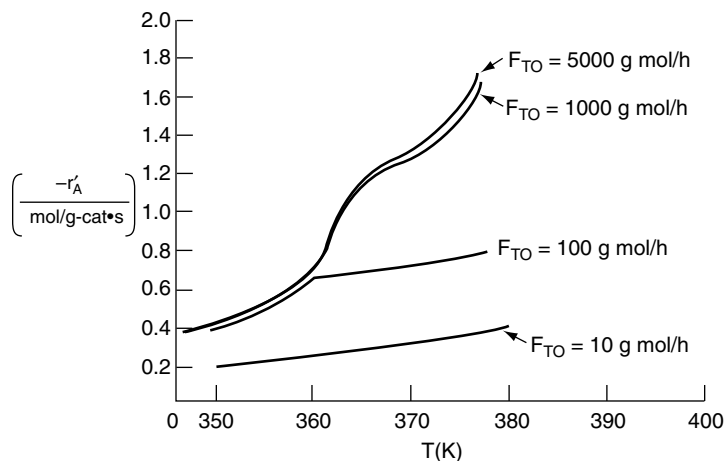


Figure P15-2B Reaction rates in a catalyst bed.

Additional information:

Gas properties:

Diffusivity: $0.1 \text{ cm}^2/\text{s}$

Density: 0.001 g/cm^3

Viscosity: $0.0001 \text{ g/cm}\cdot\text{s}$

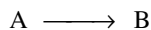
Bed properties:

Tortuosity of pellet: 1.414

Bed permeability: 1 millidarcy

Porosity = 0.3

P15-3B *Concept problem:* The reaction



is carried out in a differential packed-bed reactor at different temperatures, flow rates, and particle sizes. The results shown in Figure P15-3B were obtained.

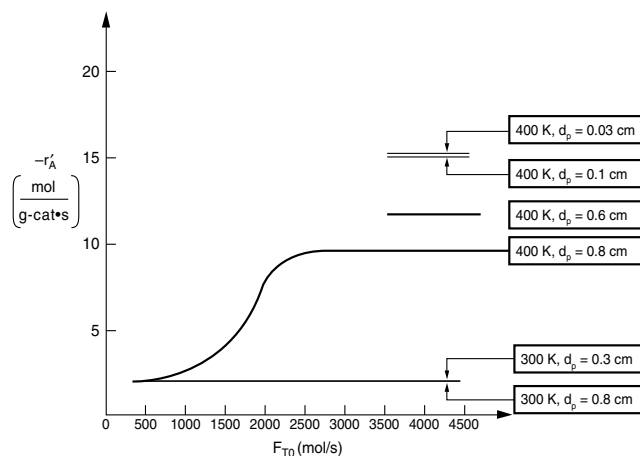


Figure P15-3B Reaction rates in a catalyst bed.

- What regions (i.e., conditions d_p , T , F_{T0}) are external mass transfer-limited?
- What regions are reaction rate-limited?
- What region is internal-diffusion-controlled?
- What is the internal effectiveness factor at $T = 400 \text{ K}$ and $d_p = 0.8 \text{ cm}$? (*Ans.:* $\eta = 0.625$)



- P15-4_A** *Concept problem:* Curves A, B, and C in Figure P15-4_A show the variations in reaction rate for three different reactions catalyzed by solid catalyst pellets. What can you say about each reaction?

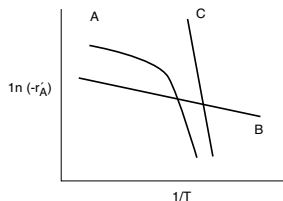


Figure P15-4_A Temperature dependence of three reactions.

- P15-5_B** A first-order heterogeneous irreversible reaction is taking place within a spherical catalyst pellet that is plated with platinum throughout the pellet (see Figure 15-3). The reactant concentration halfway between the external surface and the center of the pellet (i.e., $r = R/2$) is equal to one-tenth the concentration of the pellet's external surface. The concentration at the external surface is 0.001 g mol/dm^3 , the diameter ($2R$) is $2 \times 10^{-3} \text{ cm}$, and the diffusion coefficient is $0.1 \text{ cm}^2/\text{s}$.



- What is the concentration of reactant at a distance of $3 \times 10^{-4} \text{ cm}$ in from the external pellet surface? (*Ans.:* $C_A = 2.36 \times 10^{-4} \text{ mol/dm}^3$.)
- To what diameter should the pellet be reduced if the effectiveness factor is to be 0.8? (*Ans.:* $d_p = 6.8 \times 10^{-4} \text{ cm}$. Critique this answer!)
- If the catalyst support were not yet plated with platinum, how would you suggest that the catalyst support be plated *after* it had been reduced by grinding?

- P15-6_B** The swimming rate of a small organism [*J. Theoret. Biol.*, 26, 11 (1970)] is related to the energy released by the hydrolysis of adenosine triphosphate (ATP) to adenosine diphosphate (ADP). The rate of hydrolysis is equal to the rate of diffusion of ATP from the midpiece to the tail (see Figure P15-6_B). The diffusion coefficient of ATP in the midpiece and tail is $3.6 \times 10^{-6} \text{ cm}^2/\text{s}$. ADP is converted to ATP in the midsection, where its concentration is $4.36 \times 10^{-5} \text{ mol/cm}^3$. The cross-sectional area of the tail is $3 \times 10^{-10} \text{ cm}^2$.

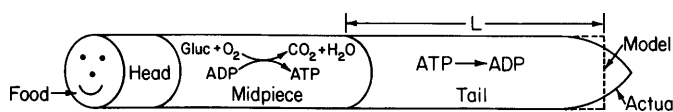


Figure P15-6_B Swimming of an organism.

Application
Pending
for Problem
Hall of
Fame

- Derive an equation for diffusion and reaction in the tail.
 - Derive an equation for the effectiveness factor in the tail.
 - Taking the reaction in the tail to be of zero order, calculate the length of the tail. The rate of reaction in the tail is $23 \times 10^{-18} \text{ mol/s}$.
 - Compare your answer with the average tail length of $41 \text{ }\mu\text{m}$. What are possible sources of error?
- P15-7_B** A first-order, heterogeneous, irreversible reaction is taking place within a catalyst pore that is plated with platinum entirely along the length of the pore (Figure P15-7_B). The reactant concentration at the plane of symmetry (i.e., equal distance from the pore mouth) of the pore is equal to one-tenth the concentration at the pore mouth. The concentration at the pore mouth is 0.001 mol/dm^3 , the pore length ($2L$) is $2 \times 10^{-3} \text{ cm}$, and the diffusion coefficient is $0.1 \text{ cm}^2/\text{s}$.
- Derive an equation for the effectiveness factor.
 - What is the concentration of reactant at $L/2$?

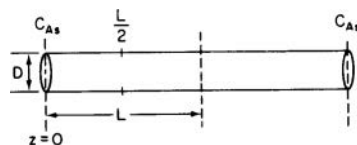
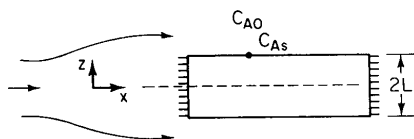


Figure P15-7B Single catalyst pore.

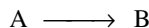
- (c) To what length should the pore length be reduced if the effectiveness factor is to be 0.8?
 (d) If the catalyst support were not yet plated with platinum, how would you suggest the catalyst support be plated *after* the pore length, L , had been reduced by grinding?

P15-8_A A first-order reaction is taking place inside a porous catalyst. Assume dilute concentrations and neglect any variations in the axial (x) direction.

- (a) Derive an equation for both the internal and overall effectiveness factors for the rectangular porous slab shown in Figure P15-8_A.
 (b) Repeat part (a) for a cylindrical catalyst pellet where the reactants diffuse inward in the radial direction. (**C-level** problem, i.e., P15-8_C(b).)

Figure P15-8_A Flow over porous catalyst slab.

P15-9_B The irreversible reaction



is taking place in the same porous catalyst slab shown in Figure P15-8_A. The reaction is zero order in A.

- (a) Show that the concentration profile using the symmetry B.C. is

$$\frac{C_A}{C_{A,s}} = 1 + \phi_0^2 \left[\left(\frac{z}{L} \right)^2 - 1 \right] \quad (\text{P15-9.1})$$

where

$$\phi_0^2 = \frac{kL^2}{2D_c C_{A,s}} \quad (\text{P15-9.2})$$

- (b) For a Thiele modulus of 1.0, at what point in the slab is the concentration zero? For $\phi_0 = 4$?
 (c) What is the concentration you calculate at $z = 0.1 L$ and $\phi_0 = 10$ using Equation (P15-9.1)? What do you conclude about using this equation?
 (d) Plot the dimensionless concentration profile $\psi = C_A/C_{A,s}$ as a function of $\lambda = z/L$ for $\phi_0 = 0.5, 1, 5$, and 10. *Hint:* there are regions where the concentration is zero. Show that $\lambda_c = (1 - 1/\phi_0)$ is the start of this region where the gradient and concentration are both zero. [L. K. Jang, R. L. York, J. Chin, and L. R. Hile, *Inst. Chem. Engr.*, 34, 319 (2003).]
 Show that $\psi = \phi_0^2 \lambda^2 - 2\phi_0(\phi_0 - 1)\lambda + (\phi_0 - 1)^2$ for $\lambda_c \leq \lambda < 1$.
 (e) The effectiveness factor can be written as

$$\eta = \frac{\int_0^L -r_A A_c dz}{-r_{A,s} A_c L} = \frac{\int_0^{z_c} -r_A A_c dz + \int_{z_c}^L -r_A A_c dz}{-r_{A,s} A_c L} \quad (\text{P15-9.3})$$

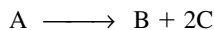


where z_c (λ_c) is the point at which both the concentration gradients and flux go to zero, and A_c is the cross-sectional area of the slab. Show for a zero-order reaction that

$$\eta = \begin{cases} 1 & \text{for } \phi_0 \leq 1.0 \\ 1 - \lambda_c = \frac{1}{\phi_0} & \text{for } \phi_0 \geq 1 \end{cases} \quad (\text{P15-9.4})$$

- (f) Make a sketch for η versus ϕ_0 similar to the one shown in Figure 15-5.
 (g) Repeat parts (a) to (f) for a spherical catalyst pellet.
 (h) What do you believe to be the point of this problem?

P15-10_C The second-order decomposition reaction



is carried out in a tubular reactor packed with catalyst pellets 0.4 cm in diameter. The reaction is internal-diffusion-limited. Pure A enters the reactor at a superficial velocity of 3 m/s, a temperature of 250°C, and a pressure of 500 kPa. Experiments carried out on smaller pellets where surface reaction is limiting yielded a specific reaction rate of 0.05 m⁶/mol·g-cat·s. Calculate the length of bed necessary to achieve 80% conversion. Critique the numerical answer. (**Ans.:** $L = 2.8 \times 10^{-5}$ m)

Additional information:

Effective diffusivity: 2.66×10^{-8} m²/s

Pellet density: 2×10^6 g/m³

Ineffective diffusivity: 0.00 m²/s

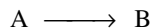
Internal surface area: 400 m²/g

Bed porosity: 0.4

P15-11_C Derive the concentration profile and effectiveness factor for cylindrical pellets 0.2 cm in diameter and 1.5 cm in length. Neglect diffusion through the ends of the pellet.

- (a) Assume that the reaction is a first-order isomerization. *Hint:* Look for a Bessel function.
 (b) Rework Problem P15-10_C for these pellets.

P15-12_B *Extension of Problem P15-7_B.* The elementary isomerization reaction



is taking place on the walls of a cylindrical catalyst pore (see Figure P15-7_B.) In one run, a catalyst poison P entered the reactor together with the reactant A. To estimate the effect of poisoning, we assume that the poison renders the catalyst pore walls near the pore mouth ineffective up to a distance z_1 , so that no reaction takes place on the walls in this entry region.

- (a) Show that before poisoning of the pore occurred, the effectiveness factor was given by

$$\eta = \frac{1}{\phi} \tanh \phi$$

where

$$\phi = L \sqrt{\frac{2k}{rD_e}}$$

with k = reaction-rate constant (length/time)

r = pore radius (length)

D_e = effective molecular diffusivity (area/time)

- (b) Derive an expression for the concentration profile and also for the molar flux of A in the ineffective region, $0 < z < z_1$, in terms of z_1 , D_{AB} , C_{A1} , and C_{As} . Without solving any further differential equations, obtain the new effectiveness factor η' for the poisoned pore.

P15-13_B *Falsified Kinetics*. The irreversible gas-phase dimerization

is carried out at 8.2 atm in a stirred contained-solids reactor to which only pure A is fed. There are 40 g of catalyst in each of the four spinning baskets. The following runs were carried out at 227°C:

Total Molar Feed Rate, F_{T0} (g mol/min)	1	2	4	6	11	20
Mole Fraction A in Exit, y_A	0.21	0.33	0.40	0.57	0.70	0.81

The following experiment was carried out at 237°C:

$$F_{T0} = 9 \text{ g mol/min} \quad y_A = 0.097$$

- What are the apparent reaction order and the apparent activation energy?
- Determine the true reaction order, specific reaction rate, and activation energy.
- Calculate the Thiele modulus and effectiveness factor.
- What pellet diameter should be used to make the catalyst more effective?
- Calculate the rate of reaction on a rotating disk made of the catalytic material when the gas-phase reactant concentration is 0.01 g mol/L and the temperature is 227°C. The disk is flat, nonporous, and 5 cm in diameter.

Additional information:

Effective diffusivity: 0.23 cm²/s

Radius of catalyst pellets: 1 cm

Surface area of porous catalyst: 49 m²/g-cat

Color of pellets: blushing peach

Density of catalyst pellets: 2.3 g/cm³

P15-14_B Derive Equation (15-39). *Hint:* Multiply both sides of Equation (15-25) for n th order reaction; that is,

$$\frac{d^2 y}{d\lambda^2} - \phi_n^2 y^n = 0$$

by $2dy/d\lambda$, rearrange to get

$$\frac{d}{d\lambda} \left(\frac{dy}{d\lambda} \right)^2 = \phi_n^2 y^n 2 \frac{dy}{d\lambda}$$

and solve using the boundary conditions $dy/d\lambda = 0$ at $\lambda = 0$.

P15-15_B *Applications of Diffusion and Reaction to Tissue Engineering*. The equations describing diffusion and reaction in porous catalysts also can be used to derive rates of tissue growth and have been studied by Professor Kristi Anseth and her students at the University of Colorado. One important area of tissue growth is in cartilage tissue in joints such as the knee. Over 200,000 patients per year receive knee joint replacements. Alternative strategies include the growth of cartilage to repair the damaged knee.

One approach is to deliver cartilage-forming cells in a hydrogel to the damaged area such as the one shown in Figure WP15-1.1 on the CRE Web site.

Here, the patient's own cells are obtained from a biopsy and embedded in a hydrogel, which is a cross-linked polymer network that is swollen in water. In order for the cells to survive and grow new tissue, many properties of the gel must be tuned to allow diffusion of important species in and out (e.g., nutrients *in* and cell-secreted extracellular molecules such as collagen *out*). Because there is no blood flow through the cartilage, oxygen transport to the cartilage cells is primarily by diffusion. Consequently, the design must be such that the gel can maintain the necessary rates of diffusion of

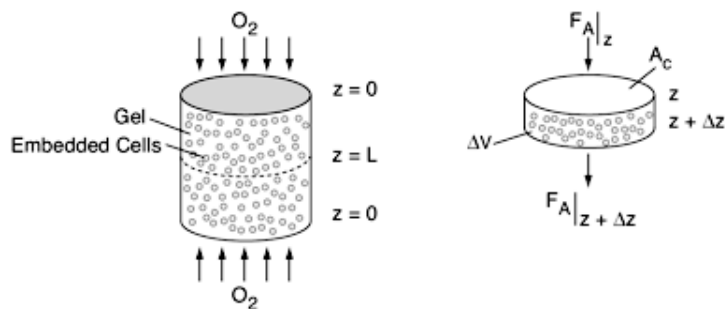


Figure P15-15_B Schematic of cartilage cell system.

nutrients (e.g., O_2) into the hydrogel. These rates of exchange in the gel depend on the geometry and the thickness of the gel. To illustrate the application of chemical reaction engineering principles to tissue engineering, we will examine the diffusion and consumption of one of the nutrients, oxygen.

Our examination of diffusion and reaction in catalyst pellets showed that in many cases the reactant concentration near the center of the particle was virtually zero. If this condition were to occur in a hydrogel, the cells at the center would die. Consequently, the gel thickness needs to be designed to allow rapid transport of oxygen.

Let's consider the simple gel geometry shown in Figure P15-15_B. We want to find the gel

thickness at which the minimum oxygen consumption rate is 10^{-13} mol/cell/h $\left(k = \frac{10^{-3} \text{ mol } O_2}{\text{dm}^3 \text{ h}} \right)$.

The cell density in the gel is 10^{10} cells/dm³, the bulk concentration of oxygen C_{A0} ($z = 0$) is 2×10^{-4} mol/dm³, and the diffusivity, D_{AB} , is 10^{-5} cm²/s.

- (a) Show that the dimensionless form of concentration and length, $\psi = C_A/C_{A0}$, and $\lambda = z/L$, differential mole balance on O_2 gives

$$\frac{d^2\psi}{d\lambda^2} - \frac{kL^2}{D_{AB}C_{A0}} = 0$$

- (b) Show the dimensionless O_2 concentration profile in the gel is

$$\psi = \frac{C_A}{C_{A0}} = \phi_0 \lambda (\lambda - 2) + 1$$

where

$$\lambda = z/L$$

$$\phi_0 = \left(\frac{k}{2D_{AB}C_{A0}} \right) L^2$$

- (c) Solve the gel thickness when the concentration at $z = 0$ and $C_A = 0.1$ mmole/dm³.
 (d) How would your answers change if the reaction kinetics were (1) first order in the O_2 concentration with $k_1 = 10^{-2}$ h⁻¹?

- (e) Carry out a quasi-steady-state analysis using Equation (E15-1.19) along with the overall balance

$$\frac{dN_w}{dt} = v_c W_{O_2} \Big|_{z=0} A_c$$

to predict the O_2 flux and collagen build-up as a function of time.

- (f) Sketch ψ versus λ at different times.
 (g) Sketch λ_c as a function of time. *Hint:* $V = A_c L$. Assume $\alpha = 10$ and the stoichiometric coefficient for oxygen to collagen, v_c , is 0.05 mass fraction of cell/mol O_2 . $A_c = 2 \text{ cm}^2$.



JOURNAL CRITIQUE PROBLEMS

- P15C-1** Use the Weisz–Prater criterion to determine if the reaction discussed in *AIChE J.*, 10, 568 (1964) is diffusion-rate-limited.
P15C-2 Use the references given in *Ind. Eng. Chem. Prod. Res. Dev.*, 14, 226 (1975) to define the iodine value, saponification number, acid number, and experimental setup. Use the slurry reactor analysis to evaluate the effects of mass transfer and determine if there are any mass transfer limitations.

SUPPLEMENTARY READING

1. There are a number of books that discuss internal diffusion in catalyst pellets; however, one of the first books that should be consulted on this and other topics on heterogeneous catalysis is

LAPIDUS, L., AND N. R. AMUNDSON, *Chemical Reactor Theory: A Review*, Upper Saddle River, N.J.: Prentice Hall, 1977.

In addition, see

ARIS, R., *Elementary Chemical Reactor Analysis*. Upper Saddle River, NJ: Prentice Hall, 1989, Chap. 6. Old, but one should find the references listed at the end of this reading particularly useful.

FOGLER, JOSEPH J., AKA JOFO, *A Chemical Reaction Engineers Guide to the Country of Jofostan*. To be self published, hopefully by 2020.

LUSS, D., “Diffusion—Reaction Interactions in Catalyst Pellets,” p. 239 in *Chemical Reaction and Reactor Engineering*. New York: Marcel Dekker, 1987.

The effects of mass transfer on reactor performance are also discussed in

COLLINS, FRANK C., AND GEORGE E. KIMBALL, “Diffusion Controlled Reaction Rates,” *Journal of Colloid Science*, Vol. 4, Issue 4, August 1949, Pages 425-437.

SATTERFIELD, C. N., *Heterogeneous Catalysis in Industrial Practice*, 2nd ed. New York: McGraw-Hill, 1991.

2. Diffusion with homogeneous reaction is discussed in

ASTARITA, G., and R. OCONE, *Special Topics in Transport Phenomena*. New York: Elsevier, 2002.

Gas-liquid reactor design is also discussed in

BUTT, JOHN B., *Reaction Kinetics and Reactor Design*, 2nd ed. Boca Raton, FL: CRC Press, 2000.

SHAH, Y. T., *Gas–Liquid–Solid Reactor Design*. New York: McGraw-Hill, 1979.

3. Modeling of CVD reactors is discussed in

DOBKIN, DANIEL, AND M. K. ZUKRAW, *Principles of Chemical Vapor Deposition*. The Netherlands: Kluwer Academic Publishers, 2003.

HESS, D. W., K. F. JENSEN, and T. J. ANDERSON, "Chemical Vapor Deposition: A Chemical Engineering Perspective," *Rev. Chem. Eng.*, 3, 97, 1985.

4. Multiphase reactors are discussed in

RAMACHANDRAN, P. A., and R. V. CHAUDHARI, *Three-Phase Catalytic Reactors*. New York: Gordon and Breach, 1983.

RODRIGUES, A. E., J. M. COLO, and N. H. SWEED, eds., *Multiphase Reactors*, Vol. 1: *Fundamentals*. Alphen aan den Rijn, The Netherlands: Sitjhoff and Noordhoff, 1981.

RODRIGUES, A. E., J. M. COLO, and N. H. SWEED, eds., *Multiphase Reactors*, Vol. 2: *Design Methods*. Alphen aan den Rijn, The Netherlands: Sitjhoff and Noordhoff, 1981.

SHAH, Y. T., B. G. KELKAR, S. P. GODBOLE, and W. D. DECKWER, "Design Parameters Estimations for Bubble Column Reactors" (journal review), *AIChE J.*, 28, 353 (1982).

The following *Advances in Chemistry Series* volume discusses a number of multiphase reactors:

FOGLER, H. S., ed., *Chemical Reactors*, ACS Symp. Ser. 168. Washington, DC: American Chemical Society, 1981, pp. 3–255.

5. Fluidization

KUNII, DAIZO, AND OCTAVE LEVENSPIEL, *Fluidization Engineering*, 2nd ed. (Butterworths Series in Chemical Engineering Deposition). Stoneham, MA: Butterworth-Heinemann, 1991.

In addition to Kunii and Levenspiel's book, many correlations can be found in

DAVIDSON, J. F., R. CLIFF, and D. HARRISON, *Fluidization*, 2nd ed. Orlando: Academic Press, 1985.

YATES, J. G., *Fundamentals of Fluidized-Bed Chemical Processes*, 3rd ed. London: Butterworth, 1983.